

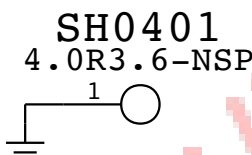
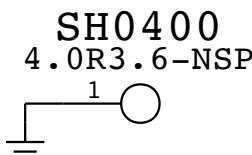
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| DESIGN: X502/MLB_CATZ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  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| LAST CHANGE: Tue Aug 9 17:02:57 2016                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   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| LAST_MODIFICATION=Tue Aug 9 17:03:06 2016                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              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                                                                                                                                                                                                                                                                                  | 6   | CPU MISC/JTAG/CFG/RSVD        | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7   | CPU LPDDR3 Interface          | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 8   | CPU & PCH Power               | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 9   | CPU & PCH Grounds             | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 10  | CPU Core Decoupling           | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 11  | CPU GT Decoupling             |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 12  | PCH Decoupling                |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 13  | PCH Audio/LPC/SPI/SMBus       |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 14  | PCH Power Management          |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 15  | PCH PCIE/USB/CLKS             |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 16  | PCH SPI/UART/GPIO             |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 18  | CPU/PCH Merged XDP            |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 22  | LPDDR3 VREF Margining         |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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                                                                                                                                                                                                                                                                                  | 23  | LPDDR3 DRAM Channel A (00-31) |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 24  | LPDDR3 DRAM Channel A (32-63) |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 25  | LPDDR3 DRAM Channel B (00-31) |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 26  | LPDDR3 DRAM Channel B (32-63) |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 27  | LPDDR3 DRAM Termination       |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 26                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 28  | USB-C HIGH SPEED 1            |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 27                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 29  | USB-C HIGH SPEED 2            |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 28                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 30  | USB-C SUPPORT                 |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 29                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 31  | USB-C PORT CONTROLLER A       |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 30                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 32  | USB-C PORT CONTROLLER B       |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 31                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 33  | USB-C CONNECTOR A             |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 32                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 34  | USB-C CONNECTOR B             |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 33                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 35  | USB-C SUPPORT 2               |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 34                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 37  | WIFI/BT MODULE                |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 41                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 52  | SMC Project Support           | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 53  | SMBus Connections             | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 43                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 54  | Power Sensors High Side       | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 44                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 55  | Power Sensors Load Side       | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 45                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 56  | Power Sensors Extended        | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 46                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 58  | Thermal Sensors               | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 47                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 61  | SPI ROM & SWD Debug           | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 48                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 62  | HDA BRIDGE                    | AHAAGE         | 03/23/2016 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 49                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 63  | JACK CODEC                    | AHAAGE         | 09/22/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 64  | SPEAKER AMP                   | AHAAGE         | 03/23/2016 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 51                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 66  | JACK TRANSLATORS              |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 52                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 69  | DC-IN & BATTERY CONNECTORS    |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 53                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 70  | PBUS Supply & Battery Charger |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 54                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 71  | VReg CPU VCC Cntl             |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 72  | CPU IMVP VCC & VCCSA          |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 56                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 73  | IMVP VCCSA                    |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 57                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 74  | GT IMVP VCCGT                 |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 58                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 76  | VR - 5V S4, 3.3V S5           |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 59                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 77  | VR - OPC (EDRAM)              |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 60                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 78  | PMIC IC & Power Control       |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 61                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| 62                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| 63                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 81  | RAIL DESENSE CAPS             |                |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 65                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| 66                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 85  | eDP Display Connector         | X362           | 06/23/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 68                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 99  | DEVELOPMENT ONLY              | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 69                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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                                                                                                                                                                                                                                                                                  | 102 | NC_ AND NO_TEST SIGNALS       | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 71                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 103 | Memory Signal Swaps           | X502-EXP       | 12/03/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 72                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 104 | FCT, ICT PROPERTIES           | PAULM          | 06/15/2015 |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| DOCUMENTS / BOARDS / ASSEMBLIES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        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| <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>051-02265</td><td>1</td><td>SCHEM,MLB_CATZ,X502</td><td>SCH</td><td></td><td></td></tr><tr><td>820-00875</td><td>1</td><td>PCBF,MLB_CATZ,X502</td><td>MLB</td><td>CRITICAL</td><td></td></tr><tr><td colspan="6">639-03266 PCBA,MLB_CATZ,XXXXX,X502</td></tr><tr><td>685-00125</td><td>1</td><td>COMMON PARTS,MLB_CATZ,X502</td><td>CBOM</td><td></td><td>CMN_PARTS_BOM</td></tr><tr><td>985-00239</td><td>1</td><td>DEV PARTS,MLB_CATZ,X502</td><td>DEV1</td><td></td><td>DEVELOPMENT_LIST</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                               |                |            |                  |   |  |   |  |   |  | PART NUMBER                               | QTY        | DESCRIPTION             | REFERENCE DES   | CRITICAL | BOM OPTION | 051-02265 | 1                 | SCHEM,MLB_CATZ,X502 | SCH |   |   | 820-00875         | 1     | PCBF,MLB_CATZ,X502 | MLB | CRITICAL |                   | 639-03266 PCBA,MLB_CATZ,XXXXX,X502 |            |   |   |          |       | 685-00125  | 1 | COMMON PARTS,MLB_CATZ,X502 | CBOM    |       | CMN_PARTS_BOM | 985-00239 | 1 | DEV PARTS,MLB_CATZ,X502 | DEV1  |            | DEVELOPMENT_LIST | <table><tr><th colspan="3">DRAWING TITLE</th><th>DRAWING NUMBER</th><th>SIZE</th></tr><tr><td colspan="3">SCHEM,MLB-CATZ,X502</td><td>051-02265</td><td>D</td></tr><tr><td colspan="3" rowspan="4"> Apple Inc.</td><td colspan="2">REVISION</td></tr><tr><td colspan="2">1.0.0</td></tr><tr><td colspan="2">BRANCH</td></tr><tr><td colspan="2">PAGE</td></tr><tr><td colspan="3">NOTICE OF PROPRIETARY PROPERTY:<br/>THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:<br/>I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE<br/>II NOT TO REPRODUCE OR COPY IT<br/>III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART<br/>IV ALL RIGHTS RESERVED</td><td colspan="2">SHEET</td></tr><tr><td colspan="3"></td><td colspan="2">1 OF 73</td></tr></table> |                      |       |            | DRAWING TITLE |   |                 | DRAWING NUMBER | SIZE       | SCHEM,MLB-CATZ,X502 |   |                   | 051-02265 | D          |  Apple Inc. |    |                     | REVISION |            | 1.0.0 |    | BRANCH            |  | PAGE |    | NOTICE OF PROPRIETARY PROPERTY:<br>THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. 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                                                                                                                                                                                                                                                                                  | 1   | SCHEM,MLB_CATZ,X502           | SCH            |            |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 820-00875                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1   | PCBF,MLB_CATZ,X502            | MLB            | CRITICAL   |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 639-03266 PCBA,MLB_CATZ,XXXXX,X502                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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| 685-00125                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1   | COMMON PARTS,MLB_CATZ,X502    | CBOM           |            | CMN_PARTS_BOM    |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| 985-00239                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1   | DEV PARTS,MLB_CATZ,X502       | DEV1           |            | DEVELOPMENT_LIST |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| DRAWING TITLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          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| SCHEM,MLB-CATZ,X502                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                               | 051-02265      | D          |                  |   |  |   |  |   |  |                                           |            |                         |                 |          |            |           |                   |                     |     |   |   |                   |       |                    |     |          |                   |                                    |            |   |   |          |       |            |   |                            |         |       |               |           |   |                         |       |            |                  |                                                                                                    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| NOTICE OF PROPRIETARY PROPERTY:<br>THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING:<br>I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE<br>II NOT TO REPRODUCE OR COPY IT<br>III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART<br>IV ALL RIGHTS RESERVED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| BOM_COST_GROUP=NO COST ITEMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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|                                                                                                                                                                                                                                                                                                                         | 8                | 7                                      | 6                                           | 5                                    | 4                          | 3                      | 2                       | 1 |  |
| D                                                                                                                                                                                                                                                                                                                       | Major ICs        |                                        |                                             |                                      | Programmables (All Builds) |                        |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | CPU              |                                        |                                             |                                      | SMC                        |                        |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | PART NUMBER      | QTY                                    | DESCRIPTION                                 | REFERENCE DES                        | CRITICAL                   | BOM OPTION             |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 998-00235        | 1                                      | IC,CPU,SKL-ULT,2+3E,42X24MM,BGA1356         | U0500                                | CRITICAL                   | CPU_SKL:BASE           |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 998-04195        | 1                                      | INTERPOSER,VTT ADAPTER,SKL-U,BGA1356        | U0500                                | CRITICAL                   | CPU_SKL:VTT_INTERPOSER |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00168        | 1                                      | CPU,SKYU,QJ8N,D0,QS,2/2,2.3,15W,BGA1356     | U0500                                | CRITICAL                   | CPU_SKL:2_2_QS_2.3     |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00170        | 1                                      | CPU,SKYU,QJ8K,D0,QS,2/2,2.6,15W,BGA1356     | U0500                                | CRITICAL                   | CPU_SKL:2_2_QS_2.6     |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00149        | 1                                      | CPU,SKYU,QJ57,J0,ES0,2/3,1.6,15W,BGA1356    | U0500                                | CRITICAL                   | CPU_SKL:2_3_ES0_2GD    |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00150        | 1                                      | CPU,SKYU,QJ58,J0,ES0,2/3,1.6,28W,BGA1356    | U0500                                | CRITICAL                   | CPU_SKL:2_3_ES0_BT     |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00219        | 1                                      | CPU,SKYU,QK2T,K1,SQS,1.8,15W,.95,BGA1356    | U0500                                | CRITICAL                   | CPU_SKL:2_3_SQS_1G8    |                         |   |                                                                                   |
| C                                                                                                                                                                                                                                                                                                                       | 337S00220        | 1                                      | CPU,SKYU,QKBY,K1,SQS,2.2,15W,1.05,BG1356    | U0500                                | CRITICAL                   | CPU_SKL:2_3_SQS_2G2    |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00222        | 1                                      | CPU,SKYU,QK33,K1,SQS,2.0,15W,1.0,BG1356     | U0500                                | CRITICAL                   | CPU_SKL:2_3_SQS_2G0    |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00233        | 1                                      | CPU,SKYU,QK32,K1,SQS,2.4,15W,BGA1356        | U0500                                | CRITICAL                   | CPU_SKL:2_3_SQS_2G4    |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00232        | 1                                      | CPU,SKY,SR2JC,K1,PRQ,1.8,15W,BGA1356        | U0500                                | CRITICAL                   | CPU_SKL:2_3_PRO_1G8    |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00239        | 1                                      | CPU,SKY,SR2JM,K1,PRQ,2.0,15W,BGA1356        | U0500                                | CRITICAL                   | CPU_SKL:2_3_PRO_2G0    |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 337S00234        | 1                                      | CPU,SKYU,SR2JL,K1,PRQ,2.4,15W,BGA1356       | U0500                                | CRITICAL                   | CPU_SKL:2_3_PRO_2G4    |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | ACE              |                                        |                                             |                                      | MICRON                     |                        |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 353S00422        | 2                                      | IC,CD3215,USB PWR SWITCH,A0,6X6MM,BGA96     | U3100,U3200                          | CRITICAL                   | ACE:A0                 |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 353S00660        | 2                                      | IC,CD3215,ACE,A1,USB PWR SWITCH,BGA96       | U3100,U3200                          | CRITICAL                   | ACE:A1                 |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 353S00807        | 2                                      | IC,CD3215,ACE,B0,USB PWR SWITCH,BGA96       | U3100,U3200                          | CRITICAL                   | ACE:B0                 |                         |   |                                                                                   |
| B                                                                                                                                                                                                                                                                                                                       | 353S00887        | 2                                      | IC,CD3215,ACE,B0,USB PWR SW,BLNK,BGA96      | U3100,U3200                          | CRITICAL                   | ACE:B0_B               | (BOOT CODE: 0002.08.07) |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 353S00888        | 2                                      | IC,CD3215,ACE,B0,USB PWR SW,OTP=2,BGA96     | U3100,U3200                          | CRITICAL                   | ACE:B0_2               | (BOOT CODE: 0002.08.07) |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 353S00926        | 2                                      | IC,CD3215,ACE,B03,USB PWR SW,BLNK,BGA96     | U3100,U3200                          | CRITICAL                   | ACE:B0_3               | (BOOT CODE: 0002.08.08) |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 353S00961        | 2                                      | IC,CD3215,ACE,C00,USB PWR SW,BLNK,BGA96     | U3100,U3200                          | CRITICAL                   | ACE:C0                 |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | TBT ALPINE RIDGE |                                        |                                             |                                      |                            |                        |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 338S00160        | 1                                      | IC,TBT,ALPINE RIDGE DP,QSJVV,B1,6X6MM,BGA96 | U2800                                | CRITICAL                   | TBT_AR:B1_QS           |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 998-04160        | 1                                      | IC,TBT,ALP-RIDGE DP,SLL44-TRAY,B1,CSP337    | U2800                                | CRITICAL                   | TBT_AR:B1_PROQ_TRAY    |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 338S00176        | 1                                      | IC,TBT,ALPN-RIDGE DP,SLL43-T&R,B1,CSP337    | U2800                                | CRITICAL                   | TBT_AR:B1_PROQ         |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 338S00229        | 1                                      | IC,TBT,ALPINE RIDGE,QSTY,QS,C0,CSP337       | U2800                                | CRITICAL                   | TBT_AR:C0_QS           |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 338S00249        | 1                                      | IC,TBT,ALPINE RIDGE,QT58,QS,C1,CSP337       | U2800                                | CRITICAL                   | TBT_AR:C1_QS           |                         |   |                                                                                   |
| A                                                                                                                                                                                                                                                                                                                       | 338S00254        | 1                                      | IC,TBT,ALPINE RIDGE,SLLSM,PRQ,C1,CSP337     | U2800                                | CRITICAL                   | TBT_AR:C1_PROQ         |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | WIRELESS MODULE  |                                        |                                             |                                      |                            |                        |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 339S0250         | 1                                      | MODULE,WIFI/BT,STELLA CIDRE,MUR,LGA80       | U3700                                | CRITICAL                   | WIRELESS:MURATA        |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 339S0251         | 1                                      | MODULE,WIFI/BT,STELLA CIDRE,USI,LGA80       | U3700                                | CRITICAL                   | WIRELESS:USI           |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | TBT ROM          |                                        |                                             |                                      |                            |                        |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 335S00024        | 1                                      | IC,FLASH,USON8,512KBIT,75MHZ                | U3770                                | CRITICAL                   | BT_ROM:BLANK           |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 335S00837        | 335S00024                              | ALT_CMN                                     | ALL                                  | ALTERNATE                  |                        |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 341S00196        | 1                                      | IC,BT ROM (V53) DVT,X261                    | U3770                                | CRITICAL                   | BT_ROM:X261            |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 341S00397        | 1                                      | IC,BT ROM (V53) PROTO0,X502                 | U3770                                | CRITICAL                   | BT_ROM:PROTO0          |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         | 341S00397        | 1                                      | IC,BT ROM (V53) PROTO0,X502                 | U3770                                | CRITICAL                   | BT_ROM:PVT             |                         |   |                                                                                   |
| WIFI ROM                                                                                                                                                                                                                                                                                                                |                  |                                        |                                             |                                      |                            |                        |                         |   |                                                                                   |
| 335S0956                                                                                                                                                                                                                                                                                                                | 1                | IC,MEMORY,EEPROM,4K,1.7V-5.5V,UDFN8    | U3780                                       | CRITICAL                             | WIFI-ROM:BLANK             |                        |                         |   |                                                                                   |
| 335S00145                                                                                                                                                                                                                                                                                                               | 335S0956         | ALT_CMN                                | ALL                                         | ALTERNATE                            |                            |                        |                         |   |                                                                                   |
| 341S00607                                                                                                                                                                                                                                                                                                               | 1                | WIFI ROM (P175) PRE-DVT,WW1,X502       | U3780                                       | CRITICAL                             | WIFI-ROM:MURATA-FCC        |                        |                         |   |                                                                                   |
| 341S00608                                                                                                                                                                                                                                                                                                               | 1                | WIFI ROM (P175) PRE-DVT,WW2,X502       | U3780                                       | CRITICAL                             | WIFI-ROM:MURATA-ETSI       |                        |                         |   |                                                                                   |
| 341S00609                                                                                                                                                                                                                                                                                                               | 1                | WIFI ROM (P175) PRE-DVT,WW3,X502       | U3780                                       | CRITICAL                             | WIFI-ROM:MURATA-APAC       |                        |                         |   |                                                                                   |
| 341S00610                                                                                                                                                                                                                                                                                                               | 1                | WIFI ROM (P175) PRE-DVT,IND,X502       | U3780                                       | CRITICAL                             | WIFI-ROM:MURATA-IND        |                        |                         |   |                                                                                   |
| 341S00636                                                                                                                                                                                                                                                                                                               | 1                | WIFI ROM (P177) USI-WW1,X502           | U3780                                       | CRITICAL                             | WIFI-ROM:USI-FCC           |                        |                         |   |                                                                                   |
| 341S00637                                                                                                                                                                                                                                                                                                               | 1                | WIFI ROM (P177) USI-WW2,X502           | U3780                                       | CRITICAL                             | WIFI-ROM:USI-ETSI          |                        |                         |   |                                                                                   |
| 341S00638                                                                                                                                                                                                                                                                                                               | 1                | WIFI ROM (P177) USI-WW3,X502           | U3780                                       | CRITICAL                             | WIFI-ROM:USI-APAC          |                        |                         |   |                                                                                   |
| 341S00639                                                                                                                                                                                                                                                                                                               | 1                | WIFI ROM (P177) USI-IND,X502           | U3780                                       | CRITICAL                             | WIFI-ROM:USI-IND           |                        |                         |   |                                                                                   |
| AR ROM                                                                                                                                                                                                                                                                                                                  |                  |                                        |                                             |                                      |                            |                        |                         |   |                                                                                   |
| 335S00133                                                                                                                                                                                                                                                                                                               | 1                | IC,SPI SERIAL FLASH,8MBITS,3.0V,USON8  | U2890                                       | CRITICAL                             | AR_ROM:BLANK               |                        |                         |   |                                                                                   |
| 341S00451                                                                                                                                                                                                                                                                                                               | 1                | IC,NVM / AR (V0.8.15.E1) PROTO 1,X502  | U2890                                       | CRITICAL                             | AR_ROM:PROTO1              |                        |                         |   |                                                                                   |
| 341S00512                                                                                                                                                                                                                                                                                                               | 1                | IC,NVM (VB1-10.11-E2.6.3) PROTO 2,X502 | U2890                                       | CRITICAL                             | AR_ROM:PROTO2              |                        |                         |   |                                                                                   |
| 341S00559                                                                                                                                                                                                                                                                                                               | 1                | IC,NVM (V16.8) EVT,X502                | U2890                                       | CRITICAL                             | AR_ROM:EVT                 |                        |                         |   |                                                                                   |
| 341S00606                                                                                                                                                                                                                                                                                                               | 1                | IC,NVM (V1.5) PRE-DVT,X502             | U2890                                       | CRITICAL                             | AR_ROM:PREDVT              |                        |                         |   |                                                                                   |
| 341S00628                                                                                                                                                                                                                                                                                                               | 1                | IC,NVM (V3.8),DVT,X502                 | U2890                                       | CRITICAL                             | AR_ROM:DVT                 |                        |                         |   |                                                                                   |
| 341S00661                                                                                                                                                                                                                                                                                                               | 1                | IC,NVM (VTBD),PVT,X502                 | U2890                                       | CRITICAL                             | AR_ROM:PVT                 |                        |                         |   |                                                                                   |
| BOM_COST_GROUP=NO COST ITEMS                                                                                                                                                                                                                                                                                            |                  |                                        |                                             |                                      |                            |                        |                         |   |                                                                                   |
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                                   |                  |                                        |                                             | LAST CHANGE: Thu Aug 4 21:00:42 2016 |                            |                        |                         |   |                                                                                   |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                  |                                        |                                             | BOM Configuration                    |                            |                        |                         |   |                                                                                   |
|  Apple Inc.                                                                                                                                                                                                                        |                  |                                        |                                             | DRAWING NUMBER<br>051-02265          |                            | SIZE<br>D              |                         |   |                                                                                   |
|                                                                                                                                                                                                                                                                                                                         |                  |                                        |                                             | REVISION<br>1.0.0                    |                            | BRANCH                 |                         |   |                                                                                   |
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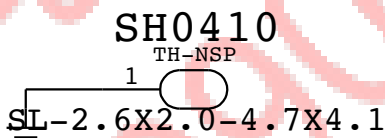
MOUNTING HOLES

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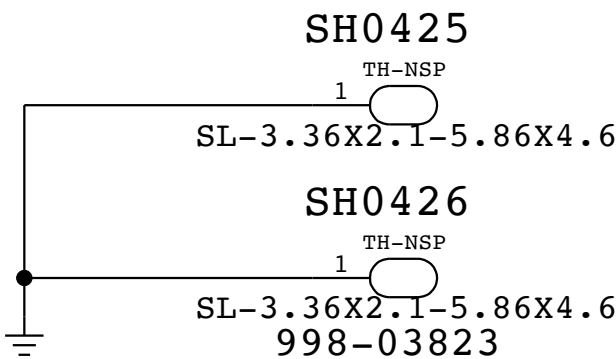
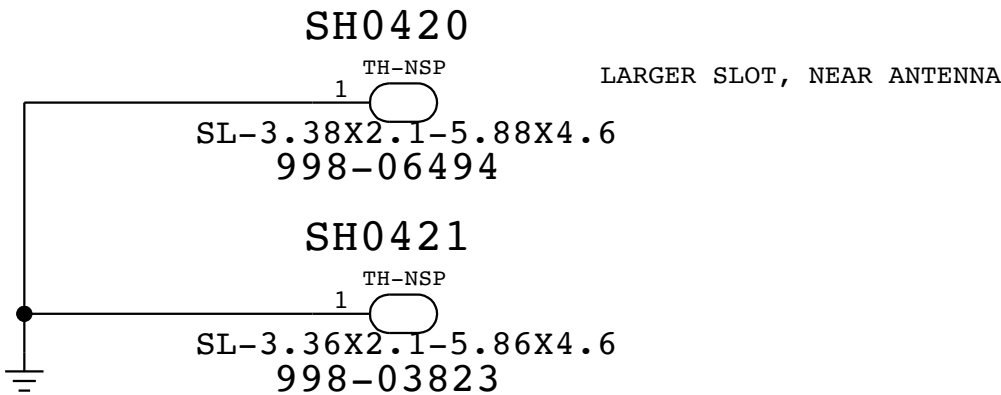


FAN MOUNTING HOLE

998-03850

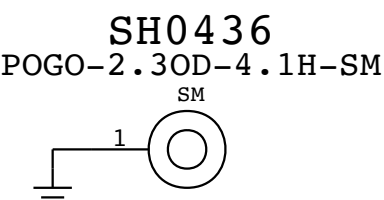
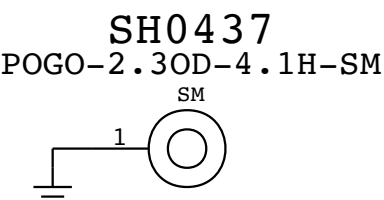
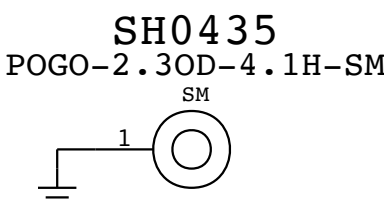
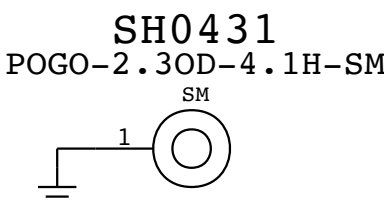
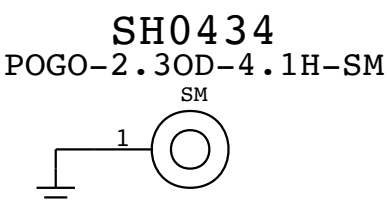
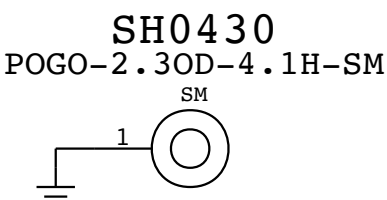


PLATED HOLES

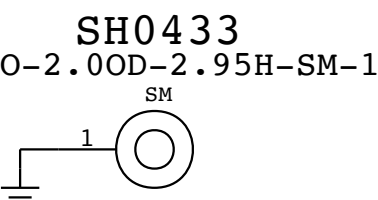
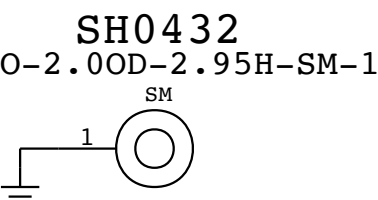


POGO PINS

870-01680

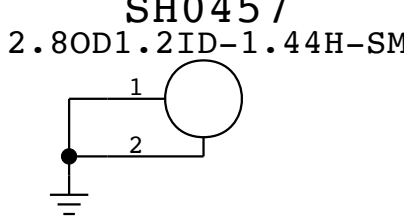
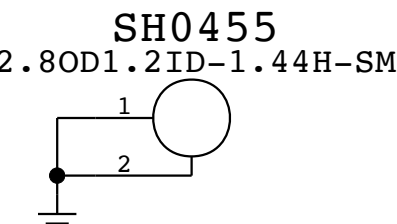
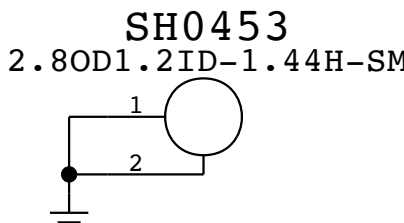
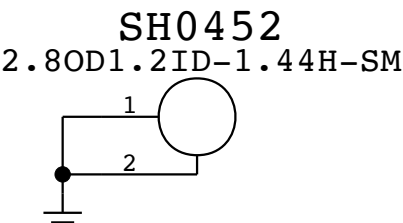
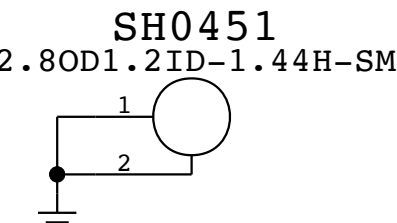
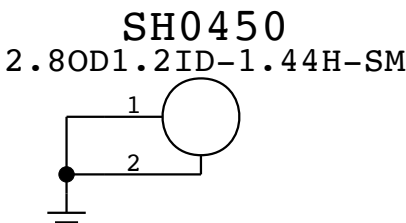


870-02068



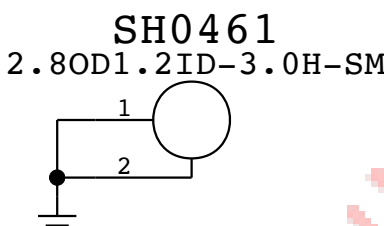
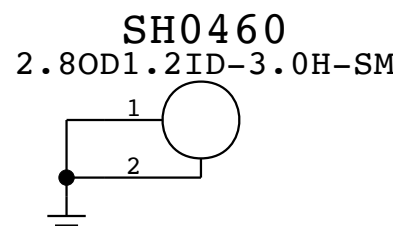
TOP STANDOFFS

860-00385



BOTTOM STANDOFFS

860-00468



SHIELD CANS

MEMORY CAN - TOP

| PART NUMBER | QTY | DESCRIPTION                   | REFERENCE DES | CRITICAL | BOM OPTION            |
|-------------|-----|-------------------------------|---------------|----------|-----------------------|
| 806-07887   | 1   | SHIELD CAN FENCE,DRAM,MN,X520 | SHLD4         | CRITICAL | SHIELD_CAN_MEMORY_TOP |

MEMORY CAN - BOTTOM

| PART NUMBER | QTY | DESCRIPTION                           | REFERENCE DES | CRITICAL | BOM OPTION            |
|-------------|-----|---------------------------------------|---------------|----------|-----------------------|
| 806-08894   | 1   | SHIELD FENCE,BOT DRAM,SUS,PRE-MN,X520 | SHLD1         | CRITICAL | SHIELD_CAN_MEMORY_BOT |

WIRELESS CAN

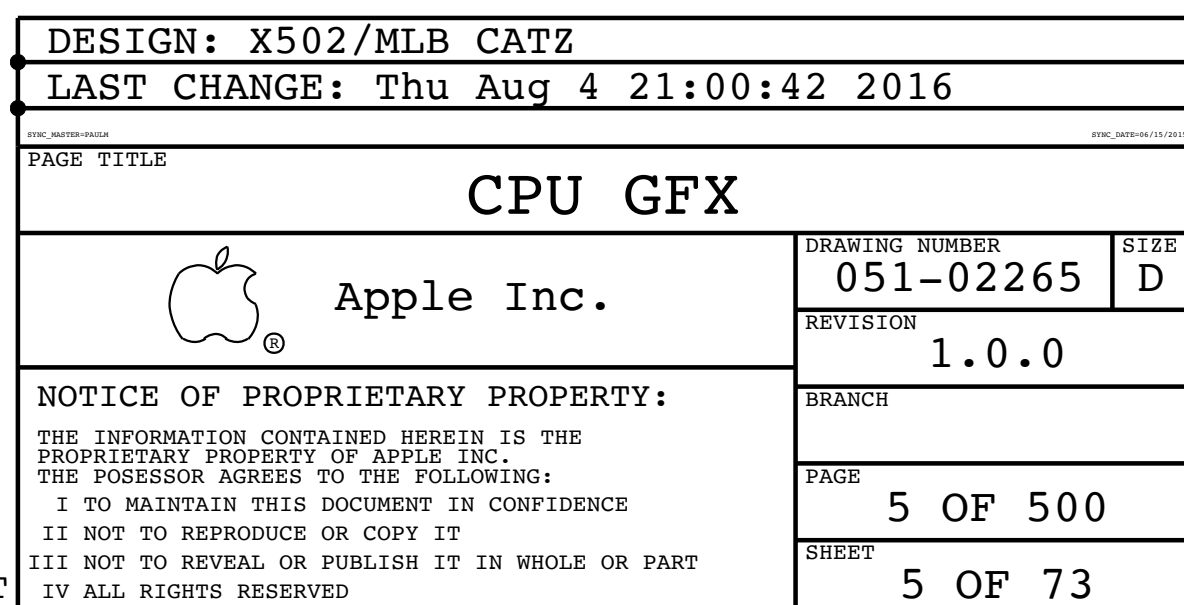
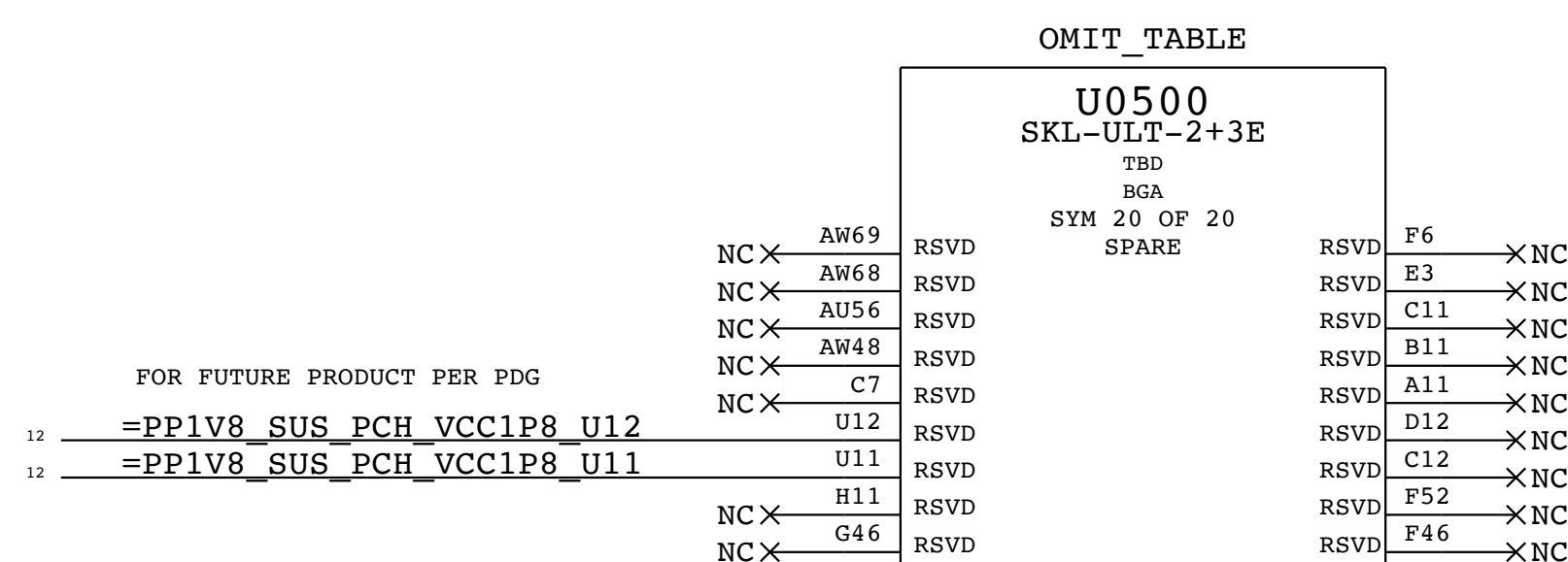
| PART NUMBER | QTY | DESCRIPTION                          | REFERENCE DES | CRITICAL | BOM OPTION      |
|-------------|-----|--------------------------------------|---------------|----------|-----------------|
| 806-07886   | 1   | SHIELD CAN,EMI,WIFI,SYM,MN,TALL,X520 | SHLD2         | CRITICAL | SHIELD_CAN_WIFI |

USB-C CAN

| PART NUMBER | QTY | DESCRIPTION                   | REFERENCE DES | CRITICAL | BOM OPTION      |
|-------------|-----|-------------------------------|---------------|----------|-----------------|
| 806-07885   | 1   | SHIELD CAN,AR,USB-C,THRU,X520 | SHLD3         | CRITICAL | SHIELD_CAN_USBC |

BOM\_COST\_GROUP=MECHANICALS

|                                                                                                                                                                                                                                                                                                                   |                |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                             |                |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                                                                                                                                                                                                                              |                |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |           |
| PD Parts                                                                                                                                                                                                                                                                                                          |                |           |
|  Apple Inc.                                                                                                                                                                                                                  | DRAWING NUMBER | 051-02265 |
|                                                                                                                                                                                                                                                                                                                   | REVISION       | 1.0.0     |
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|                                                                                                                                                                                                                                                                                                                   |                | PAGE      |
|                                                                                                                                                                                                                                                                                                                   |                | SHEET     |
|                                                                                                                                                                                                                                                                                                                   |                | 4 OF 500  |
|                                                                                                                                                                                                                                                                                                                   |                | 4 OF 73   |





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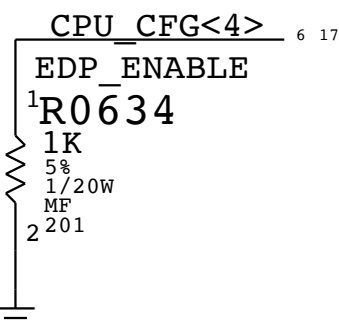
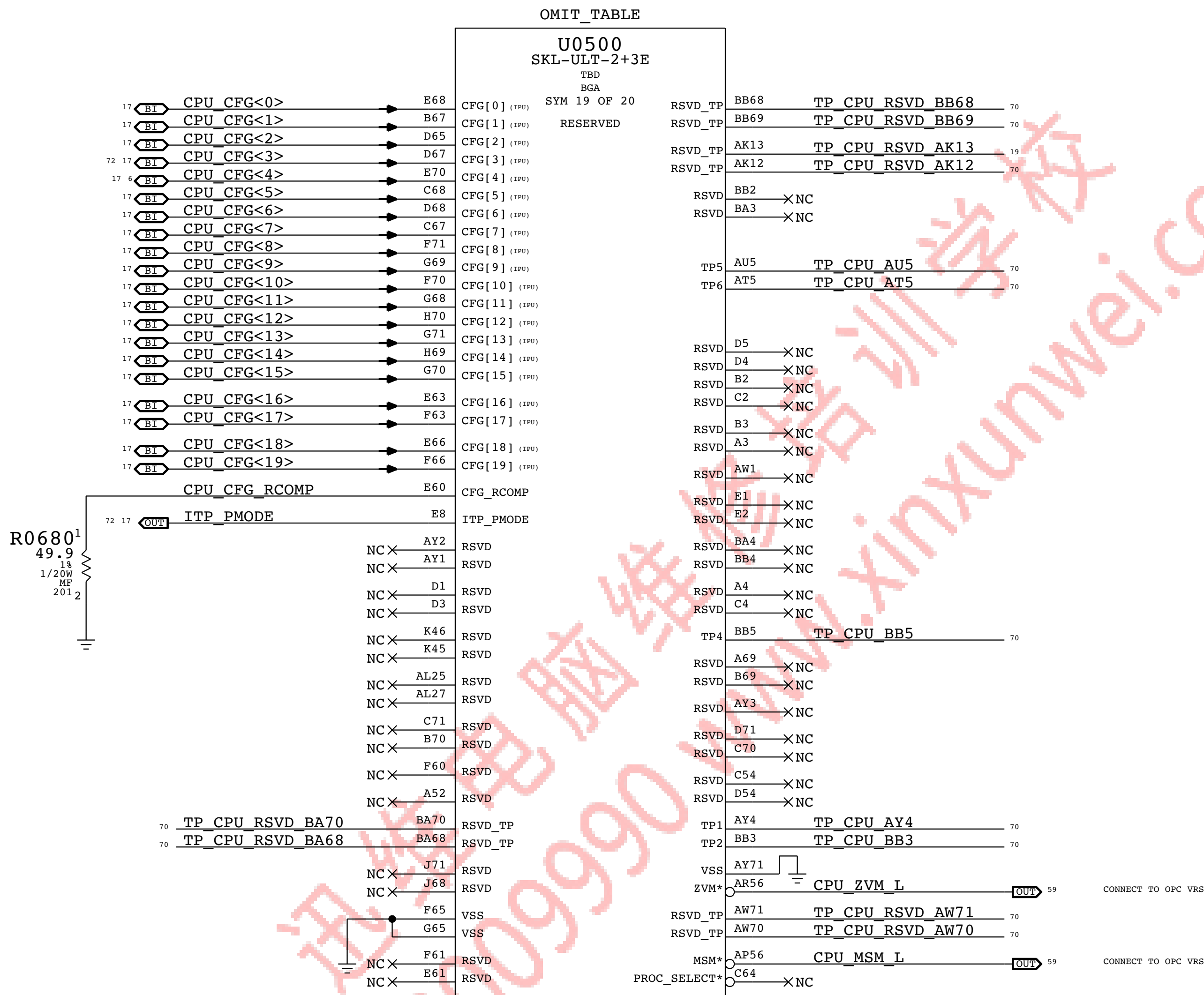
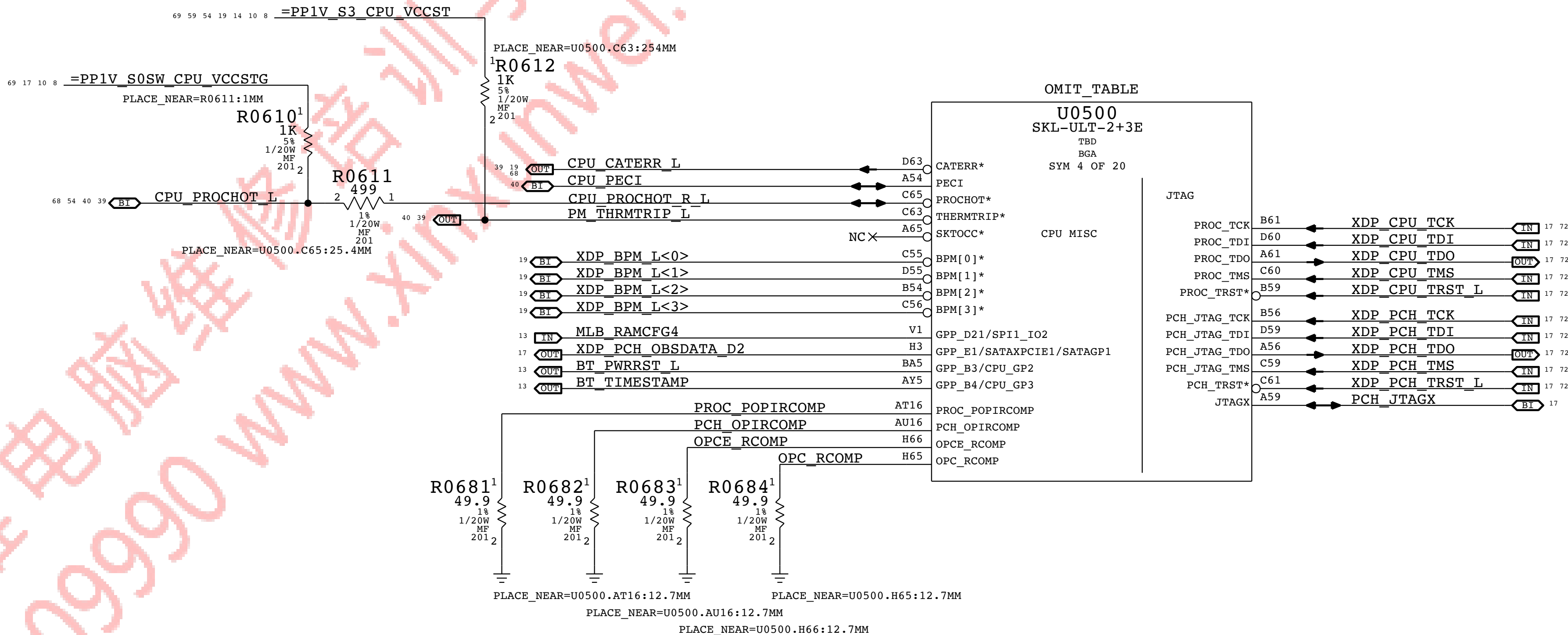
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BOM\_COST\_GROUP=CPU & CHIPSET

|                                                                                                                                                                                                                                                                                                                    |                |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                              |                |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                                                                                                                                                                                                                               |                |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                         |                |           |
| CPU MISC/JTAG/CFG/RSVD                                                                                                                                                                                                                                                                                             |                |           |
|                                                                                                                                                                                                                                                                                                                    | DRAWING NUMBER | 051-02265 |
|                                                                                                                                                                                                                                                                                                                    | REVISION       | 1.0.0     |
|                                                                                                                                                                                                                                                                                                                    | PAGE           | 6 OF 500  |
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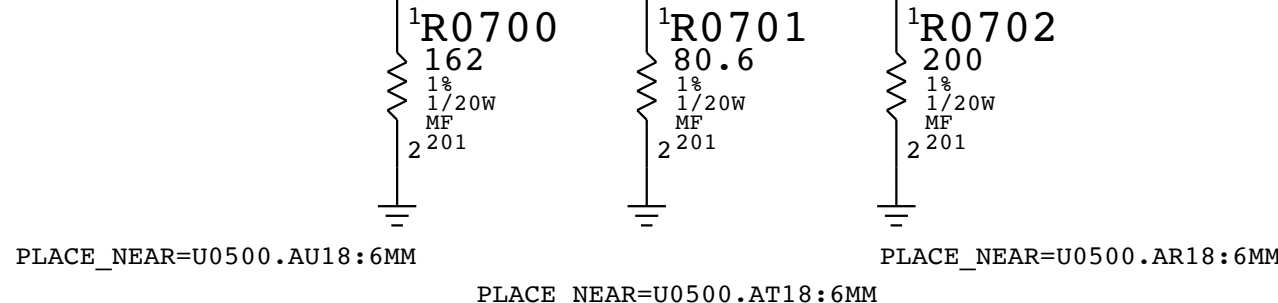
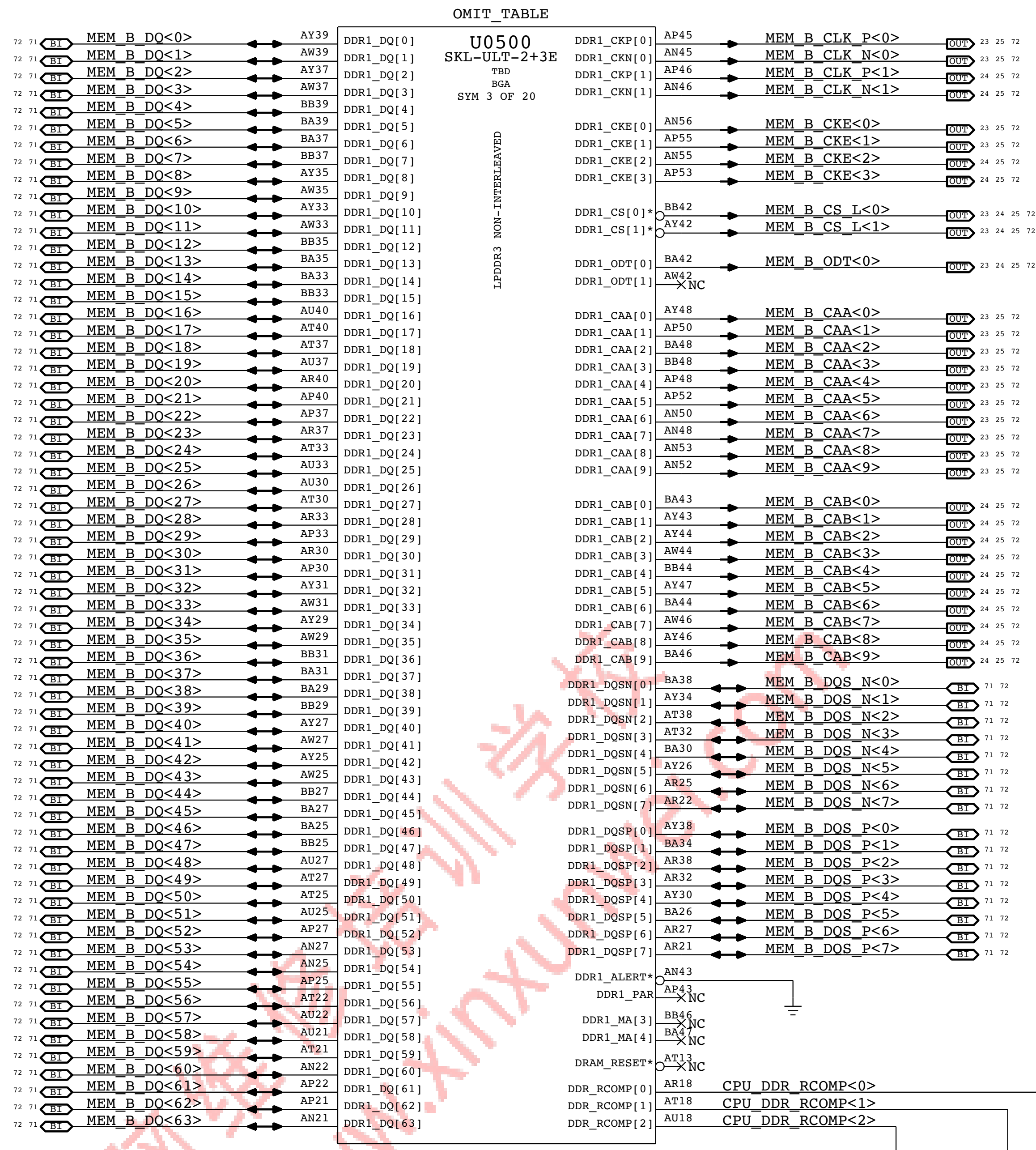
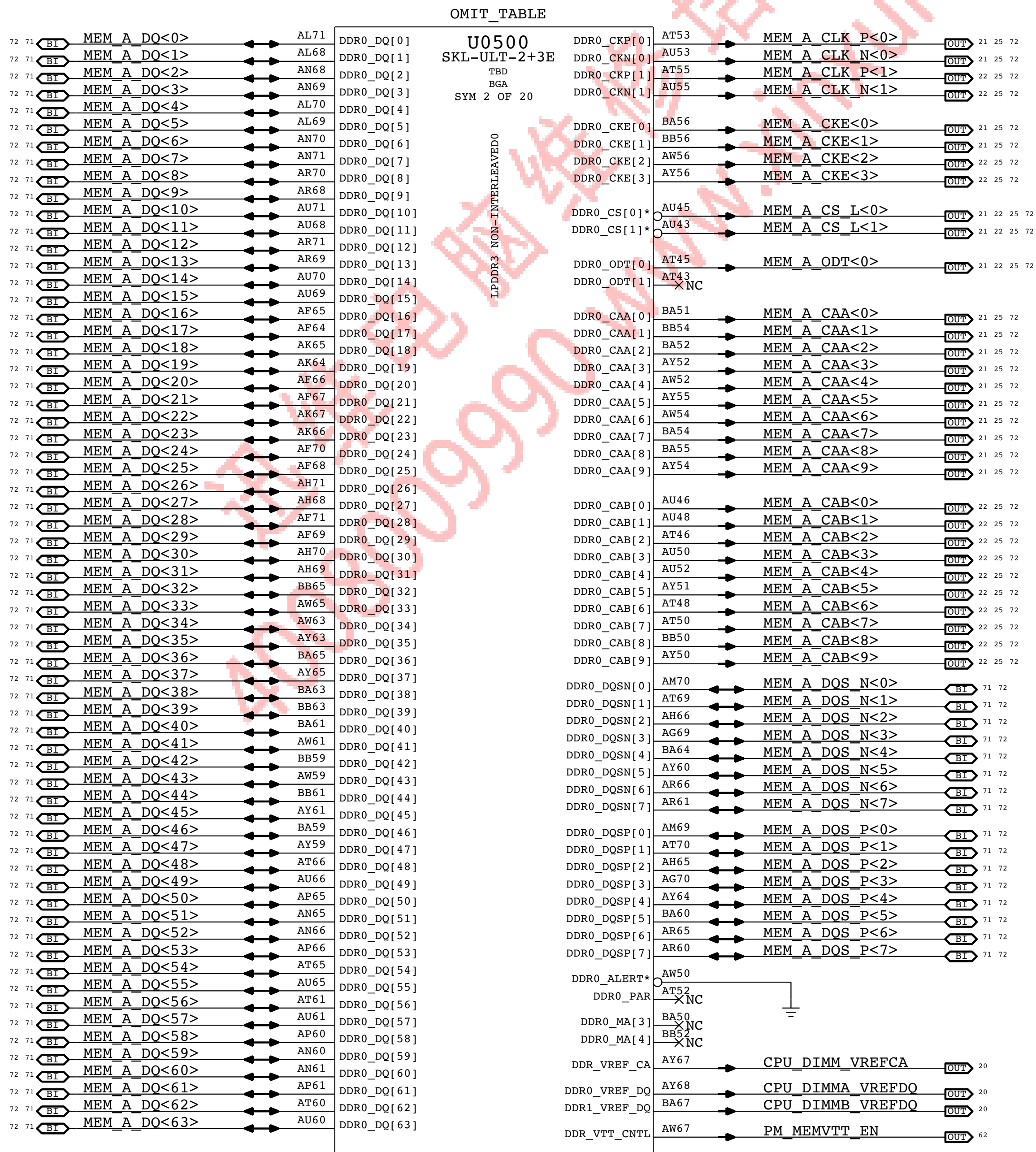
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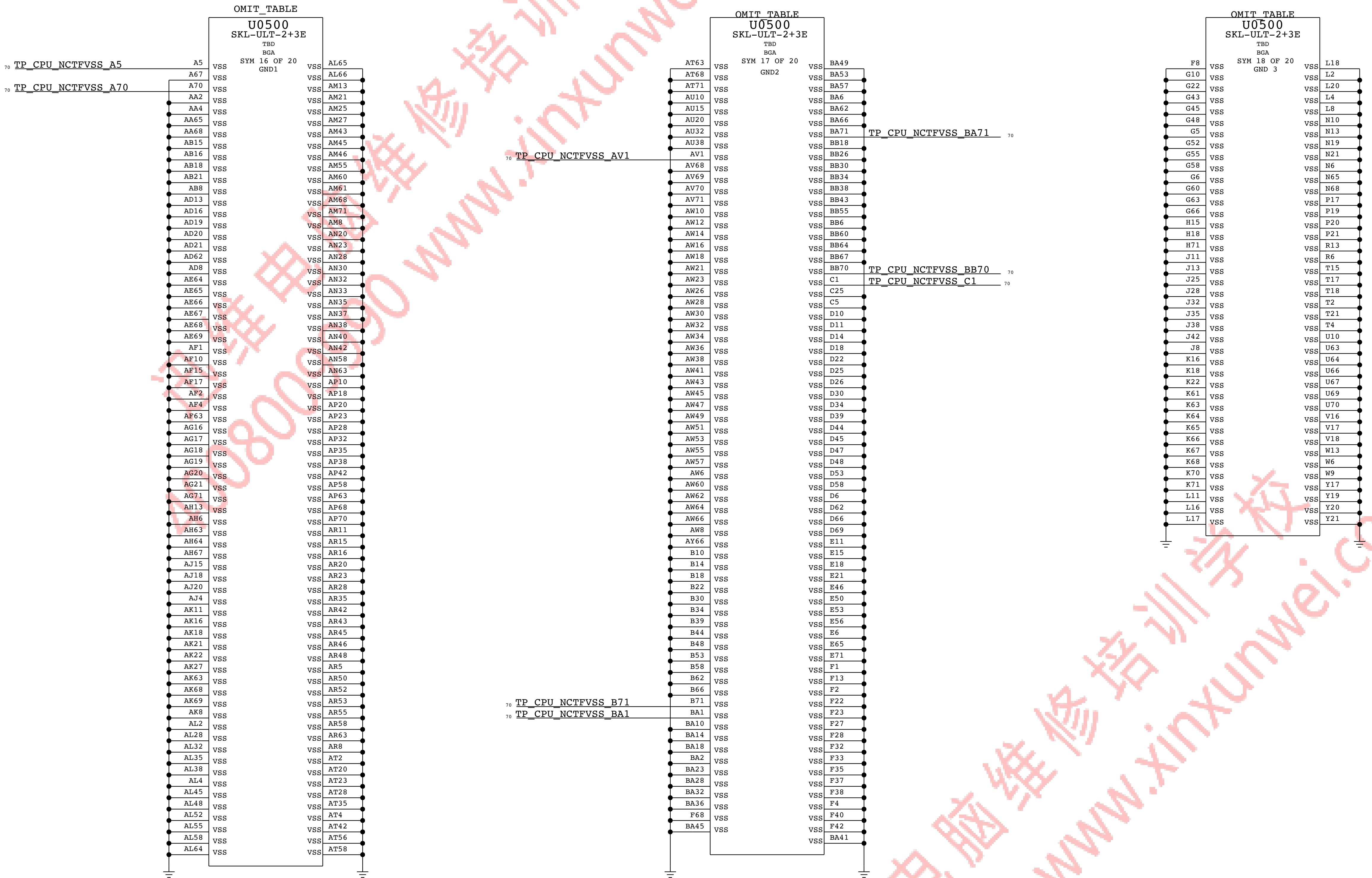
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|                                                                                                                         |                             |                      |           |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------|-----------|
| SYNC_MASTER=PAULM                                                                                                       |                             | SYNC_DATE=06/15/2015 |           |
| PAGE TITLE                                                                                                              |                             |                      |           |
| CPU LPDDR3 Interface                                                                                                    |                             |                      |           |
|  Apple Inc.                        | DRAWING NUMBER<br>051-02265 |                      | SIZE<br>D |
|                                                                                                                         | REVISION<br>1.0.0           |                      |           |
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|                                                                                                                         |                             | PAGE<br>7 OF 500     |           |
|                                                                                                                         |                             | SHEET<br>7 OF 73     |           |

BOM\_COST\_GROUP=CPU & CHIPSET

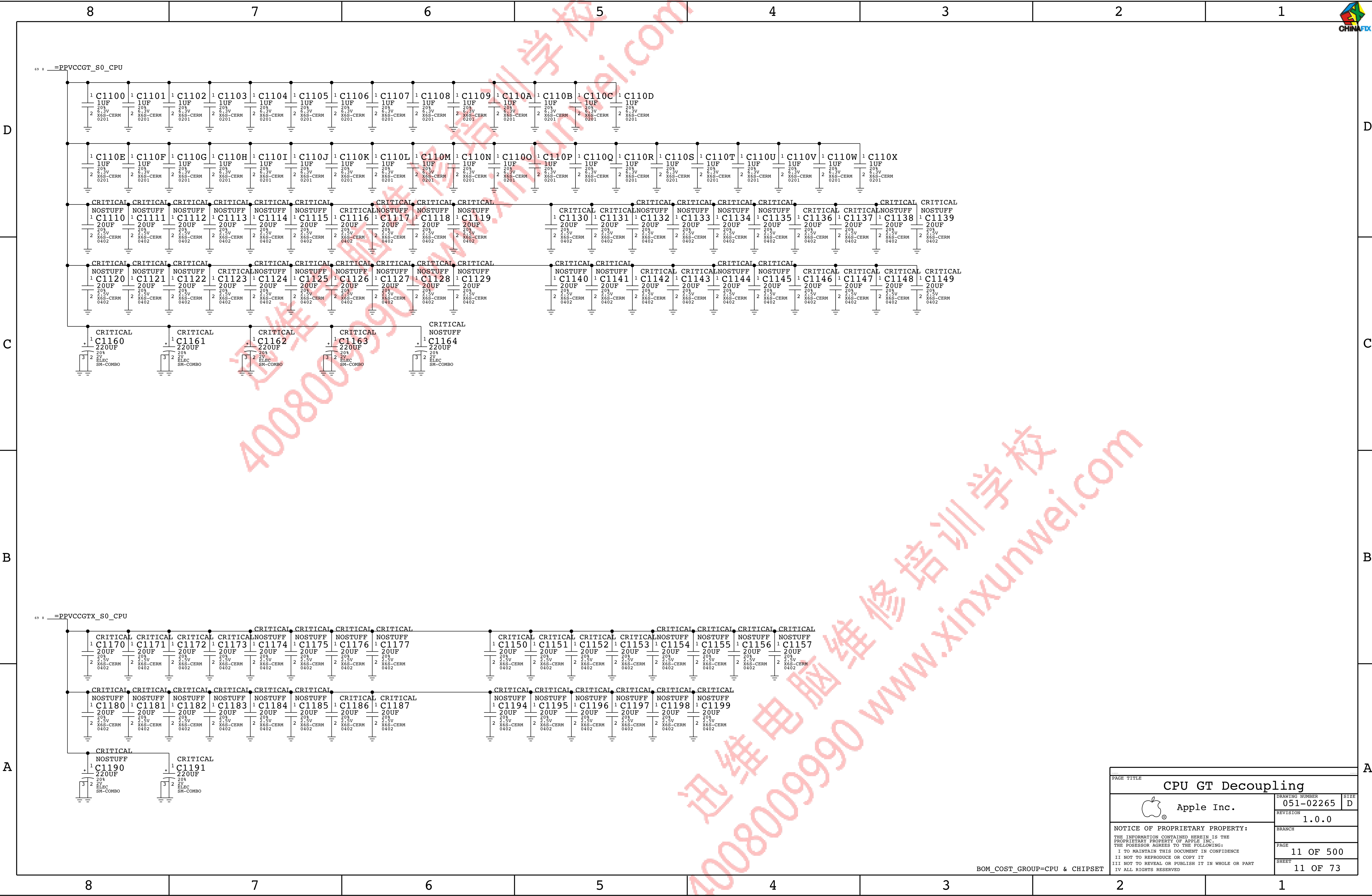




|                                                                                                                                                                                                                                                                                                                      |            |                      |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------|----------|
| SYMC_MASTER=PAULM                                                                                                                                                                                                                                                                                                    |            | SYMC_DATE=06/15/2015 |          |
| PAGE TITLE                                                                                                                                                                                                                                                                                                           |            |                      |          |
| CPU & PCH Grounds                                                                                                                                                                                                                                                                                                    |            |                      |          |
|                                                                                                                                                                                                                                 | Apple Inc. | DRAWING NUMBER       | SIZE     |
|                                                                                                                                                                                                                                                                                                                      |            | 051-02265            | D        |
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|                                                                                                                                                                                                                                                                                                                      |            | BRANCH               |          |
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BOM\_COST\_GROUP=CPU & CHIPSET





| PAGE TITLE                                                                                                                                                                                                                                                                                                         |                |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| CPU GT Decoupling                                                                                                                                                                                                                                                                                                  |                |           |
|  Apple Inc.                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02265 |
|                                                                                                                                                                                                                                                                                                                    | REVISION       | 1.0.0     |
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|                                                                                                                                                                                                                                                                                                                    | PAGE           | 11 OF 500 |
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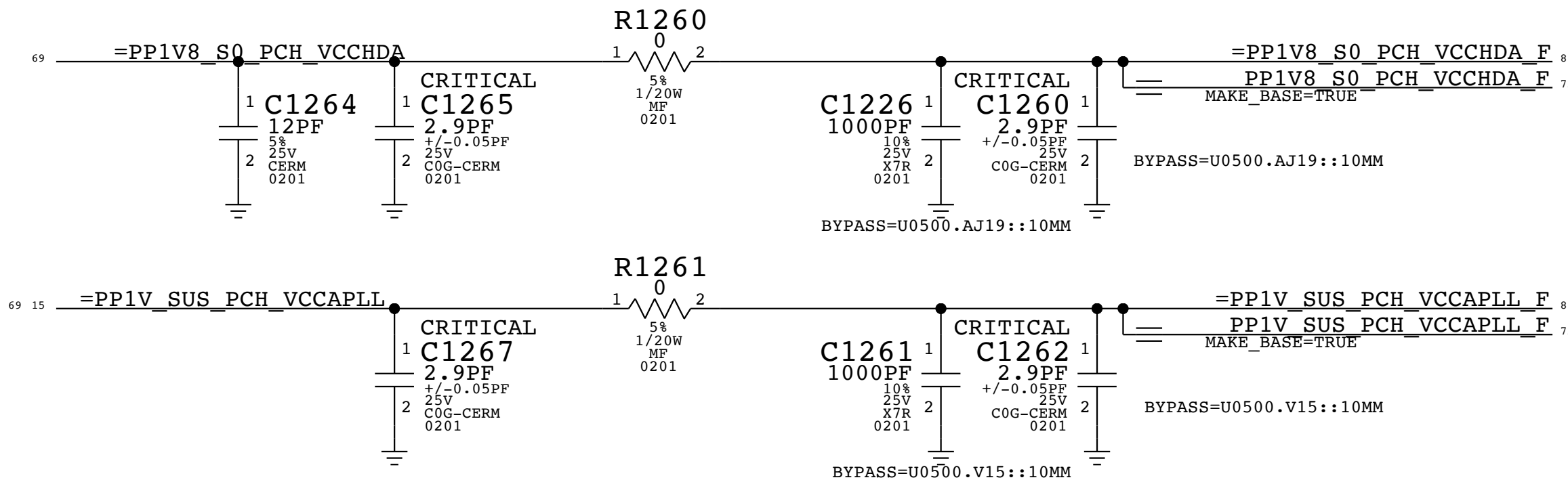
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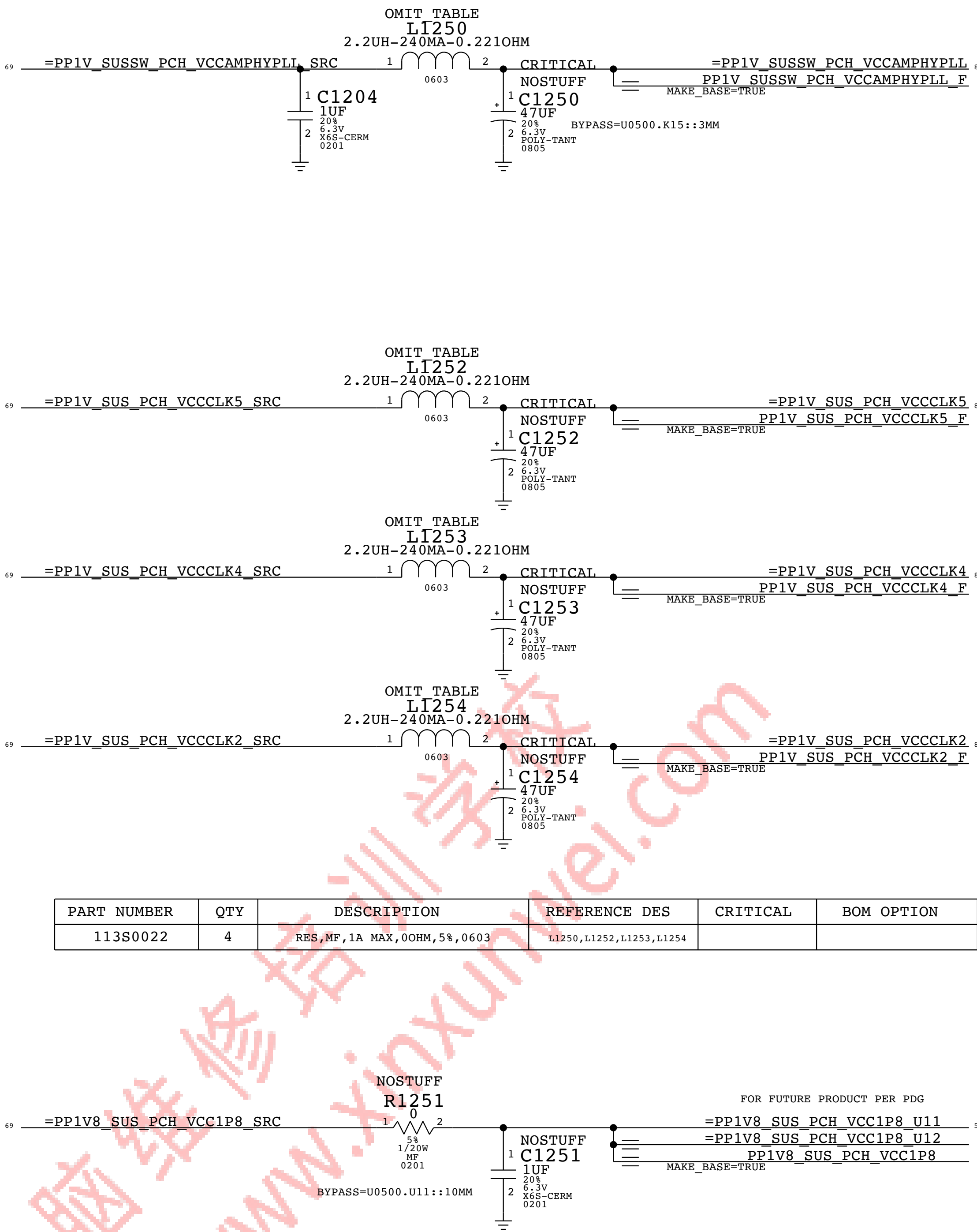
CPU CIRCUITS GENERATE NOISE AT WIFI BAND FREQUENCIES.  
USE SPECIFIC 3PF CAPS FOR BEST FILTERING OF THOSE FREQUENCIES.



FILTER PLACEHOLDERS ONLY

RAIL SIDE

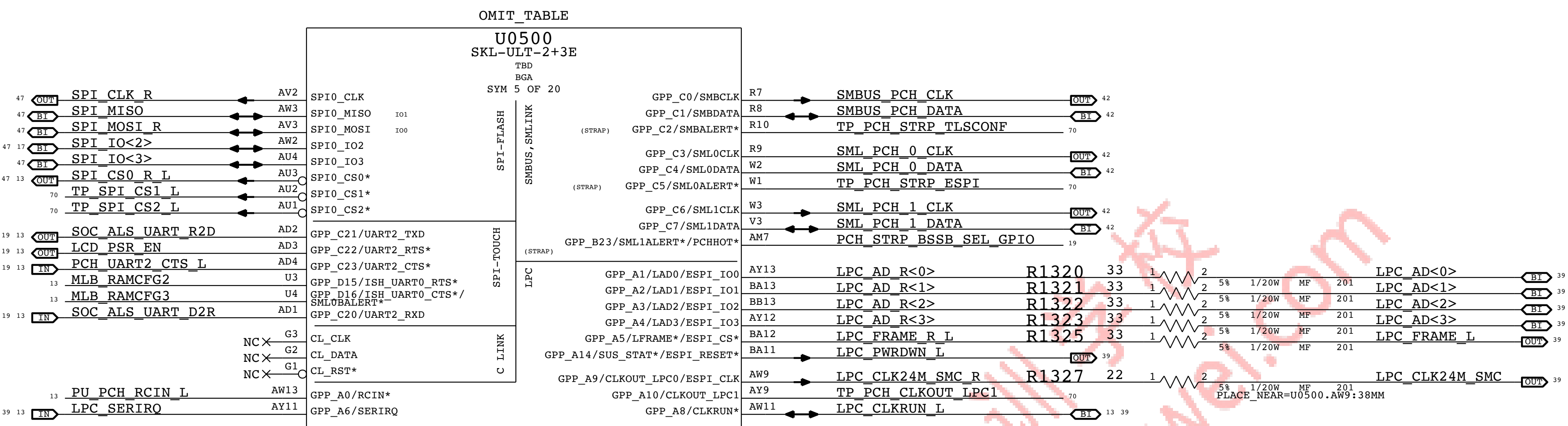
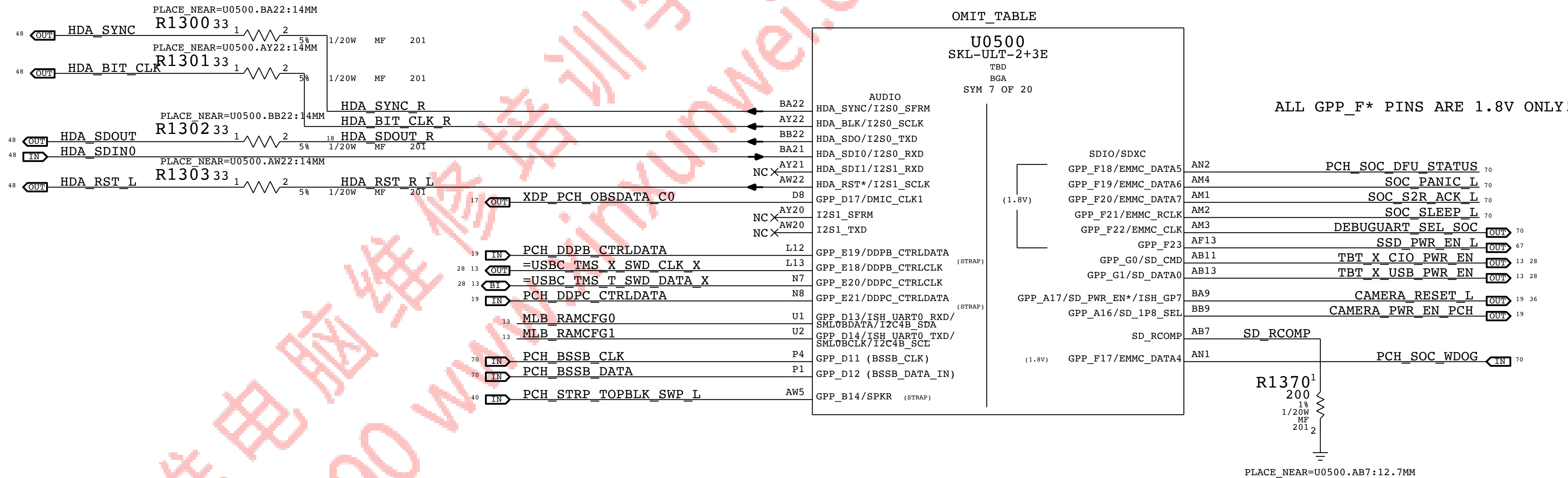
PCH SIDE



| PART NUMBER | QTY | DESCRIPTION                | REFERENCE DES           | CRITICAL | BOM OPTION |
|-------------|-----|----------------------------|-------------------------|----------|------------|
| 113S0022    | 4   | RES,MF,1A MAX,0OHM,5%,0603 | L1250,L1252,L1253,L1254 |          |            |

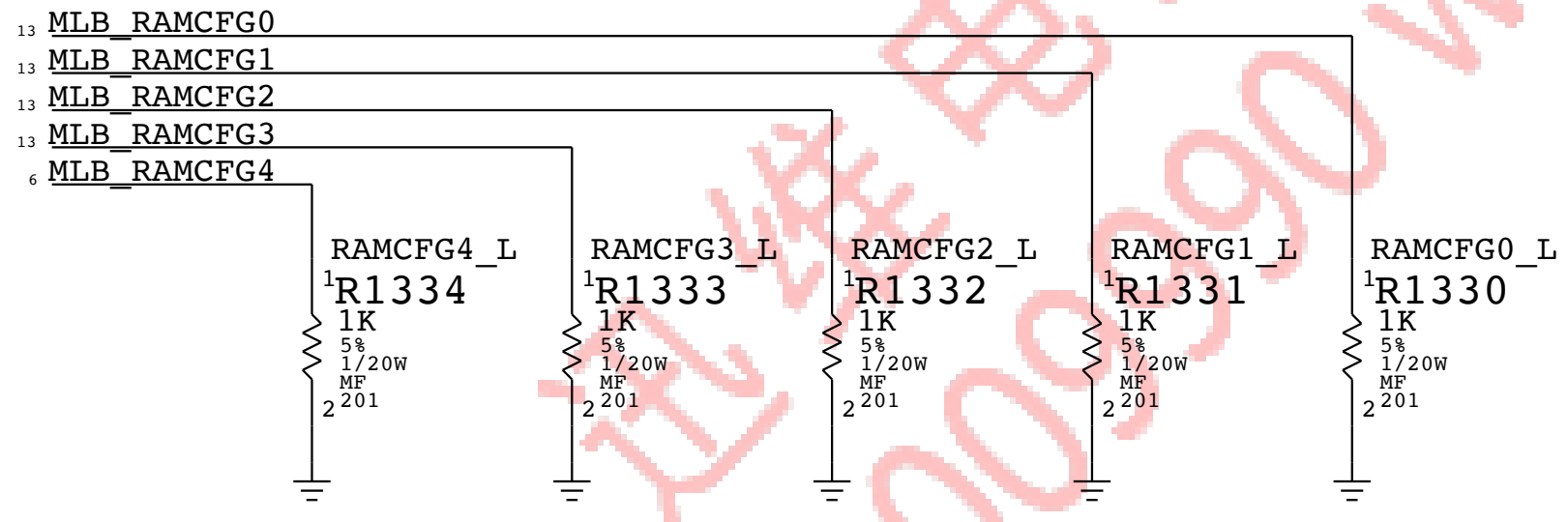
BOM\_COST\_GROUP=CPU & CHIPSET

|                                                                                                                   |                |           |
|-------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| PAGE TITLE                                                                                                        |                |           |
| PCH Decoupling                                                                                                    |                |           |
|                                                                                                                   | DRAWING NUMBER | 051-02265 |
|                                                                                                                   | REVISION       | 1.0.0     |
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MEMORY CONFIGURATION STRAPS.

PCH INTERNAL PULL-UPS ARE TO 3.3V.



| BOM GROUP   | BOM OPTIONS                                           |
|-------------|-------------------------------------------------------|
| RAMCFG_SLOT | RAMCFG4_L, RAMCFG3_L, RAMCFG2_L, RAMCFG1_L, RAMCFG0_L |

BOM\_COST\_GROUP=CPU & CHIPSET

DESIGN: X502/MLB CATZ  
LAST CHANGE: Thu Aug 4 21:00:42 2016

PAGE TITLE  
**PCH Audio/LPC/SPI/SMBus**

 Apple Inc.

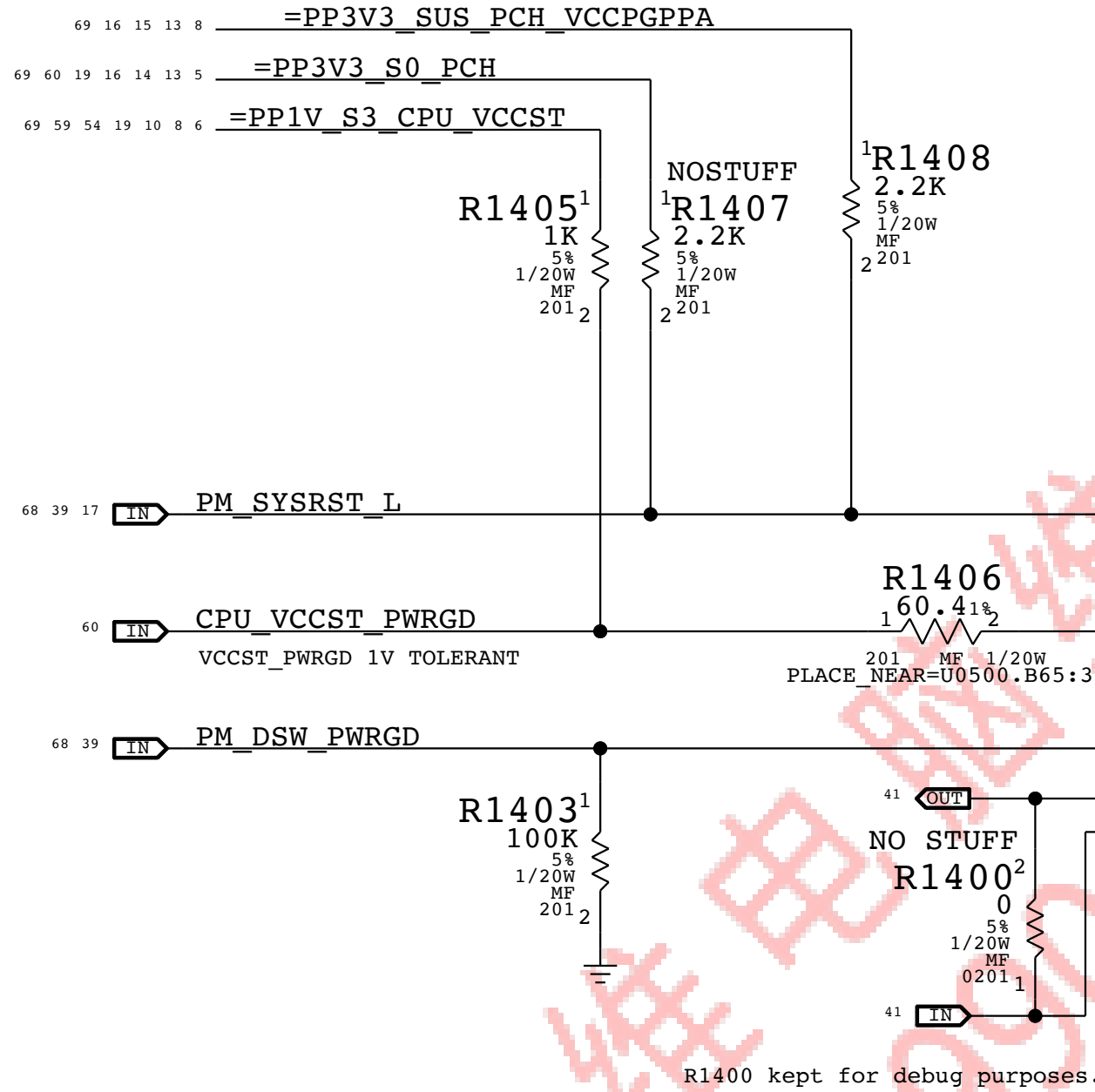
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DRAWING NUMBER  
051-02265  
REVISION  
1.0.0  
BRANCH  
PAGE  
13 OF 500  
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13 OF 73

SIZE  
D



PCH Reset Button

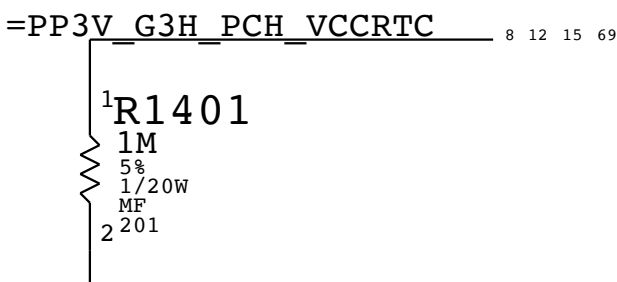


R1400 kept for debug purposes.

OMIT TABLE

U0500  
SKL-ULT-2+3E  
TBD  
BGA  
SYM 11 OF 20  
SYSTEM POWER MANAGEMENT

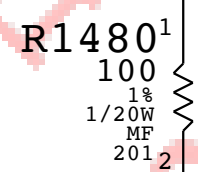
|                       |      |                   |                      |
|-----------------------|------|-------------------|----------------------|
| GPP_B12/SLP_S0*       | AT11 | PM_SLP_S0_L       | 14 39 60 72          |
| GPD4/SLP_S3*          | AP15 | PM_SLP_S3_L       | 14 26 39 59 60 66 72 |
| GPD5/SLP_S4*          | BA16 | PM_SLP_S4_L       | 14 39 60 68 72       |
| GPD10/SLP_S5*         | AY16 | PM_SLP_S5_L       | 14 39 60 72          |
| SLP_SUS*              | AN15 | PM_SLP_SUS_L      | 14 60 72             |
| SLP_LAN*              | AW15 |                   |                      |
| GPD9/SLP_WLAN*        | BB17 | TP_PCH_SLP_WLAN_L | 70                   |
| GPD6/SLP_A*           | AN16 | TP_PCH_SLP_A_L    | 72                   |
| GPD3/PWRBTN*          | BA15 | PCH_PWRBTN_L      | 14 41                |
| GPD1/ACPRESENT*       | AY15 | SSD_SR_EN_L       | 14 67                |
| GPD0/BATLOW*          | AU13 | PM_BATLOW_L       | 14 28 39             |
| GPP_A11/PME*          | AU11 | TP_PCH_PME_L      | 70                   |
| INTRUDER*             | AP16 | PCH_INTRUDER_L    |                      |
| GPP_B11/EXT_PWR_GATE* | AM10 | PCH_HSIO_PWR_EN   | 60                   |
| GPP_B2/VRALER*        | AM11 | BT_LOW_PWR_L      | 14 34 35             |



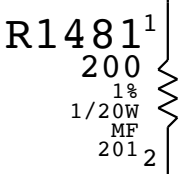
OMIT TABLE

U0500  
SKL-ULT-2+3E  
TBD  
BGA  
SYM 9 OF 20

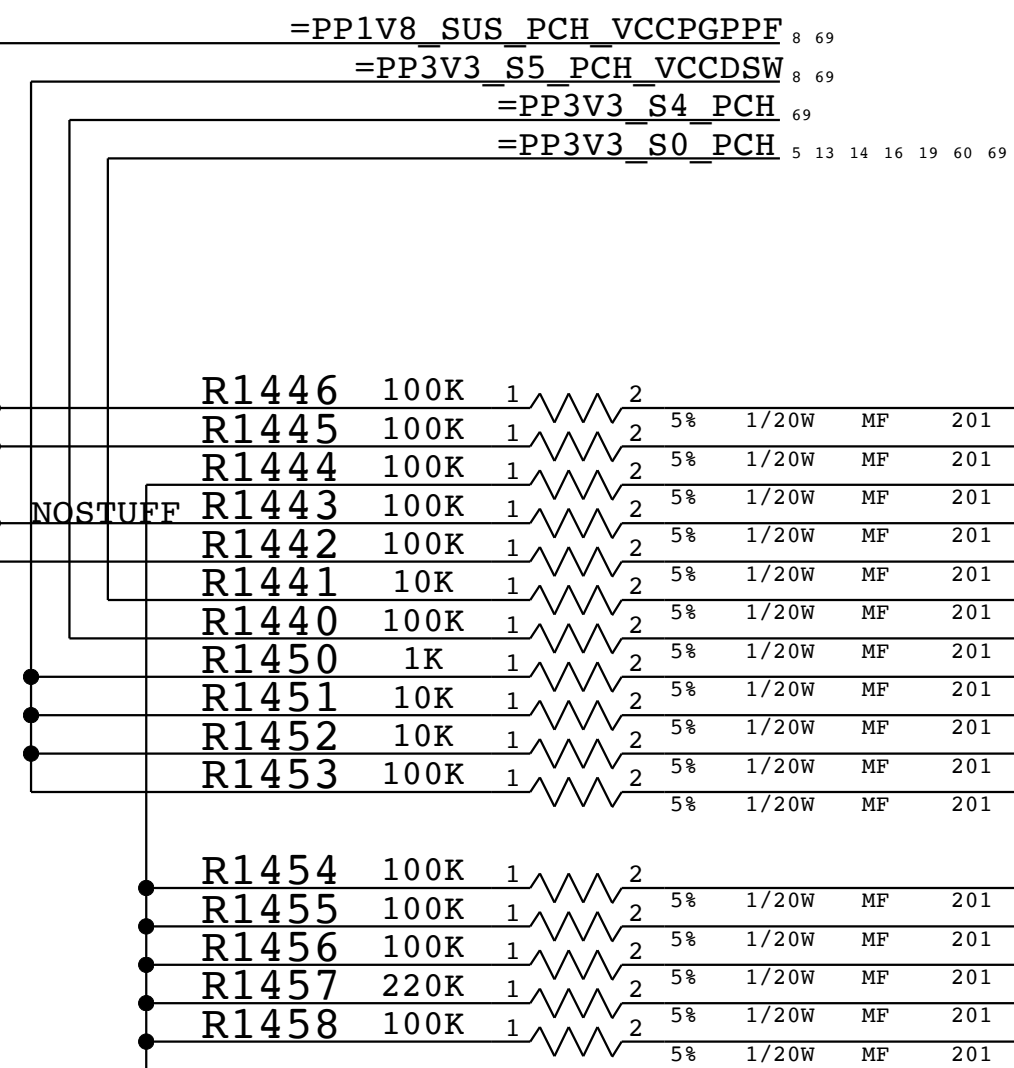
|         |           |                               |      |                   |
|---------|-----------|-------------------------------|------|-------------------|
| NCX A36 | CSI2_DN0  | CSI2_CLKN0                    | C37  | XNC               |
| NCX B36 | CSI2_DP0  | CSI2_CLKP0                    | D37  | XNC               |
| NCX C38 | CSI2_DN1  | CSI2_CLKP1                    | C32  | XNC               |
| NCX D38 | CSI2_DP1  | CSI2_CLKN2                    | D32  | XNC               |
| NCX C36 | CSI2_DN2  | CSI2_CLKP2                    | C29  | XNC               |
| NCX D36 | CSI2_DP2  | CSI2_CLKN3                    | D29  | XNC               |
| NCX A38 | CSI2_DN3  | CSI2_CLKP3                    | B26  | XNC               |
| NCX B38 | CSI2_DP3  | CSI2_CLKP3                    | A26  | XNC               |
| NCX C31 | CSI2_DN4  | CSI2_COMP                     | E13  | CSI2_COMP         |
| NCX D31 | CSI2_DP4  | GPP_D0/SPI1_CS*               | M1   | TP_PCH_GPP_D0     |
| NCX C33 | CSI2_DN5  |                               |      |                   |
| NCX D33 | CSI2_DP5  |                               |      |                   |
| NCX A31 | CSI2_DN6  | GPP_F7/I2C3_SCL               | AH12 | SSD_BOOT_L        |
| NCX B31 | CSI2_DP6  | GPP_F8/I2C4_SDA               | AF11 | TP_PCH_GPP_F8     |
| NCX A33 | CSI2_DN7  | GPP_F9/I2C4_SCL               | AF12 | TP_PCH_GPP_F9     |
| NCX B33 | CSI2_DP7  | GPP_F10/I2C5_SDA/ISH_I2C2_SDA | AD11 | TP_PCH_GPP_F10    |
|         |           | GPP_F11/I2C5_SCL/ISH_I2C2_SCL | AD12 | PCH_BT_ROM_BOOT   |
|         |           | GPP_F12/EMMC_CMD              | AP4  | PCH_SWD_CLK       |
| NCX A29 | CSI2_DN8  | GPP_F13/EMMC_DATA0            | AP2  | PCH_SWD_IO        |
| NCX B29 | CSI2_DP8  | GPP_F14/EMMC_DATA1            | AP1  | PCH_SWD_MUX_SEL   |
| NCX C28 | CSI2_DN9  |                               |      |                   |
| NCX D28 | CSI2_DP9  | GPP_F15/EMMC_DATA2            | AP3  | SOC_S2R_L         |
| NCX A27 | CSI2_DN10 | GPP_F16/EMMC_DATA3            | AN3  | PCH_SOC_FORCE_DFU |
| NCX B27 | CSI2_DP10 | GPP_F6/I2C3_SDA               | AH11 | UPC_I2C_INT_L     |
| NCX C27 | CSI2_DN11 | EMMC_RCOMP                    | AT1  | EMMC_RCOMP        |
| NCX D27 | CSI2_DP11 |                               |      |                   |



ALL GPP F\* PINS ARE 1.8V ONLY!



PLACE\_NEAR=U0500.AT1:12.7MM



NOTE: PM\_SLP\_S0\_L HAS INTERNAL PULL-UP BEFORE RSMRST\_L IS RELEASED.  
THIS CAUSES A VOLTAGE DIVIDER WITH THE PULL-DOWN HERE.  
THE SIGNAL IS DRIVEN HI AFTER RSMRST\_L IS RELEASED.

|                                                                                                                                                                                                                                                                                                                         |                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                                   |                                                  |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                                                                                                                                                                                                                                    |                                                  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                                                  |
| PCH Power Management                                                                                                                                                                                                                                                                                                    |                                                  |
| Apple Inc.                                                                                                                                                                                                                                                                                                              | DRAWING NUMBER<br>051-02265<br>REVISION<br>1.0.0 |
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| BOM_COST_GROUP=CPU & CHIPSET                                                                                                                                                                                                                                                                                            |                                                  |
| PAGE<br>14 OF 500                                                                                                                                                                                                                                                                                                       |                                                  |
| SHEET<br>14 OF 73                                                                                                                                                                                                                                                                                                       |                                                  |

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PCIE Port Assignments:

SSD LANE 0

SSD LANE 1

SSD LANE 2

SSD LANE 3

Thunderbolt X lane 0

Thunderbolt X lane 1

Thunderbolt X lane 2

Thunderbolt X lane 3

AirPort

CAMERA

PLACE\_NEAR=U0500.F5:12.7mm  
R1504  
100  
1%  
1/20W  
MF  
2012

Thunderbolt T lane 0

Thunderbolt T lane 1

ANY CLKREQ CAN MAP TO ANY CLK.  
ANY CLKREQ OR CLK CAN MAP TO ANY PCIE PORT.  
UNUSED CLKREQS AND CLKS SHOULD BE DISABLED.  
PER SKYLAKE PDG, SKYLAKE PCH EDS.

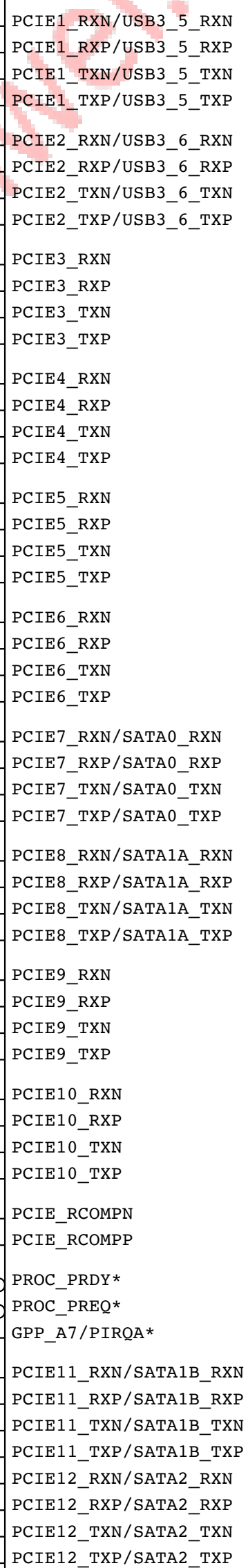
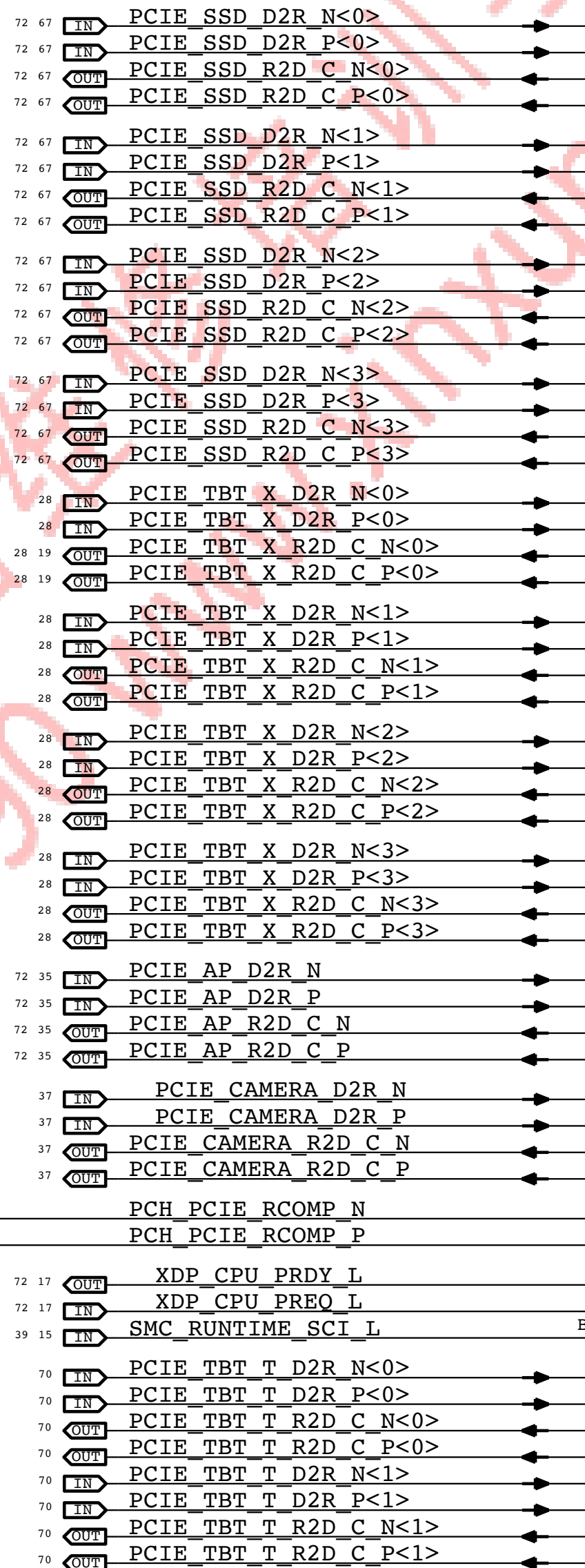
=PP3V3\_SUS\_PCH\_VCCPGPPA 8 13 14 16 69

R1550 100K 1 5% 1/20W MF 201 SMC\_RUNTIME\_SCI\_L 15 39

OMIT TABLE

U0500  
SKL-ULT-2+3E

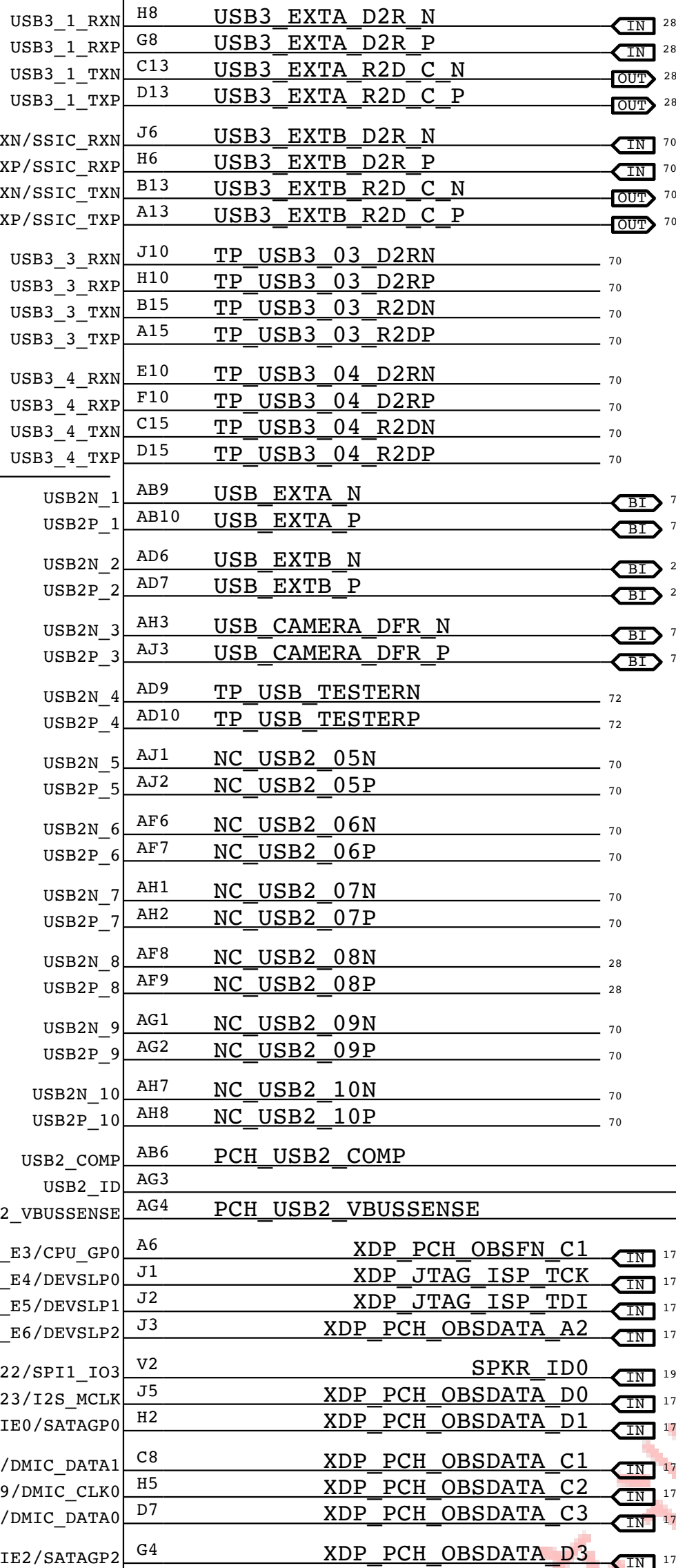
TBD  
BGA  
SYM 8 OF 20



SSIC/USB3

USB2

PCIE/USB3/SATA



EXT A (SS,DCI)

EXT B (SS)

EXT A (LS/FS/HS)

EXT B (LS/FS/HS)

PLACE\_NEAR=U0500.AG4:12.7MM  
R1503  
1K  
5%  
1/20W  
MF  
201

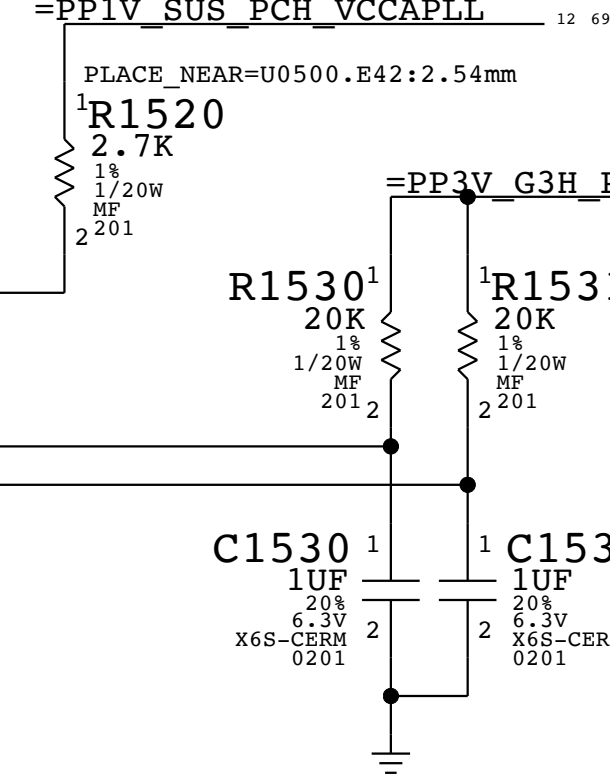
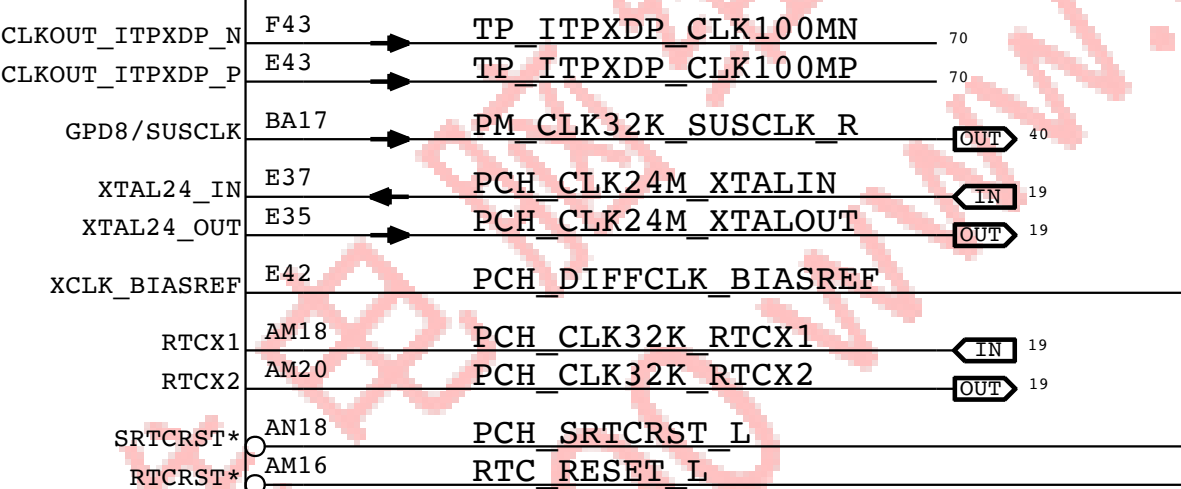
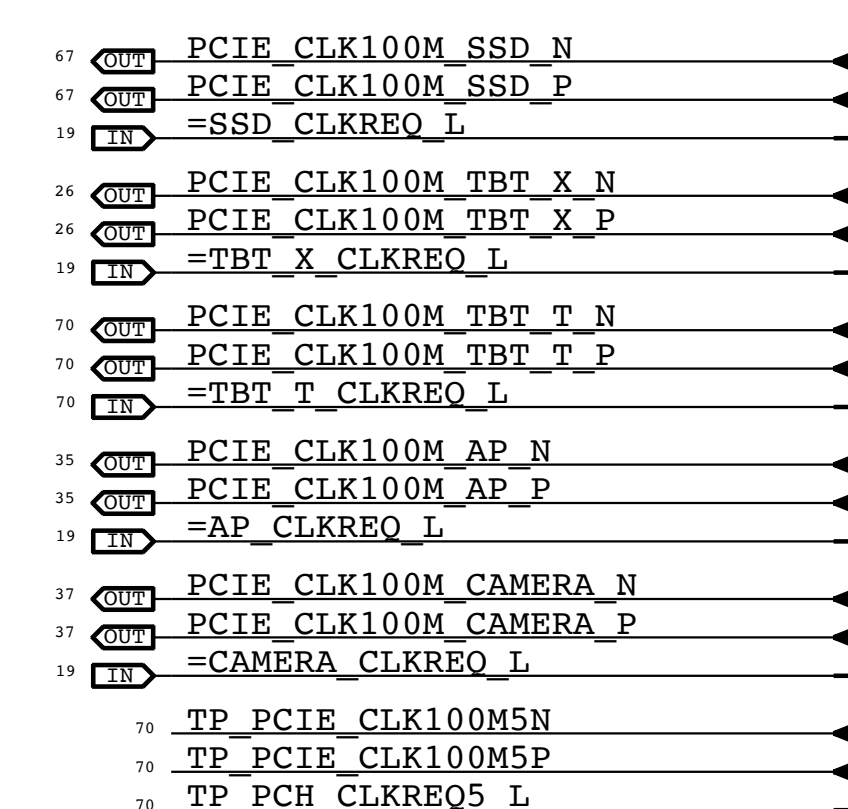
R1501  
113  
1%  
1/20W  
MF  
201

OMIT TABLE

U0500  
SKL-ULT-2+3E

TBD  
BGA  
SYM 10 OF 20

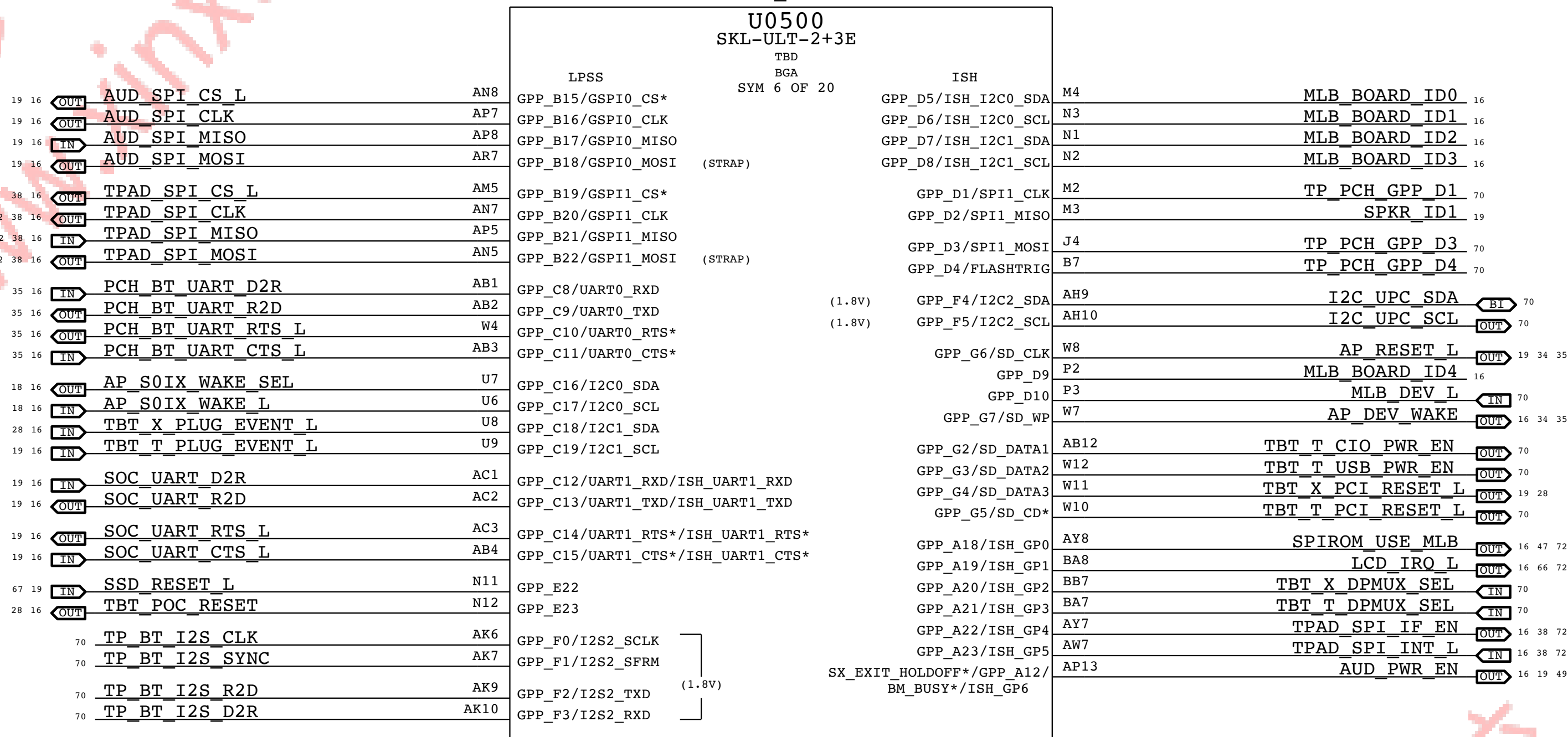
CLOCK SIGNALS



|                                                                                                                                                                                                                                                                                                                  |  |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                            |  |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                                                                                                                                                                                                                             |  |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                       |  |           |
| PCH PCIE/USB/CLKS                                                                                                                                                                                                                                                                                                |  |           |
| Apple Inc.                                                                                                                                                                                                                                                                                                       |  | D         |
| 051-02265                                                                                                                                                                                                                                                                                                        |  | SIZE      |
| 1.0.0                                                                                                                                                                                                                                                                                                            |  | REVISION  |
| BRANCH                                                                                                                                                                                                                                                                                                           |  |           |
| PAGE                                                                                                                                                                                                                                                                                                             |  | 15 OF 500 |
| SHEET                                                                                                                                                                                                                                                                                                            |  | 15 OF 73  |
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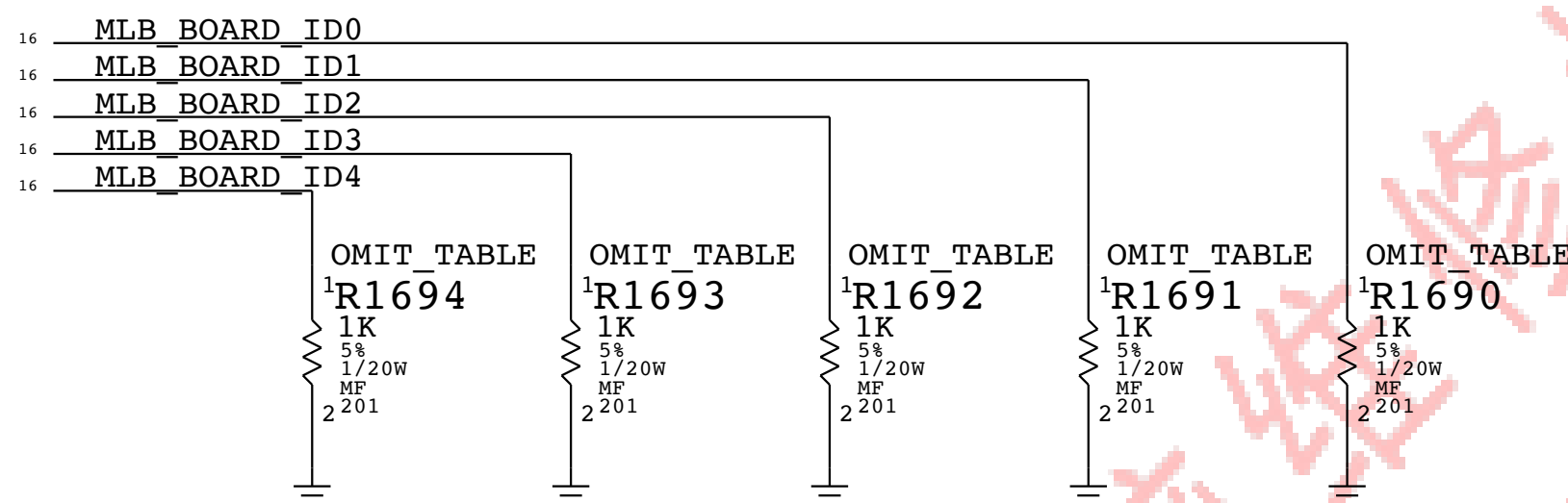
BOM\_COST\_GROUP=CPU & CHIPSET

ALL GPP\_F\* PINS ARE 1.8V ONLY!  
OMIT TABLE



MLB ID STRAPS.

PCH INTERNAL PULL-UPS ARE TO VCCGPPD = 3.3V.



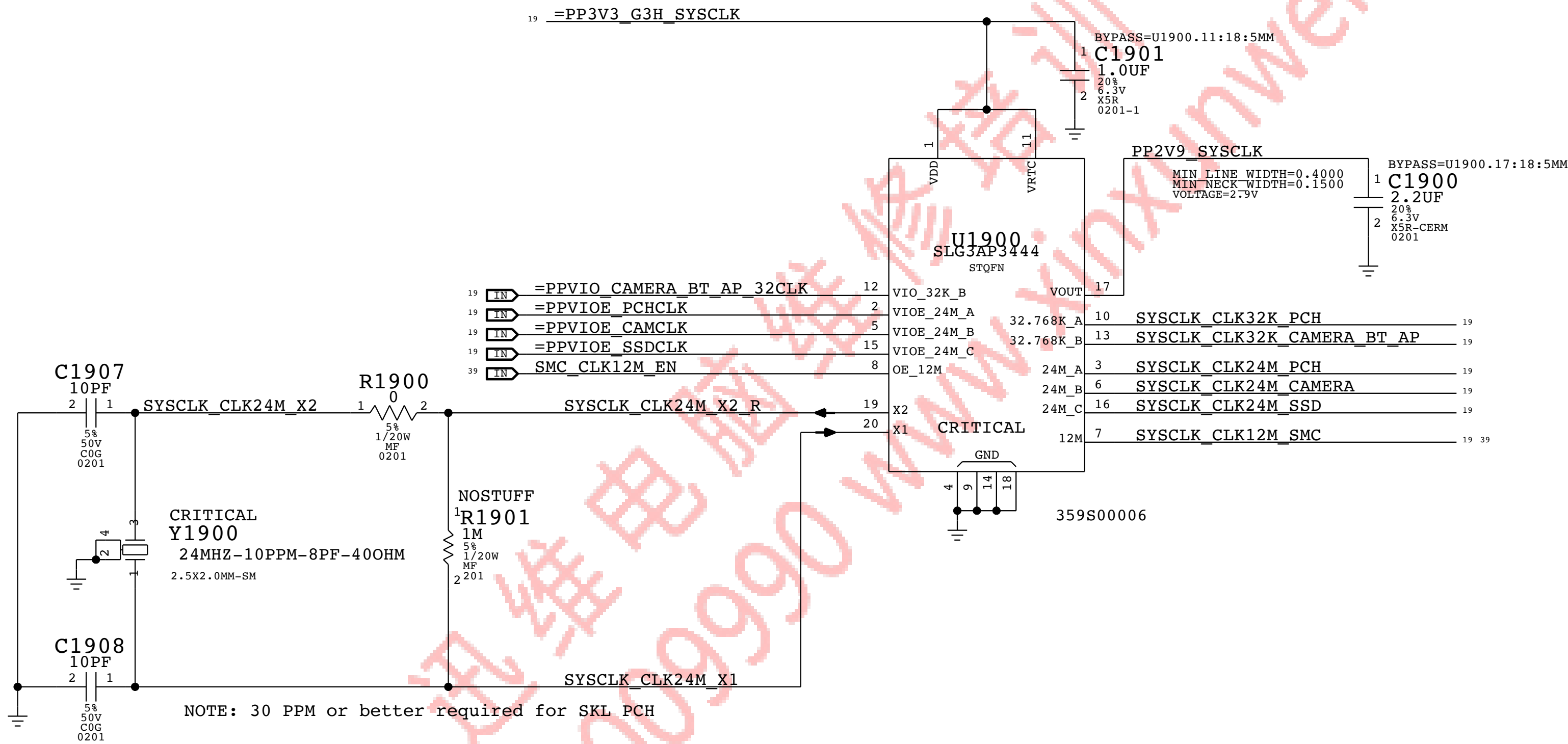
| PART#    | QTY | DESCRIPTION                    | REFERENCE DESIGNATOR(S) | BOM OPTION  | CODE    |
|----------|-----|--------------------------------|-------------------------|-------------|---------|
| 117S0006 | 0   | RES,MP,1/20W/1K OHM,5,0201,SMD |                         | BOARD_ID:0  | <11111> |
| 117S0006 | 1   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1690                   | BOARD_ID:1  | <11110> |
| 117S0006 | 1   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1691                   | BOARD_ID:2  | <11101> |
| 117S0006 | 2   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1691,R1690             | BOARD_ID:3  | <11100> |
| 117S0006 | 1   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1692                   | BOARD_ID:4  | <11011> |
| 117S0006 | 2   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1692,R1690             | BOARD_ID:5  | <11010> |
| 117S0006 | 2   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1692,R1691             | BOARD_ID:6  | <11001> |
| 117S0006 | 3   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1692,R1691,R1690       | BOARD_ID:7  | <11000> |
| 117S0006 | 1   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1693                   | BOARD_ID:8  | <10111> |
| 117S0006 | 2   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1693,R1690             | BOARD_ID:9  | <10110> |
| 117S0006 | 2   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1693,R1691             | BOARD_ID:10 | <10101> |
| 117S0006 | 3   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1693,R1691,R1690       | BOARD_ID:11 | <10100> |
| 117S0006 | 1   | RES,MP,1/20W/1K OHM,5,0201,SMD | R1694                   | BOARD_ID:16 | <01111> |

BOM\_COST\_GROUP=CPU & CHIPSET

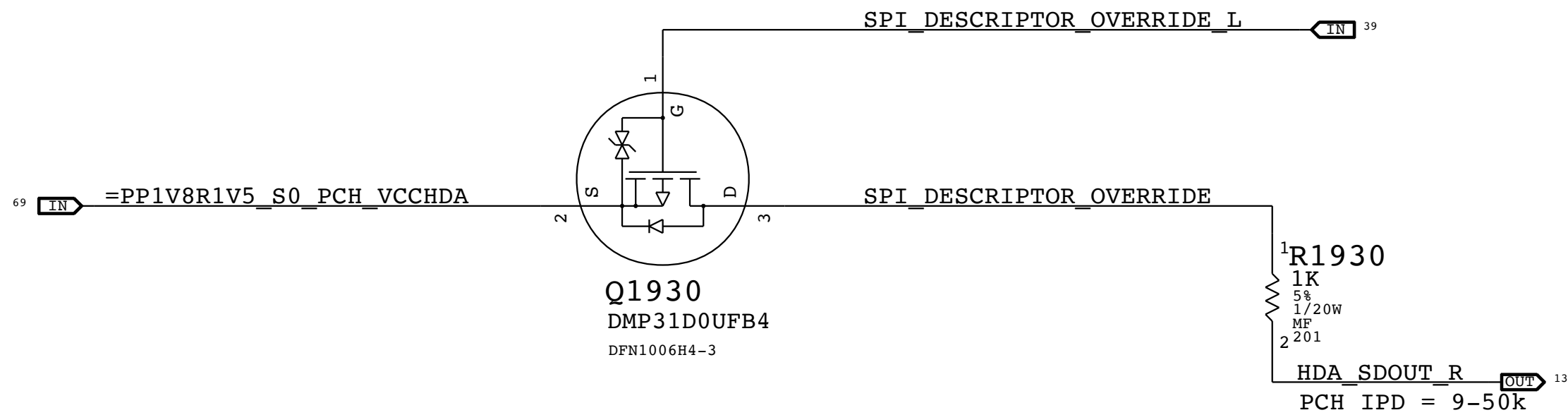
|                                                                                                                                                                                                                                                                                                                         |                |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                                   |                |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                                                                                                                                                                                                                                    |                |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                |           |
| PCH SPI/UART/GPIO                                                                                                                                                                                                                                                                                                       |                |           |
|                                                                                                                                                                                                                                                                                                                         | DRAWING NUMBER | 051-02265 |
|                                                                                                                                                                                                                                                                                                                         | REVISION       | 1.0.0     |
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| PAGE                                                                                                                                                                                                                                                                                                                    |                | 16 OF 500 |
| SHEET                                                                                                                                                                                                                                                                                                                   |                | 16 OF 73  |



System 32kHz / 12MHz / 24MHz Clock Generator

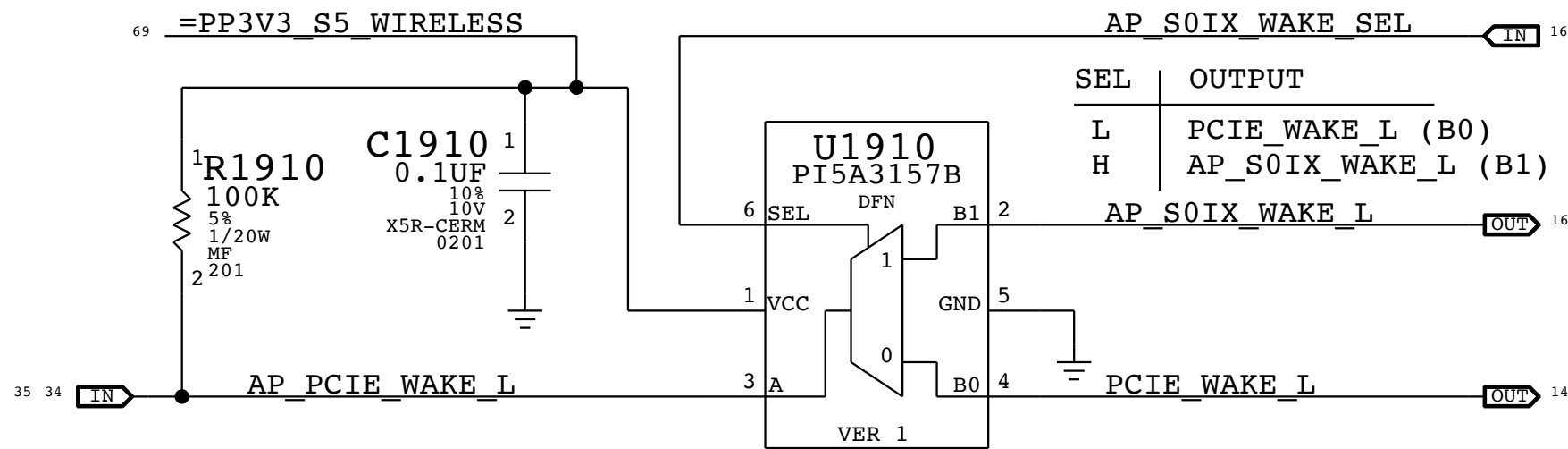


PCH ME Disable Strap



PCH uses HDA\_SDO as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. \*\*\*\*\* Circuit does not support HDA voltage >3.3V.

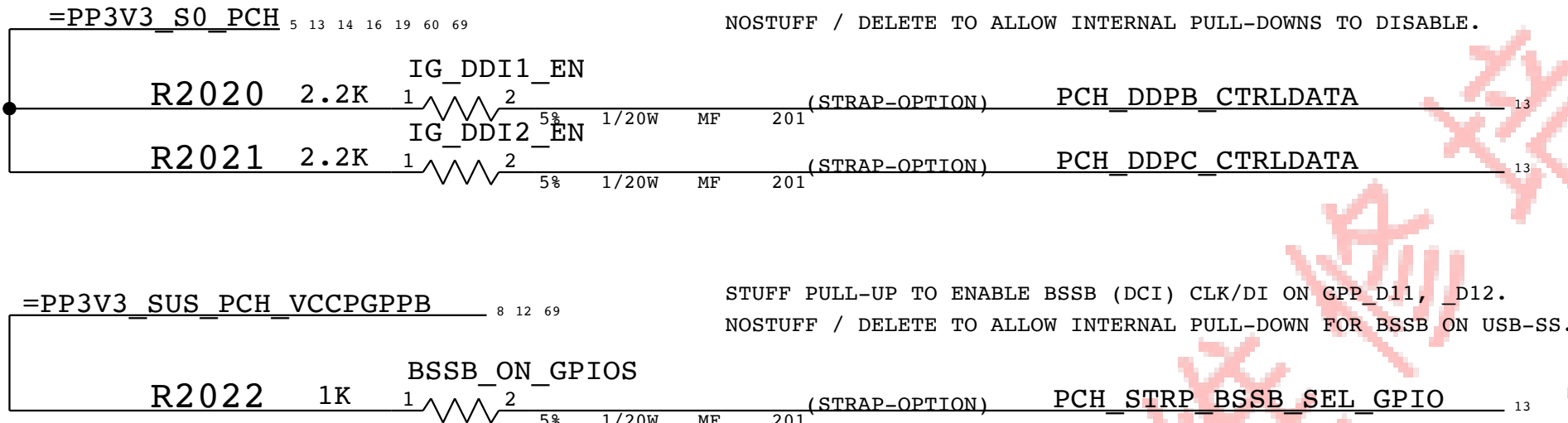
PCIe Wake Muxing



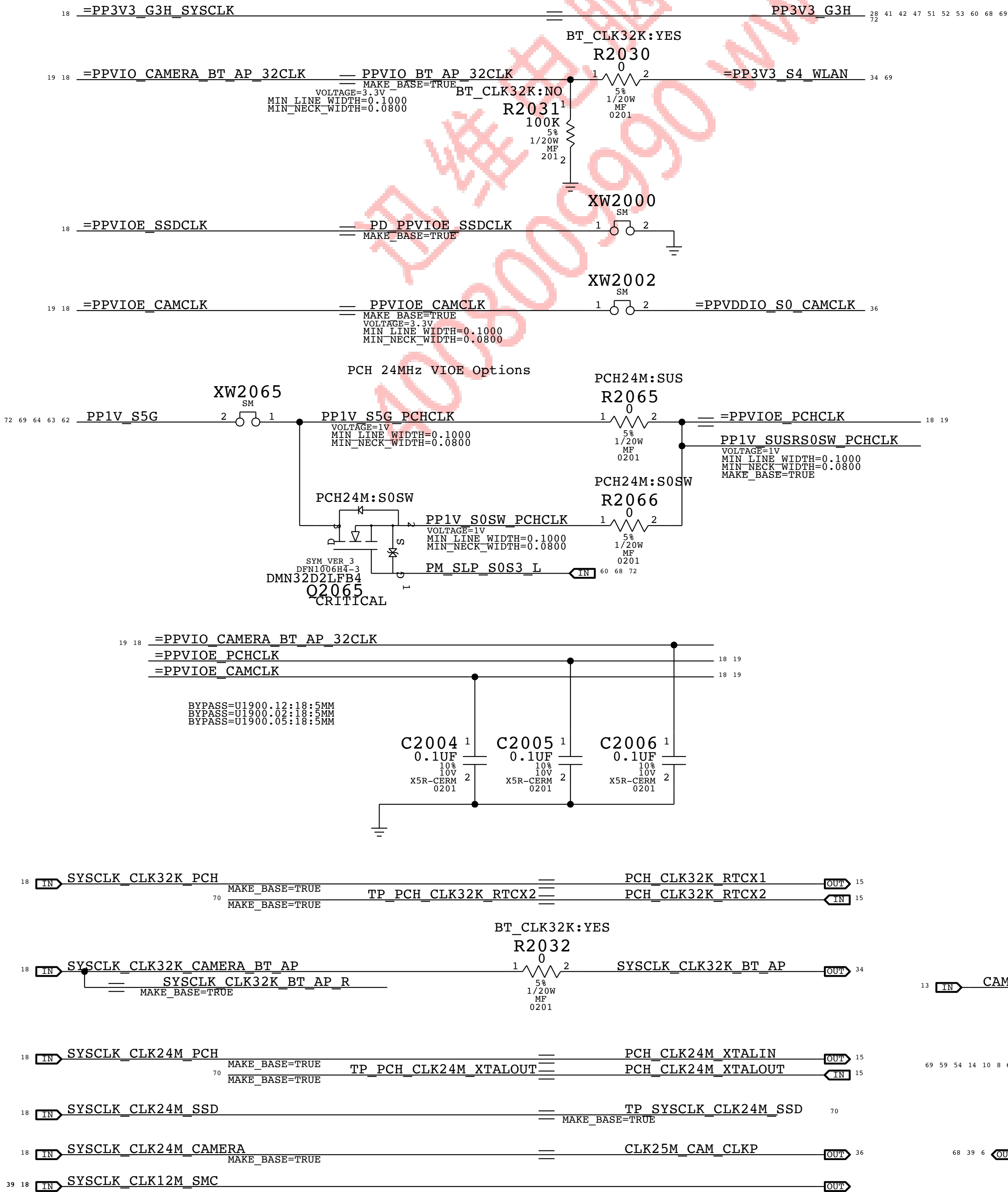
OPTION STRAPS

PROJECT DEPENDANT

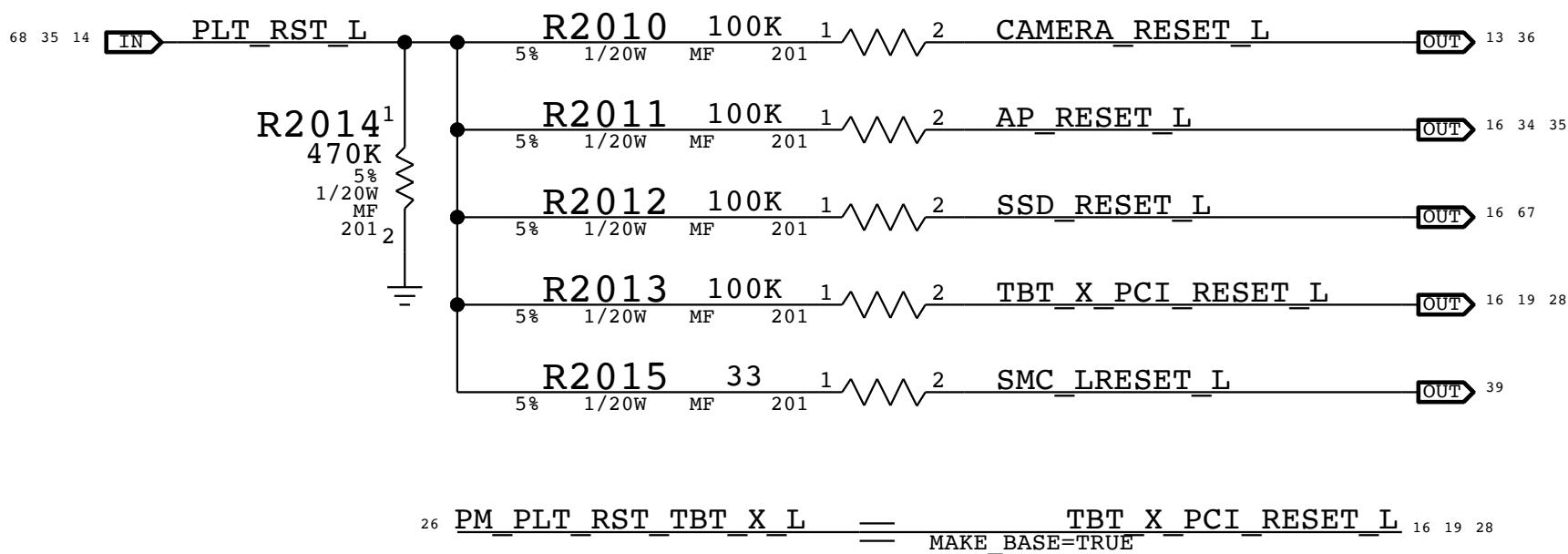
STUFF PULL-UPS TO ENABLE INTERNAL GPU DP PORTS.  
NOSTUFF / DELETE TO ALLOW INTERNAL PULL-DOWNS TO DISABLE.



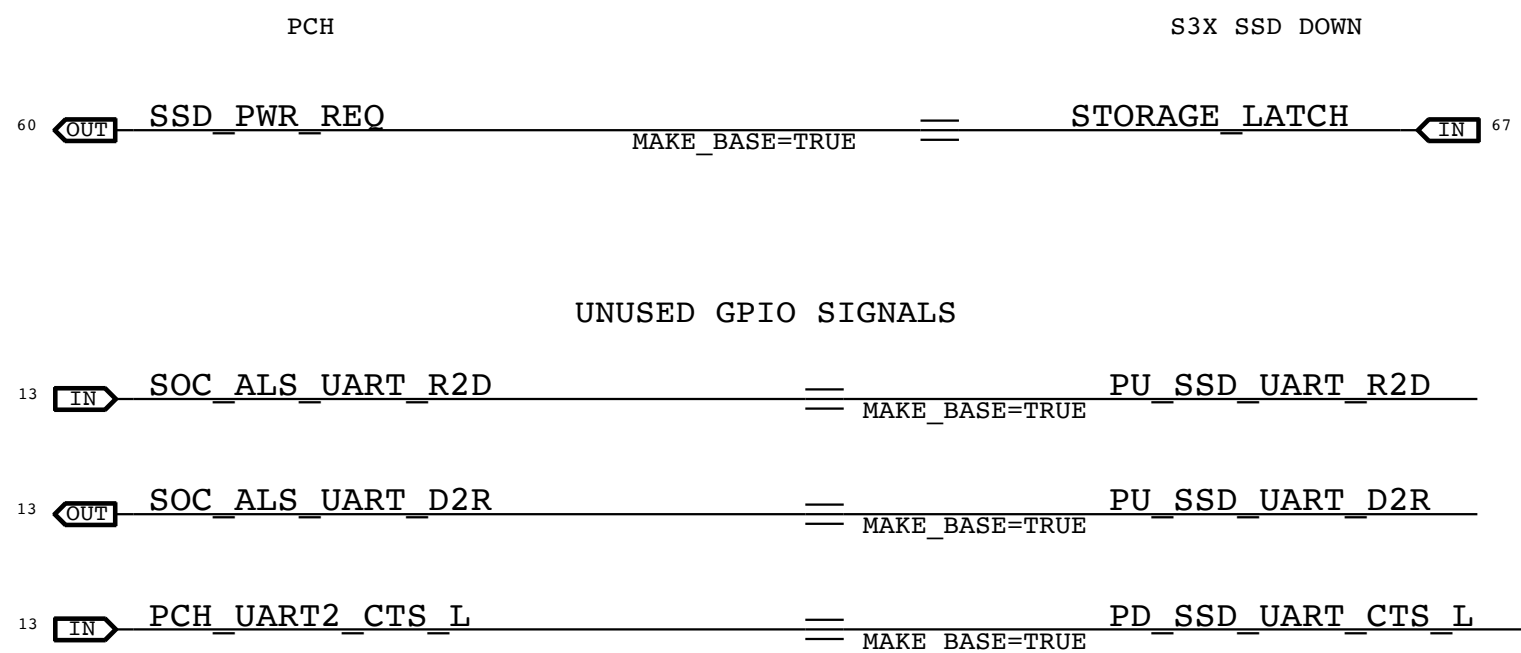
GREENCLK CLOCK CONNECTIONS.



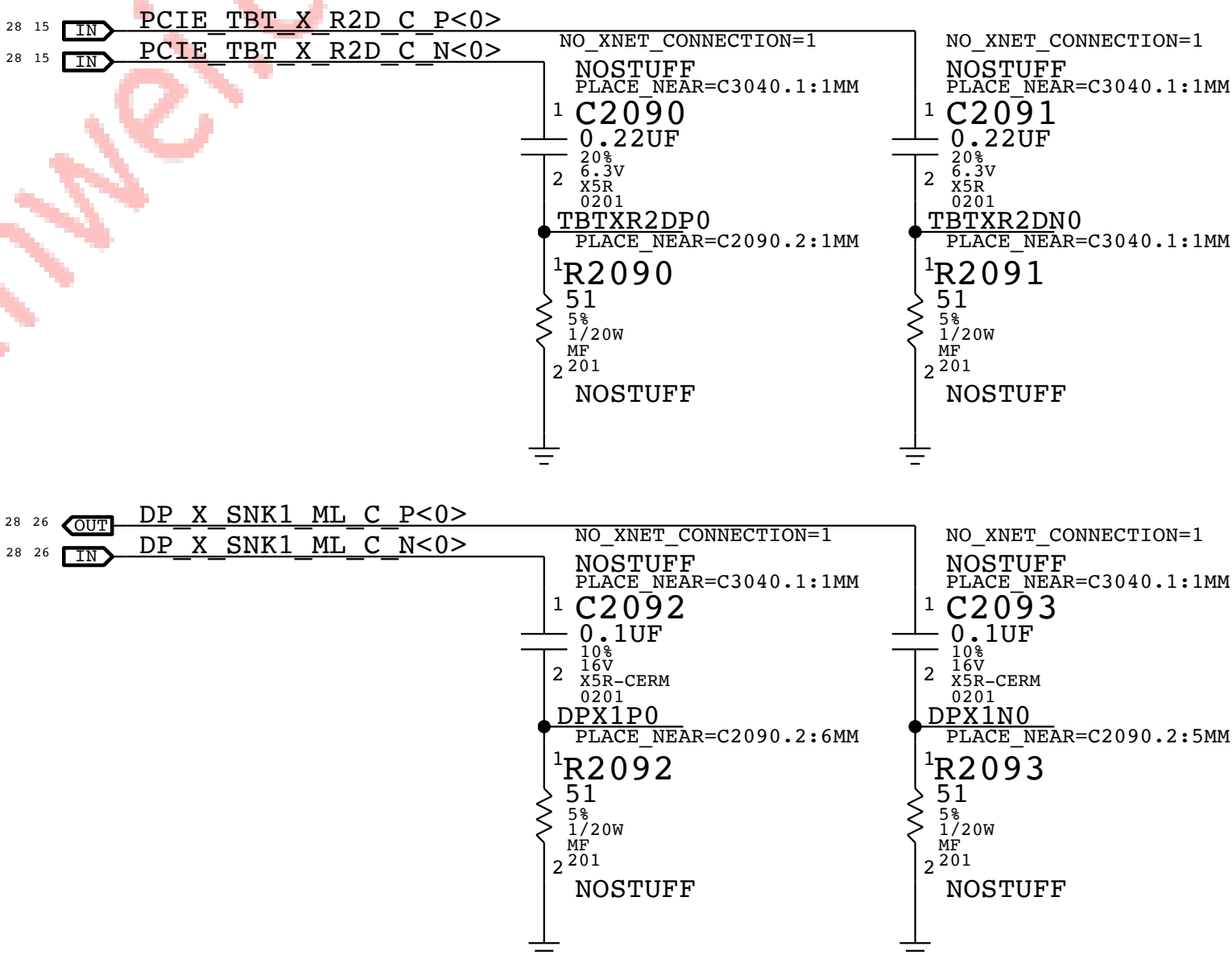
RESETS



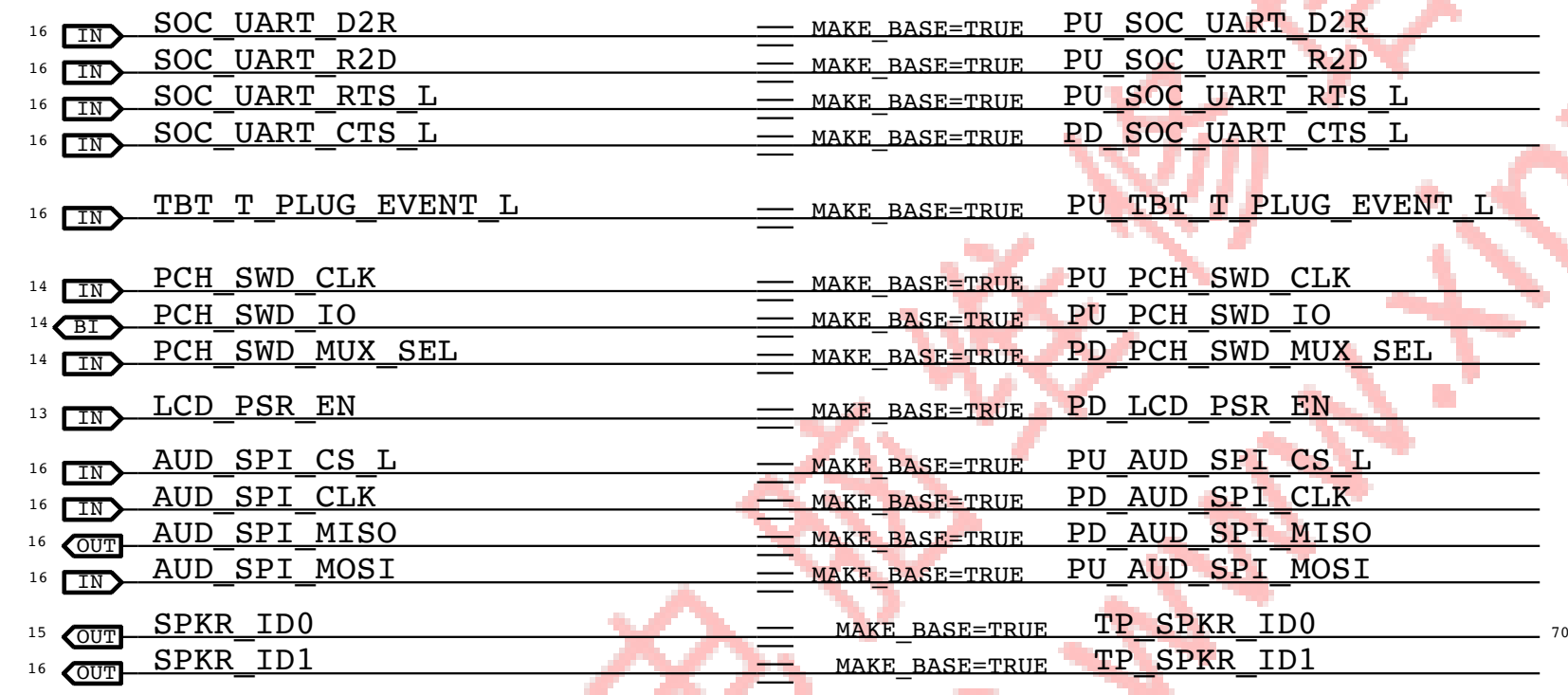
S3X SSD CONTROL



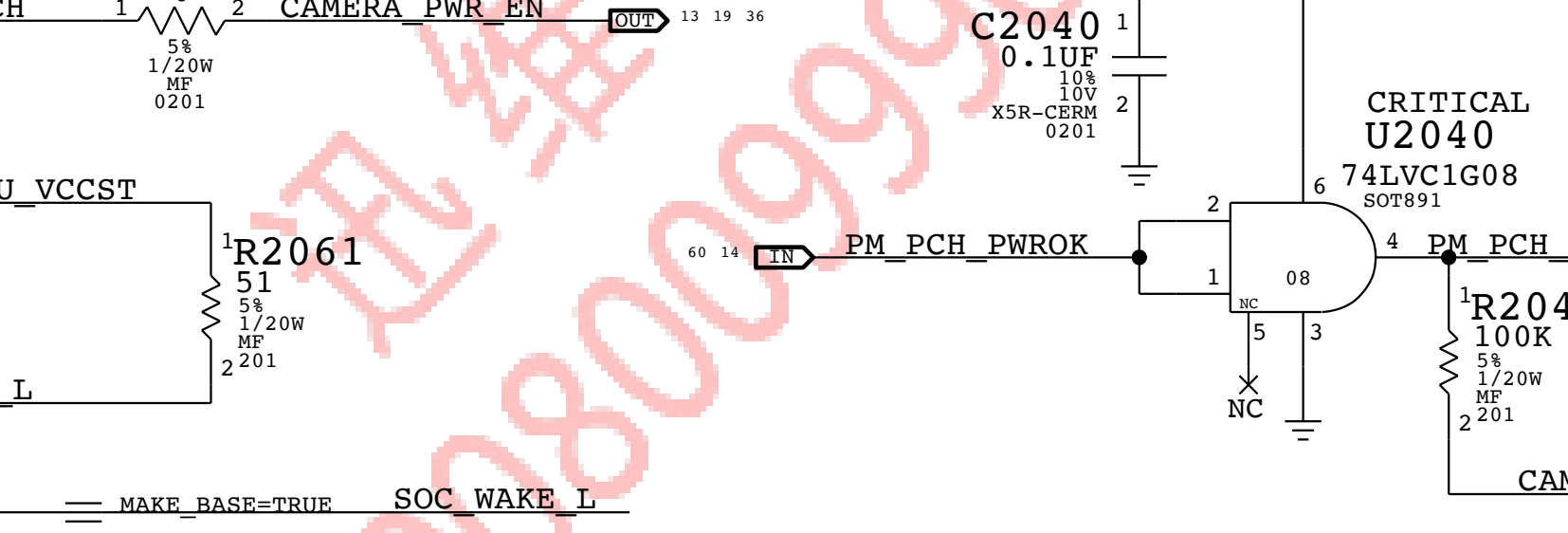
PROBE POINTS



UNUSED GPIO SIGNALS



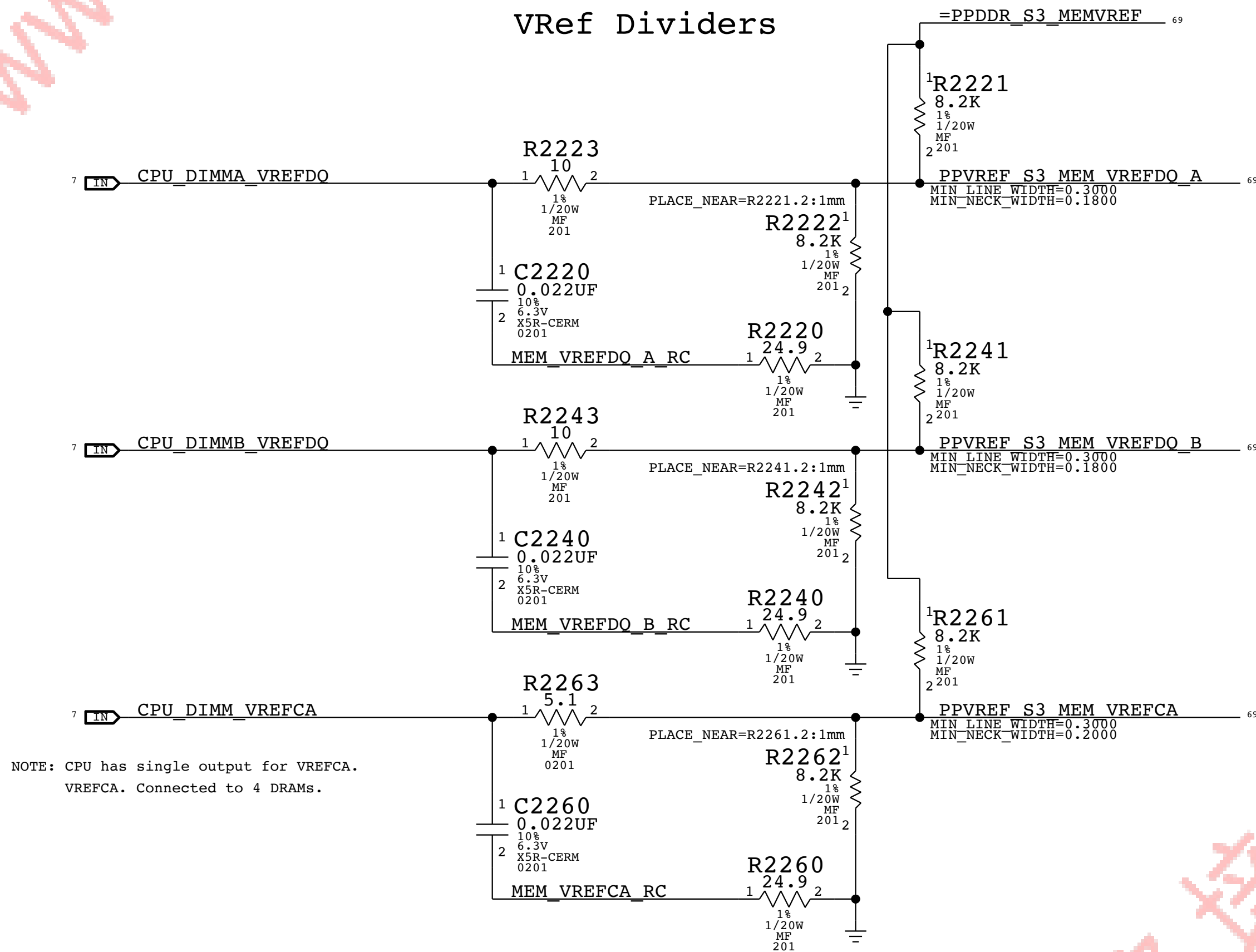
UNUSED PCH XDP SIGNALS NEED TEST POINTS.



BOM\_COST\_GROUP=CPU & CHIPSET

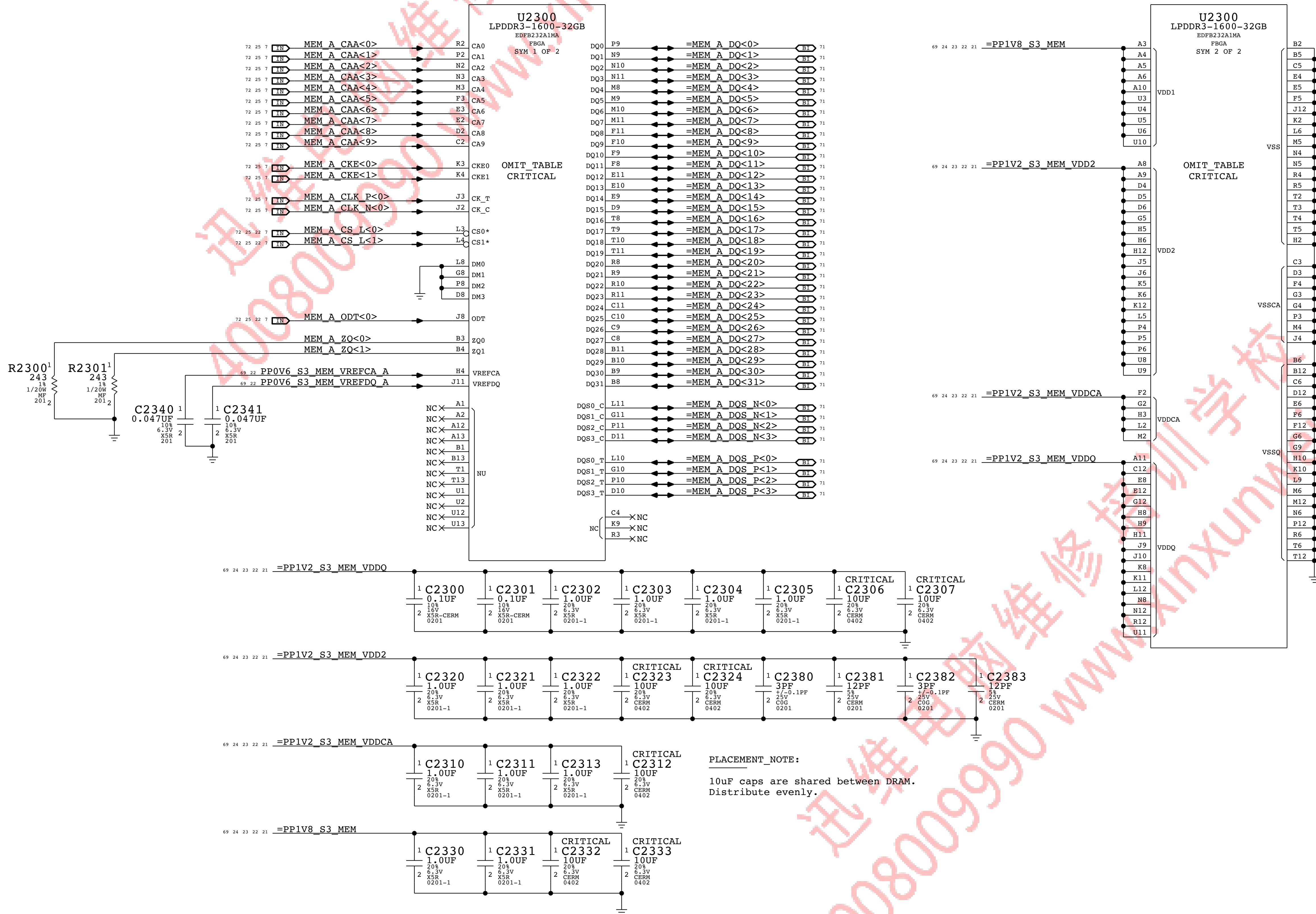
|                                                                                                                    |  |                          |
|--------------------------------------------------------------------------------------------------------------------|--|--------------------------|
| DESIGN: X502/MLB CATZ                                                                                              |  |                          |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                               |  |                          |
| PAGE TITLE                                                                                                         |  |                          |
| Chipset Support 2                                                                                                  |  |                          |
| Apple Inc.                                                                                                         |  | DRAWING NUMBER 051-02265 |
| REVISION 1.0.0                                                                                                     |  | SIZE D                   |
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CPU-Based Margining

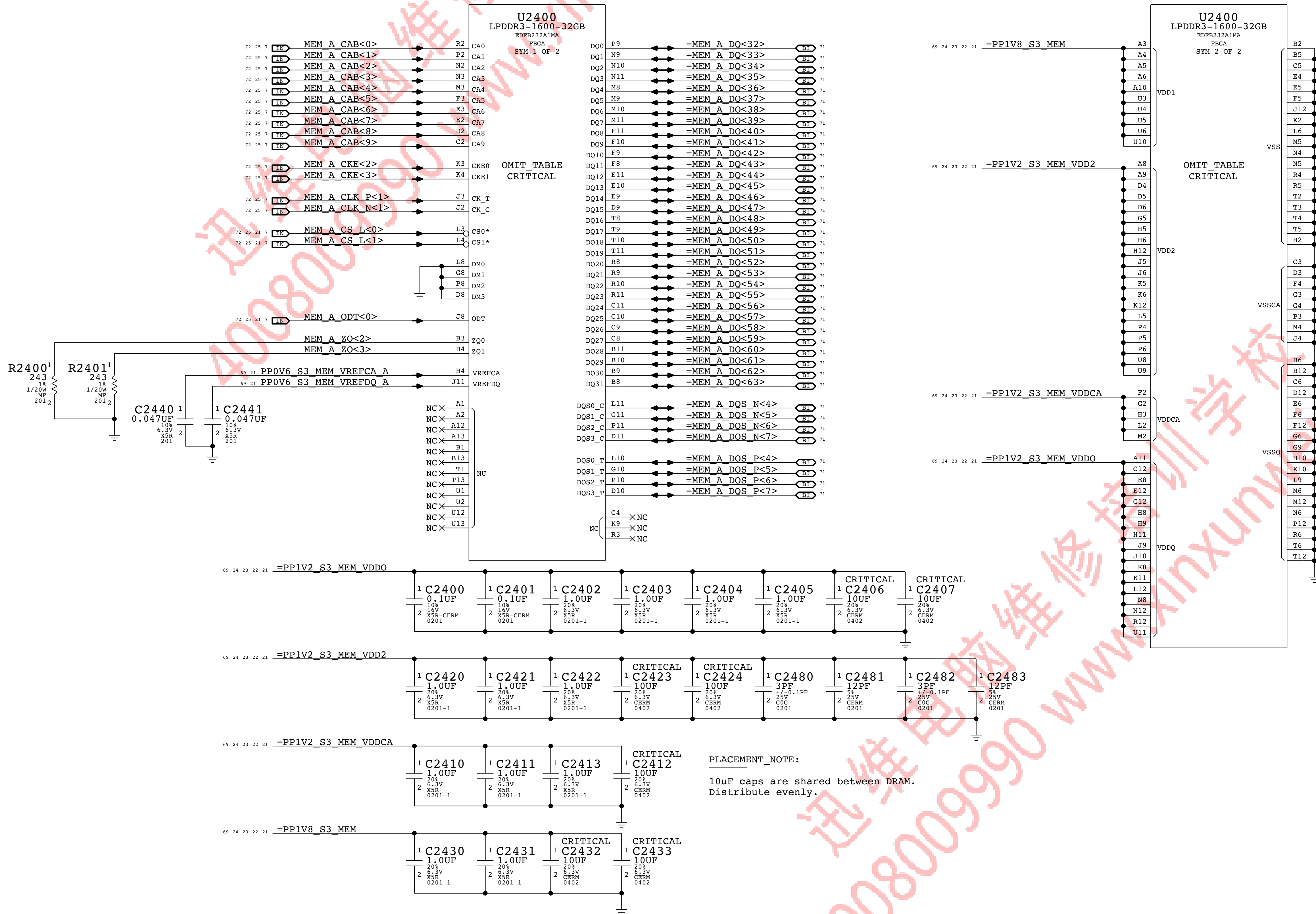


|                                                                                                                   |                |                      |           |
|-------------------------------------------------------------------------------------------------------------------|----------------|----------------------|-----------|
| SYNC MASTER=X502-EXP                                                                                              |                | SYNC DATE=12/03/2015 |           |
| PAGE TITLE                                                                                                        |                |                      |           |
| LPDDR3 VREF Margining                                                                                             |                |                      |           |
|                              | DRAWING NUMBER | 051-02265            | SIZE<br>D |
|                                                                                                                   | REVISION       | 1.0.0                |           |
|                                                                                                                   | BRANCH         |                      |           |
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| PAGE                                                                                                              | 22 OF 500      |                      |           |
| SHEET                                                                                                             | 20 OF 73       |                      |           |

# LPDDR3 CHANNEL A (0-31)



# LPDDR3 CHANNEL A (32-63)

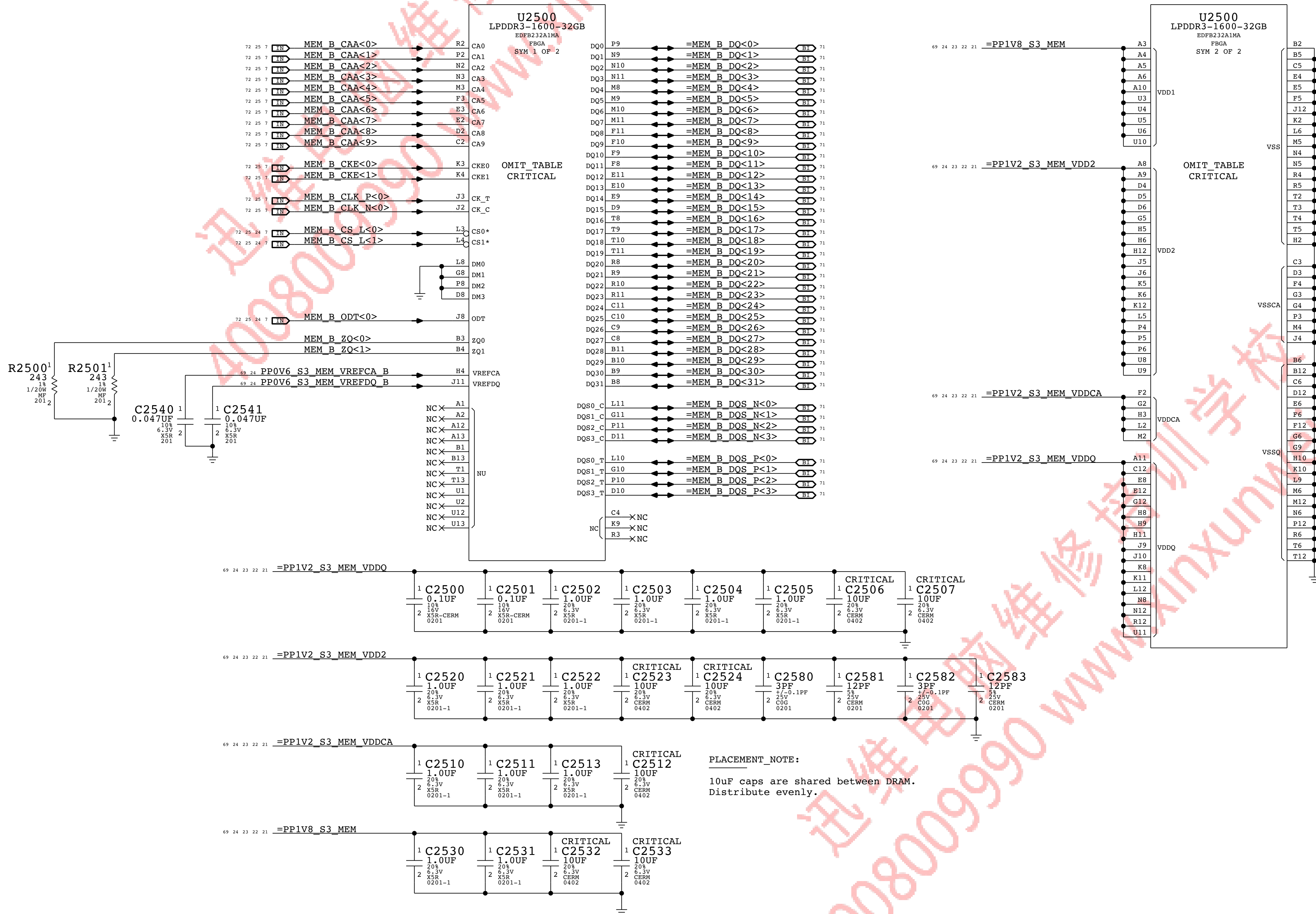


PLACEMENT NOTE:  
10uF caps are shared between DRAM.  
Distribute evenly.

|                                                                                                                   |                |                      |       |
|-------------------------------------------------------------------------------------------------------------------|----------------|----------------------|-------|
| SYNC_MASTER=X502-EXP                                                                                              |                | SYNC_DATE=12/03/2015 |       |
| PAGE TITLE                                                                                                        |                |                      |       |
| LPDDR3 DRAM Channel A                                                                                             |                | (32-63)              |       |
|                              | DRAWING NUMBER |                      | SIZE  |
|                                                                                                                   | 051-02265      |                      | D     |
|                                                                                                                   | REVISION       |                      | 1.0.0 |
|                                                                                                                   | BRANCH         |                      |       |
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BOM\_COST\_GROUP=DRAM

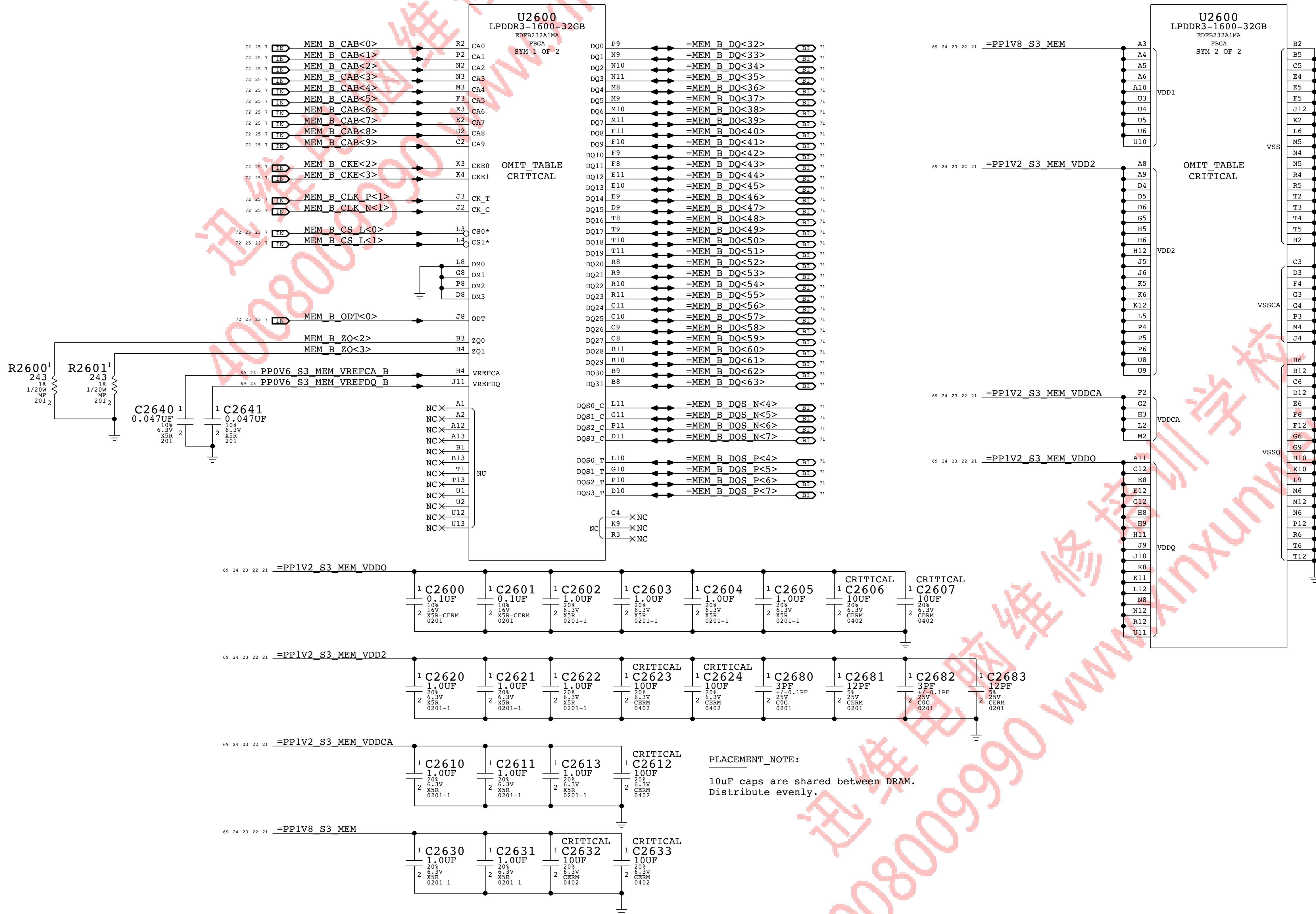
# LPDDR3 CHANNEL B (0-31)



BOM\_COST\_GROUP=DRAM

|                                                                                                                   |                |                      |      |
|-------------------------------------------------------------------------------------------------------------------|----------------|----------------------|------|
| SYNC MASTER=X502-EXP                                                                                              |                | SYNC DATE=12/03/2015 |      |
| PAGE TITLE                                                                                                        |                |                      |      |
| LPDDR3 DRAM Channel B                                                                                             |                | (00-31)              |      |
|                              | DRAWING NUMBER | 051-02265            | SIZE |
|                                                                                                                   | REVISION       | 1.0.0                | D    |
|                                                                                                                   | BRANCH         |                      |      |
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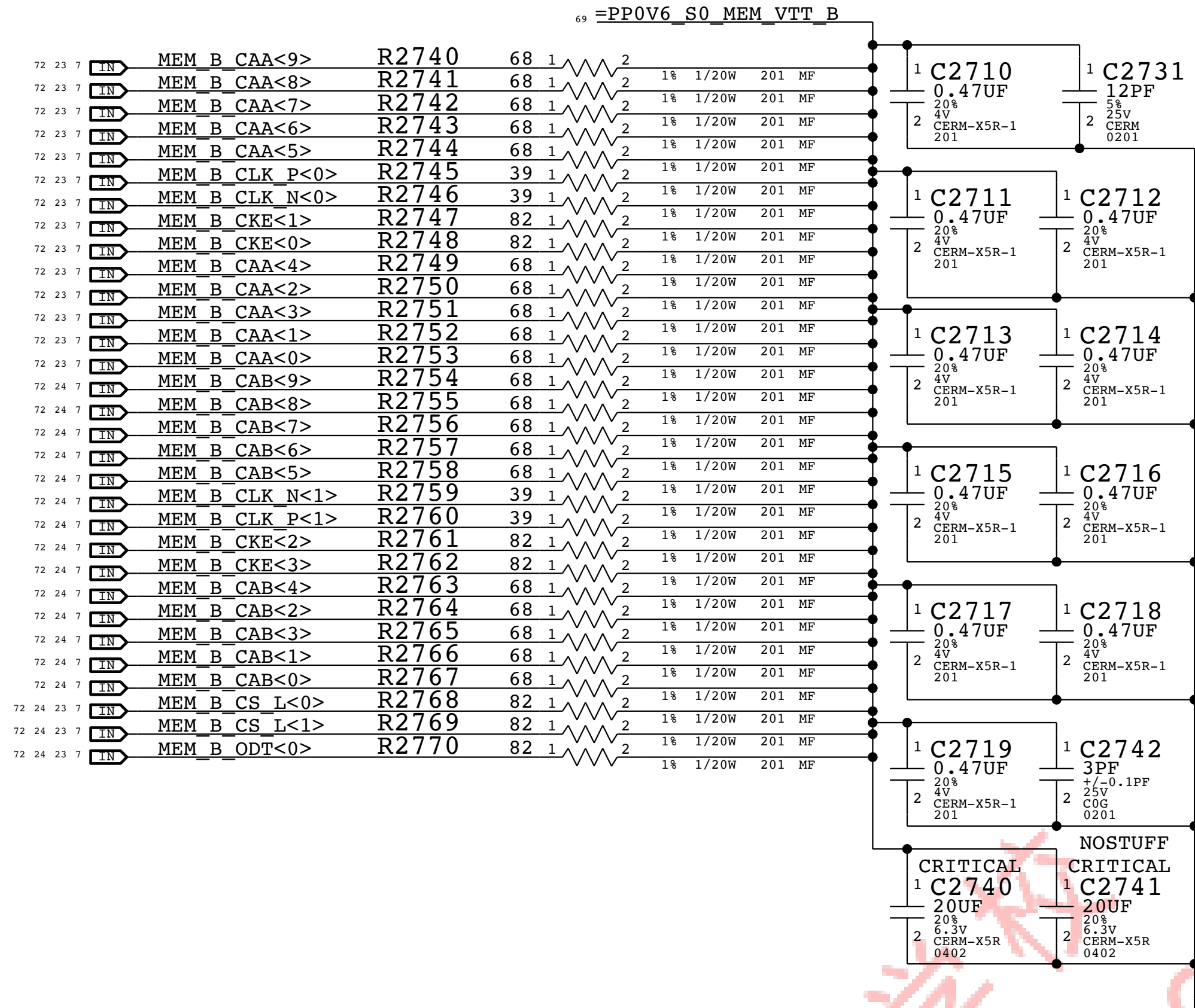
# LPDDR3 CHANNEL B (32-63)



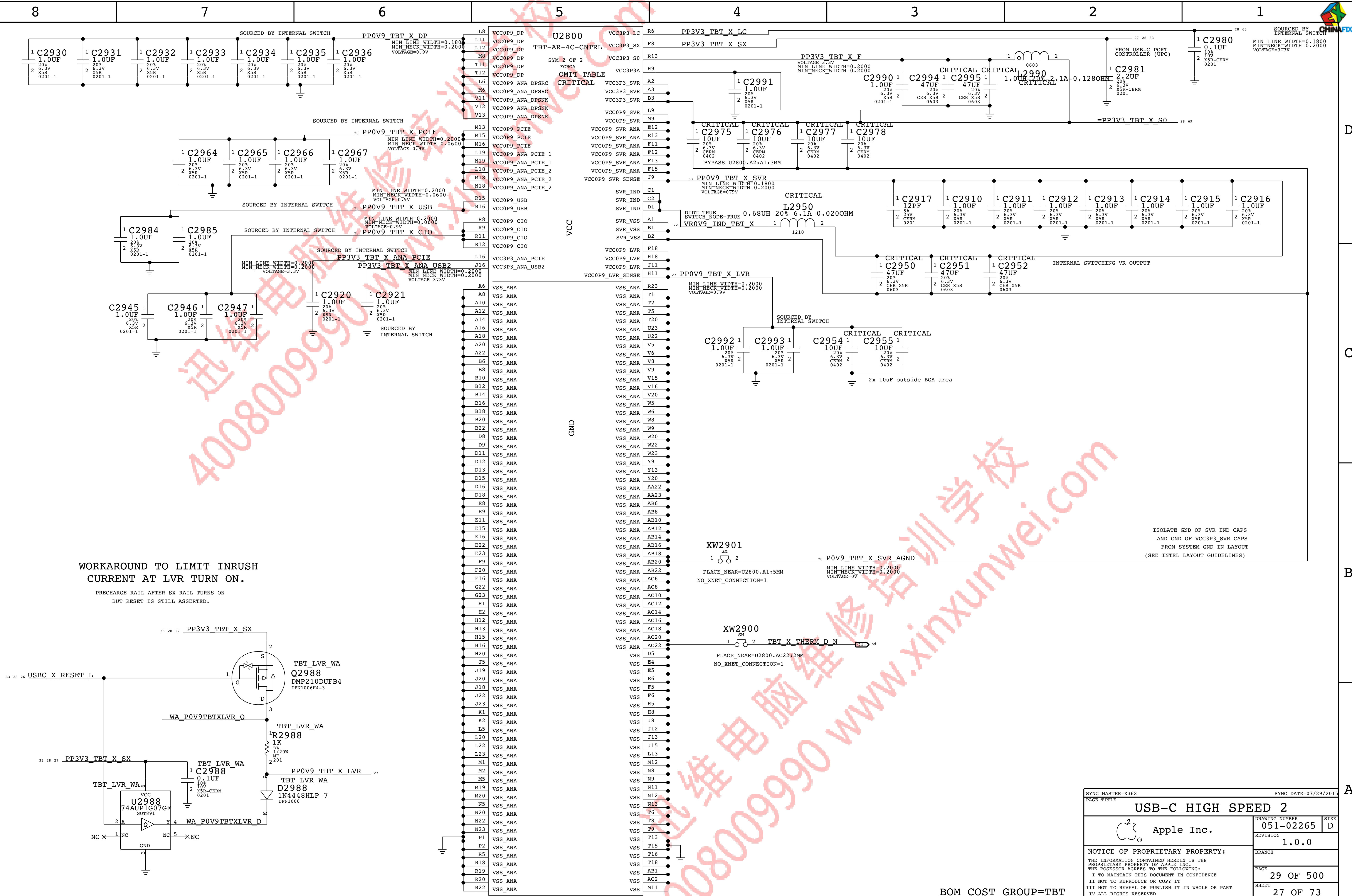
PLACEMENT NOTE:  
10uF caps are shared between DRAM.  
Distribute evenly.

|                                                                                                                                                                                                                                                                                                                         |                             |                      |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------|-----------|
| SYNC MASTER=X502-EXP                                                                                                                                                                                                                                                                                                    |                             | SYNC DATE=12/03/2015 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                              |                             |                      |           |
| LPDDR3 DRAM Channel B (32-63)                                                                                                                                                                                                                                                                                           |                             |                      |           |
|  Apple Inc.®                                                                                                                                                                                                                       | DRAWING NUMBER<br>051-02265 |                      | SIZE<br>D |
|                                                                                                                                                                                                                                                                                                                         | REVISION<br>1.0.0           |                      |           |
|                                                                                                                                                                                                                                                                                                                         | BRANCH                      |                      |           |
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| PAGE                                                                                                                                                                                                                                                                                                                    |                             | 26 OF 500            |           |
| SHEET                                                                                                                                                                                                                                                                                                                   |                             | 24 OF 73             |           |

BOM\_COST\_GROUP=DRAM

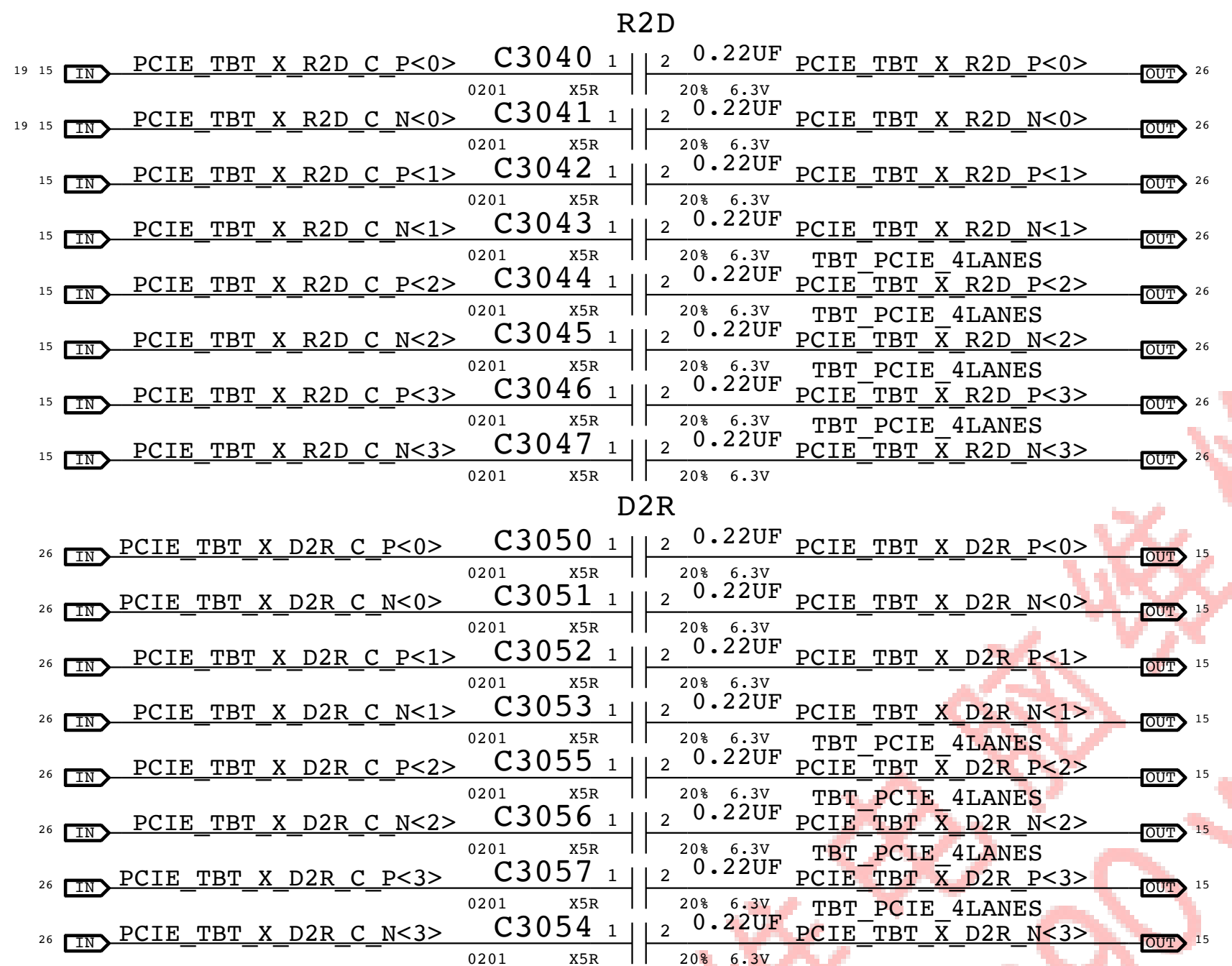






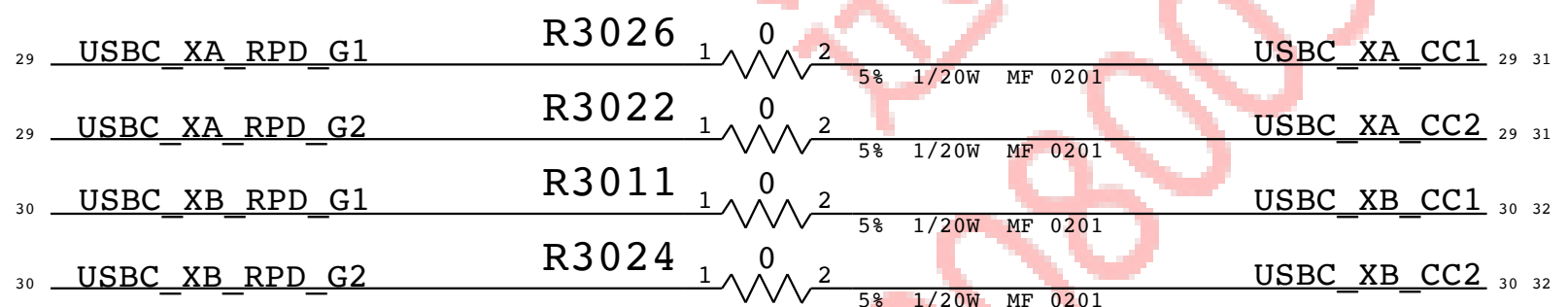


## RIDGE AC COUPLING

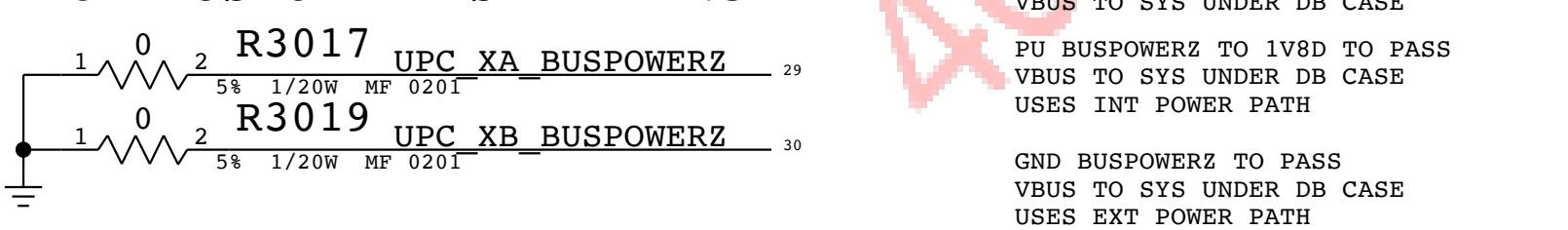


## ACE RPD STRAPPING

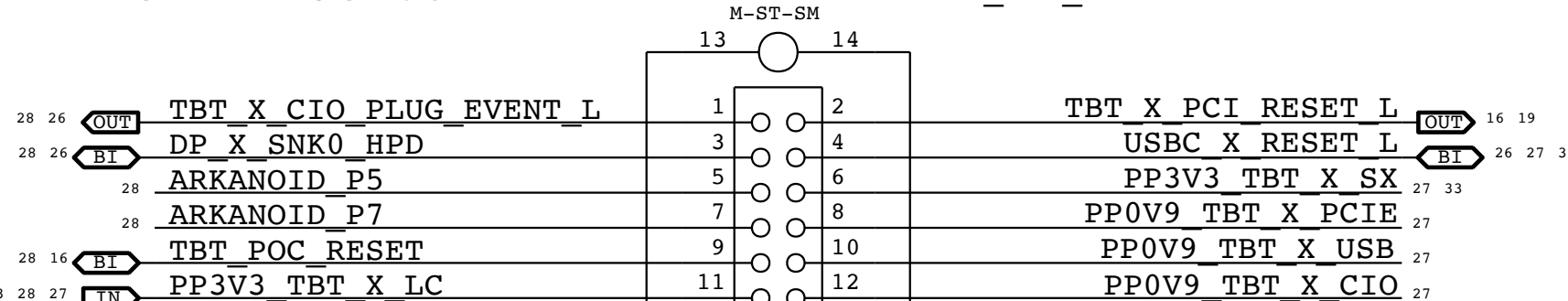
CONNECT G1/G2 TO CC1/CC2 TO RECEIVE POWER UNDER DB CASE  
CONNECT G1/G2 TO GND TO NOT RECEIVE POWER UNDER DB CASE



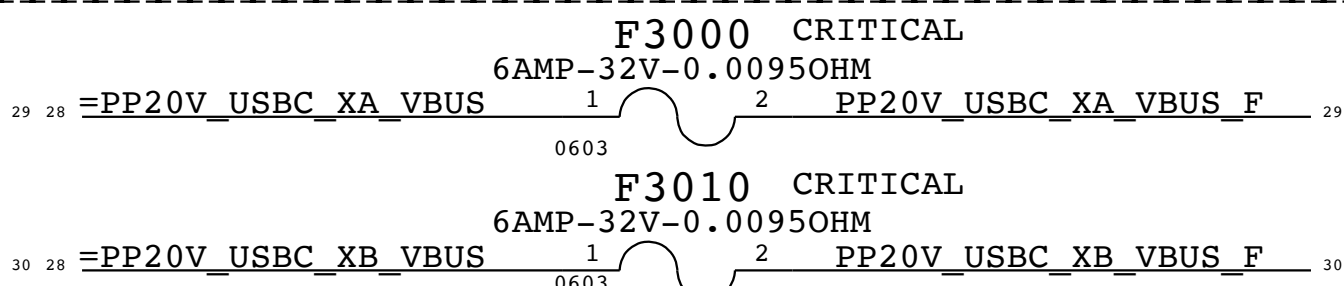
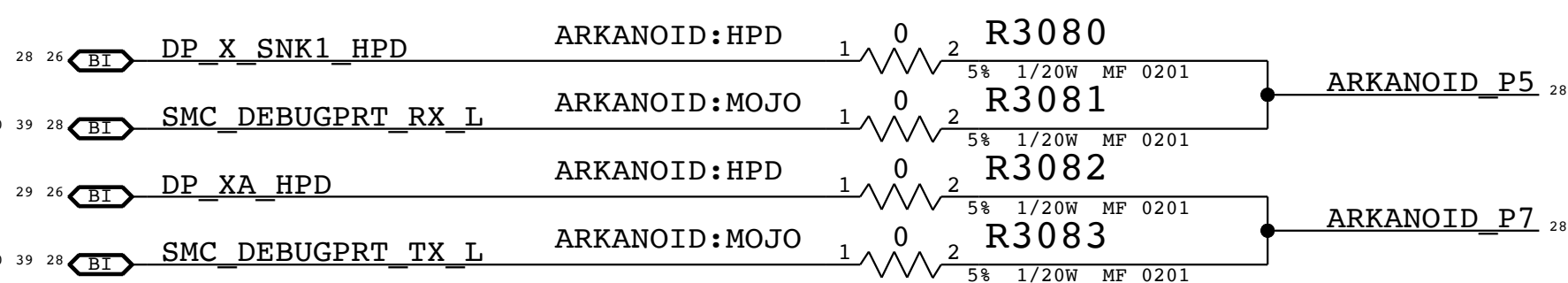
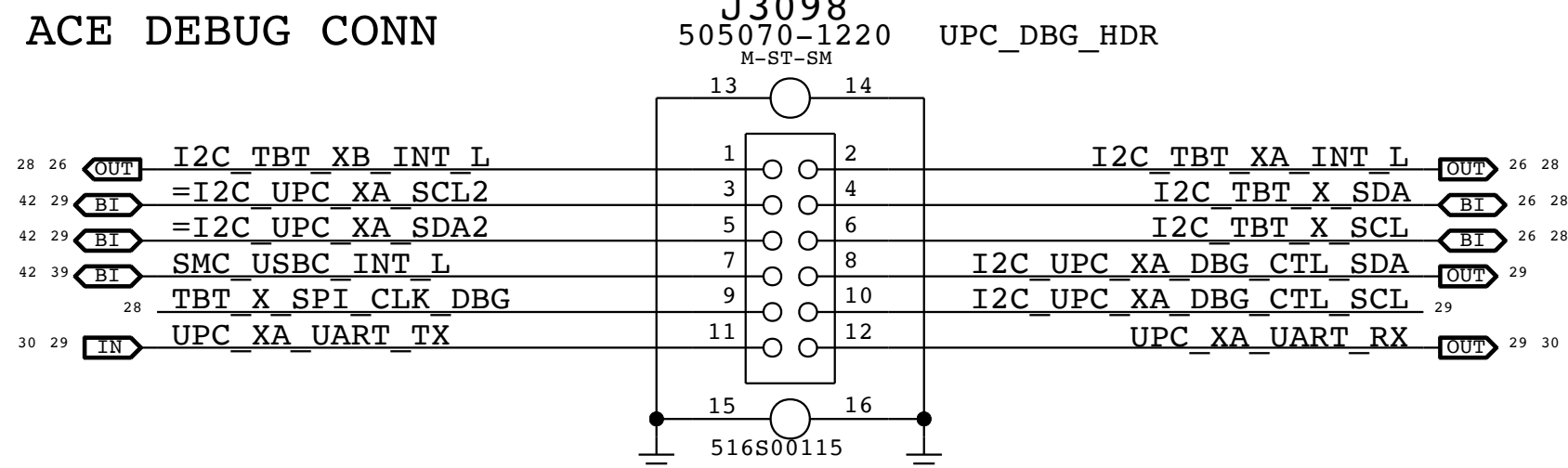
## ACE BUSPOWERZ STRAPPING



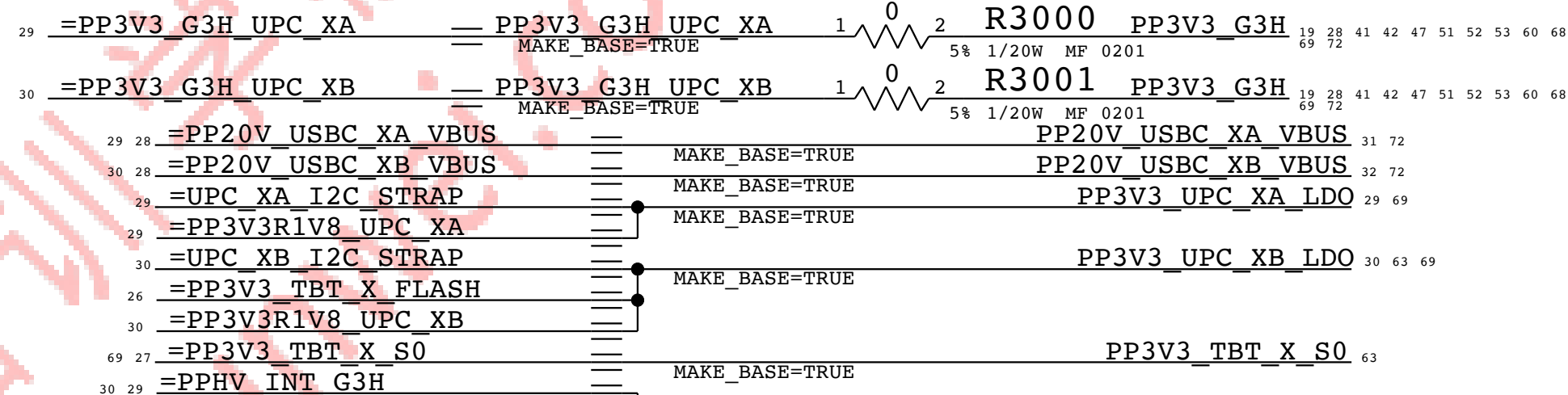
## RIDGE DEBUG CONN



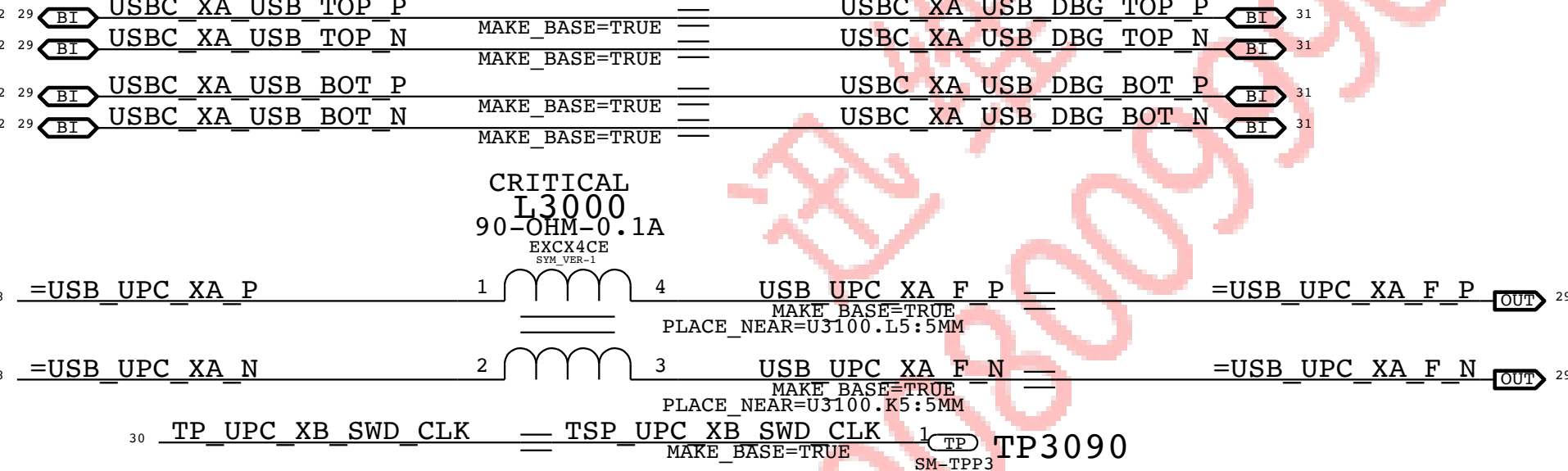
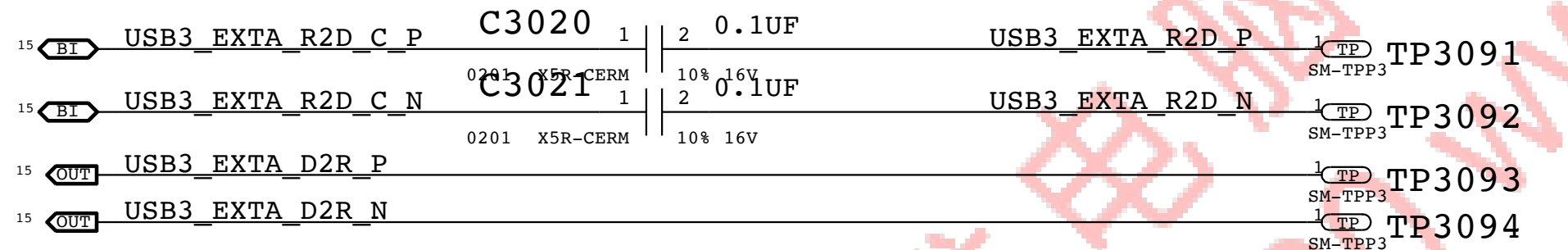
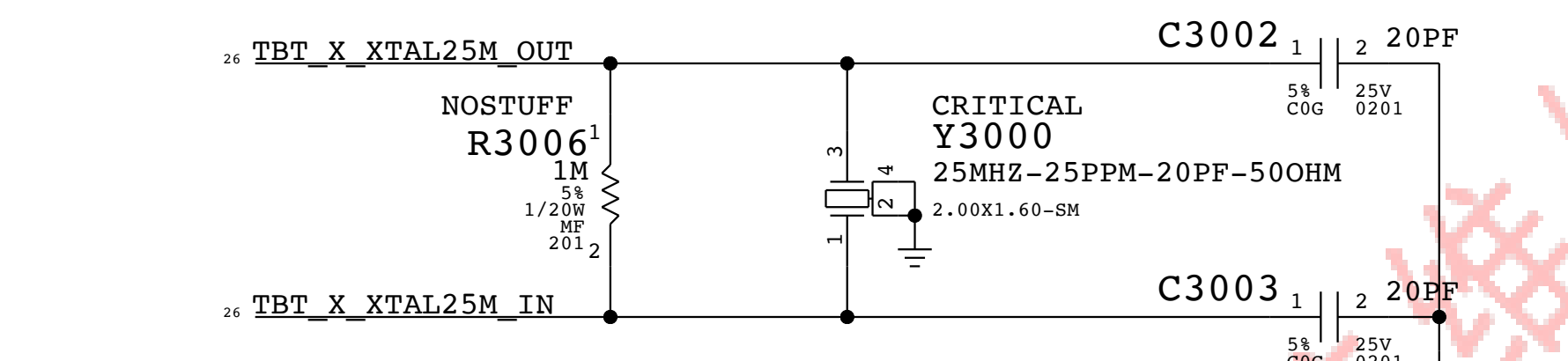
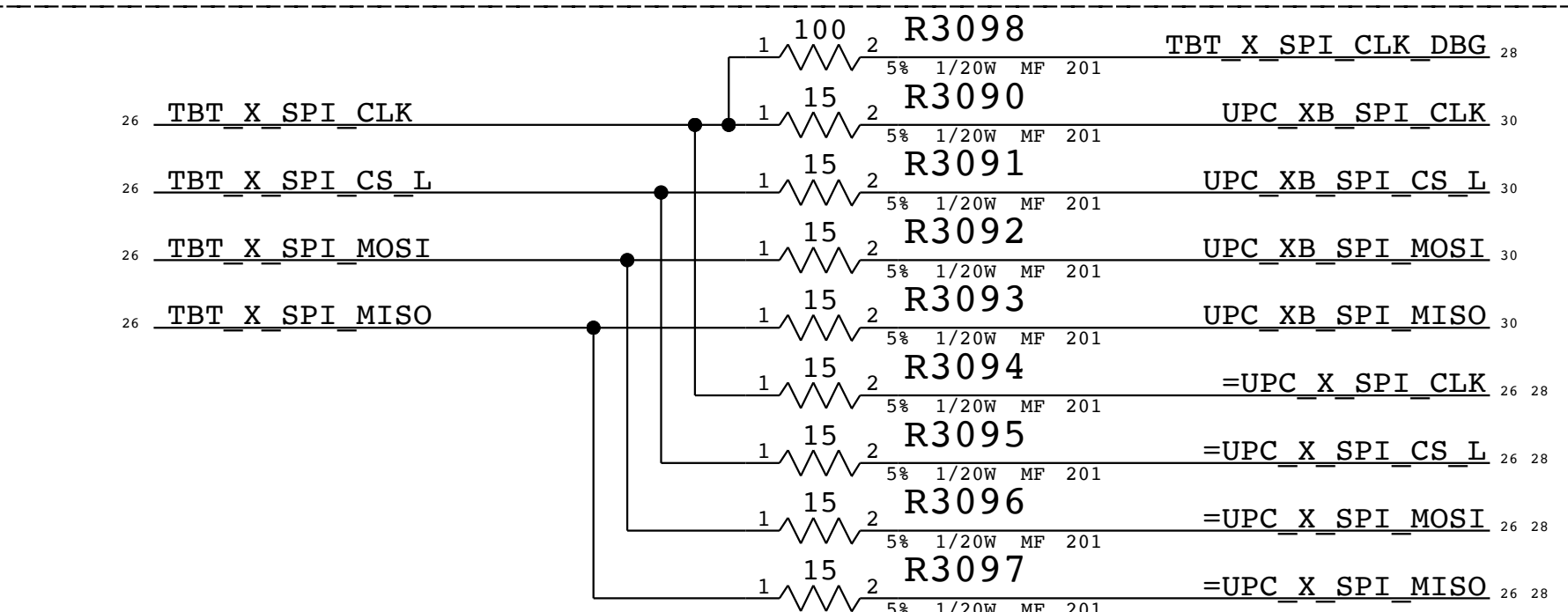
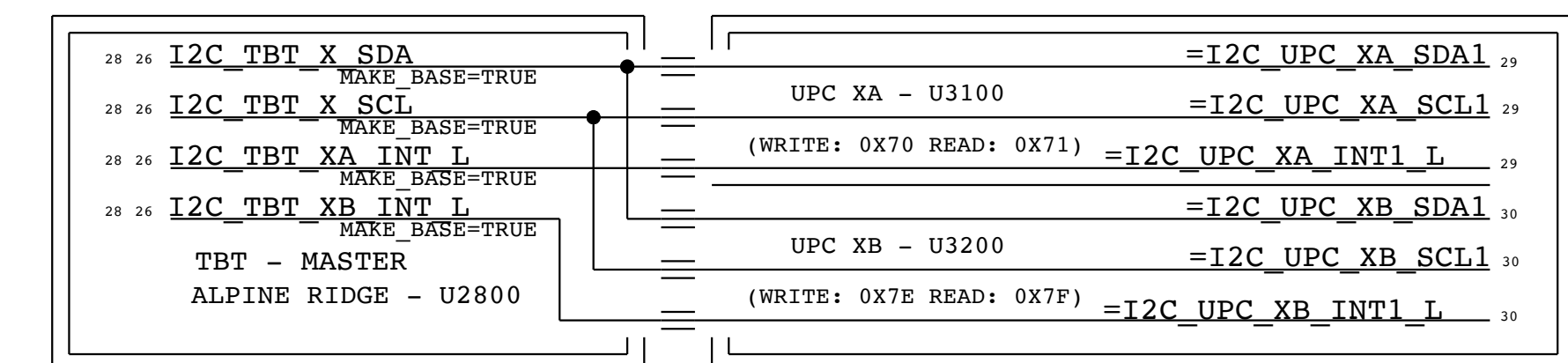
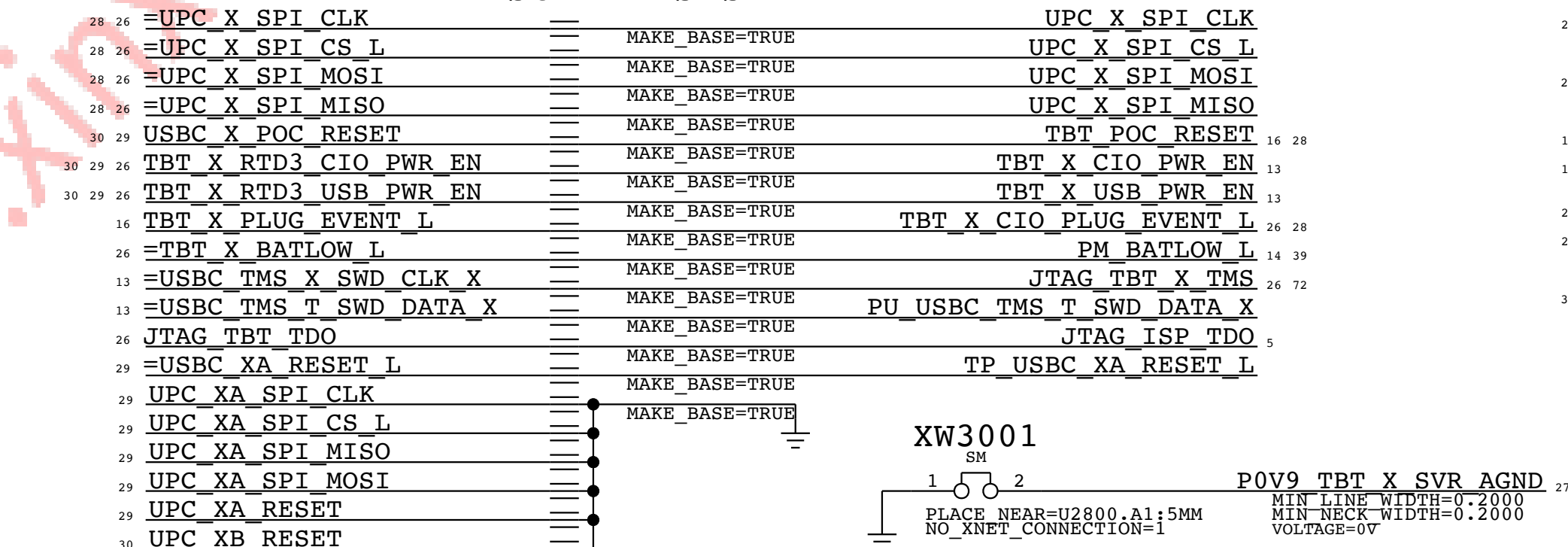
## ACE DEBUG CONN



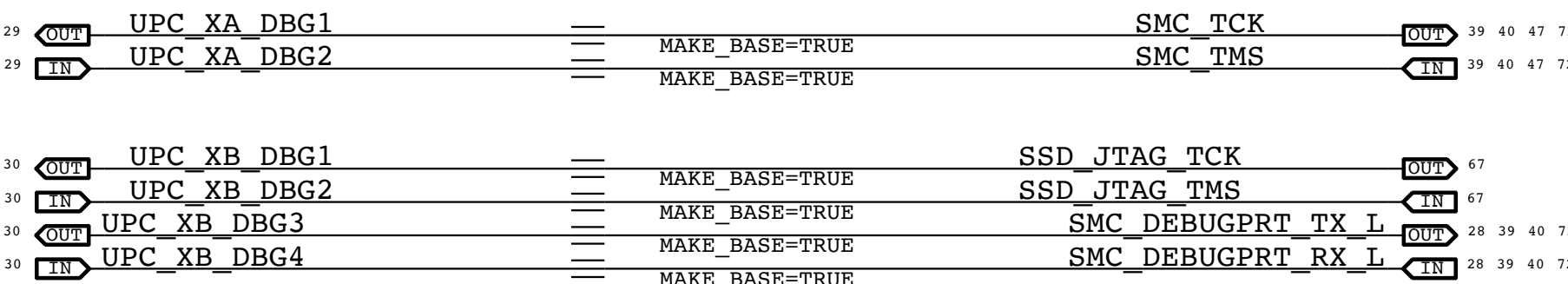
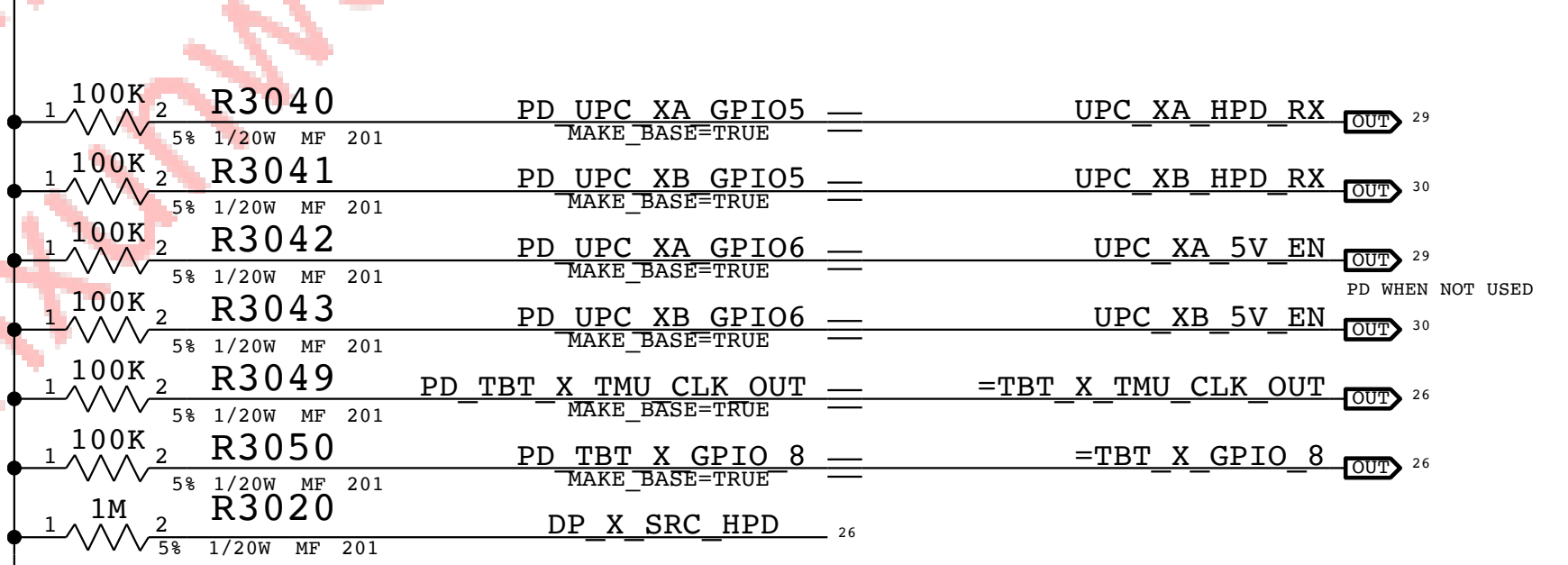
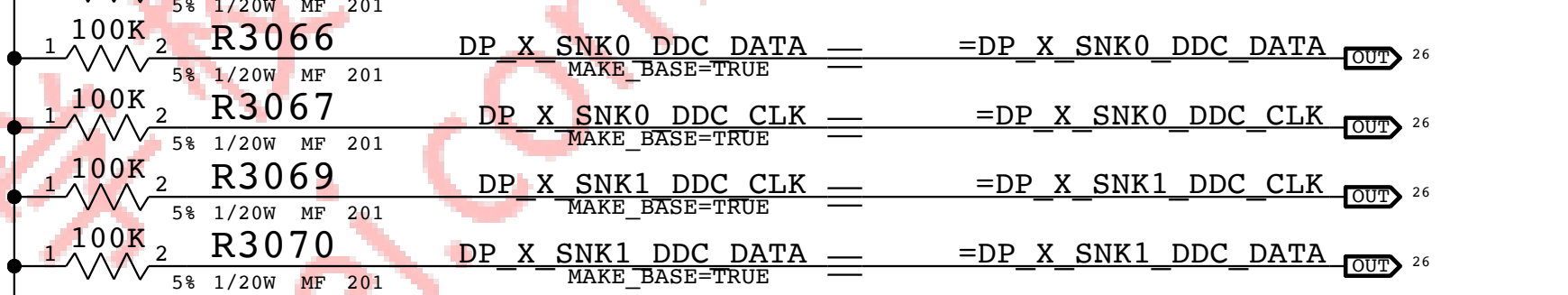
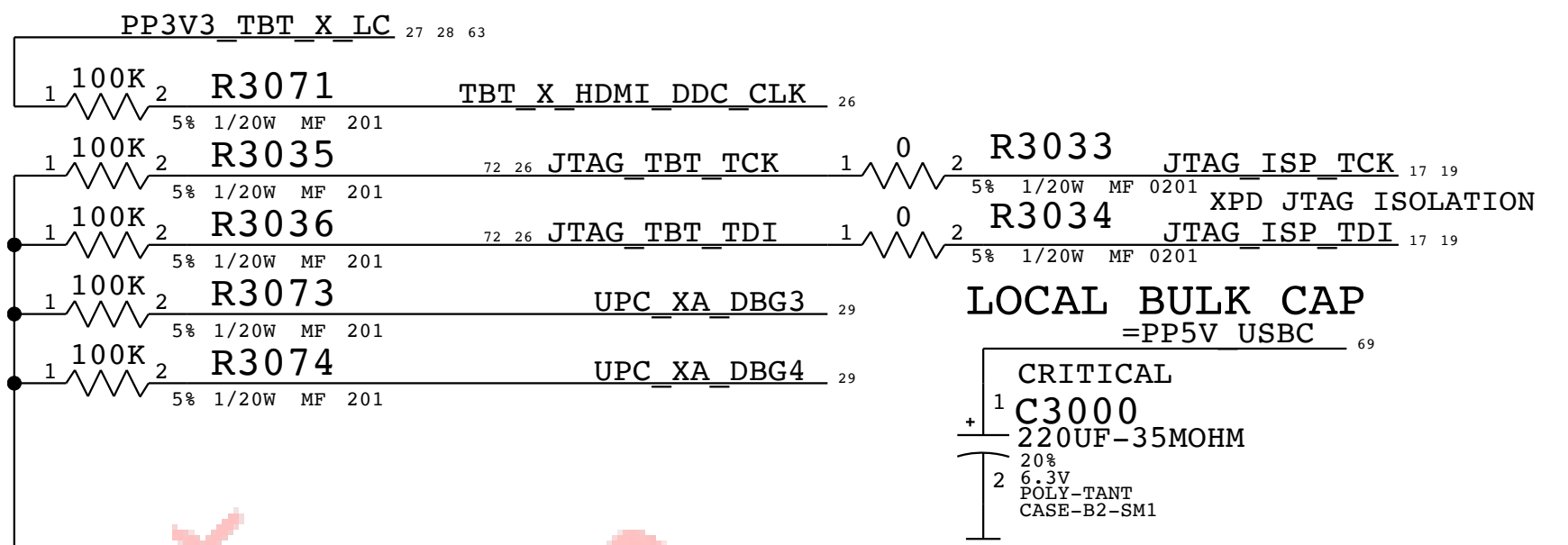
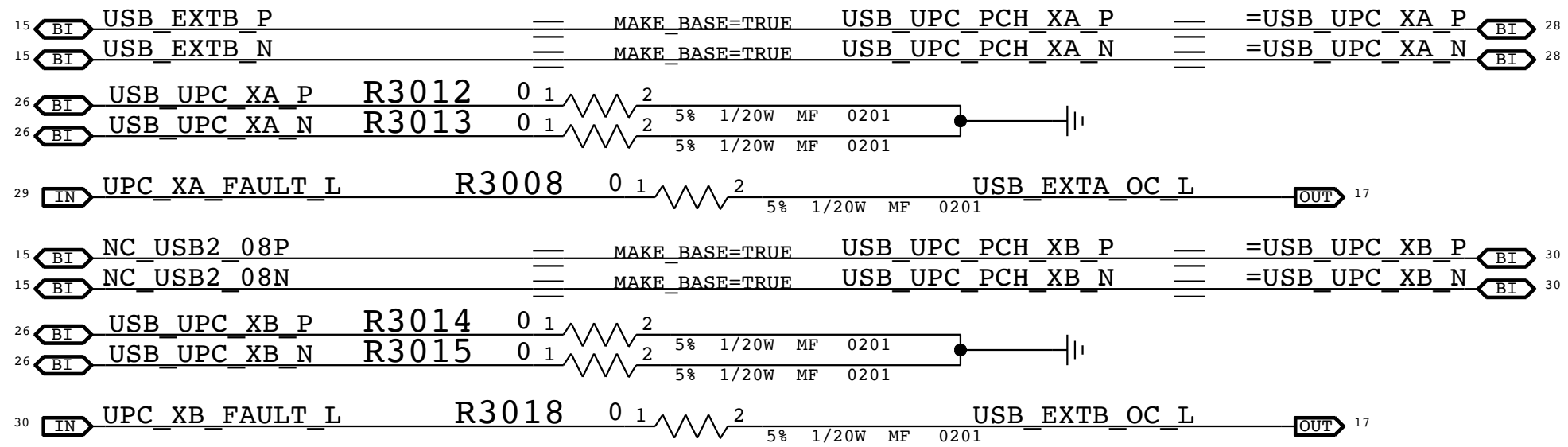
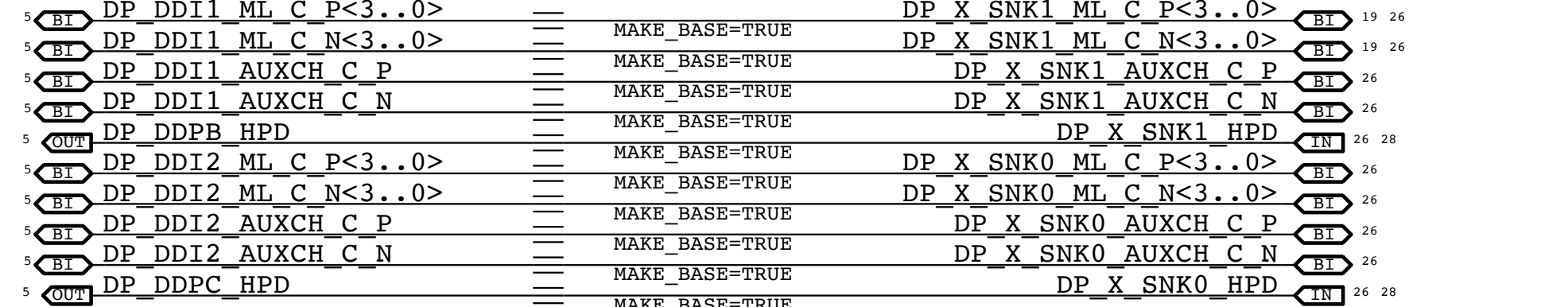
## POWER ALIASES



## MISC ALIASES



## DP / USB SOURCE ALIASES

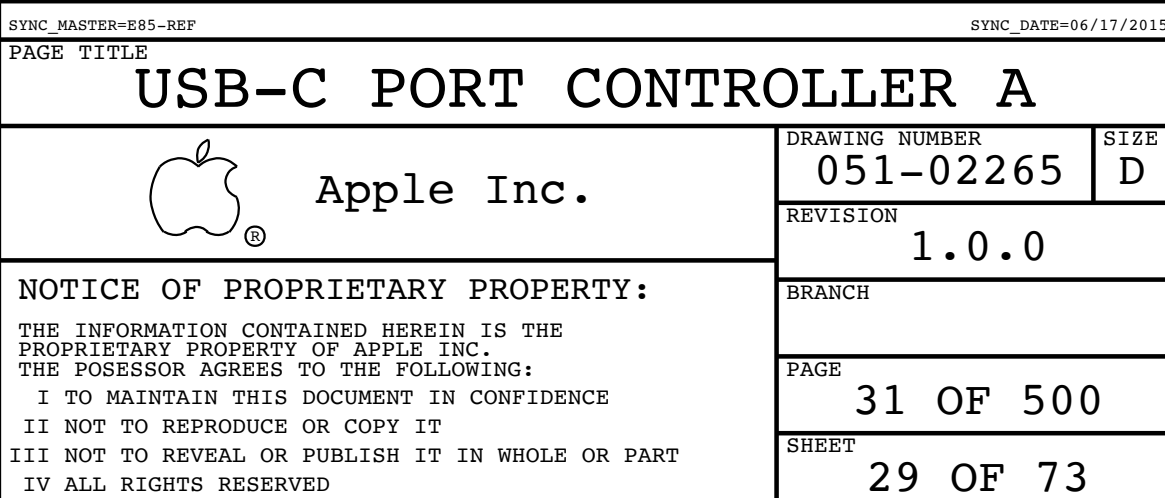


DESIGN: X502/MLB CATZ  
LAST CHANGE: Fri Aug 5 13:34:33 2016

| PAGE TITLE                                                                                                                                                                                                                                                                                                         |                |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| USB-C SUPPORT                                                                                                                                                                                                                                                                                                      |                |           |
|                                                                                                                                                                                                                                                                                                                    | DRAWING NUMBER | 051-02265 |
|                                                                                                                                                                                                                                                                                                                    | REVISION       | 1.0.0     |
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BOM\_COST\_GROUP=USB-C

## BOM COST GROUP=USB-C



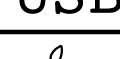
SECONDARY ACE USB-C PORT CONTROLLER (UPC)

PULL R3209 AND R3208 UP TO ACES\_LDOs FOR 1ST RIDGE'S ACES  
PULL R3209 AND R3208 DOWN TO GND FOR 2ND RIDGE'S ACES

TESTPOINTS MUST BE PRESENT FOR GPIO0, GPIO1 (EVEN IN PRODUCTION)  
USE GPIO2 FOR USB-C ANALOG AUDIO SUPPORT ON DESIGNS WITHOUT AN AUDIO JACK CONNECTOR  
USE GPIO3 FOR POWER GATE EN ON BANSURI DESIGNS

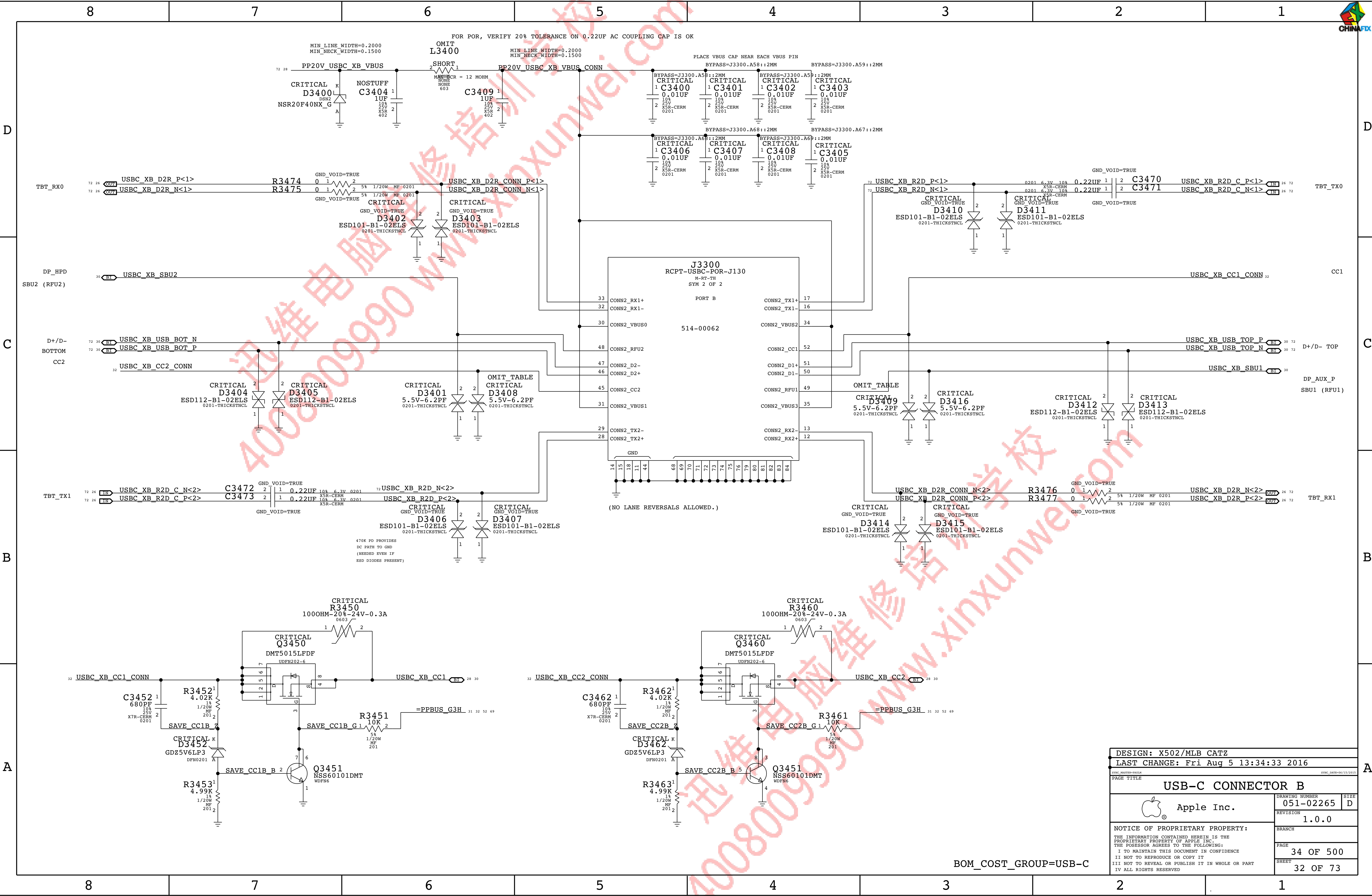
PU to PP3V3\_S4 if convenient for layout  
Otherwise PU to PP3V3\_UPC\_XA\_LDO

PIN D6 IS UNDOCUMENTED RESET  
CAN GROUND PIN D6 IN PRODUCTION

|                                                                                                                                                                                                                                                                                                                             |                |                      |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------|------|
| SYNC_MASTER=B85-REF                                                                                                                                                                                                                                                                                                         |                | SYNC_DATE=06/17/2015 |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |                |                      |      |
| USB-C PORT CONTROLLER B                                                                                                                                                                                                                                                                                                     |                |                      |      |
|  Apple Inc.                                                                                                                                                                                                                            | DRAWING NUMBER |                      | SIZE |
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BOM COST GROUP=USB-C

|                                                                                                                                                                                                                                                                                                                             |  |                     |  |           |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------|--|-----------|--|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                                       |  |                     |  |           |  |
| LAST CHANGE: Mon Aug 8 12:54:34 2016                                                                                                                                                                                                                                                                                        |  |                     |  |           |  |
| ERIC_MUZZEPA                                                                                                                                                                                                                                                                                                                |  | JERRY_HANSEN-17-049 |  |           |  |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |                     |  |           |  |
| <b>USB-C CONNECTOR A</b>                                                                                                                                                                                                                                                                                                    |  |                     |  |           |  |
|  Apple Inc.                                                                                                                                                                                                                            |  | DRAWING NUMBER      |  | SIZE      |  |
|                                                                                                                                                                                                                                                                                                                             |  | 051-02265           |  | D         |  |
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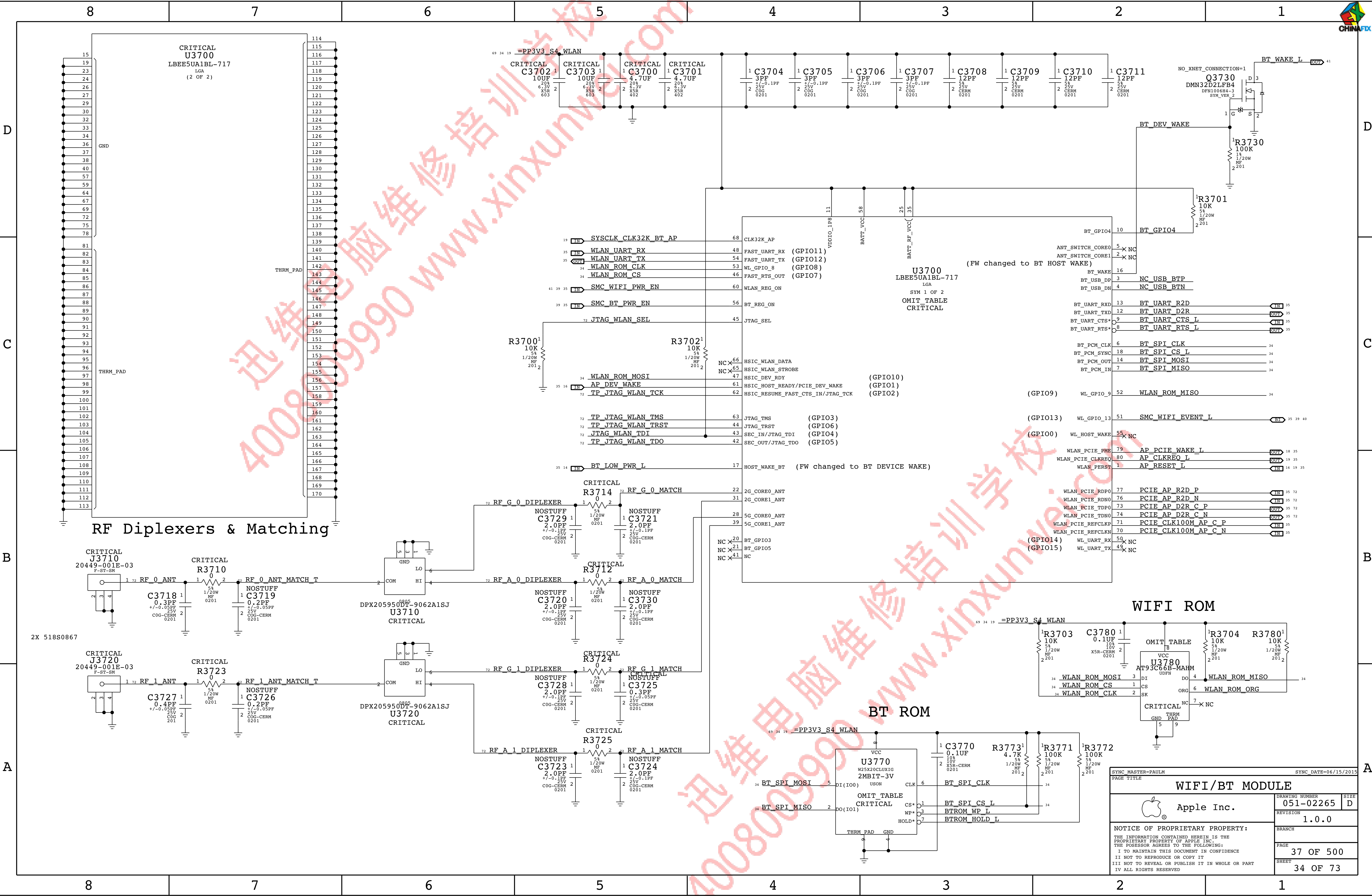


|                                                                                                                    |                      |           |
|--------------------------------------------------------------------------------------------------------------------|----------------------|-----------|
| DESIGN: X502/MLB CATZ                                                                                              |                      |           |
| LAST CHANGE: Fri Aug 5 13:34:33 2016                                                                               |                      |           |
| SYNCHMASTER:DAULD                                                                                                  | SYNCHDATE:06/15/2015 |           |
| PAGE TITLE                                                                                                         |                      |           |
| USB-C CONNECTOR B                                                                                                  |                      |           |
|  Apple Inc. ®                 | DRAWING NUMBER       | 051-02265 |
|                                                                                                                    | REVISION             | 1.0.0     |
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BOM\_COST\_GROUP=USB-C



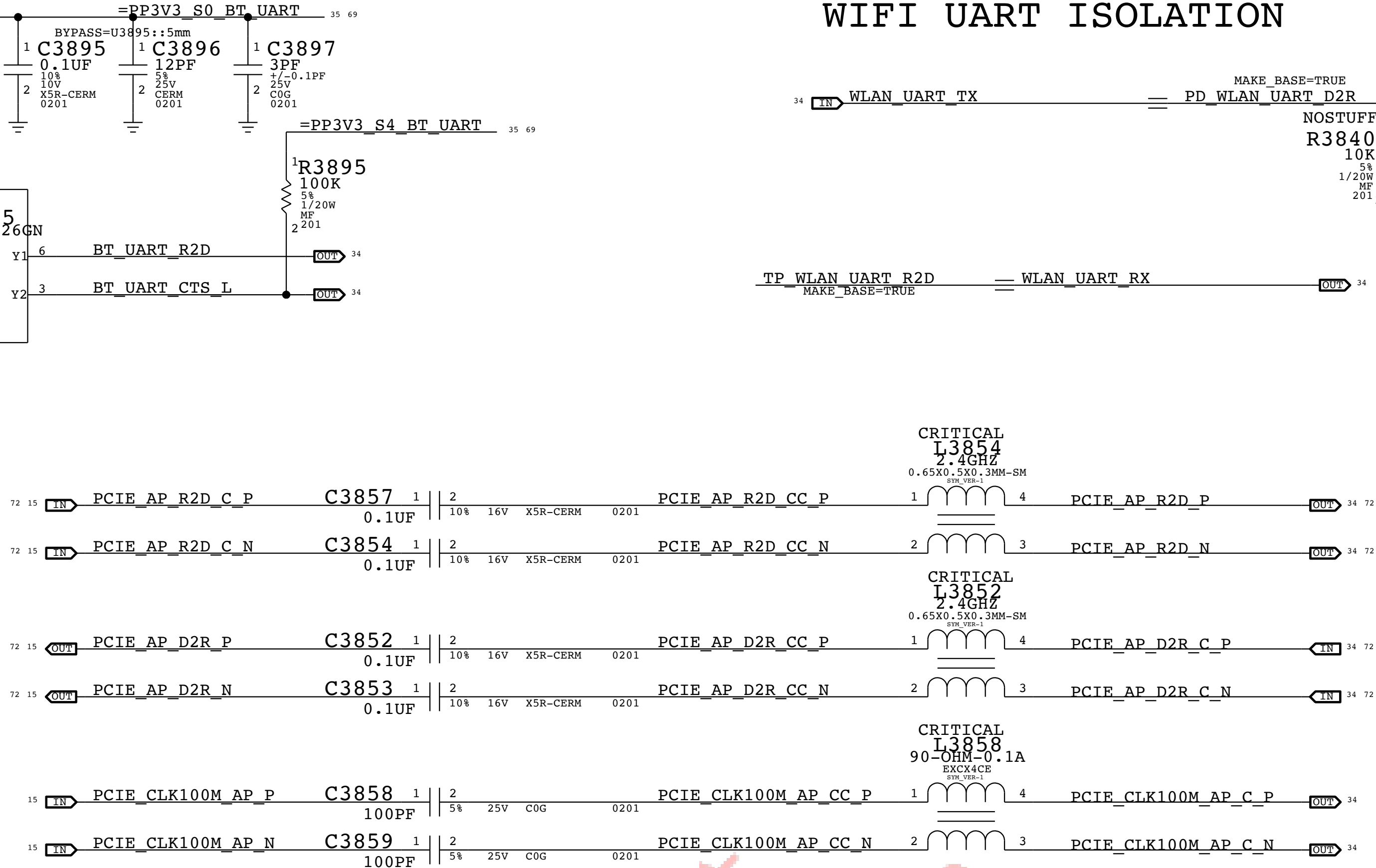
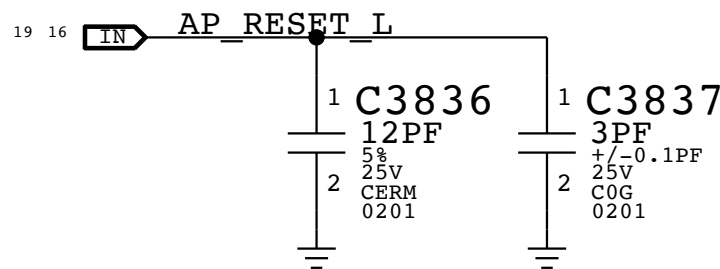
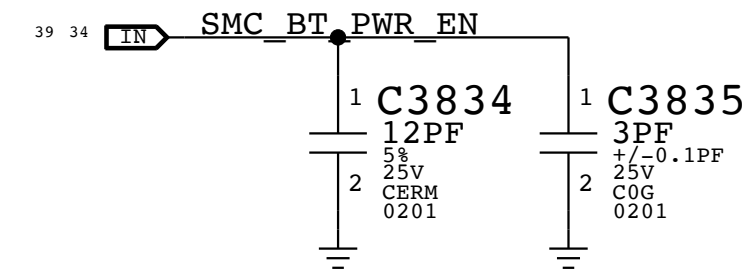
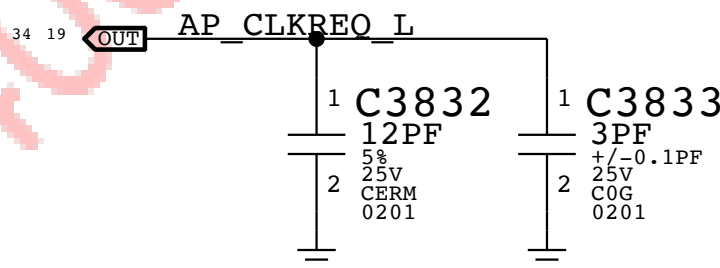
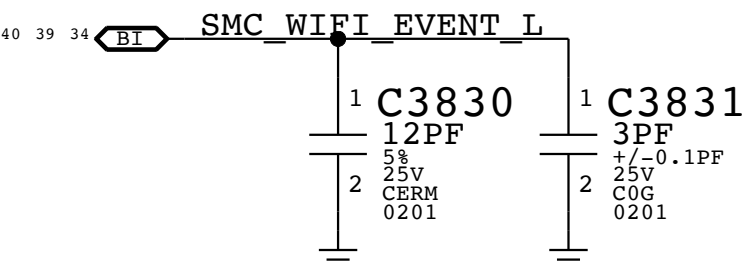
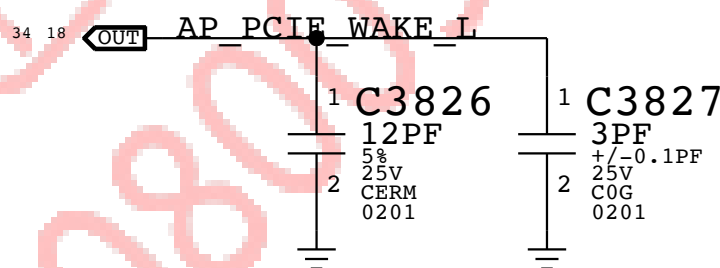
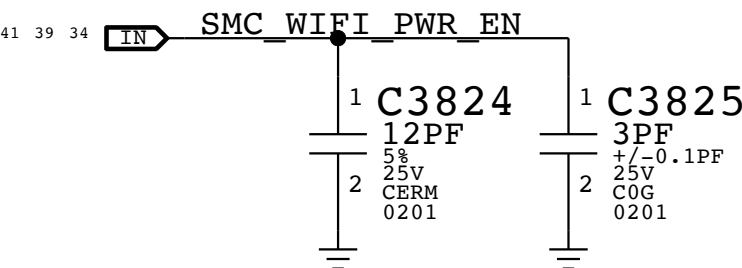
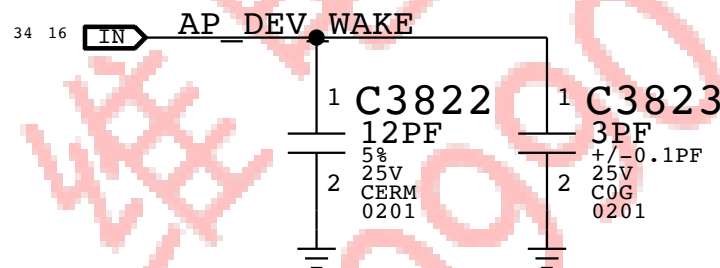
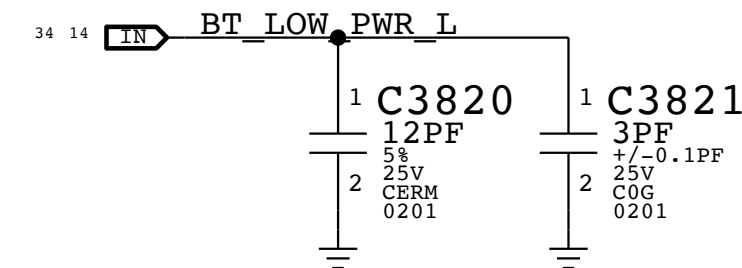
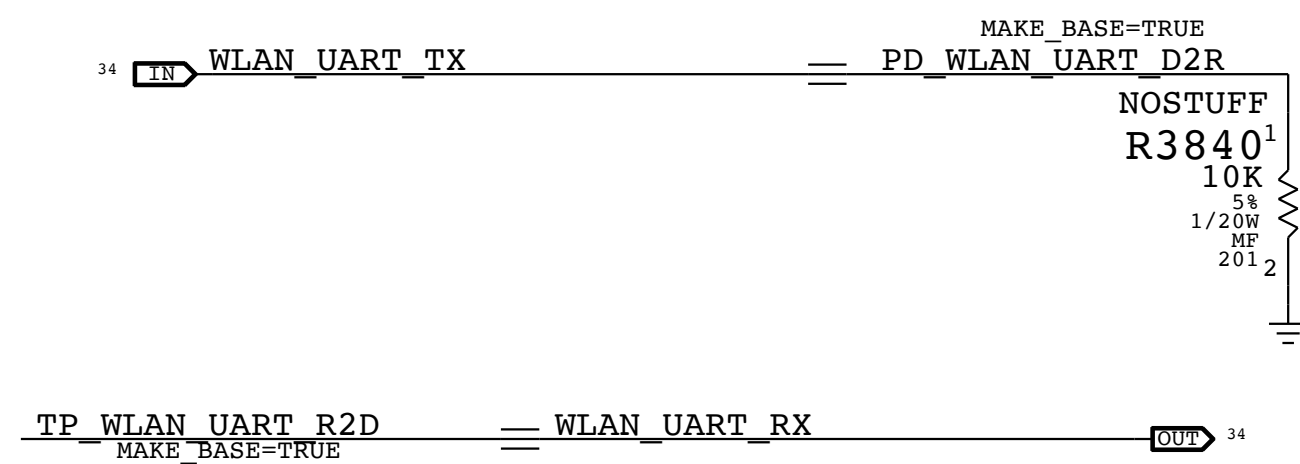
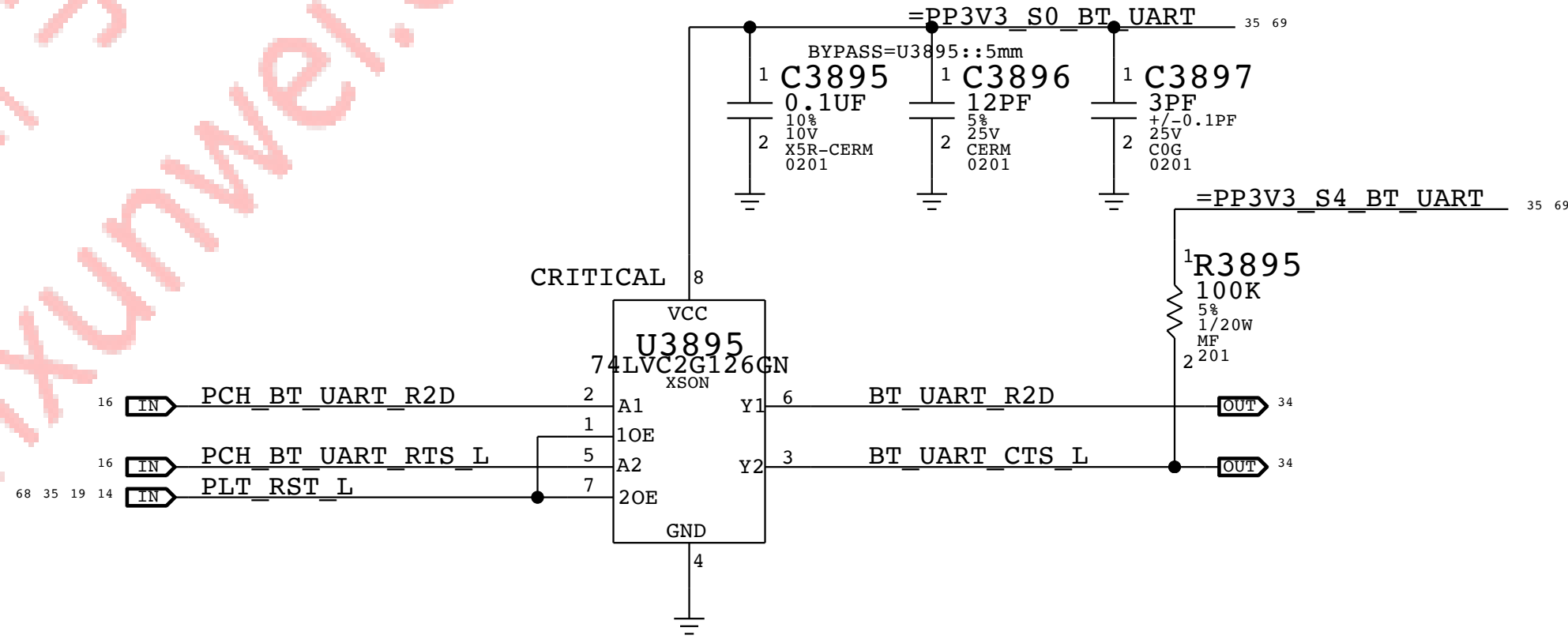
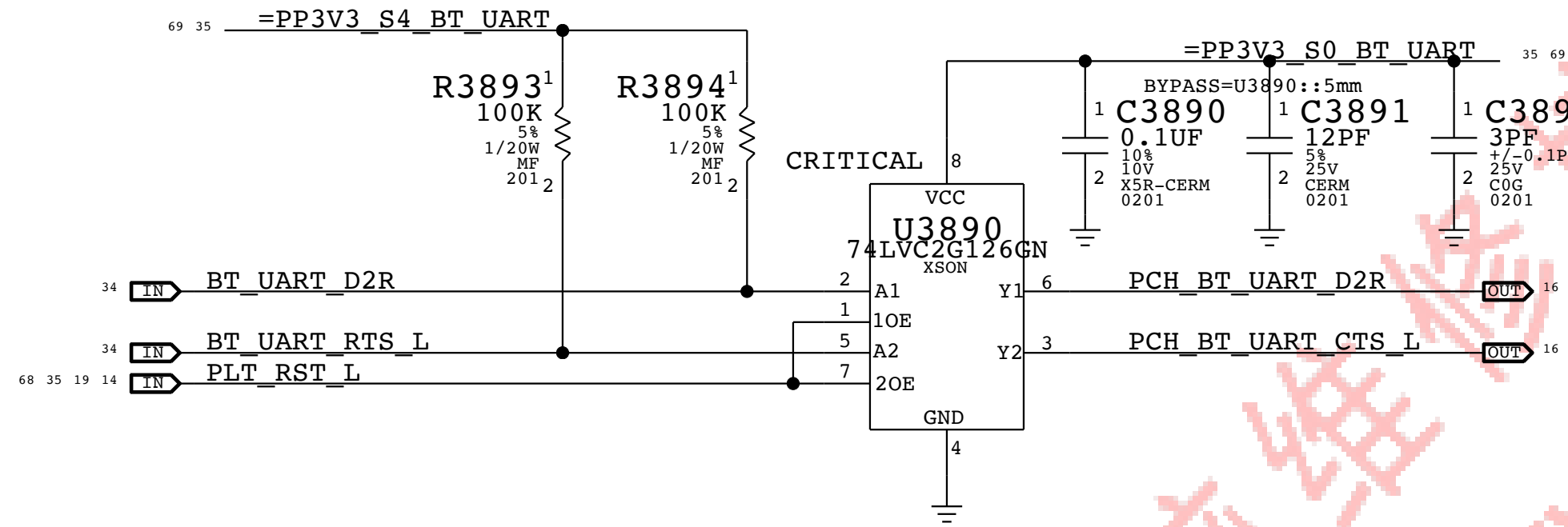
|                                                                                                                                                                                                                                                                                                                           |                |                 |      |
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| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                                     |                |                 |      |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                                                                                                                                                                                                                                      |                |                 |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                |                | USB-C SUPPORT 2 |      |
|  Apple Inc.                                                                                                                                                                                                                          | DRAWING NUMBER |                 | SIZE |
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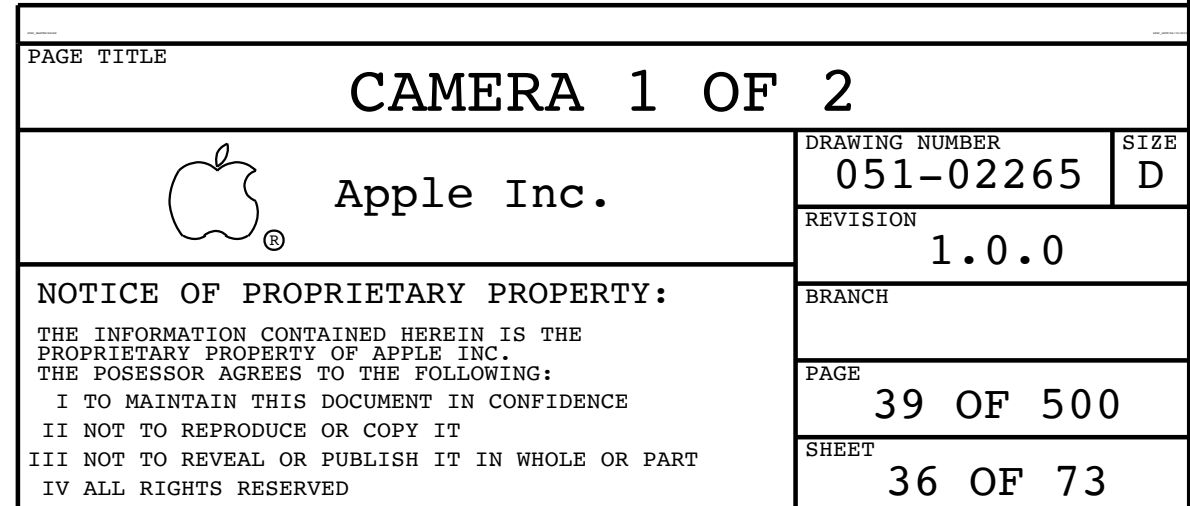
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| SYNC MASTER=PAULM                                                                                                                                                                                                                                                                                                        |                | SYNC DATE=06/15/2015 |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                               |                |                      |      |
| WIFI/BT MODULE                                                                                                                                                                                                                                                                                                           |                |                      |      |
|  Apple Inc.®                                                                                                                                                                                                                        | DRAWING NUMBER |                      | SIZE |
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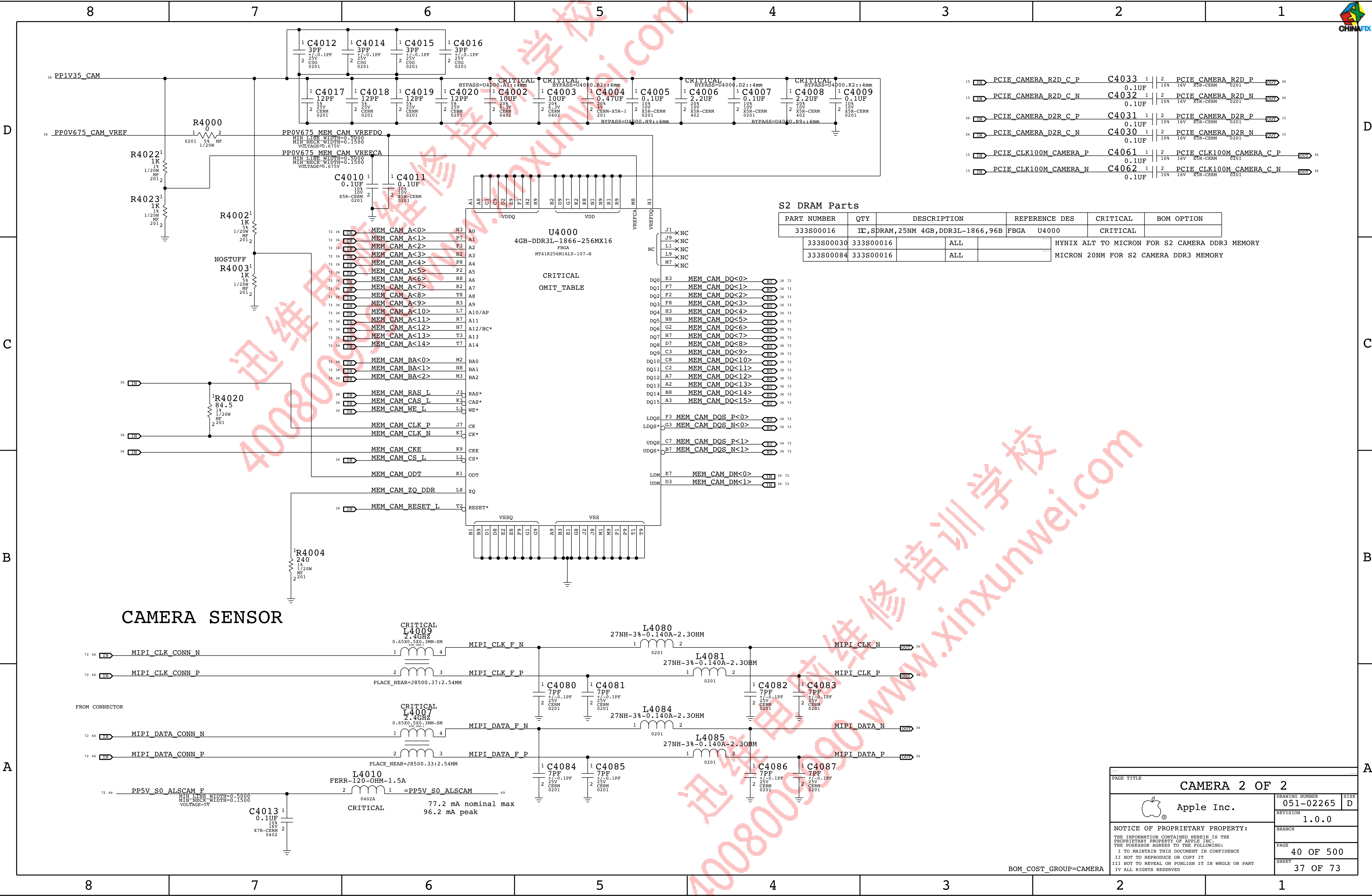
BT UART Isolation

WIFI UART ISOLATION



| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| WIFI/BT Module Support                                                                                                                                                                                                                                                                                            |                |           |
|                                                                                                                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02265 |
|                                                                                                                                                                                                                                                                                                                   | REVISION       | 1.0.0     |
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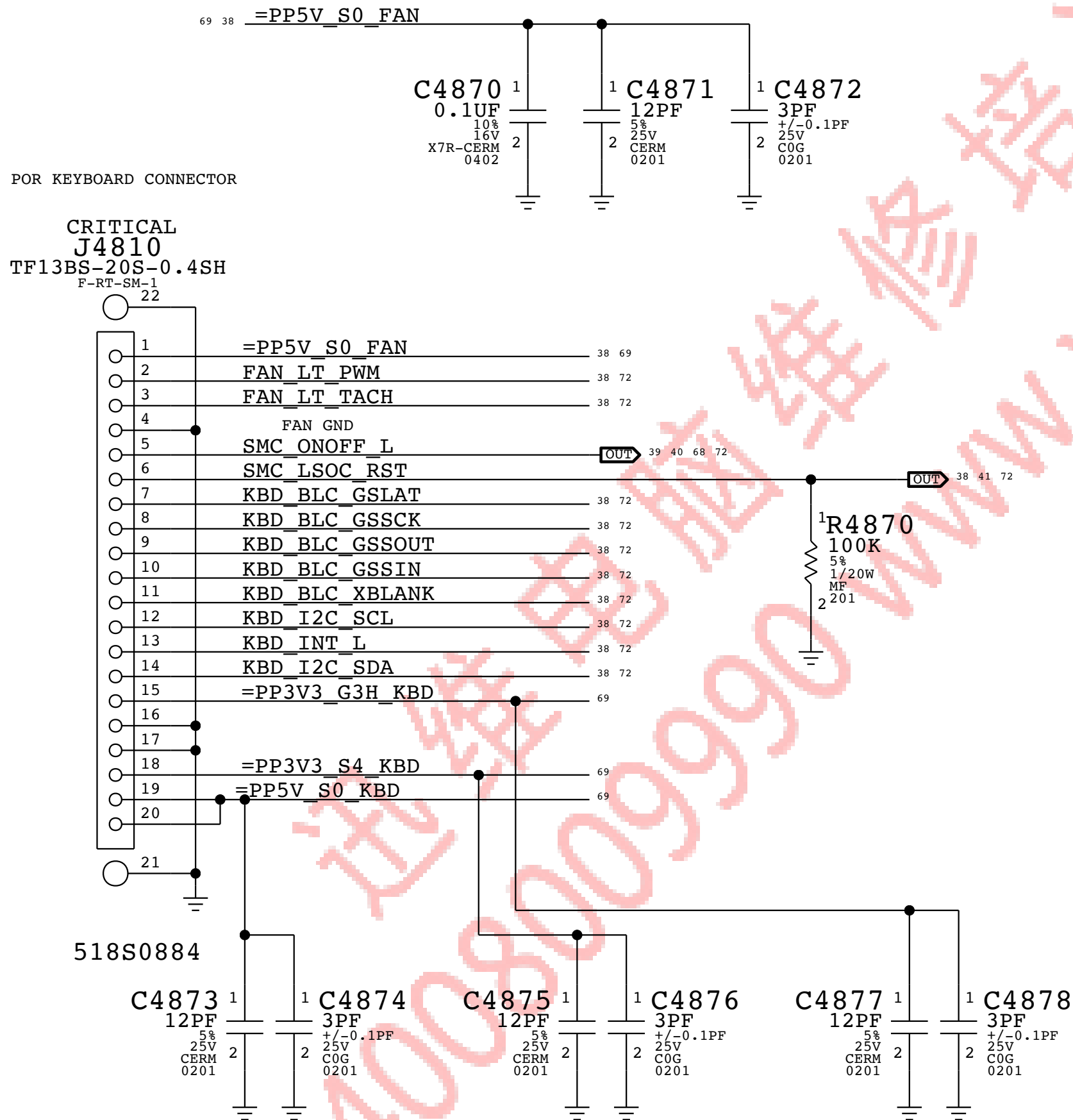
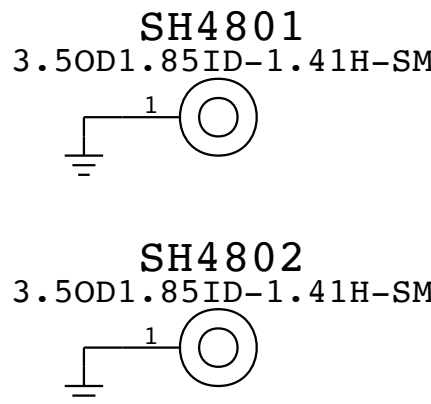


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| PAGE TITLE                                                                                                                                                                                                                                                                                                         |  |                |           |
| CAMERA 2 OF 2                                                                                                                                                                                                                                                                                                      |  |                |           |
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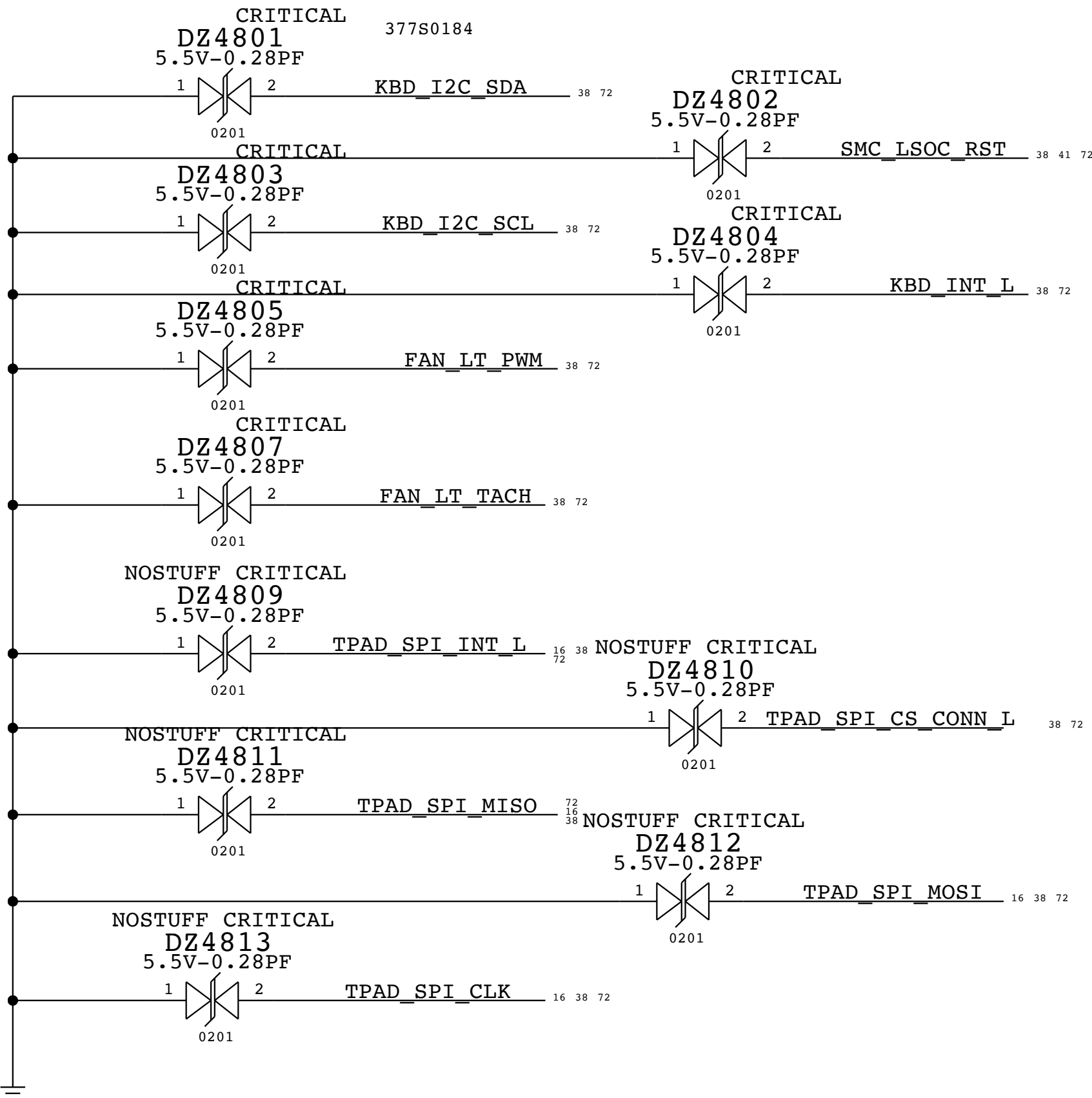
KEYBOARD CONNECTOR

TPAD CONNECTOR

COWLING BOSES  
860-00381

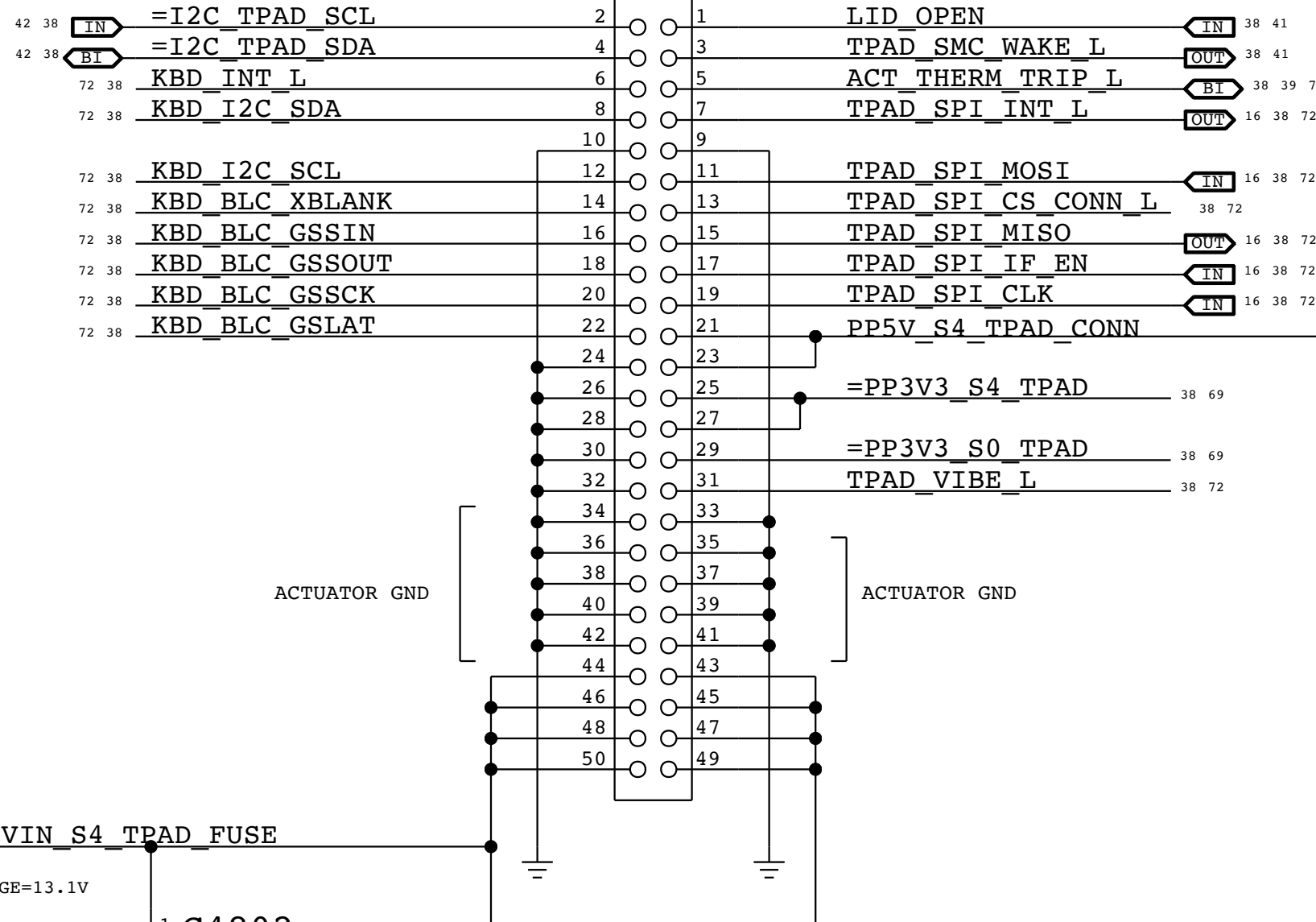


ESD DIODES

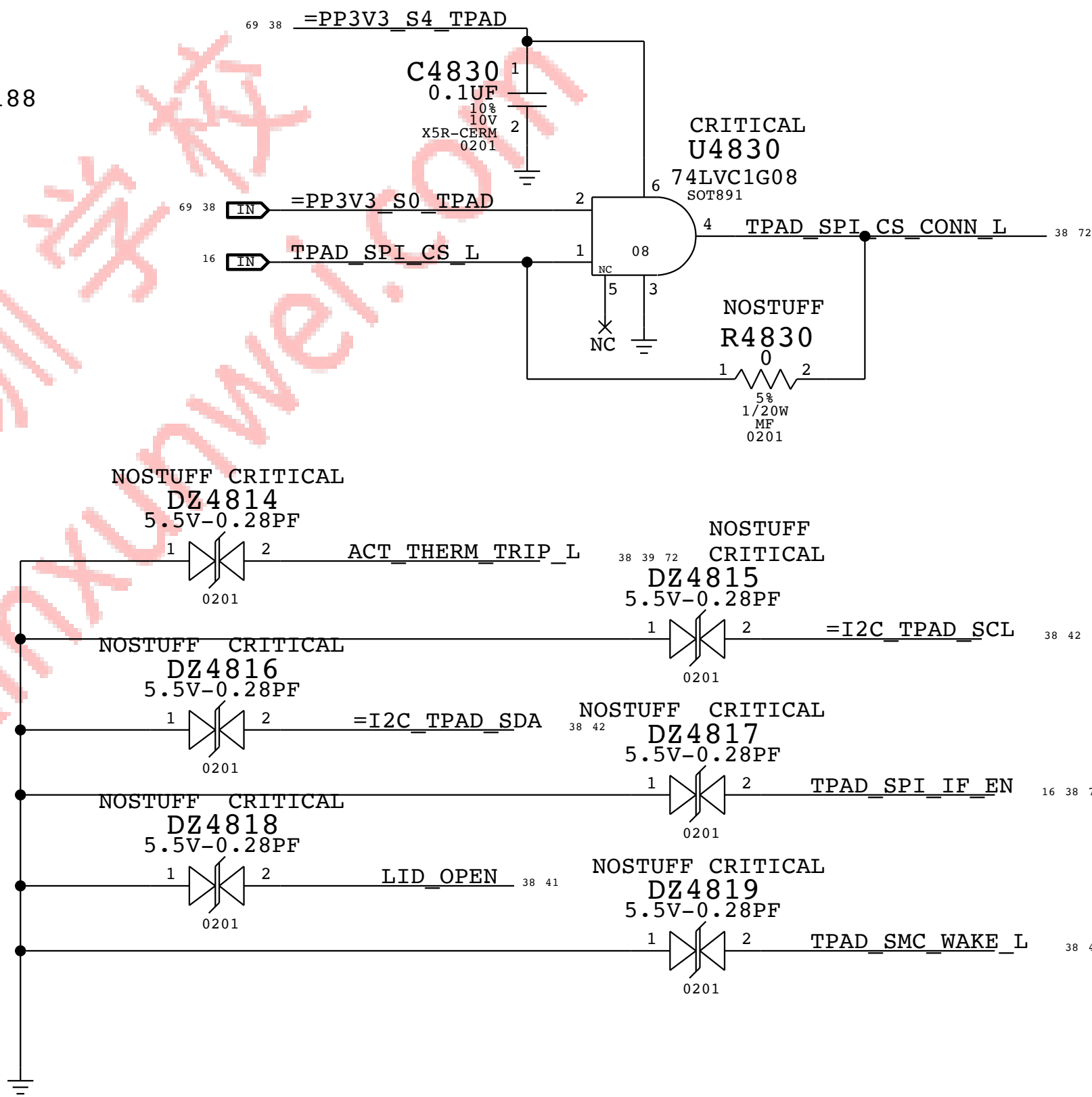


CRITICAL  
J4801

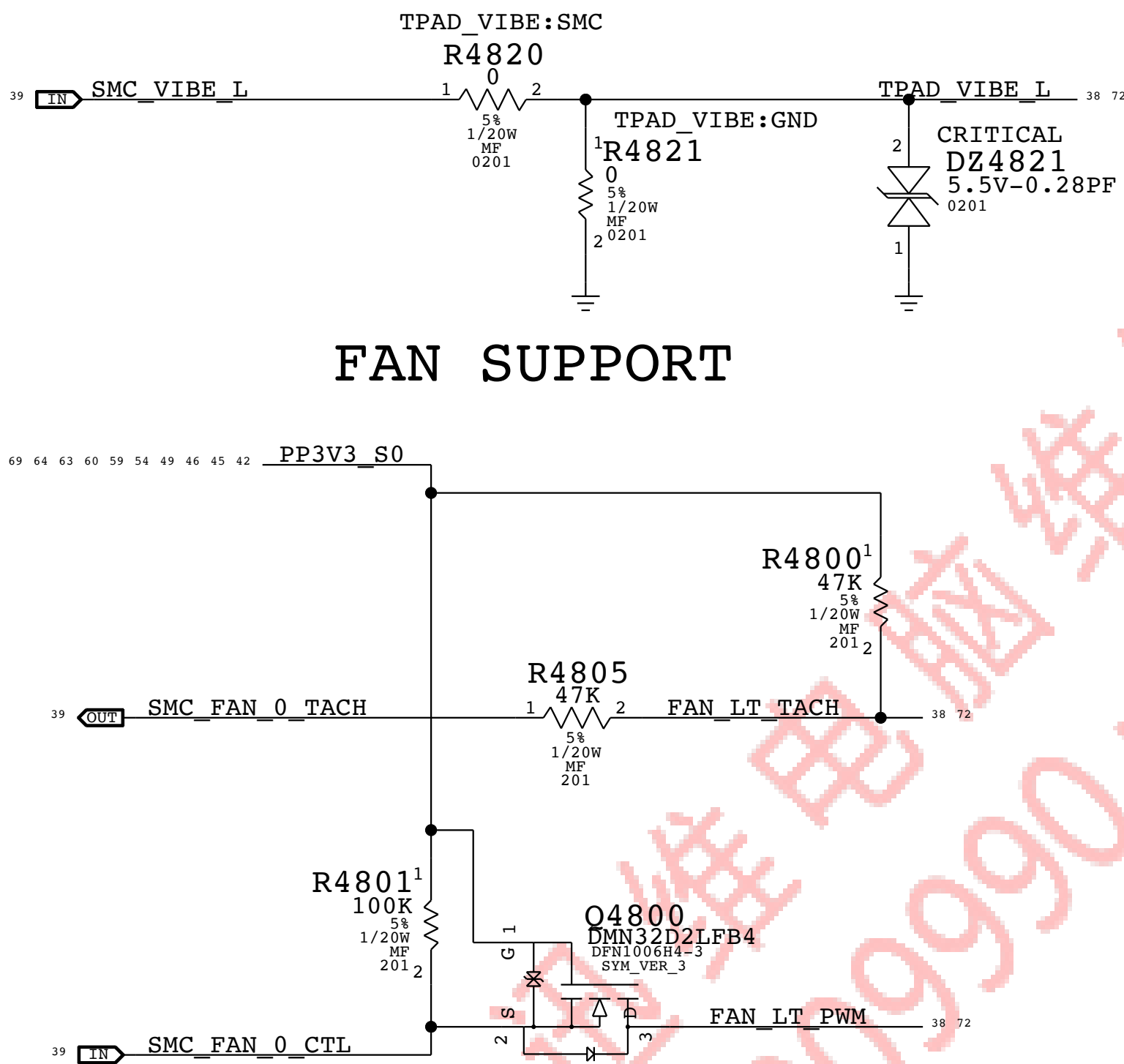
DF40C-50DS-0.4V-51  
F-SY-SM



CS ISOLATION



FAN SUPPORT



| PAGE TITLE                                                                                                                                                                                                                                                                                                         |                |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| P1:KEYBOARD & TRACKPAD CONN                                                                                                                                                                                                                                                                                        |                |           |
|  Apple Inc.                                                                                                                                                                                                                   | DRAWING NUMBER | 051-02265 |
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| SIZE D                                                                                                                                                                                                                                                                                                             |                |           |

BOM\_COST\_GROUP=KEYBOARD

D

D

C

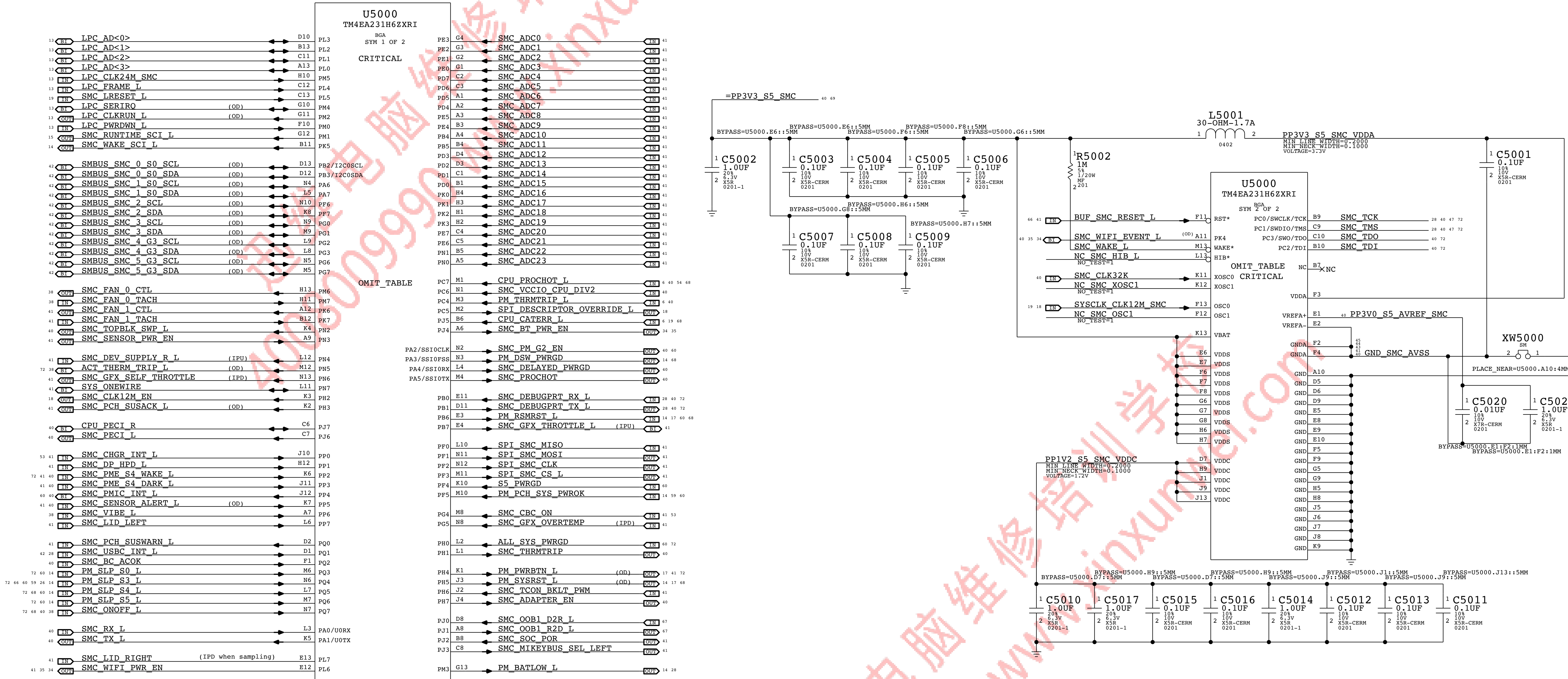
C

B

B

A

A



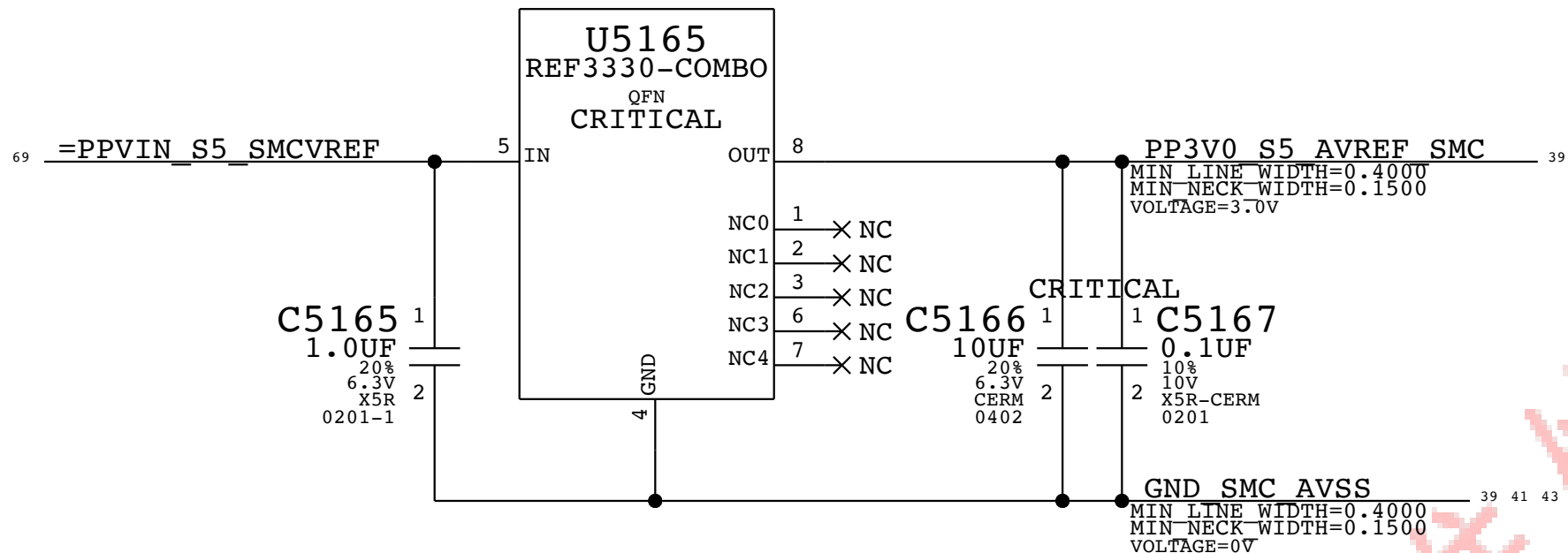
NOTE:  
SMS Interrupt can be active high or low, rename net accordingly.  
If SMS interrupt is not used, pull up to SMC rail.

NOTE:  
Unused pins have "SMC\_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

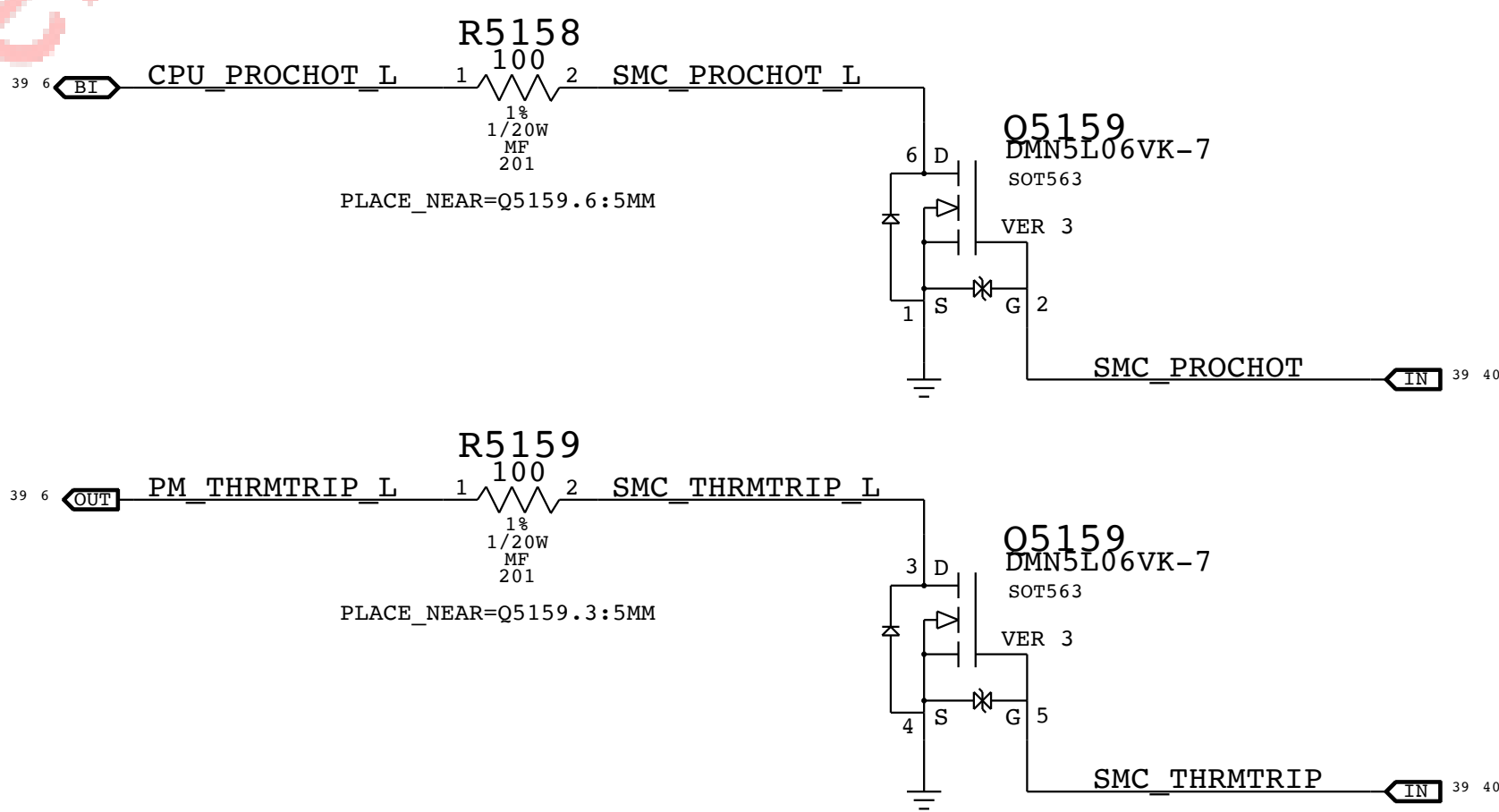
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| SYNC_MASTER=PAULM                                                                                                                                                                                                                                                                                                |  | SYNC_DATE=06/15/2015 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                       |  |                      |           |
| SMC                                                                                                                                                                                                                                                                                                              |  |                      |           |
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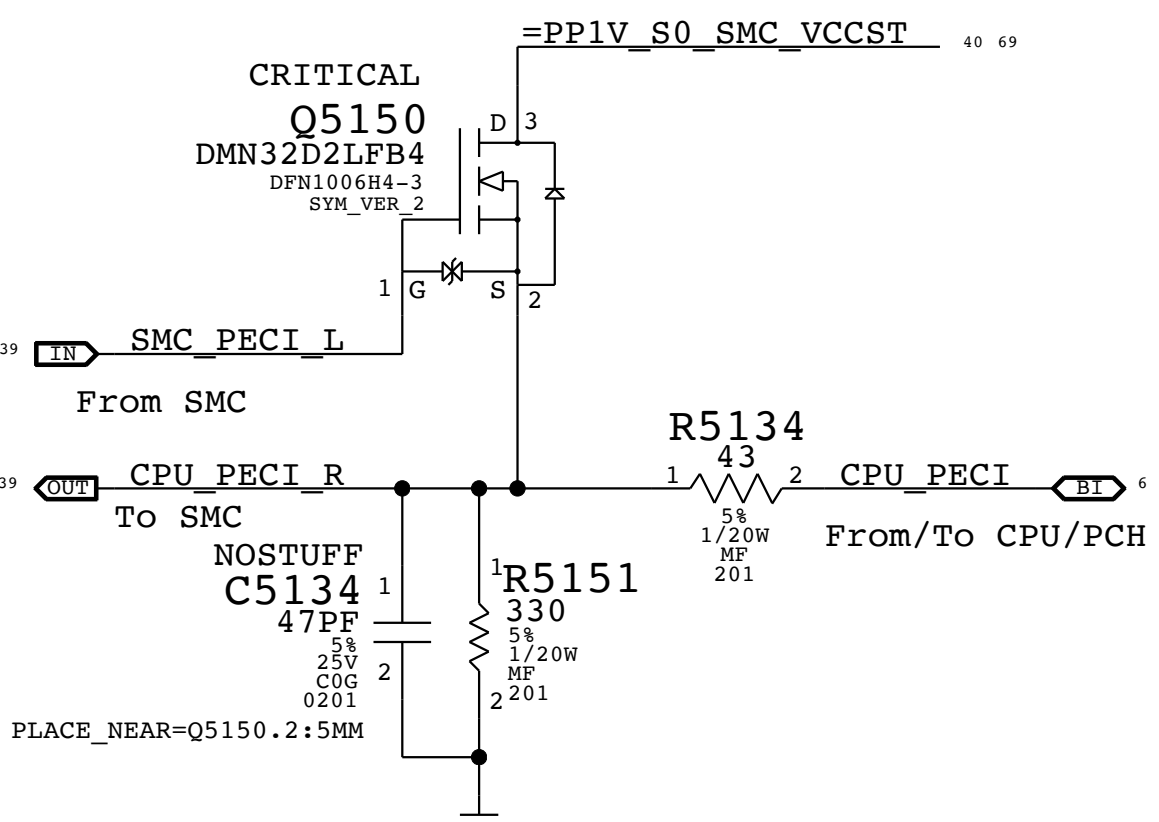
SMC AVREF Supply



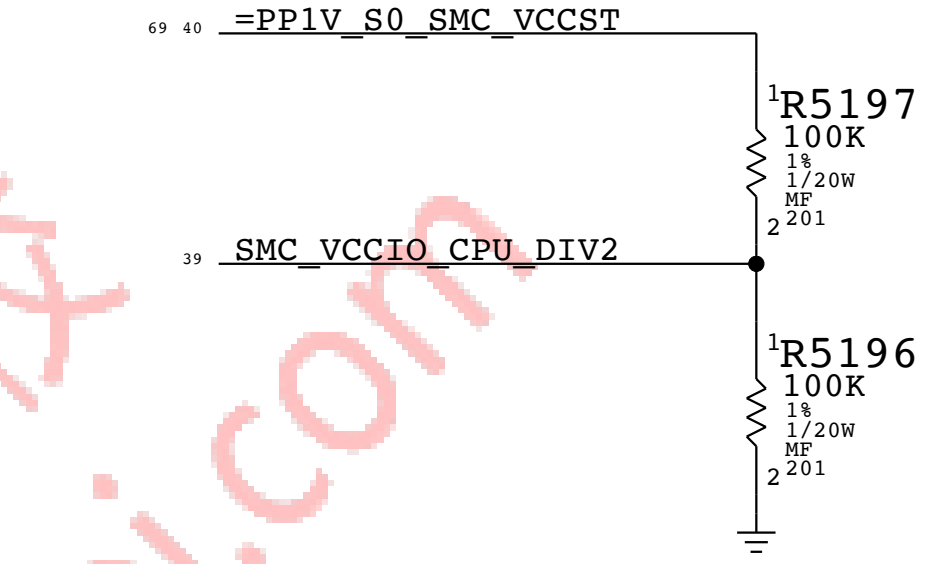
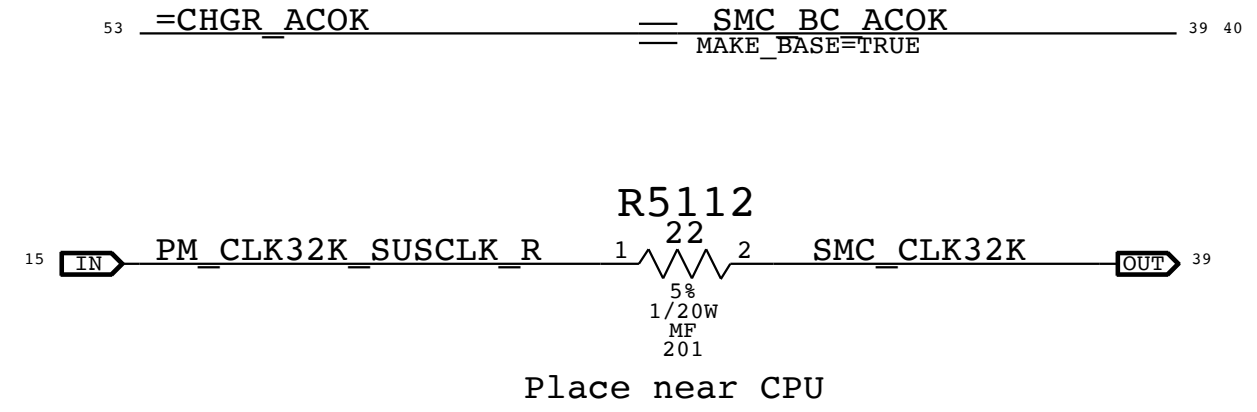
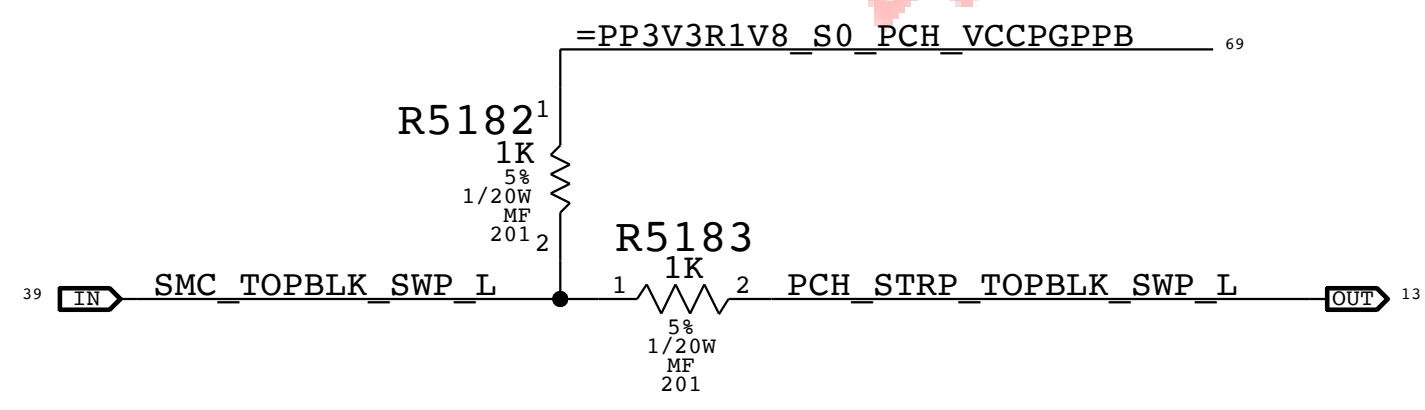
PROCHOT/THRMTRIP Support



PECI Support



Top-Block Swap



|    |    |               |                    |               |      |               |   |    |              |
|----|----|---------------|--------------------|---------------|------|---------------|---|----|--------------|
|    |    | =PP3V3_S5_SMC |                    | =PP3V3_S4_SMC |      | =PP3V3_S0_SMC |   |    |              |
| 72 | 41 | 39            | SMC PME_S4_WAKE_L  | R5166         | 100K | 1             | 2 |    |              |
| 41 | 39 |               | SMC PME_S4_DARK_L  | R5167         | 100K | 1             | 2 | 5% | 1/20W MF 201 |
| 39 | 35 | 34            | SMC WIFI_EVENT_L   | R5168         | 100K | 1             | 2 | 5% | 1/20W MF 201 |
| 60 | 39 |               | SMC PMIC_INT_L     | R5169         | 100K | 1             | 2 | 5% | 1/20W MF 201 |
| 72 | 68 | 39            | SMC_ONOFF_L        | R5170         | 10K  | 1             | 2 | 5% | 1/20W MF 201 |
| 41 | 39 |               | SMC_SENSOR_ALERT_L | R5172         | 10K  | 1             | 2 | 5% | 1/20W MF 201 |
| 39 |    |               | SMC_TX_L           | R5173         | 10K  | 1             | 2 |    |              |
| 39 |    |               | SMC_RX_L           | R5174         | 100K | 1             | 2 | 5% | 1/20W MF 201 |
| 72 | 39 | 28            | SMC_DEBUGPRT_TX_L  | R5175         | 20K  | 1             | 2 | 5% | 1/20W MF 201 |
| 72 | 39 | 28            | SMC_DEBUGPRT_RX_L  | R5176         | 20K  | 1             | 2 | 5% | 1/20W MF 201 |
| 72 | 47 | 39            | SMC_TMS            | R5177         | 10K  | 1             | 2 | 5% | 1/20W MF 201 |
| 72 | 39 |               | SMC_TDO            | R5178         | 10K  | 1             | 2 | 5% | 1/20W MF 201 |
| 72 | 39 |               | SMC_TDI            | R5179         | 10K  | 1             | 2 | 5% | 1/20W MF 201 |
| 72 | 47 | 39            | SMC_TCK            | R5180         | 10K  | 1             | 2 | 5% | 1/20W MF 201 |
| 40 | 39 |               | SMC_BC_ACOK        | R5187         | 100K | 1             | 2 | 5% | 1/20W MF 201 |
| 39 |    |               | SMC_ADAPTER_EN     | R5185         | 100K | 1             | 2 | 5% | 1/20W MF 201 |
| 40 | 39 |               | SMC_THRMTRIP       | R5186         | 10K  | 1             | 2 | 5% | 1/20W MF 201 |
| 39 |    |               | SMC_DELAYED_PWRGD  | R5191         | 100K | 1             | 2 | 5% | 1/20W MF 201 |
| 60 | 39 |               | SMC_PM_G2_EN       | R5198         | 100K | 1             | 2 | 5% | 1/20W MF 201 |
| 40 | 39 |               | SMC_PROCHOT        | R5199         | 100K | 1             | 2 | 5% | 1/20W MF 201 |

|                                                                                                                                                                                                                                                                                                                   |                |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |                |           |
| SMC Shared Support                                                                                                                                                                                                                                                                                                |                |           |
|  Apple Inc.                                                                                                                                                                                                                  | DRAWING NUMBER | 051-02265 |
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BOM\_COST\_GROUP=SMC

SMC ANALOG INPUTS

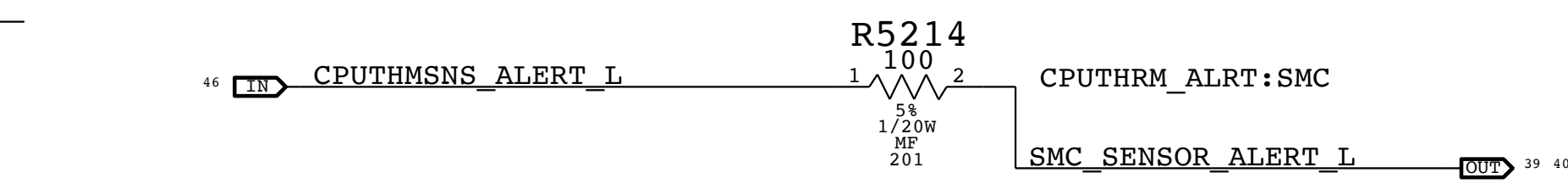
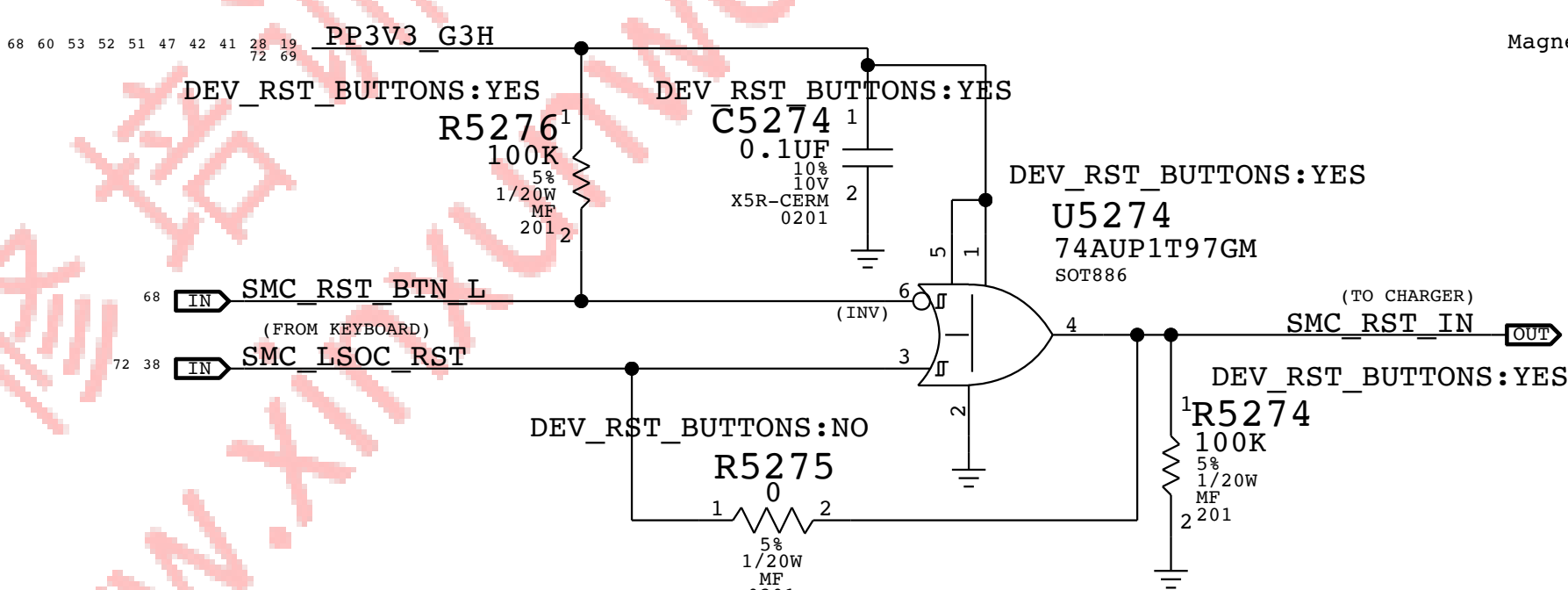
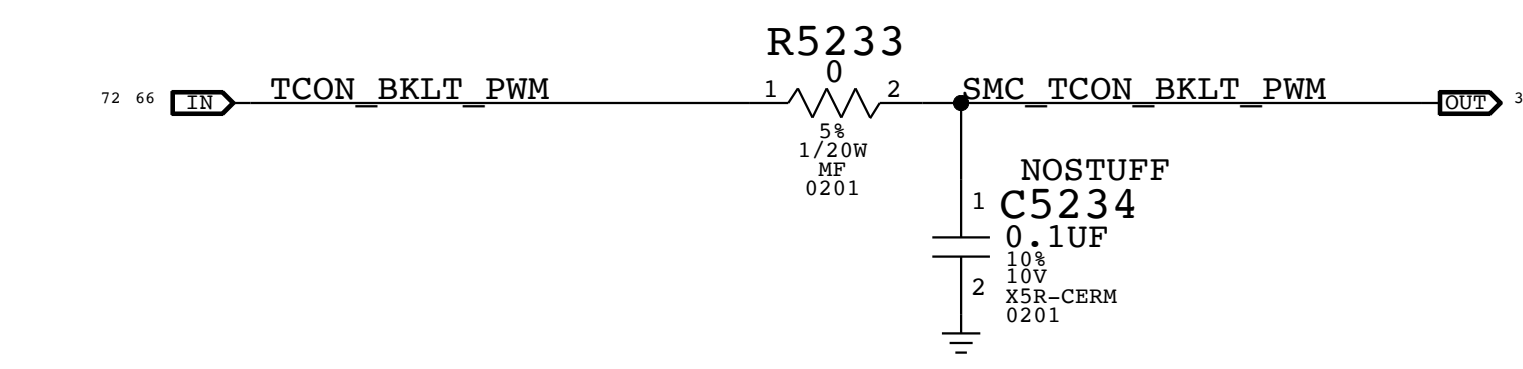
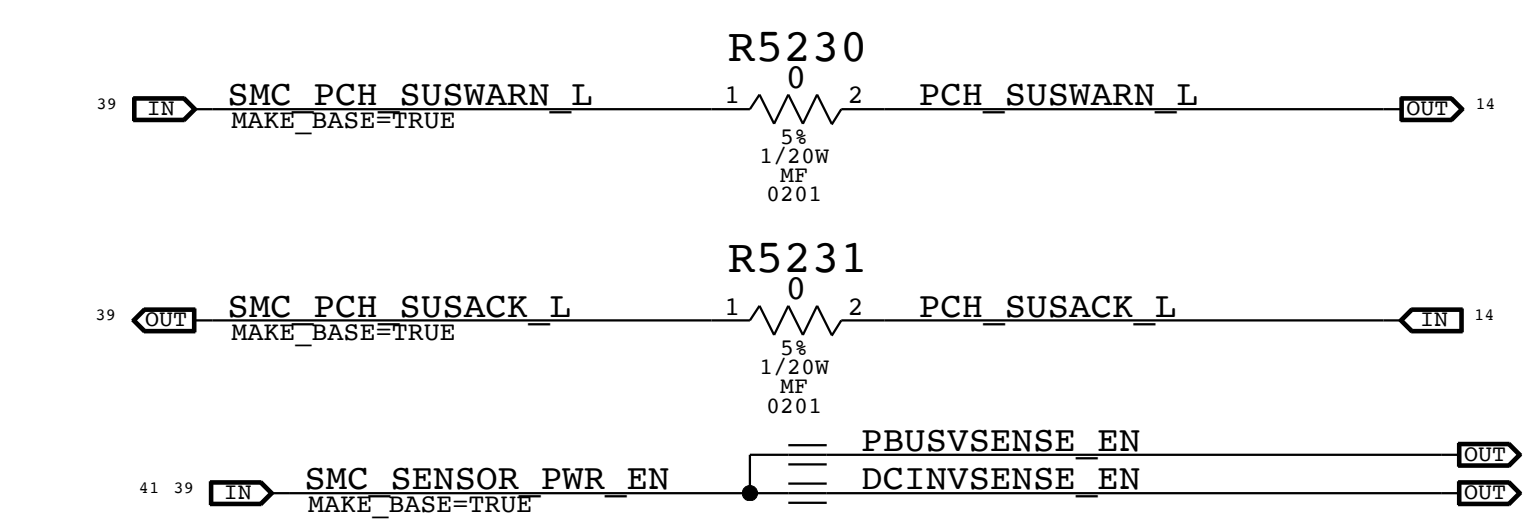
|           |                     |  |
|-----------|---------------------|--|
| SMC_ADC0  | SMC_DCIN_VSENSE     |  |
| SMC_ADC1  | SMC_DCIN_ISENSE     |  |
| SMC_ADC2  | SMC_PBUS_VSENSE     |  |
| SMC_ADC3  | SMC_BMON_ISENSE     |  |
| SMC_ADC4  | SMC_TPAD_ISENSE     |  |
| SMC_ADC5  | SMC_CPU_HI_ISENSE   |  |
| SMC_ADC6  | SMC_CPU_VSENSE      |  |
| SMC_ADC7  | SMC_CPU_ISENSE      |  |
| SMC_ADC8  | SMC_GT_VSENSE       |  |
| SMC_ADC9  | SMC_GT_ISENSE       |  |
| SMC_ADC10 | SMC_SA_ISENSE       |  |
| SMC_ADC11 | SMC_1VS5G_ISENSE    |  |
| SMC_ADC12 | SMC_SSD_ISENSE      |  |
| SMC_ADC13 | SMC_3V3SSD_ISENSE   |  |
| SMC_ADC14 | SMC_1V2S3_ISENSE    |  |
| SMC_ADC15 | SMC_CPU_IMON_ISENSE |  |
| SMC_ADC16 | SMC_GT_IMON_ISENSE  |  |
| SMC_ADC17 | SMC_3V3LCD_ISENSE   |  |
| SMC_ADC18 | SMC_3V3WLS_ISENSE   |  |
| SMC_ADC19 | SMC_LCDBKLT_ISENSE  |  |
| SMC_ADC20 | SMC_SA_IMON_ISENSE  |  |
| SMC_ADC21 | SMC_5VS4_ISENSE     |  |
| SMC_ADC22 | SMC_3V3S5_ISENSE    |  |
| SMC_ADC23 | SMC_TBT_ISENSE      |  |

|                    |                |           |                       |
|--------------------|----------------|-----------|-----------------------|
| SMC_GFX_THROTTLE_L | MAKE_BASE=TRUE | NO_TEST=1 | NC_SMC_GFX_THROTTLE_L |
| SMC_GFX_OVERTEMP   | MAKE_BASE=TRUE | NO_TEST=1 | NC_SMC_GFX_OVERTEMP   |
| SMC_FAN_1_CTL      | MAKE_BASE=TRUE | NO_TEST=1 | NC_SMC_FAN_1_CTL      |
| SMC_FAN_1_TACH     | MAKE_BASE=TRUE | NO_TEST=1 | NC_SMC_FAN_1_TACH     |

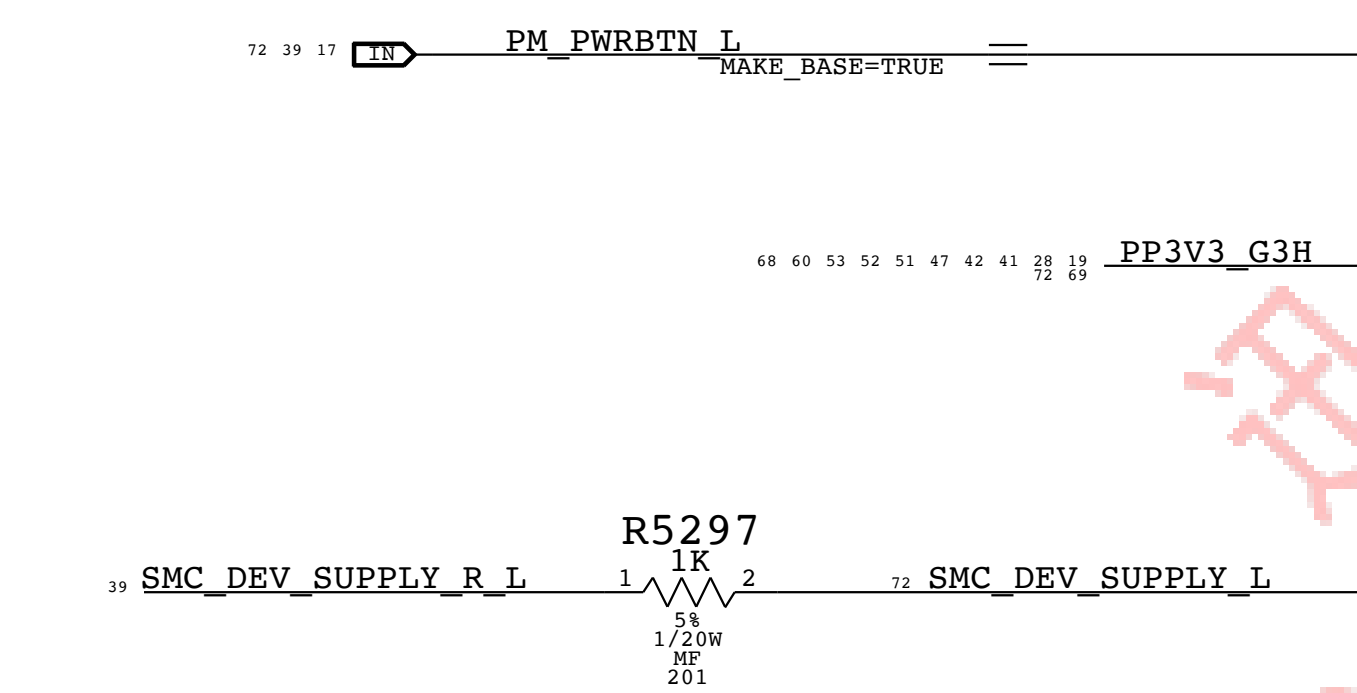
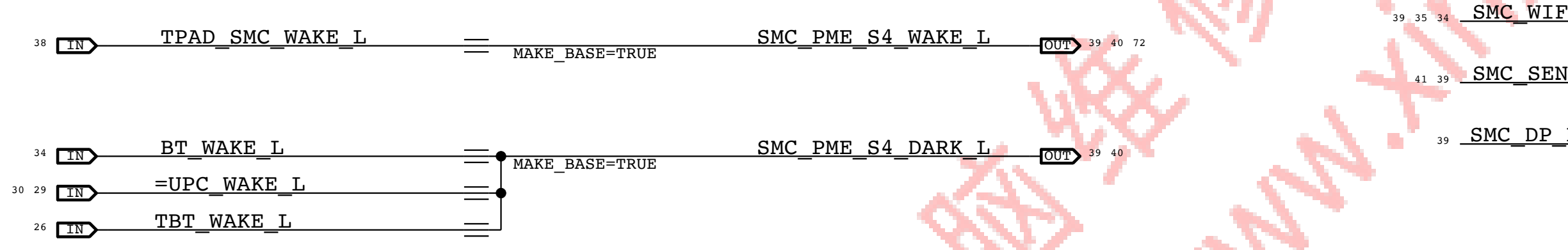
|                       |                |           |                          |
|-----------------------|----------------|-----------|--------------------------|
| SMC_SOC_POR           | MAKE_BASE=TRUE | NO_TEST=1 | NC_SMC_SOC_POR           |
| SMC_GFX_SELF_THROTTLE | MAKE_BASE=TRUE | NO_TEST=1 | NC_SMC_GFX_SELF_THROTTLE |

|              |                |           |                 |
|--------------|----------------|-----------|-----------------|
| SPI_SMC_CLK  | MAKE_BASE=TRUE | NO_TEST=1 | NC_SPI_SMC_CLK  |
| SPI_SMC_CS_L | MAKE_BASE=TRUE | NO_TEST=1 | NC_SPI_SMC_CS_L |
| SPI_SMC_MISO | MAKE_BASE=TRUE | NO_TEST=1 | NC_SPI_SMC_MISO |
| SPI_SMC_MOSI | MAKE_BASE=TRUE | NO_TEST=1 | NC_SPI_SMC_MOSI |

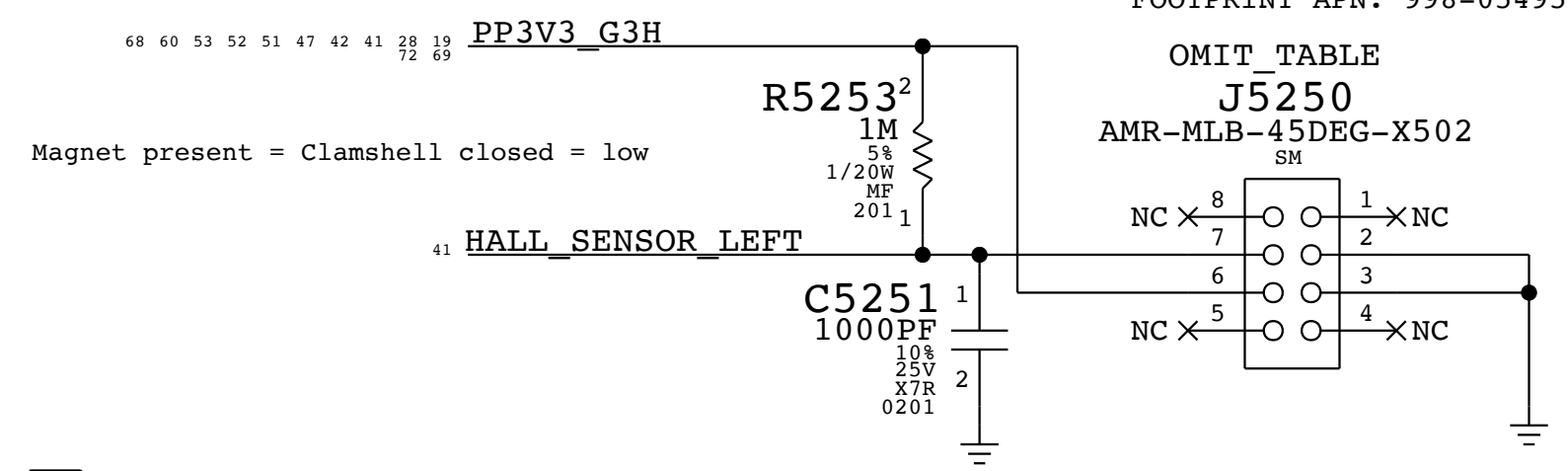
|                       |                |           |                          |
|-----------------------|----------------|-----------|--------------------------|
| SMC_MIKEYBUS_SEL_LEFT | MAKE_BASE=TRUE | NO_TEST=1 | NC_SMC_MIKEYBUS_SEL_LEFT |
| SYS_ONEWIRE           | MAKE_BASE=TRUE | NO_TEST=1 | NC_SYS_ONEWIRE           |



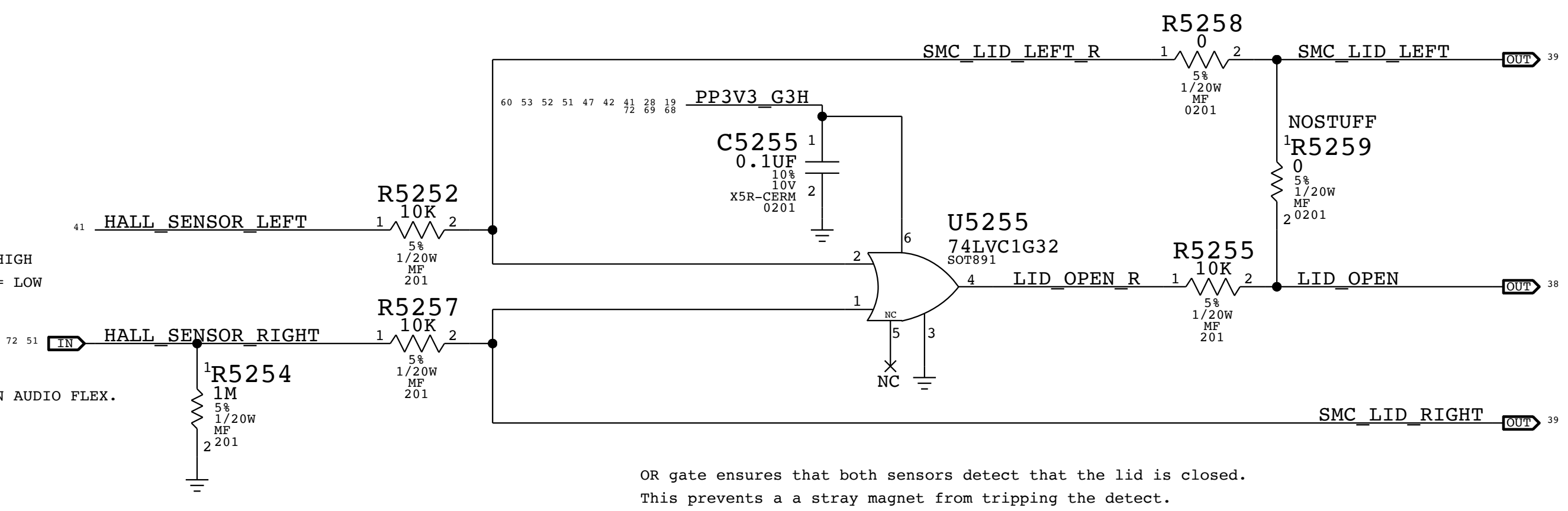
S4 SMC Wake Sources



AMR SENSOR PADS

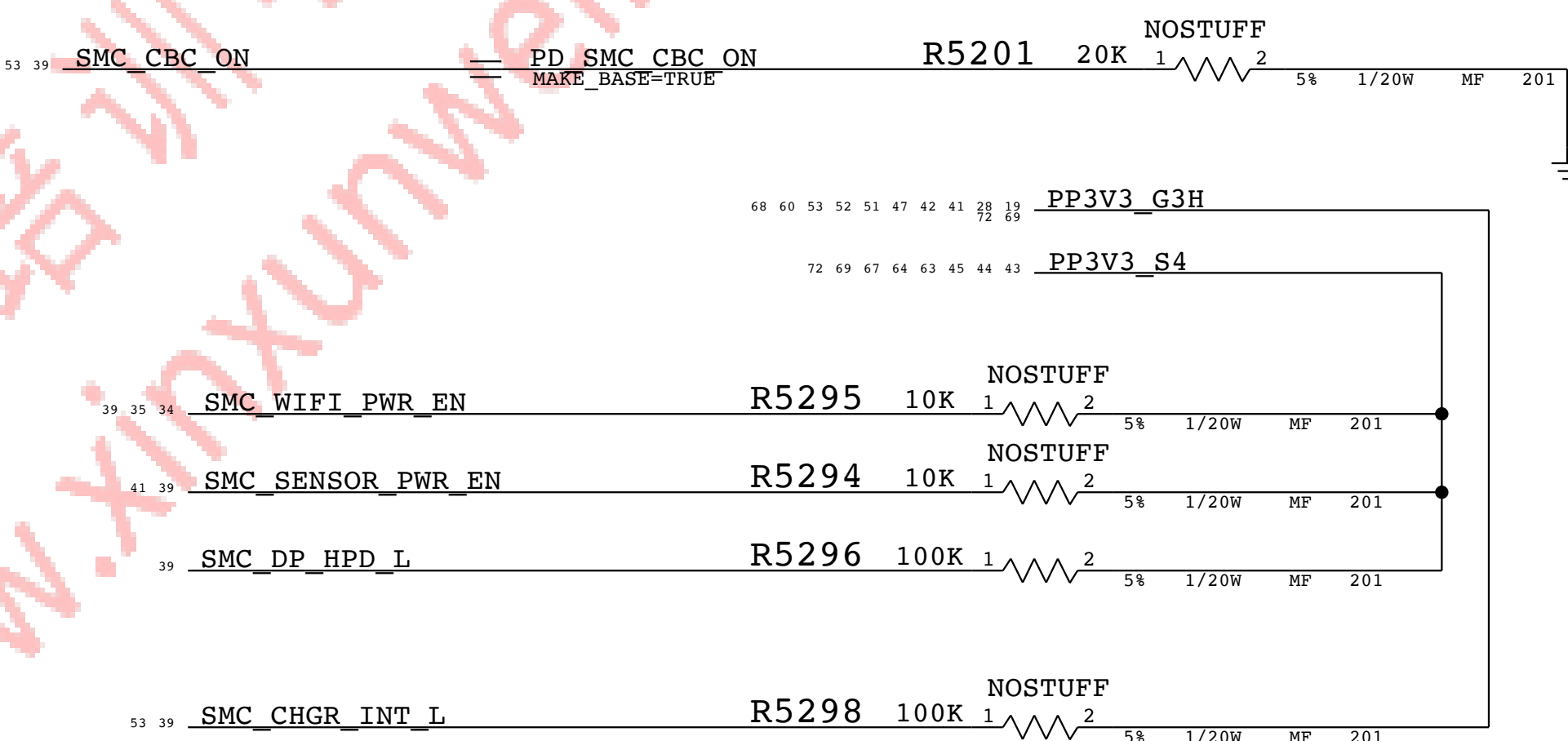


| PART NUMBER | QTY | DESCRIPTION                               | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------------------------|---------------|----------|------------|
| 677-05143   | 1   | SUBASSY (T&R) PCBA,HES INTERPOSER 45,X502 | J5250         | CRITICAL |            |



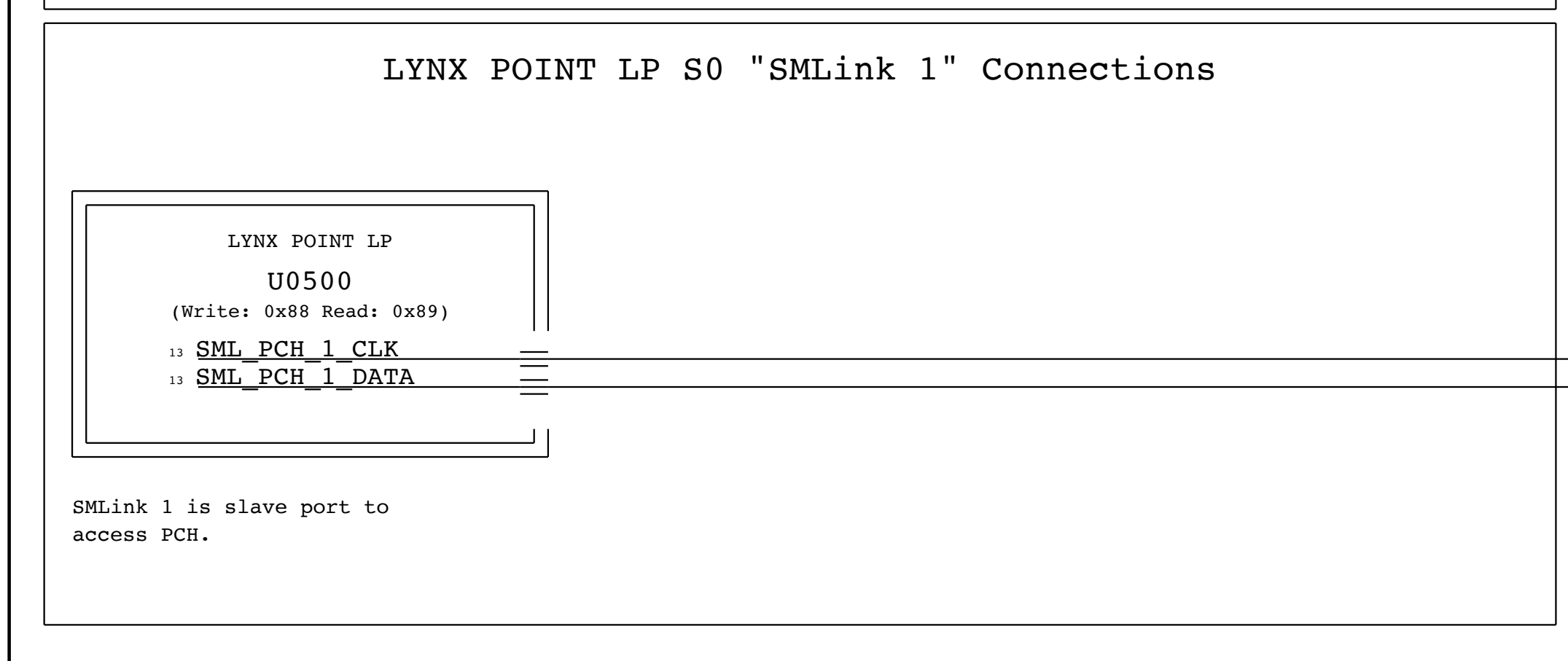
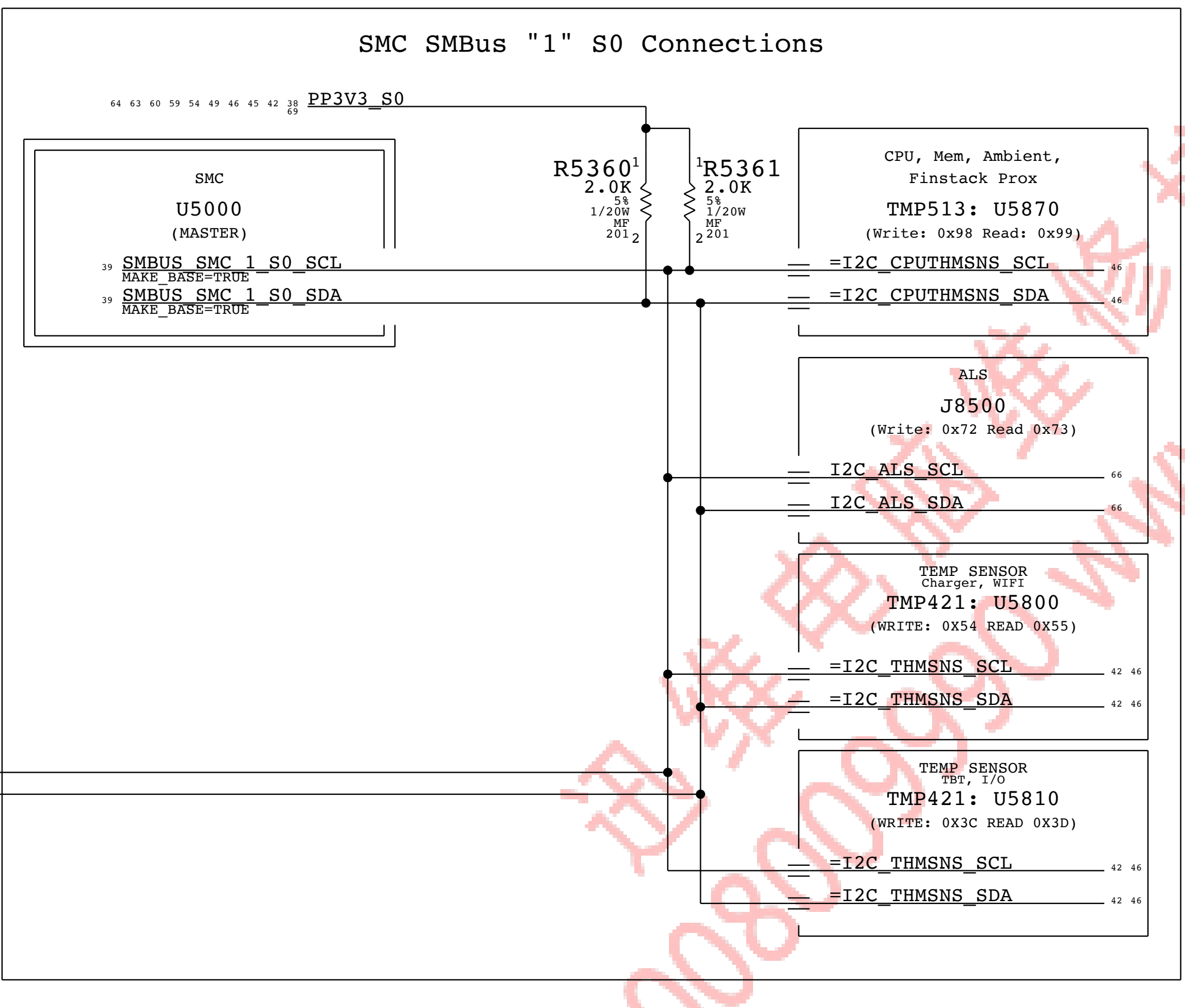
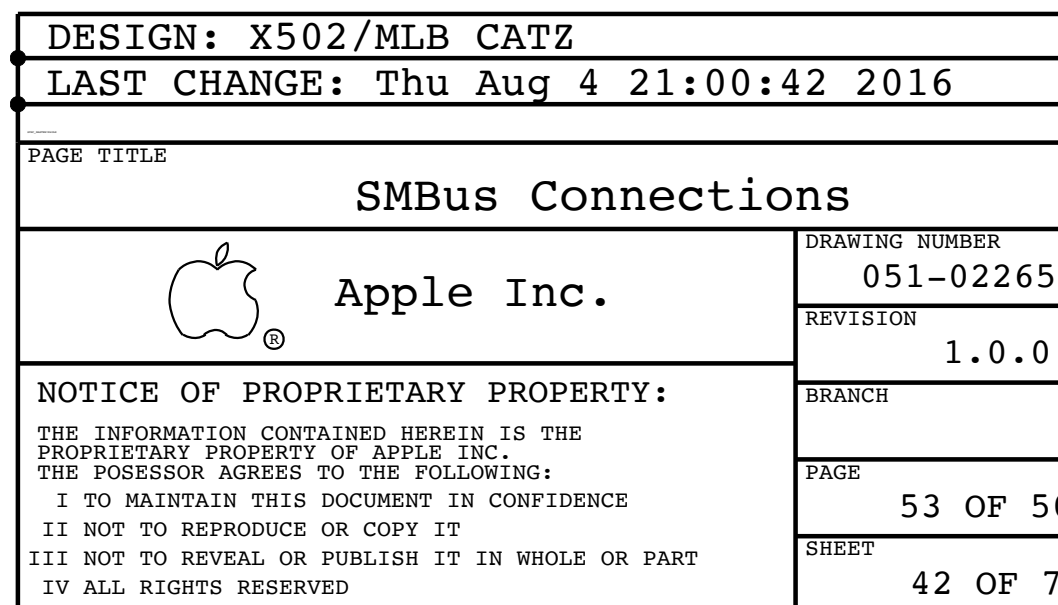
OR gate ensures that both sensors detect that the lid is closed. This prevents a stray magnet from tripping the detect.

Thermal Alerts



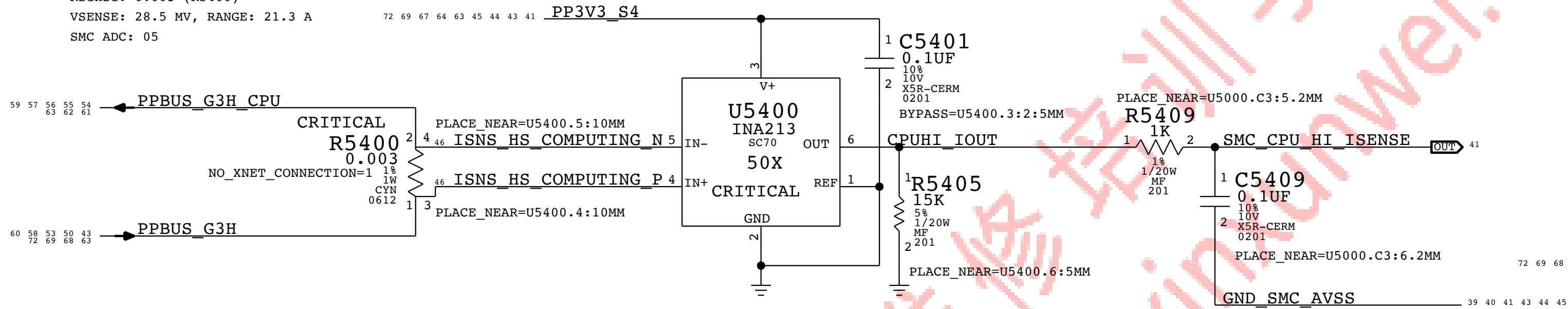
|                                                                                                                                                                                                                                                                                                               |  |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                         |  |                          |
| SYNC MASTER=PAULM                                                                                                                                                                                                                                                                                             |  |                          |
| PAGE TITLE                                                                                                                                                                                                                                                                                                    |  |                          |
| SMC Project Support                                                                                                                                                                                                                                                                                           |  |                          |
| Apple Inc.                                                                                                                                                                                                                                                                                                    |  | DRAWING NUMBER 051-02265 |
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| PAGE 52 OF 500                                                                                                                                                                                                                                                                                                |  | SIZE D                   |
| SHEET 41 OF 73                                                                                                                                                                                                                                                                                                |  | BRANCH                   |

BOM\_COST\_GROUP=SMC



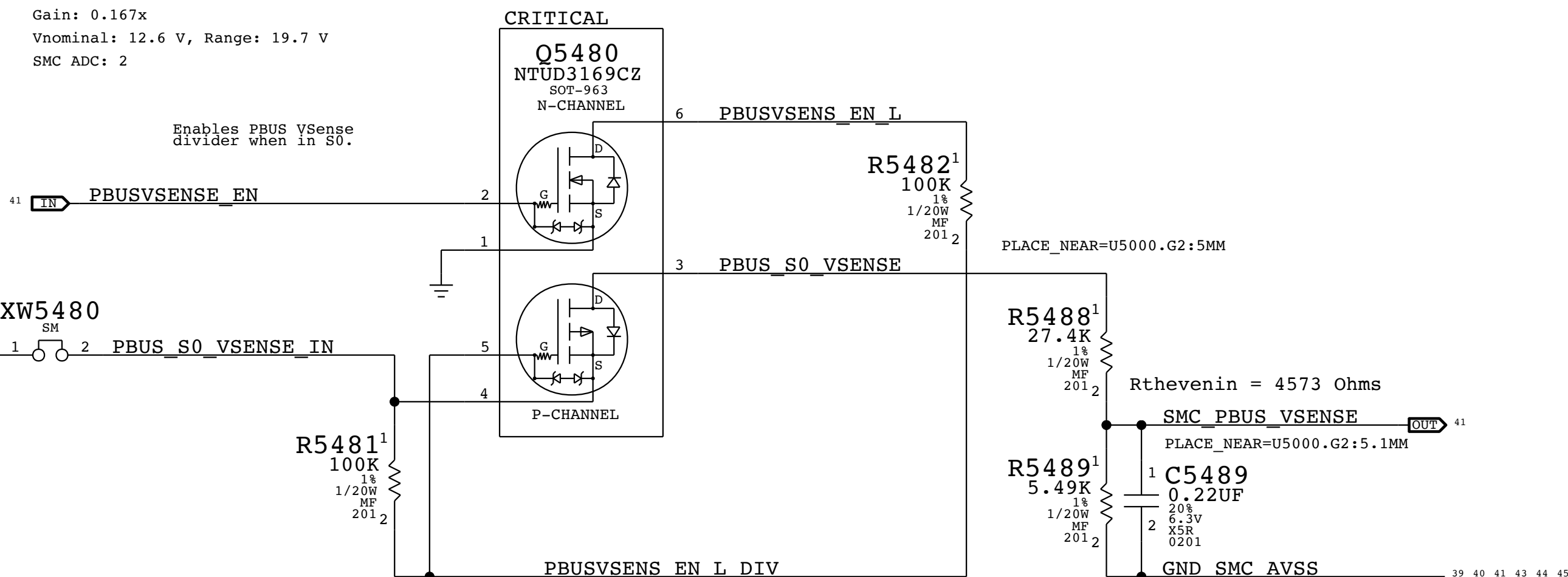
CPU High Side Current Sense (IC0R)

GAIN: 50X, EDP: 9.5 A  
Rsense: 0.003 (R5400)  
VSENSE: 28.5 MV, RANGE: 21.3 A  
SMC ADC: 05



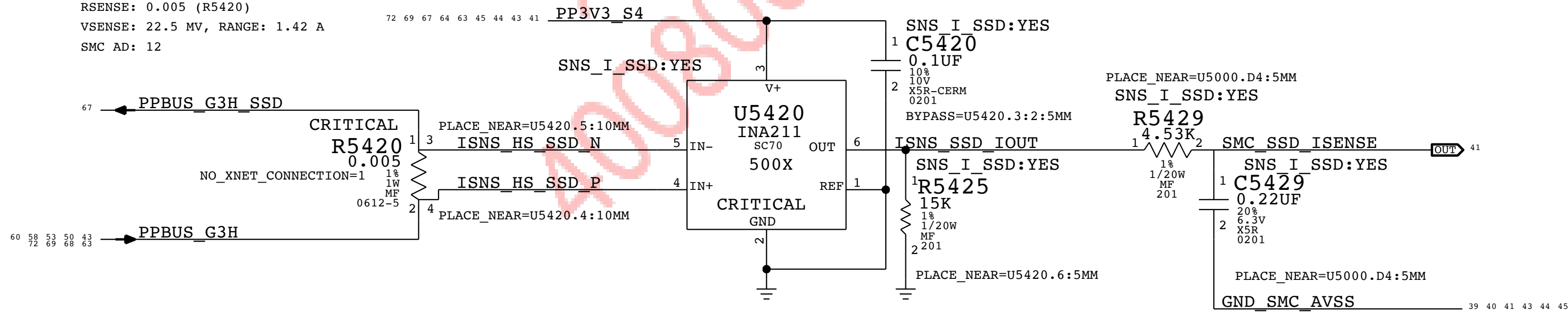
PBUS Voltage Sense & Enable (VP0R)

Gain: 0.167x  
Vnominal: 12.6 V, Range: 19.7 V  
SMC ADC: 2



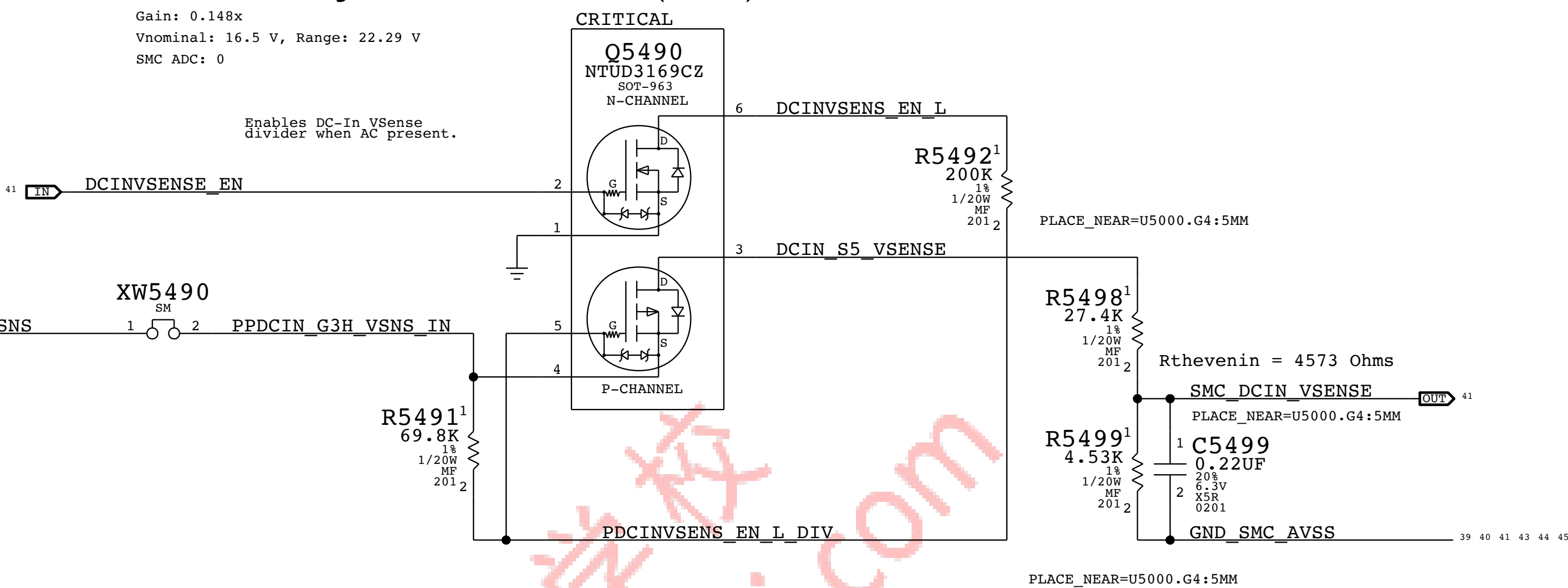
SSD NAND CURRENT SENSE (IHNC)

GAIN: 500X, EDP: 0.9 A  
RSense: 0.005 (R5420)  
VSENSE: 22.5 MV, RANGE: 1.42 A  
SMC AD: 12



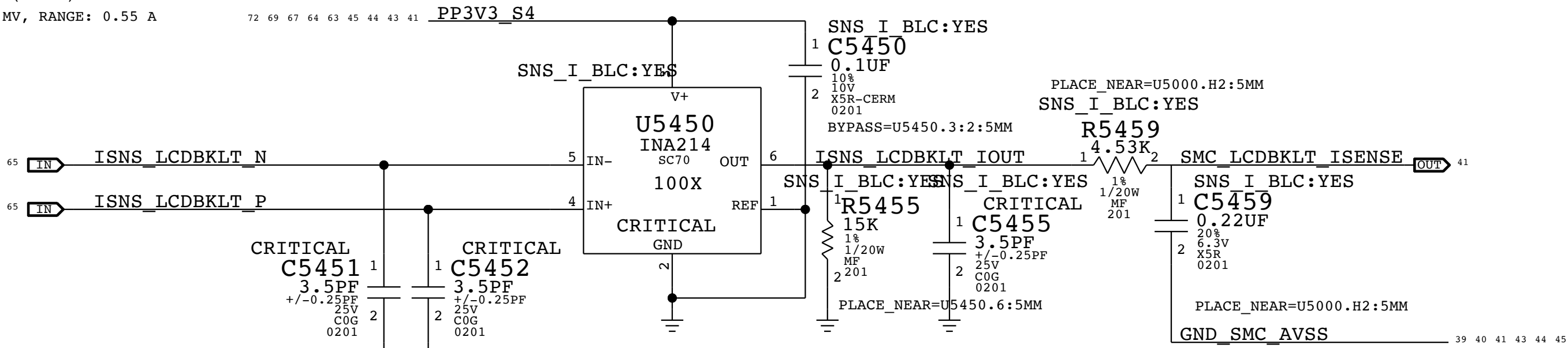
DC In Voltage Sense & Enable (VD0R)

Gain: 0.148x  
Vnominal: 16.5 V, Range: 22.29 V  
SMC ADC: 0



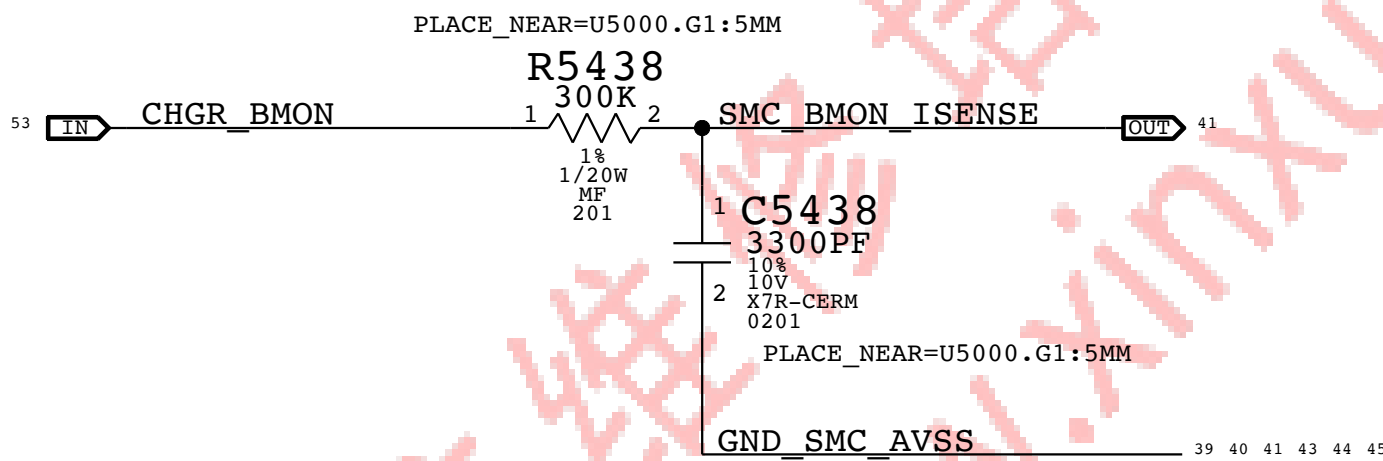
LCD BACKLIGHT CURRENT SENSE (IBLR)

GAIN: 100X, EDP: 0.9 A  
Rsense: 0.025 (R7700)  
VSENSE: 22.5 MV, RANGE: 0.55 A  
SMC AD: 19



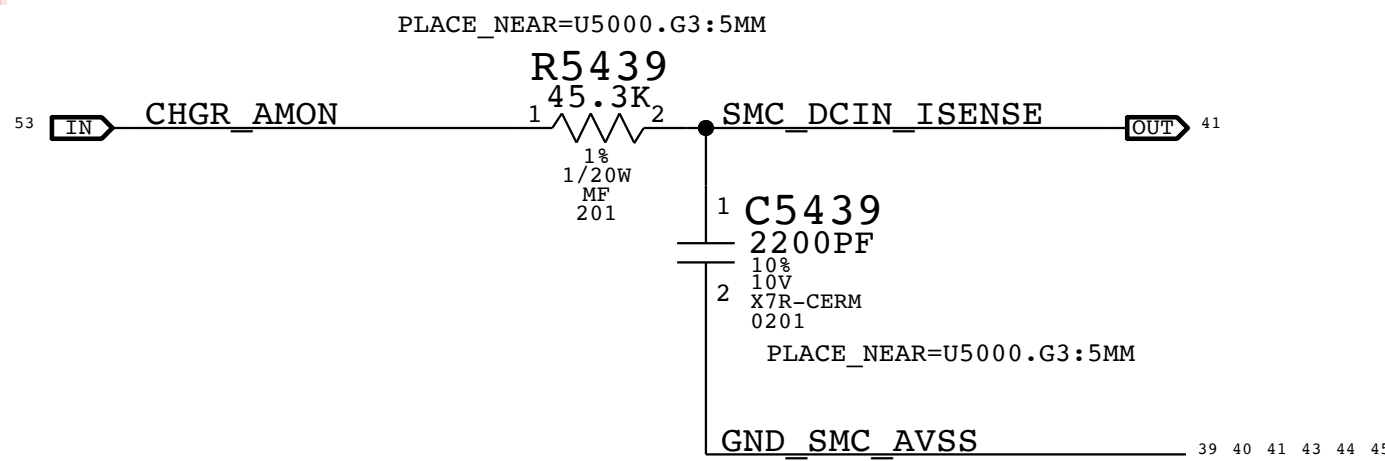
Charger (BMON) Current Sense (IPBR)

Charger Gain: 36x, EDP: 8 A  
RSense: 0.005 (R7160)  
SMC ADC: 03



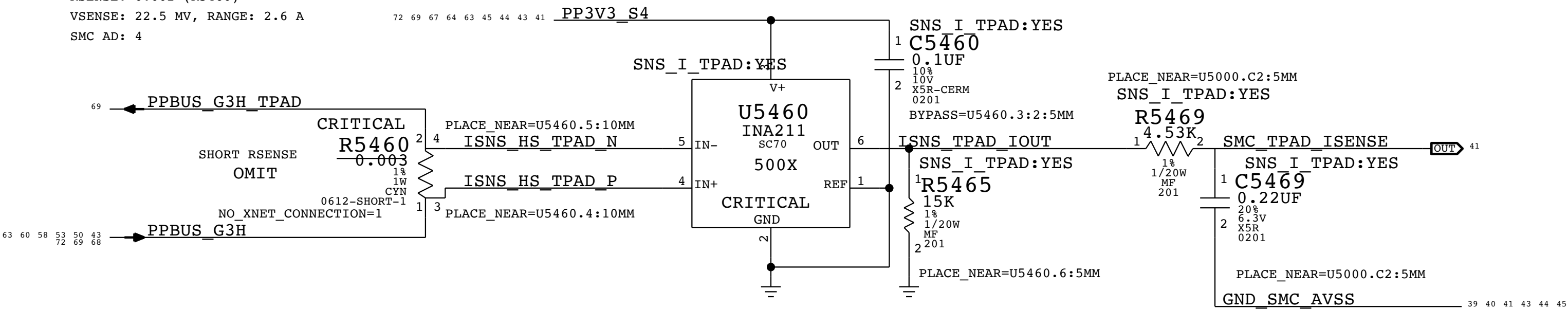
DC-IN (AMON) Current Sense (ID0R)

Charger Gain: 20x, EDP: 4.6 A  
RSense: 0.010 (R7120)  
SMC ADC: 01



TRACKPAD ACTUATOR X239 CURRENT SENSE (ITAR)

GAIN: 200X, EDP: 0.9 A  
RSense: 0.002 (R5460)  
VSENSE: 22.5 MV, RANGE: 2.6 A  
SMC AD: 4



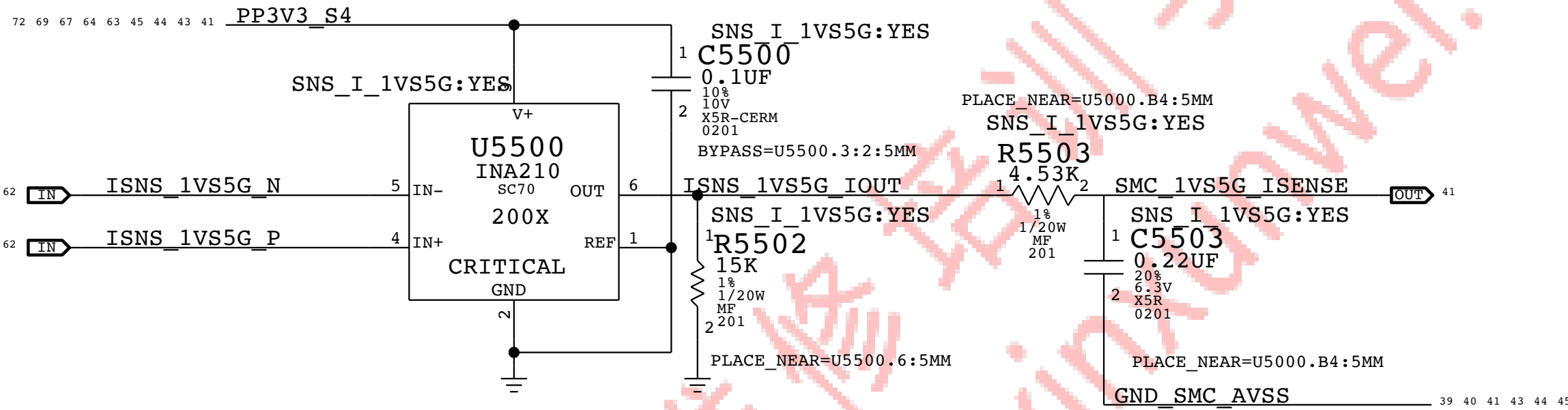
| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION    |
|-------------|-----|-------------------------------------|---------------|----------|---------------|
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5429         |          | SNS_I_SSD:NO  |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5459         |          | SNS_I_BLC:NO  |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5469         |          | SNS_I_TPAD:NO |

|                                                                                                                    |                |           |
|--------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| DESIGN: X502/MLB CATZ                                                                                              |                |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                               |                |           |
| PAGE TITLE                                                                                                         |                |           |
| Power Sensors High Side                                                                                            |                |           |
|                                                                                                                    | DRAWING NUMBER | 051-02265 |
|                                                                                                                    | REVISION       | 1.0.0     |
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BOM\_COST\_GROUP=SENSORS

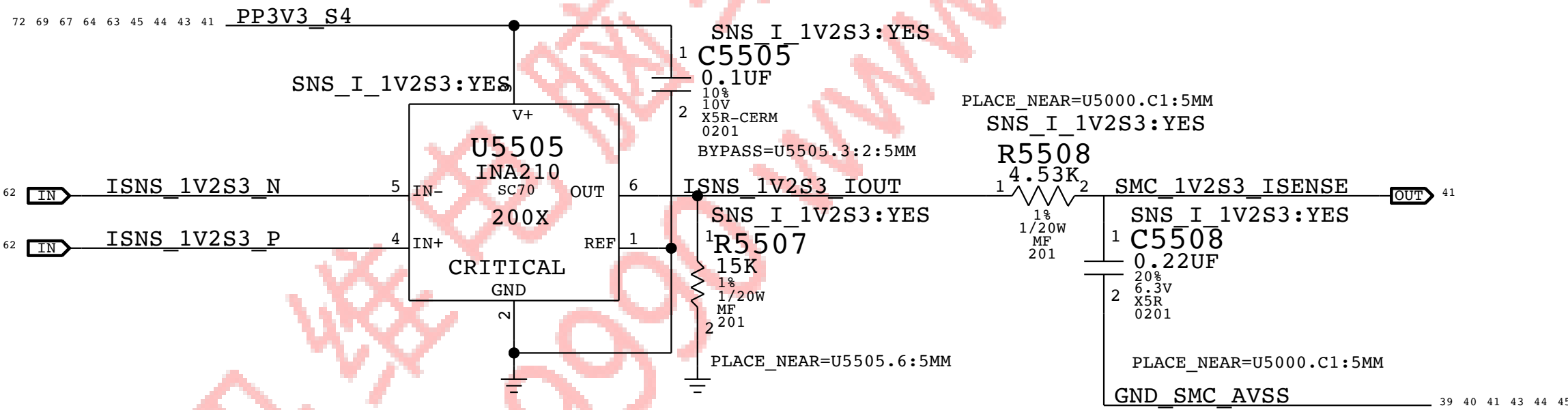
1V\_S5G CURRENT SENSE (ISIC)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.003 (R7700)  
VSENSE: 22.5 MV, RANGE: 3.0 A  
SMC AD: 11



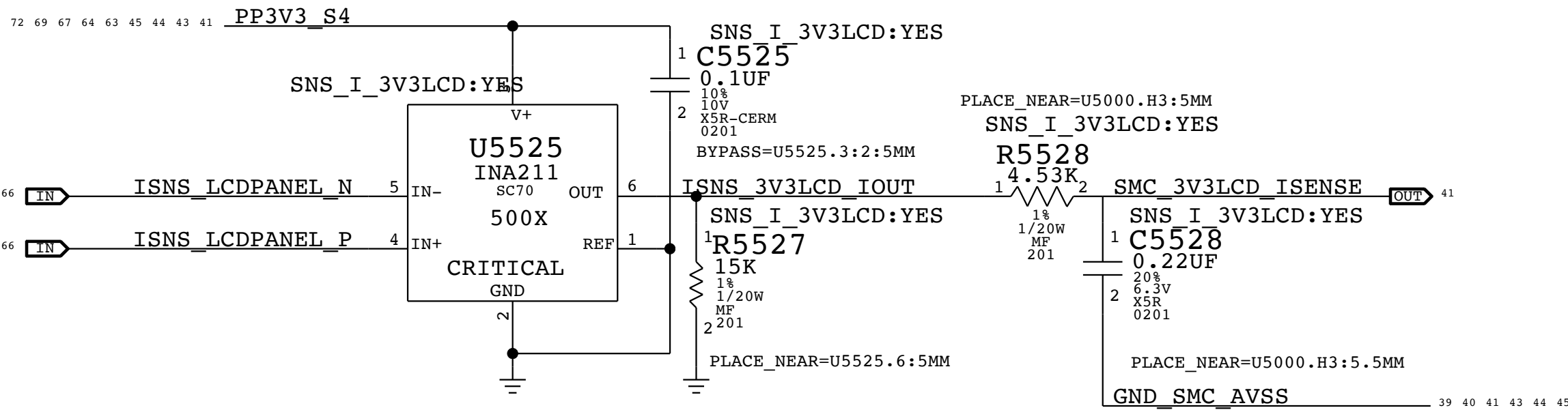
1V2\_S3 CURRENT SENSE (IMOC)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.002 (R7700)  
VSENSE: 22.5 MV, RANGE: 7.67 A  
SMC AD: 14



3V3\_S0 LCD CURRENT SENSE (ILDC)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.005 (R5525)  
VSENSE: 22.5 MV, RANGE: 1.6 A  
SMC AD: 17

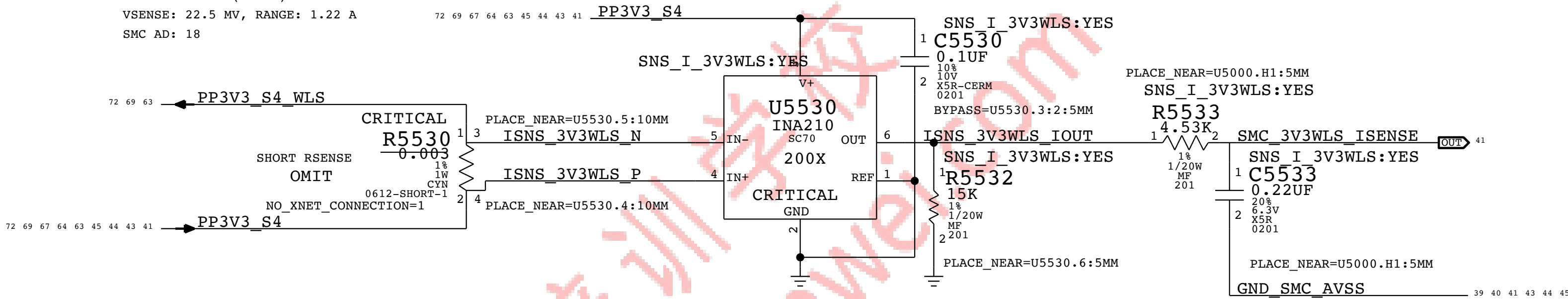


5V\_S0 LCD CURRENT SENSE (I\_\_\_)

GAIN: 500X. EDP: 0.9 A  
RSENSE: 0.005 (R8520)  
VSENSE: 22.5 MV, RANGE: 1.6 A  
SMC AD: 17

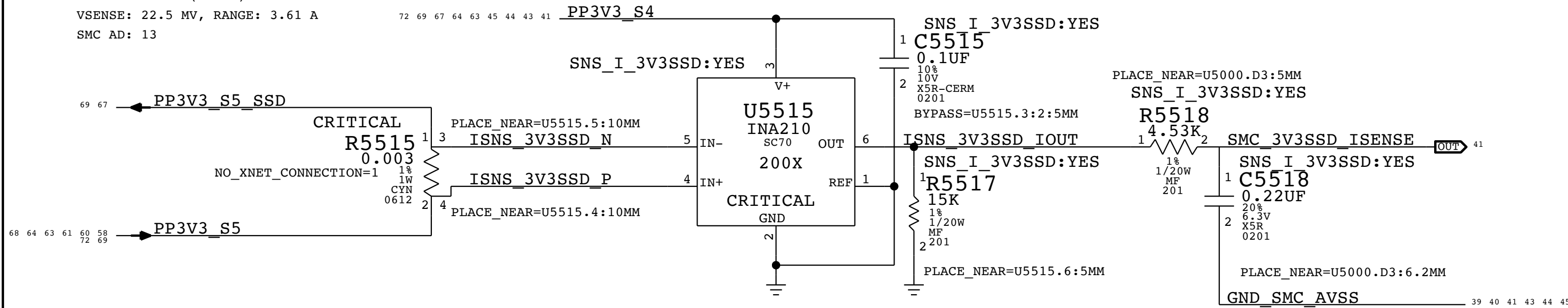
3V3\_S4 WIRELESS CURRENT SENSE (IAPC)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.010 (R7700)  
VSENSE: 22.5 MV, RANGE: 1.22 A  
SMC AD: 18



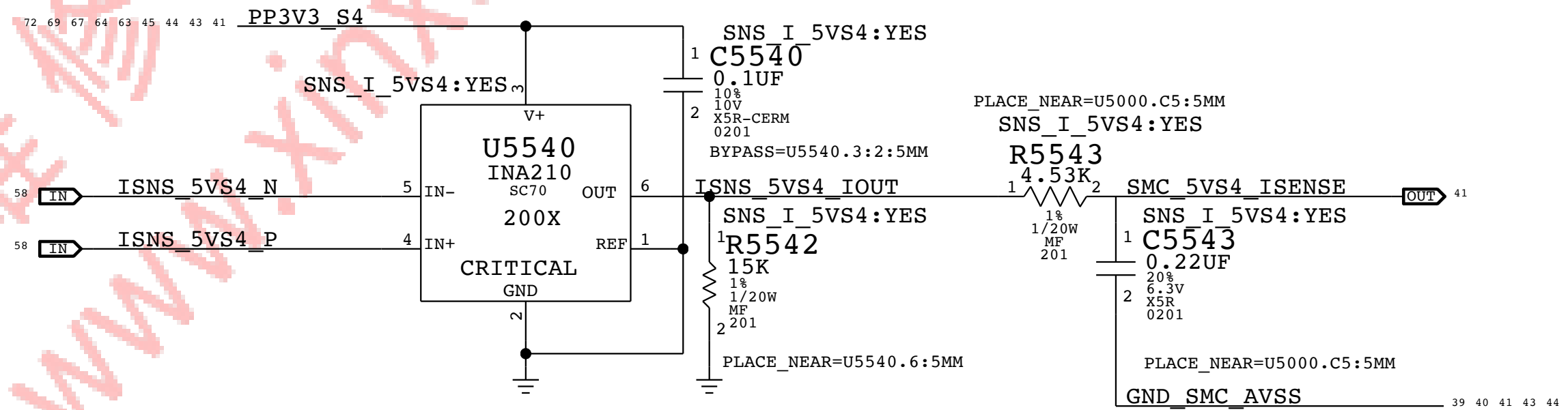
3V3\_SSD CURRENT SENSE (IHCC)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.003 (R5515)  
VSENSE: 22.5 MV, RANGE: 3.61 A  
SMC AD: 13



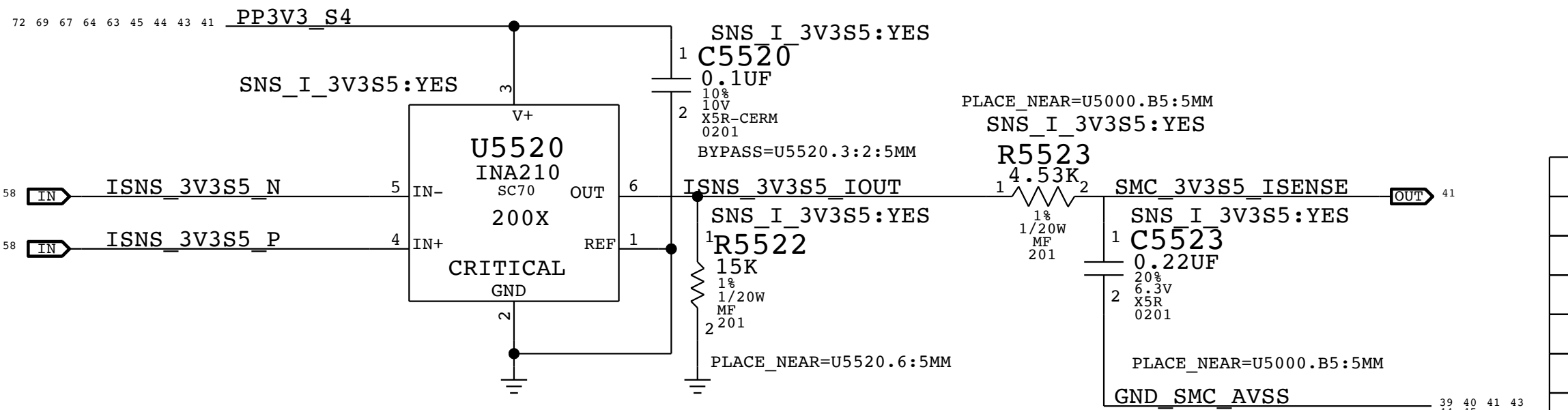
5V\_S4 CURRENT SENSE (IO5R) (HIGH-SIDE)

GAIN: 200X. EDP: ?? A  
RSENSE: 0.003 (R7900)  
VSENSE: 22.5 MV ?, RANGE: 6.8 A  
SMC AD: 21



3V3\_S5 CURRENT SENSE (IO3R) (HIGH-SIDE)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.003 (R7710)  
VSENSE: 22.5 MV ?, RANGE: 4 A  
SMC AD: 22



| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION      |
|-------------|-----|-------------------------------------|---------------|----------|-----------------|
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5503         |          | SNS_I_1VS5G:NO  |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5508         |          | SNS_I_1V2S3:NO  |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5518         |          | SNS_I_3V3SSD:NO |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5523         |          | SNS_I_3V3S5:NO  |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5528         |          | SNS_I_3V3LCD:NO |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5533         |          | SNS_I_3V3WLS:NO |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5543         |          | SNS_I_5VS4:NO   |

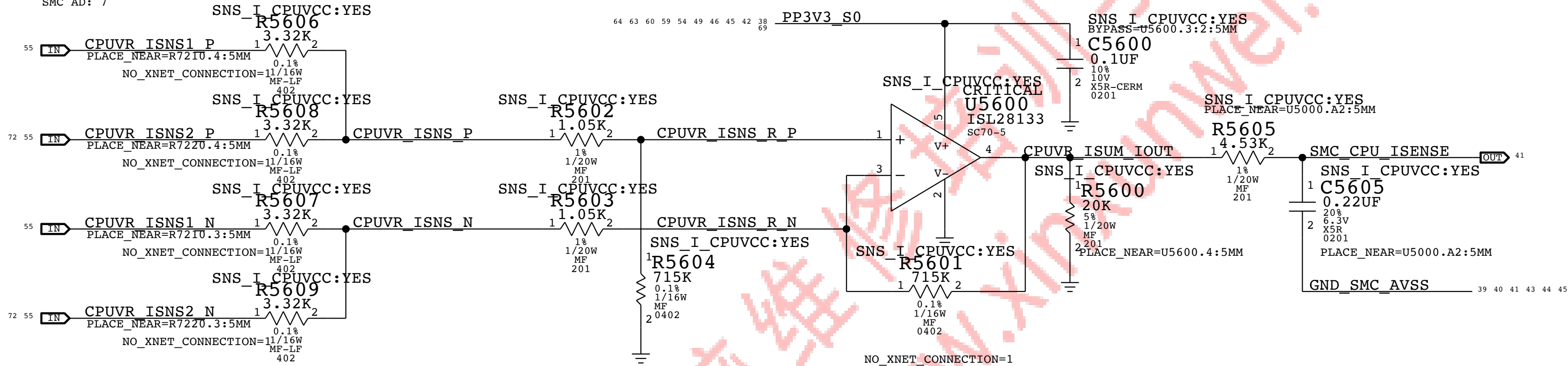
|                                                                                                                                                                                                                                                                                                                   |  |                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                             |  |                |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                                                                                                                                                                                                                              |  |                |
| PAGE TITLE                                                                                                                                                                                                                                                                                                        |  |                |
| Power Sensors Load Side                                                                                                                                                                                                                                                                                           |  |                |
|                                                                                                                                                                                                                                                                                                                   |  | DRAWING NUMBER |
|                                                                                                                                                                                                                                                                                                                   |  | 051-02265      |
|                                                                                                                                                                                                                                                                                                                   |  | REVISION       |
|                                                                                                                                                                                                                                                                                                                   |  | 1.0.0          |
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|                                                                                                                                                                                                                                                                                                                   |  | 55 OF 500      |
|                                                                                                                                                                                                                                                                                                                   |  | SHEET          |
|                                                                                                                                                                                                                                                                                                                   |  | 44 OF 73       |

BOM\_COST\_GROUP=SENSORS



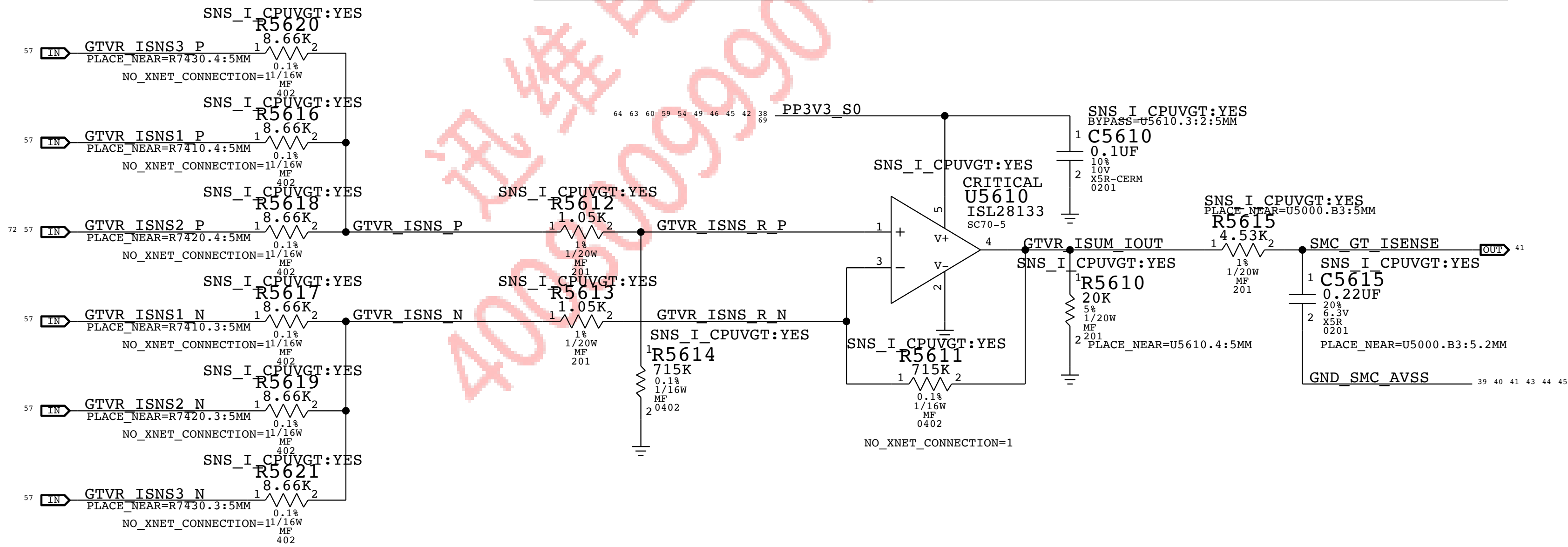
### CPU VCC CURRENT SENSE (ICAC)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.005 (R7700)  
VSENSE: 22.5 MV, RANGE: 2.27 A  
SMC AD: 7



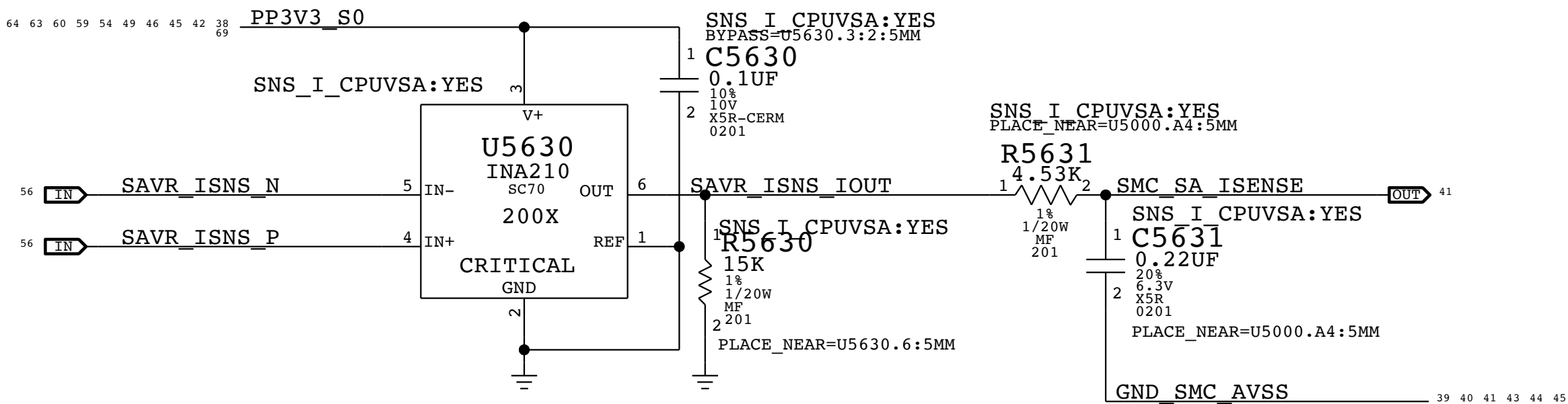
### CPU GT CURRENT SENSE (ICGC)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.005 (R7700)  
VSENSE: 22.5 MV, RANGE: 2.27 A  
SMC AD: 9



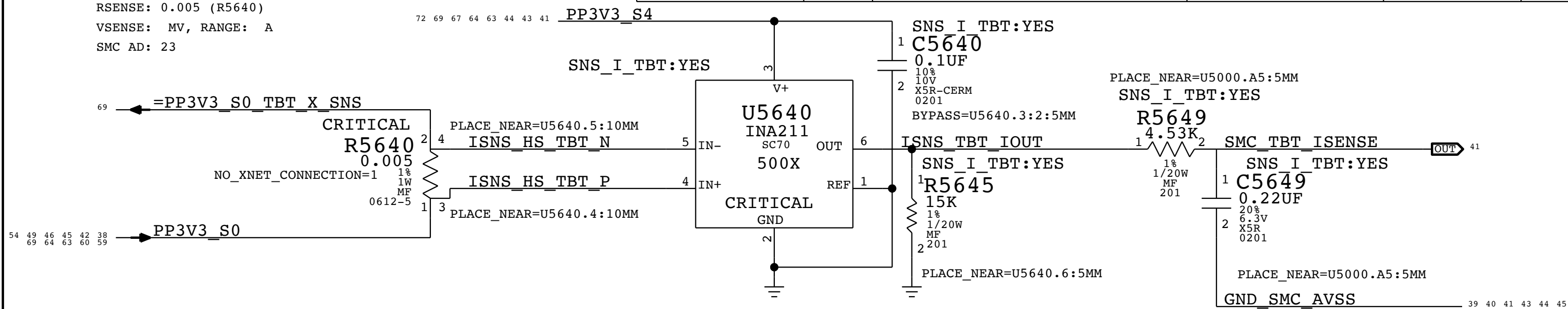
### CPU SA CURRENT SENSE (ICSC)

GAIN: 200X. EDP: 0.9 A  
RSENSE: 0.002 (R7700)  
VSENSE: 22.5 MV, RANGE: 7.67 A  
SMC AD: 10



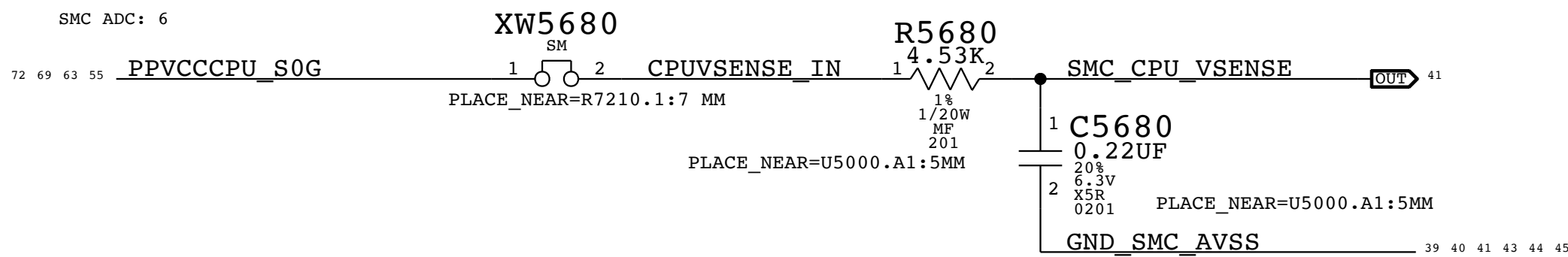
### TBT CURRENT SENSE (IULC)

GAIN: 500X. EDP: A  
RSENSE: 0.005 (R5640)  
VSENSE: MV, RANGE: A  
SMC AD: 23



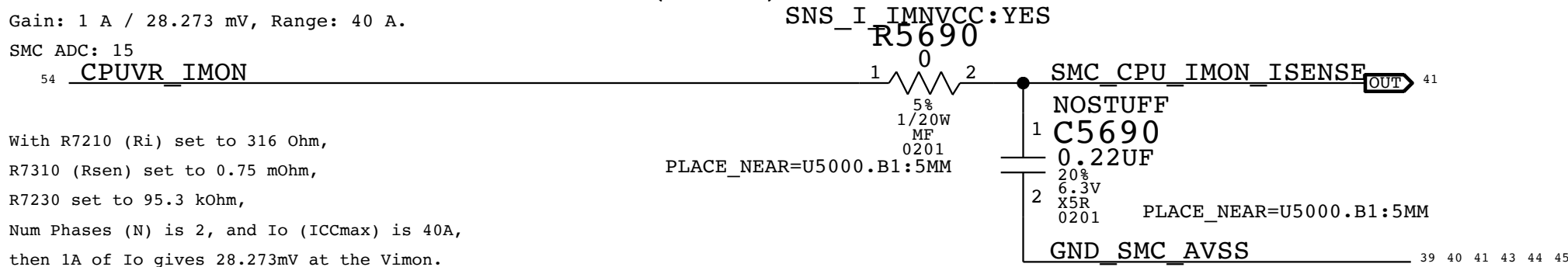
### CPU CORE VOLTAGE SENSE (VCAC)

SMC ADC: 6



### CPU CORE IMON CURRENT SENSE (ICAM)

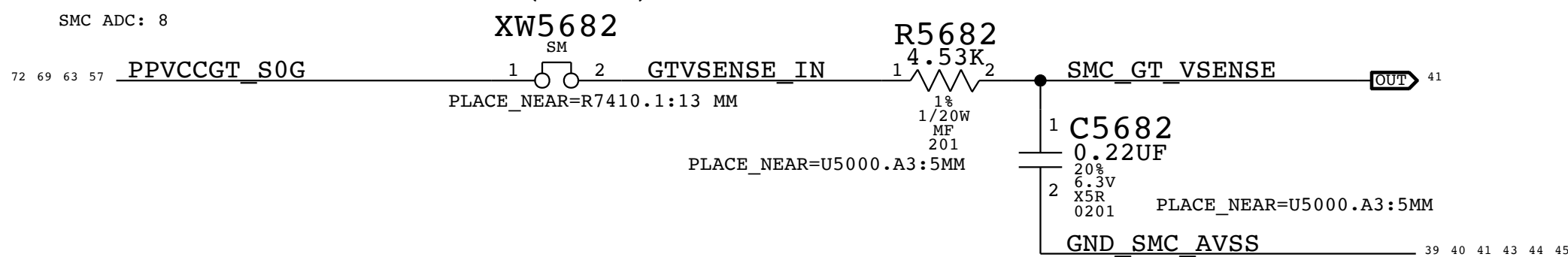
Gain: 1 A / 28.273 mV, Range: 40 A.  
SMC ADC: 15



| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION      |
|-------------|-----|-------------------------------------|---------------|----------|-----------------|
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5605         |          | SNS_I_CPUVCC:NO |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5615         |          | SNS_I_CPUVGT:NO |
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5631         |          | SNS_I_CPUVSA:NO |

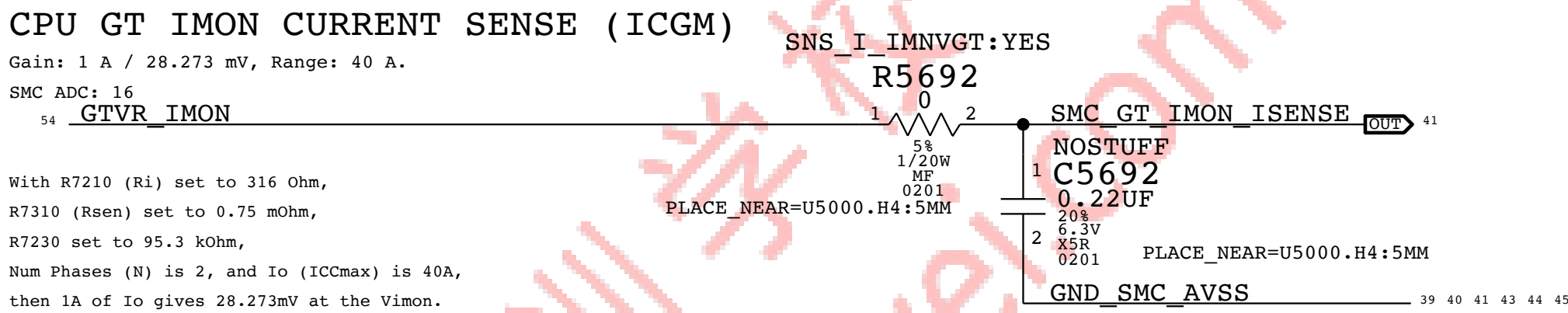
### CPU GT VOLTAGE SENSE (VCGC)

SMC ADC: 8



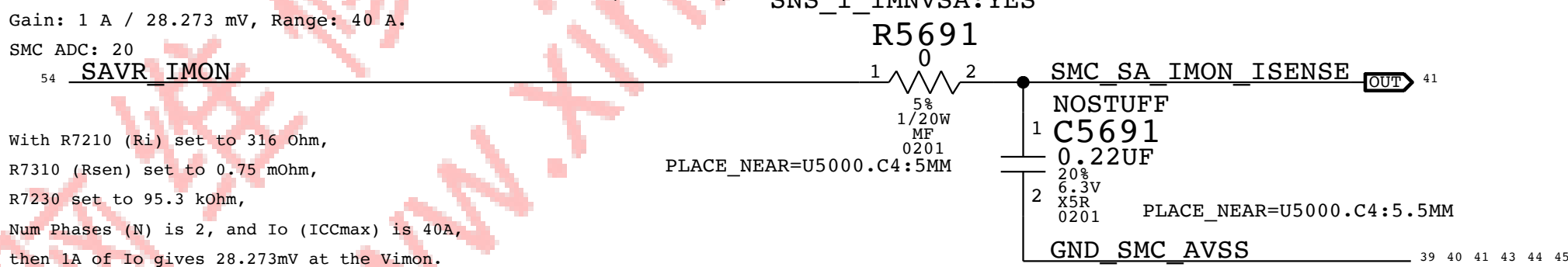
### CPU GT IMON CURRENT SENSE (ICGM)

Gain: 1 A / 28.273 mV, Range: 40 A.  
SMC ADC: 16



### CPU SA IMON CURRENT SENSE (ICSM)

Gain: 1 A / 28.273 mV, Range: 40 A.  
SMC ADC: 20



| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION   |
|-------------|-----|-------------------------------------|---------------|----------|--------------|
| 117S0008    | 1   | RES,MTL FLIM,100K,1/16W,0201,SMD,LF | C5649         |          | SNS_I_TBT:NO |

|                                                                                                                   |                |           |
|-------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| DESIGN: X502/MLB CATZ                                                                                             |                |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                              |                |           |
| PAGE TITLE                                                                                                        |                |           |
| Power Sensors Extended                                                                                            |                |           |
|                                                                                                                   | DRAWING NUMBER | 051-02265 |
|                                                                                                                   | REVISION       | 1.0.0     |
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BOM\_COST\_GROUP=SENSORS



D

C

B

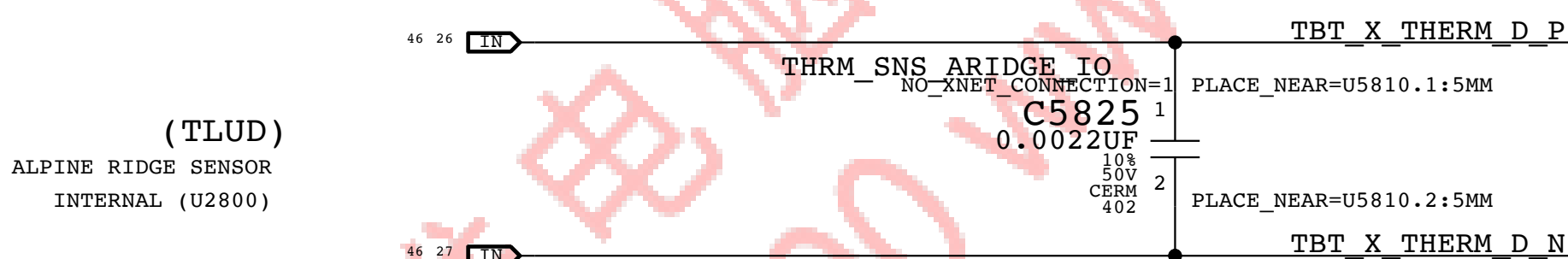
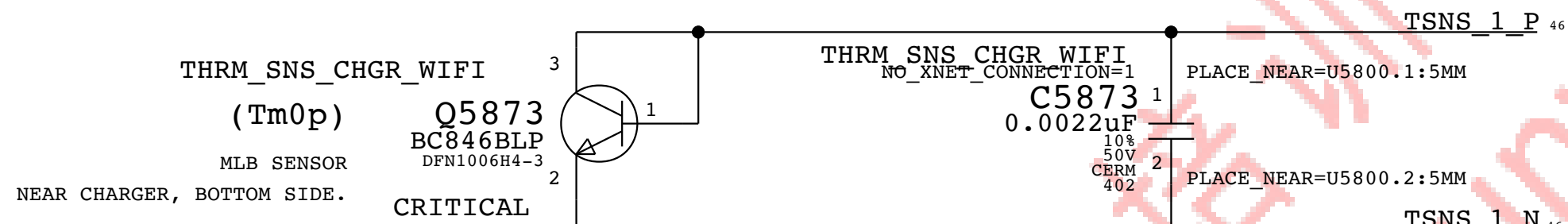
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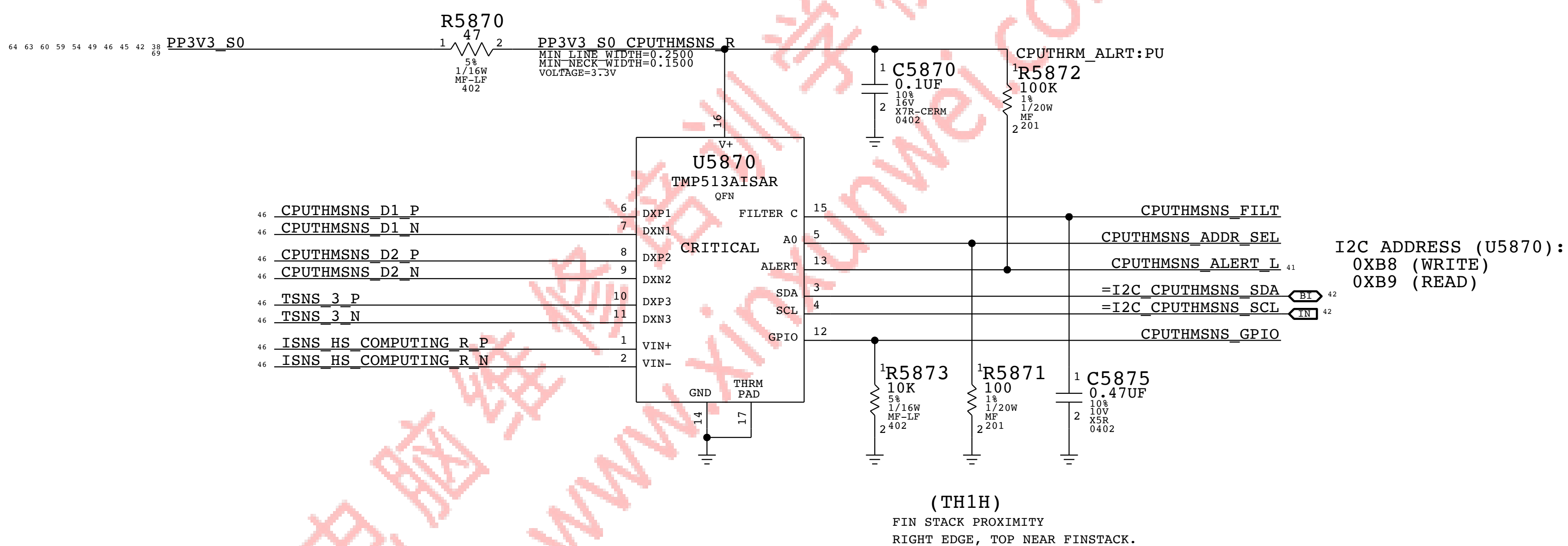
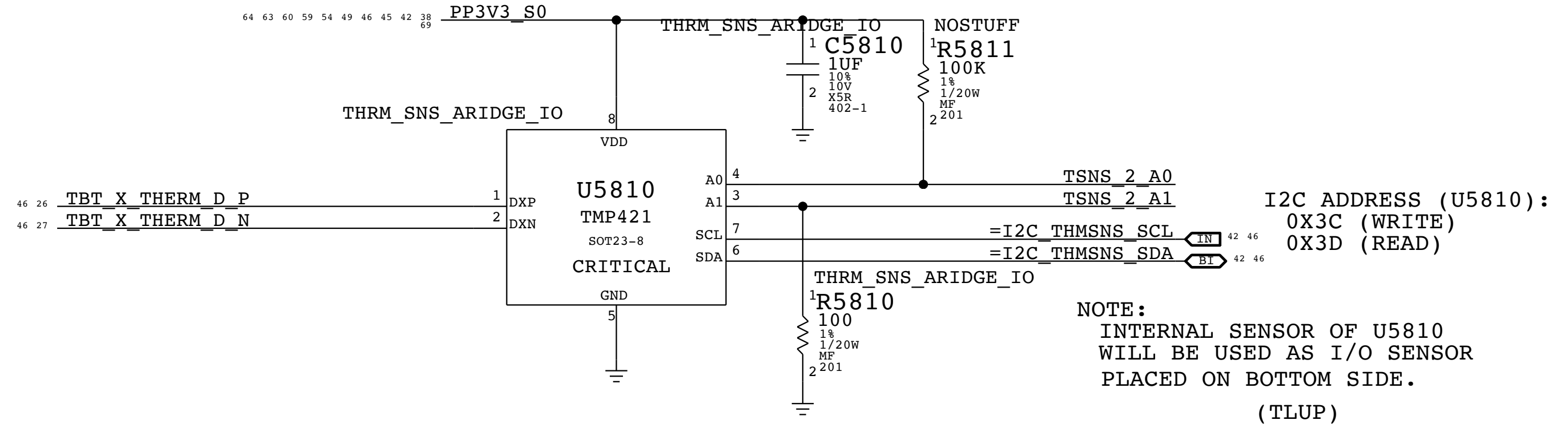
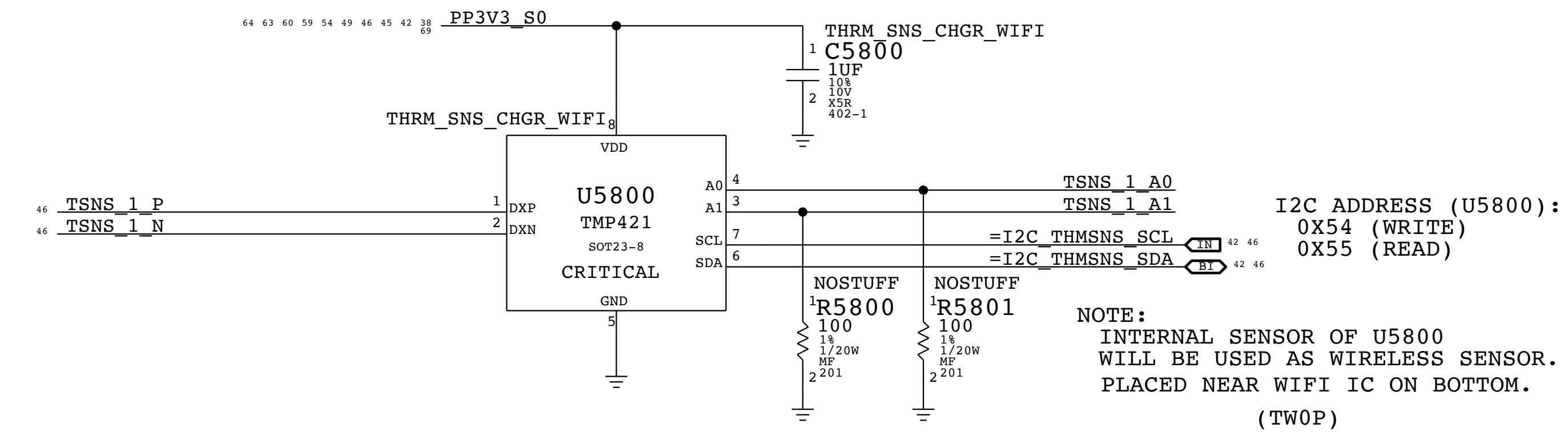
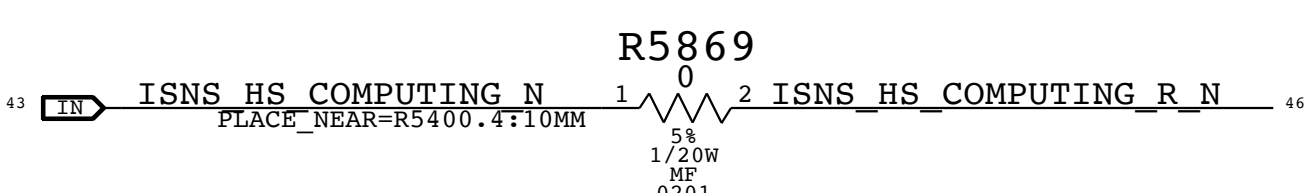
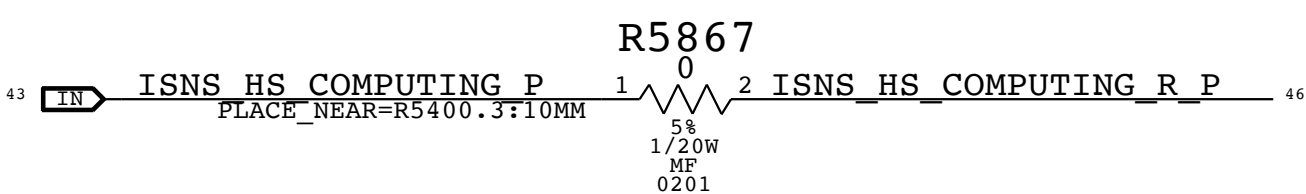
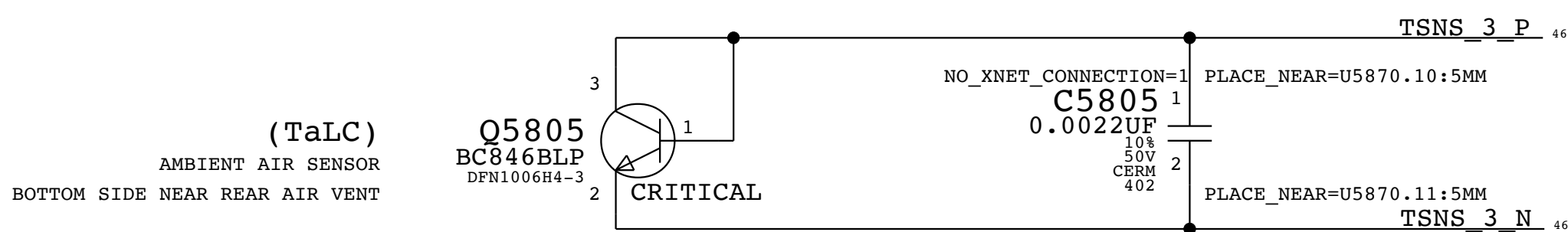
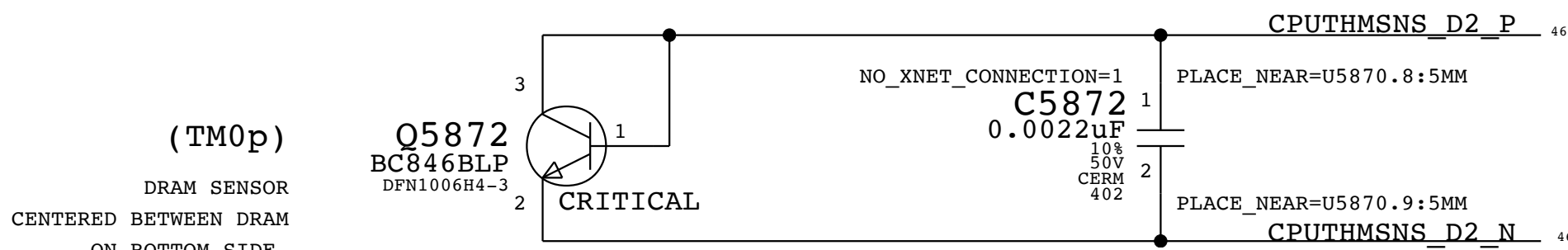
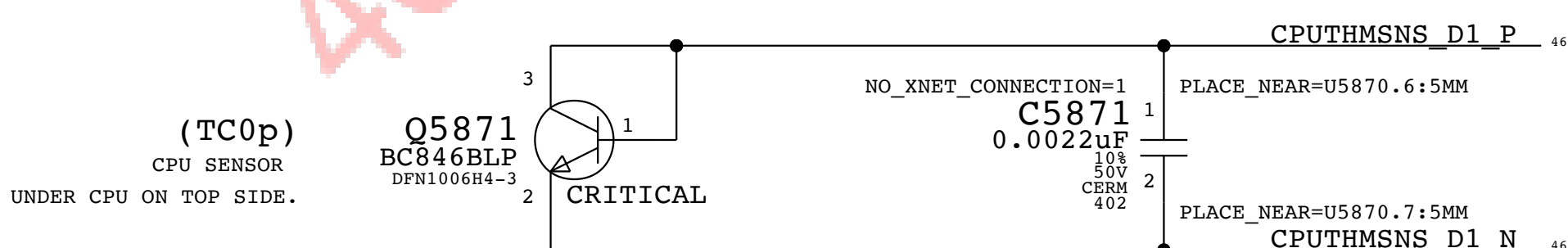
C

B

A



Thermal Sensor B & CPU High Peak Detection:  
CPU Proximity, Memory Proximity, Airflow, Fin Stack Proximity  
I2C Write: 0x98, I2C Read: 0x99

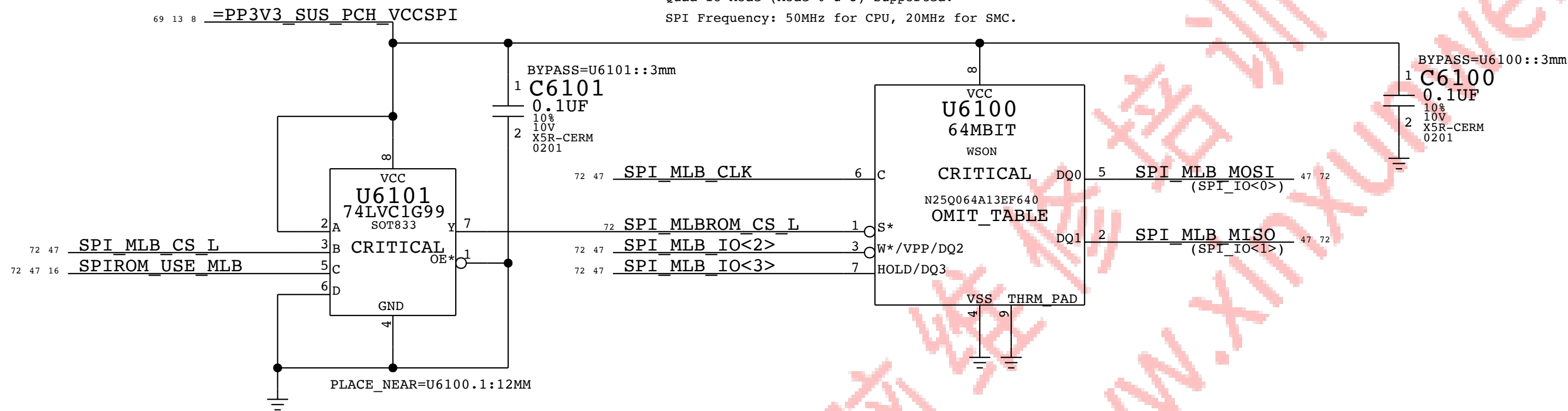


BOM\_COST\_GROUP=SENSORS

|                                                                                                                                                                                                                                                                                                                    |                |           |
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| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                              |                |           |
| LAST CHANGE: Mon Aug 8 12:54:34 2016                                                                                                                                                                                                                                                                               |                |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                         |                |           |
| Thermal Sensors                                                                                                                                                                                                                                                                                                    |                |           |
|                                                                                                                                                                                                                                                                                                                    | DRAWING NUMBER | 051-02265 |
|                                                                                                                                                                                                                                                                                                                    | REVISION       | 1.0.0     |
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|                                                                                                                                                                                                                                                                                                                    |                | PAGE      |
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SPI ROM - Combo BGA Footprint (3 vendors)

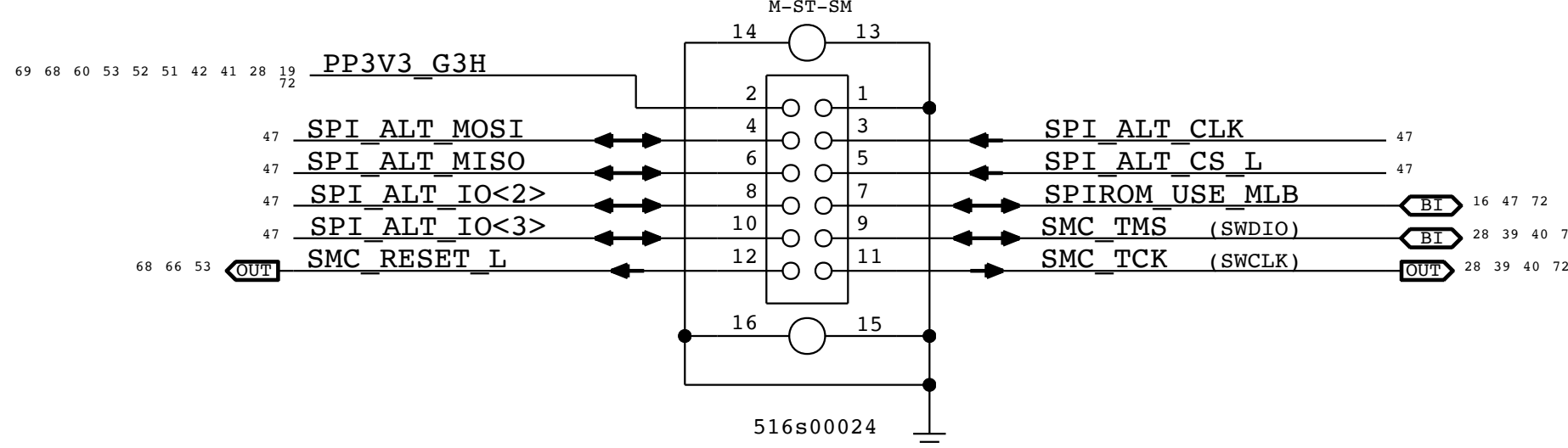
Quad-IO Mode (Mode 0 & 3) supported.  
SPI Frequency: 50MHz for CPU, 20MHz for SMC.



SPI+SWD SAM Connector

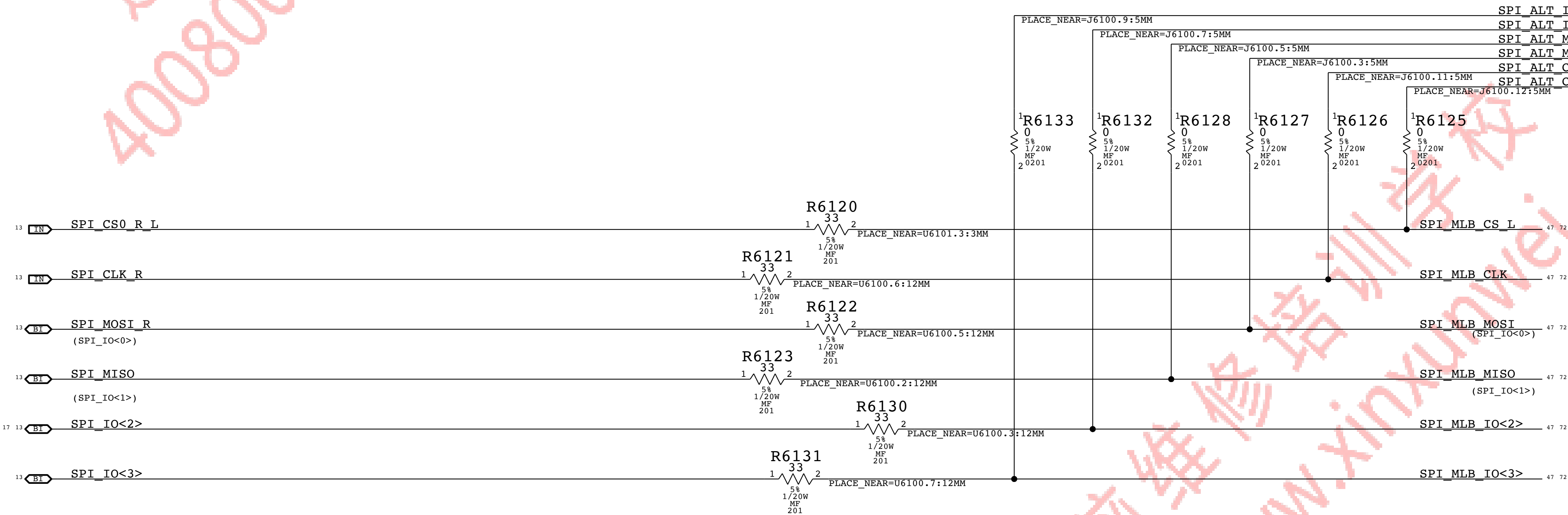
CRITICAL

J6100  
DF40PC-12DP-0.4V-51  
M-SW-SW



BootROM SPI Bus Series Termination

CPU Master



Sam Card ROM Slave

SPI ROM Slave

|                                                                                                                                                                                                                                                                                                                    |                |           |
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| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                              |                |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                                                                                                                                                                                                                               |                |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                         |                |           |
| SPI ROM & SWD Debug                                                                                                                                                                                                                                                                                                |                |           |
|                                                                                                                                                                                                                                                                                                                    | DRAWING NUMBER | 051-02265 |
|                                                                                                                                                                                                                                                                                                                    | REVISION       | 1.0.0     |
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|                                                                                                                                                                                                                                                                                                                    | SHEET          | 47 OF 73  |

BOM\_COST\_GROUP=CPU SUPPORT

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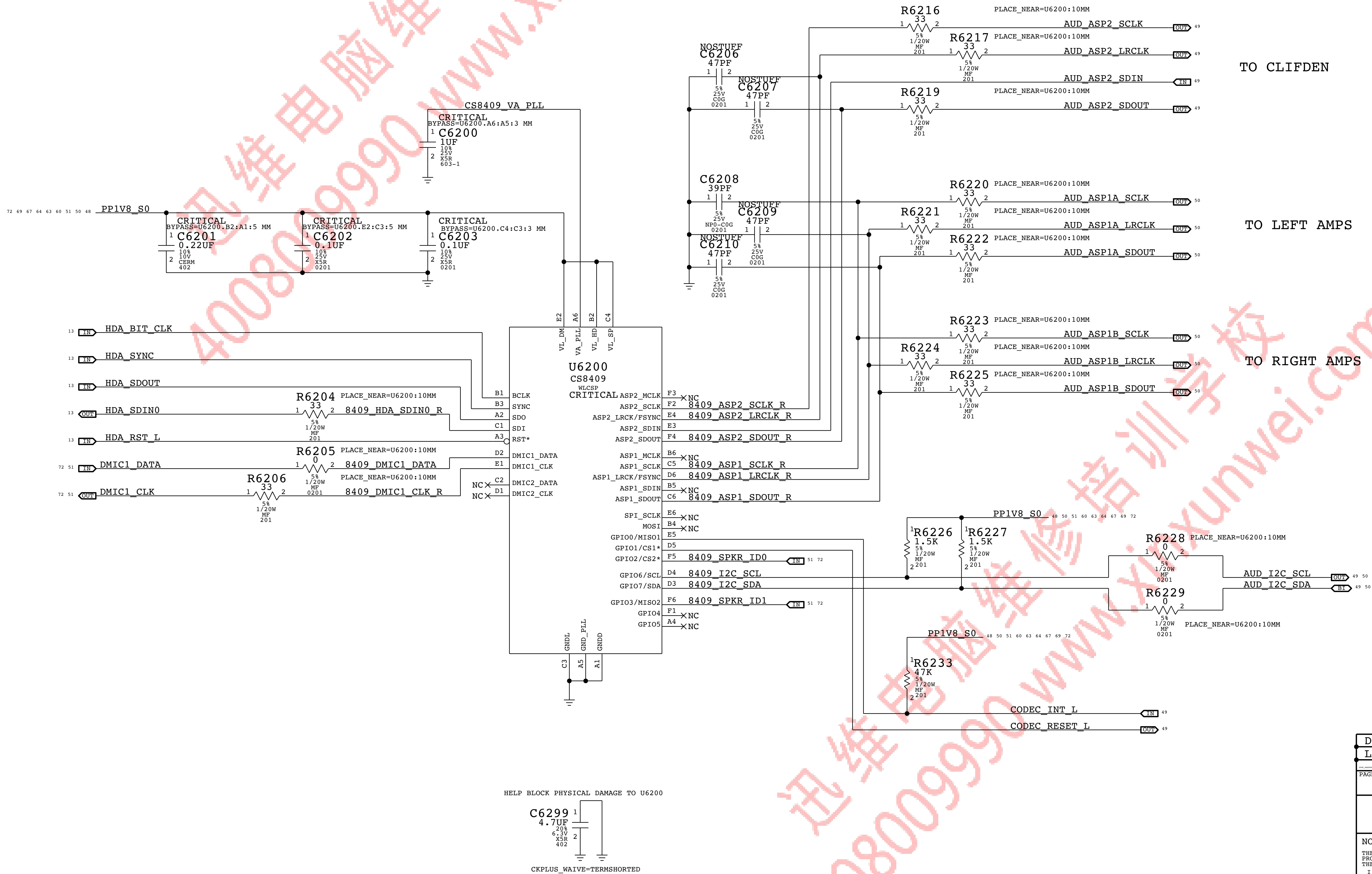
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| DESIGN: X502/MLB CATZ                                                                                             |                |           |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                              |                |           |           |
| PAGE TITLE                                                                                                        |                |           |           |
| HDA BRIDGE                                                                                                        |                |           |           |
|                                                                                                                   | DRAWING NUMBER | 051-02265 | SIZE      |
|                                                                                                                   | REVISION       | 1.0.0     | D         |
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BOM\_COST\_GROUP=AUDIO



|                                                                                                                                                                                                                                                                                                                                   |                |           |      |
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| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                                             |                |           |      |
| LAST CHANGE: Mon Aug 8 12:54:34 2016                                                                                                                                                                                                                                                                                              |                |           |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                        |                |           |      |
| JACK CODEC                                                                                                                                                                                                                                                                                                                        |                |           |      |
|  Apple Inc.                                                                                                                                                                                                                                  | DRAWING NUMBER |           | SIZE |
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4X MONO SPEAKER AMPLIFIERS

APN: 353S00685  
GAIN: 0DBFS = 7VRMS

D

D

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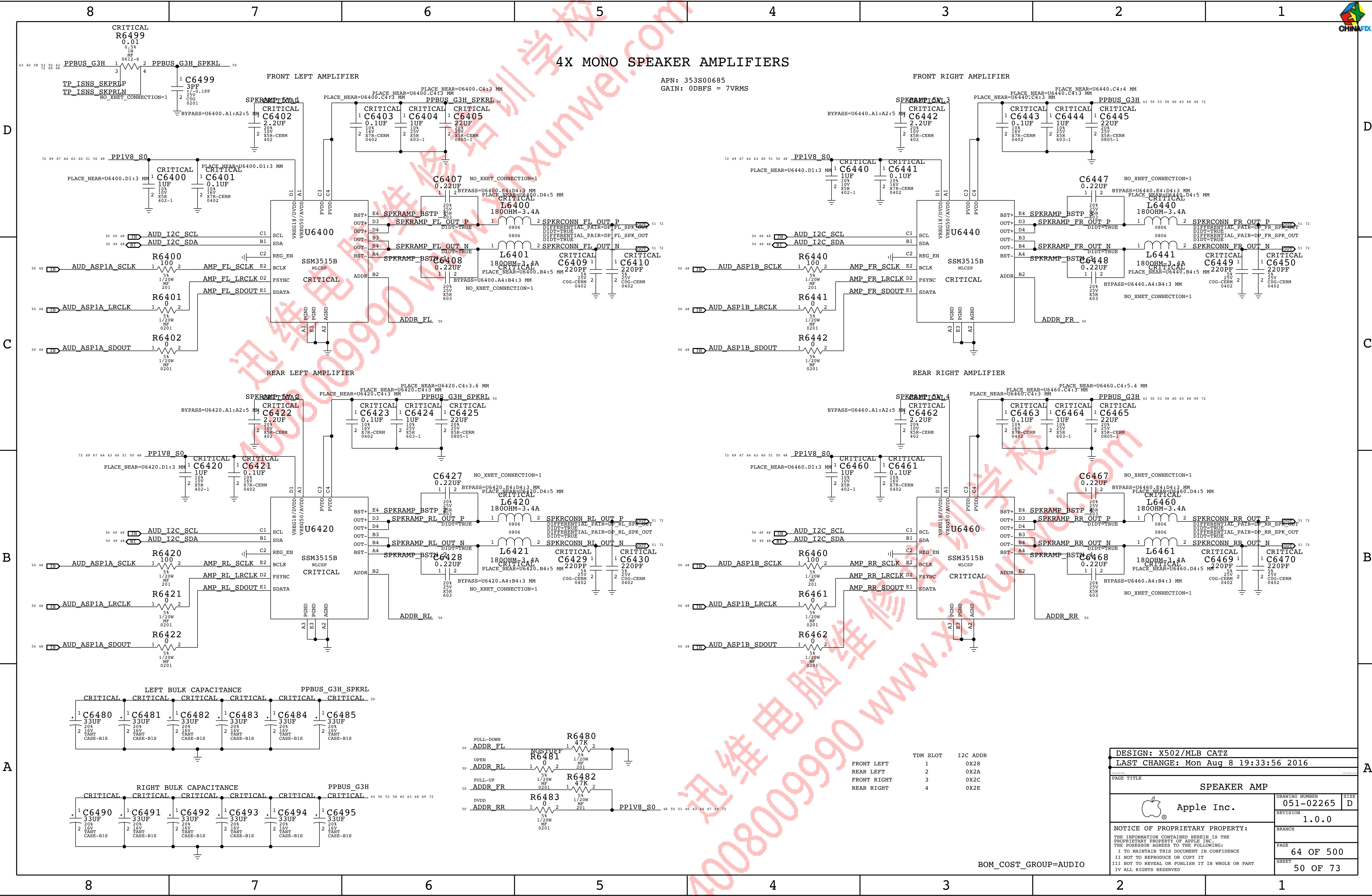
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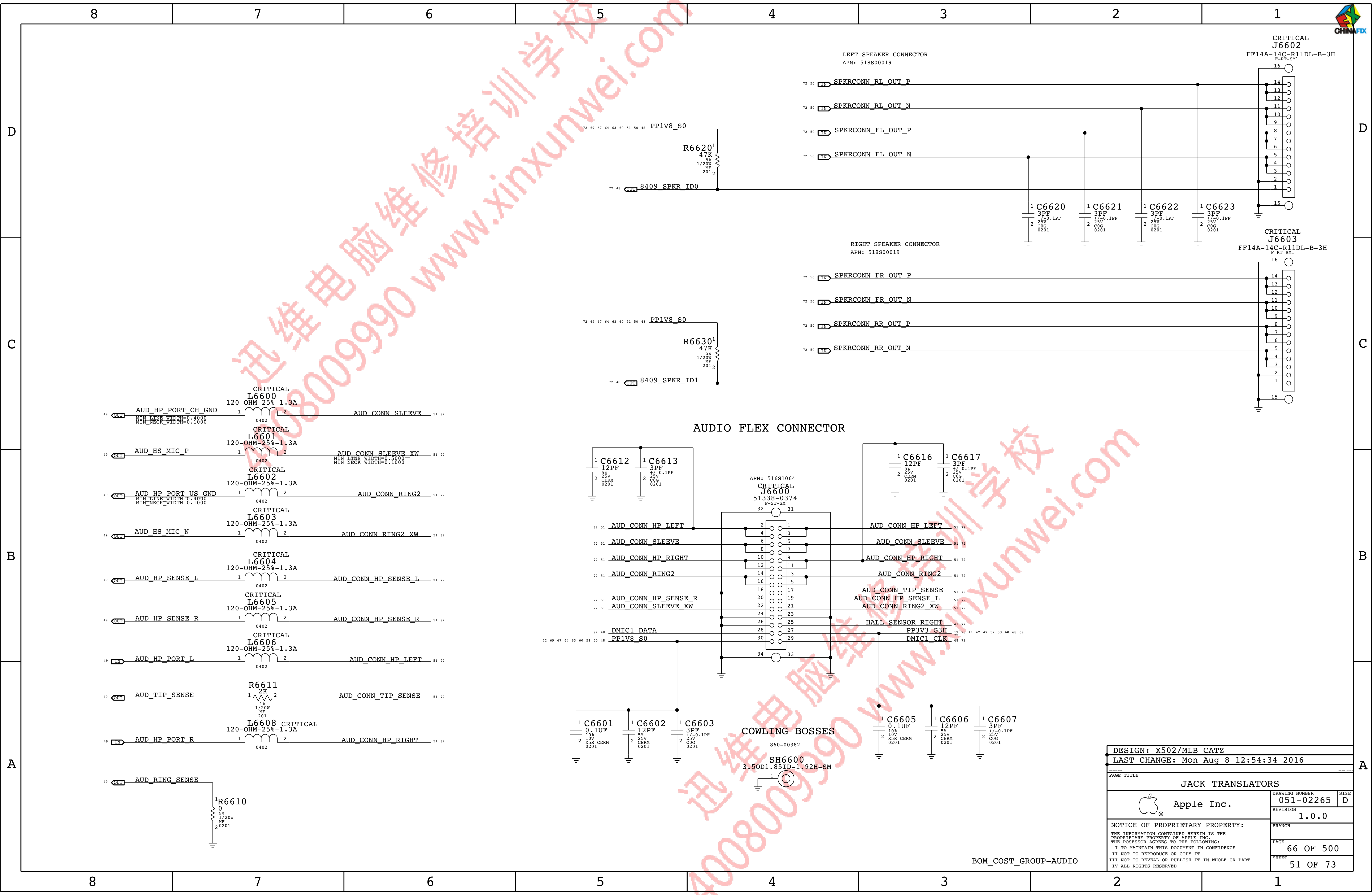
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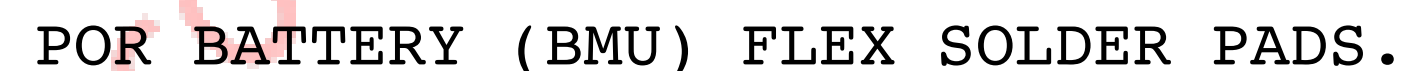


| TDM SLOT    | I2C ADDR |
|-------------|----------|
| FRONT LEFT  | 1 0X28   |
| REAR LEFT   | 2 0X2A   |
| FRONT RIGHT | 3 0X2C   |
| REAR RIGHT  | 4 0X2E   |

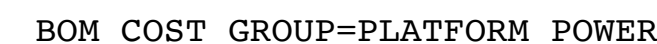
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| DESIGN: X502/MLB CATZ                                                                                              |  |                          |
| LAST CHANGE: Mon Aug 8 19:33:56 2016                                                                               |  |                          |
| PAGE TITLE                                                                                                         |  |                          |
| SPEAKER AMP                                                                                                        |  |                          |
| Apple Inc.                                                                                                         |  | DRAWING NUMBER 051-02265 |
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BOM\_COST\_GROUP=AUDIO





BMU power flex is soldered to MLB.



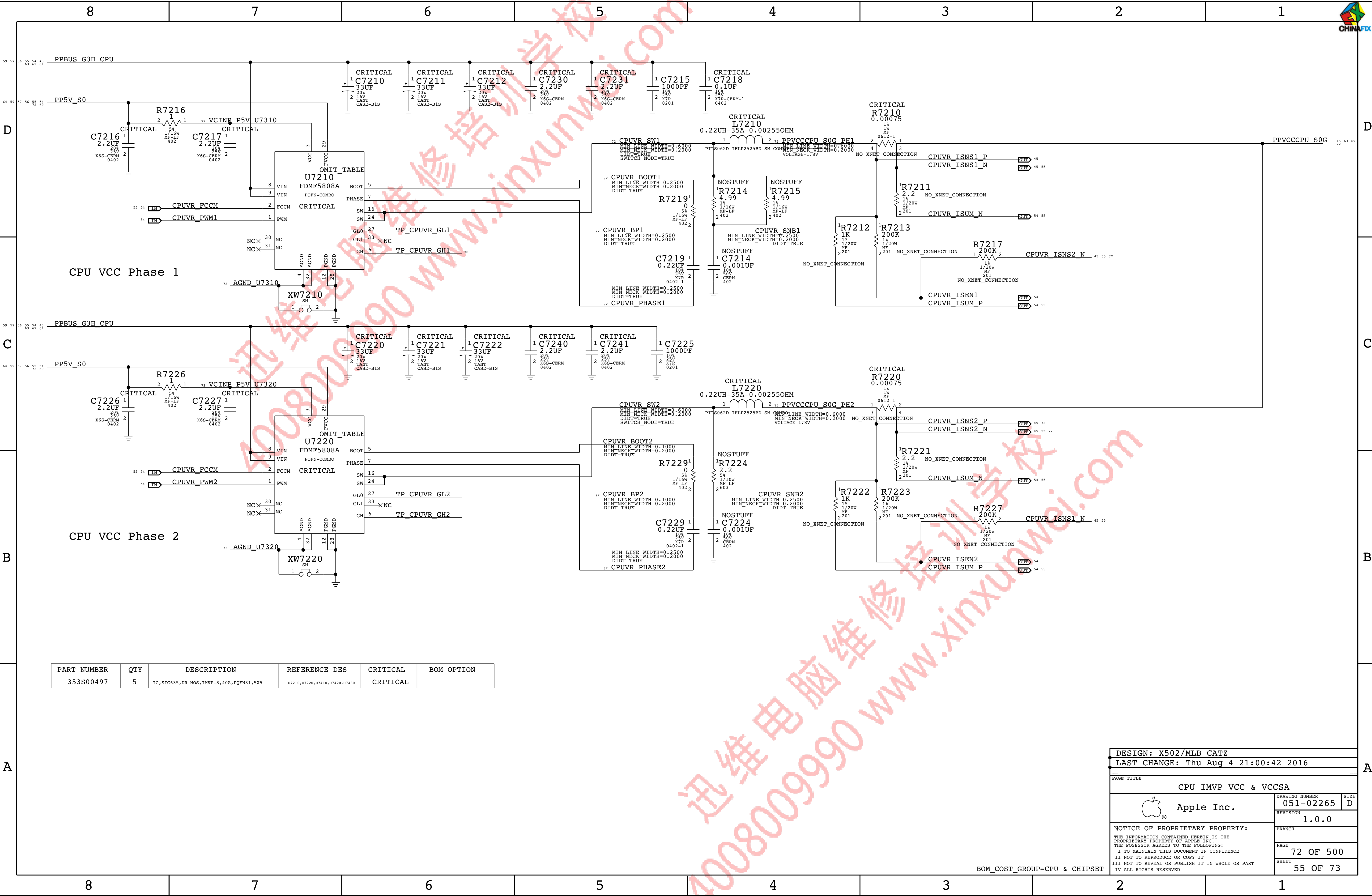
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| PAGE TITLE                                                                                                                                                                                                                                                                                                             |                |      |
| DC-IN & BATTERY CONNECTORS                                                                                                                                                                                                                                                                                             |                |      |
|  Apple Inc.                                                                                                                                                                                                                       | DRAWING NUMBER | SIZE |
|                                                                                                                                                                                                                                                                                                                        | 051-02265      | D    |
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BOM\_COST\_GROUP=PLATFORM POWER

|                                                                                                                                                                                                                                                                                                                             |  |                       |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------|-----------|
| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                                       |  |                       |           |
| LAST CHANGE: Mon Aug 8 12:54:34 2016                                                                                                                                                                                                                                                                                        |  |                       |           |
| SYNCH MASTER-PAULK                                                                                                                                                                                                                                                                                                          |  | SYNCH DATE=06/15/2015 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |                       |           |
| PBUS Supply & Battery Charger                                                                                                                                                                                                                                                                                               |  |                       |           |
|  Apple Inc.                                                                                                                                                                                                                            |  | DRAWING NUMBER        | SIZE      |
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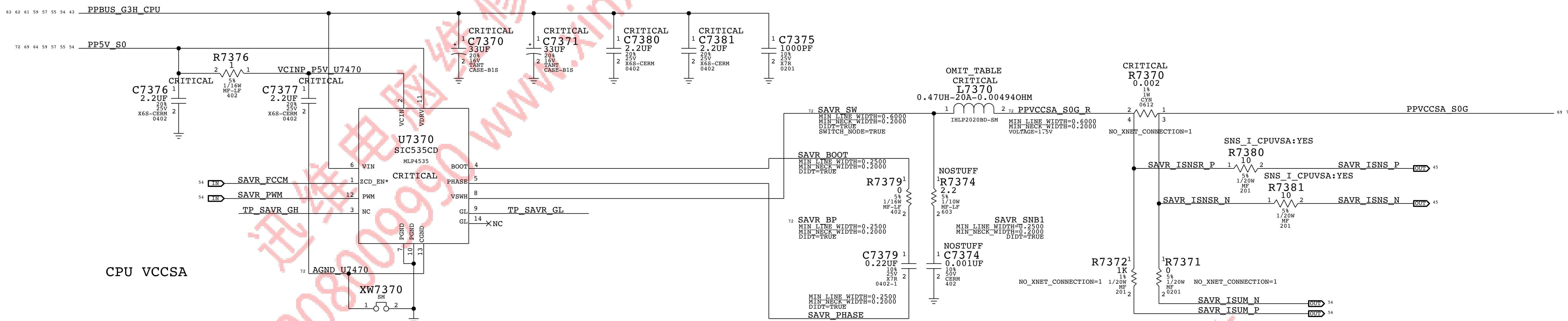


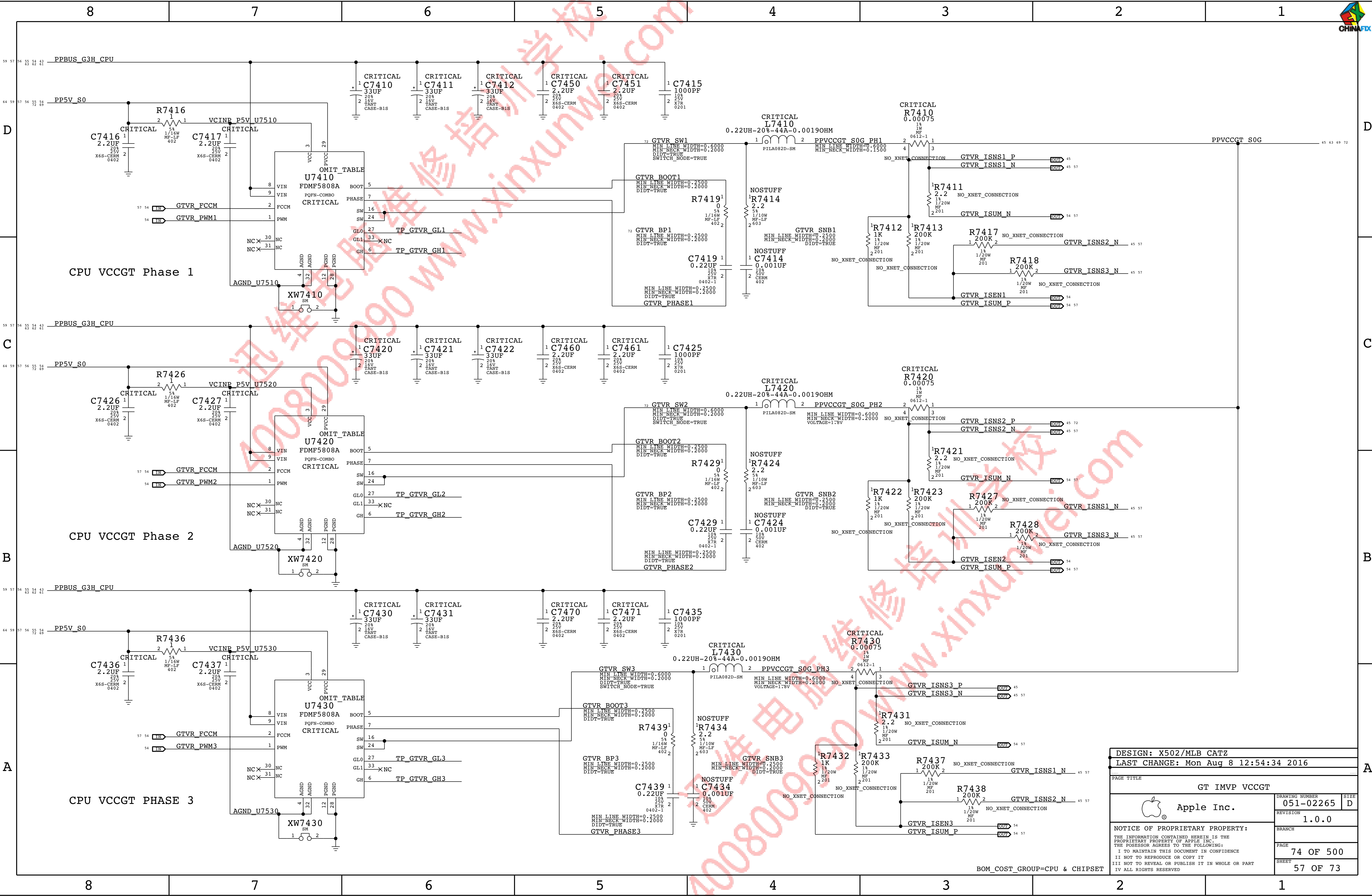
| PART NUMBER | QTY | DESCRIPTION                                  | REFERENCE DES                     | CRITICAL | BOM OPTION |
|-------------|-----|----------------------------------------------|-----------------------------------|----------|------------|
| 353S00497   | 5   | IC, SIC635, DR MOS, IMVP-B, 40A, PQFN31, 5X5 | U7210, U7220, U7410, U7420, U7430 | CRITICAL |            |

|                                                                                                                    |                |           |           |
|--------------------------------------------------------------------------------------------------------------------|----------------|-----------|-----------|
| DESIGN: X502/MLB CATZ                                                                                              |                |           |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                               |                |           |           |
| PAGE TITLE                                                                                                         |                |           |           |
| CPU IMVP VCC & VCCSA                                                                                               |                |           |           |
|                                                                                                                    | DRAWING NUMBER | 051-02265 | SIZE D    |
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BOM\_COST\_GROUP=CPU & CHIPSET

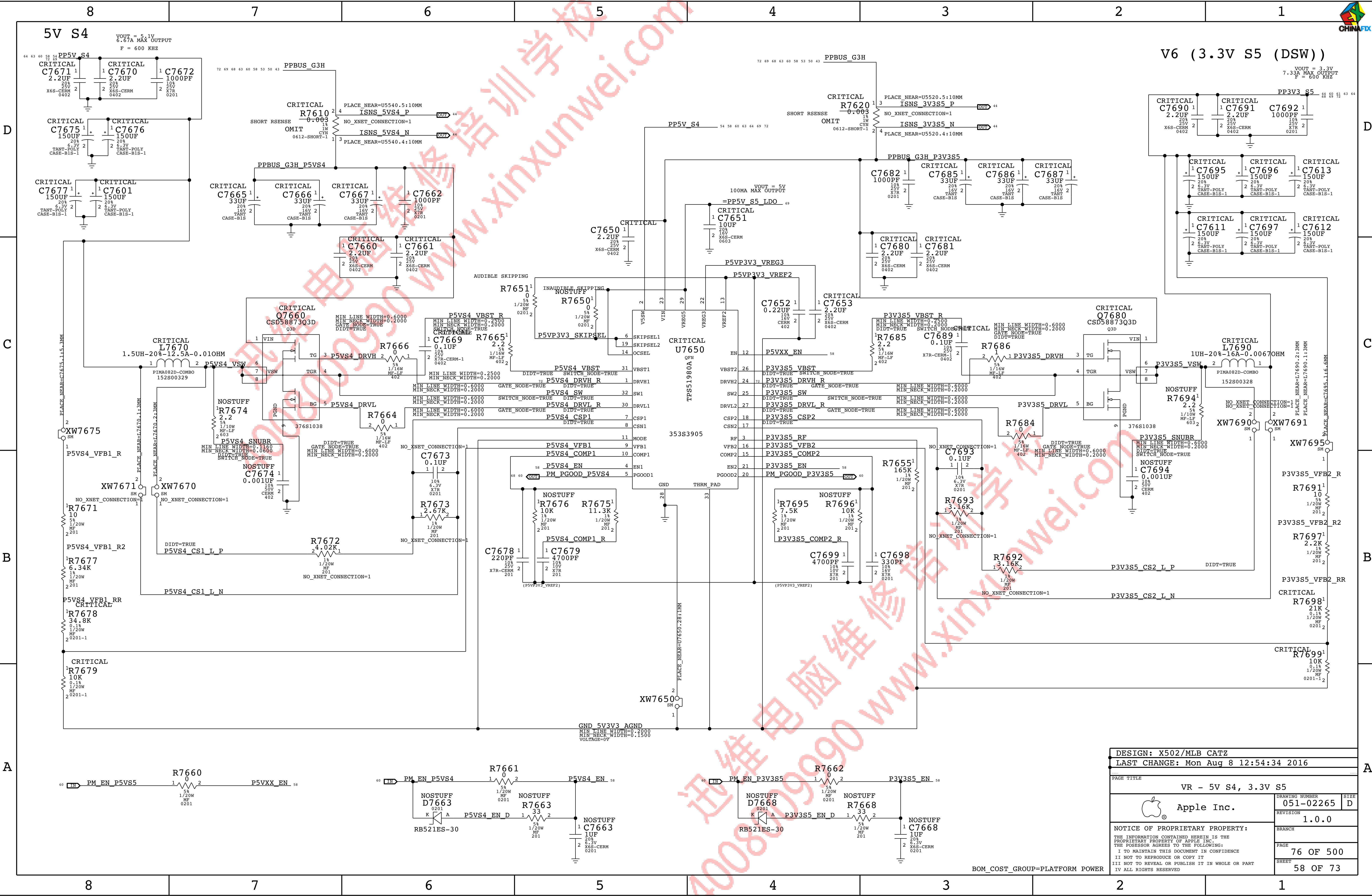
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| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                  |                   |                   |      |
| LAST CHANGE: Mon Aug 8 12:54:34 2016                                                                                                                                                                                                                                                   |                   |                   |      |
| PAGE TITLE                                                                                                                                                                                                                                                                             |                   |                   |      |
| IMVP VCCSA                                                                                                                                                                                                                                                                             |                   |                   |      |
|  Apple Inc.                                                                                                                                                                                       | DRAWING NUMBER    |                   | SIZE |
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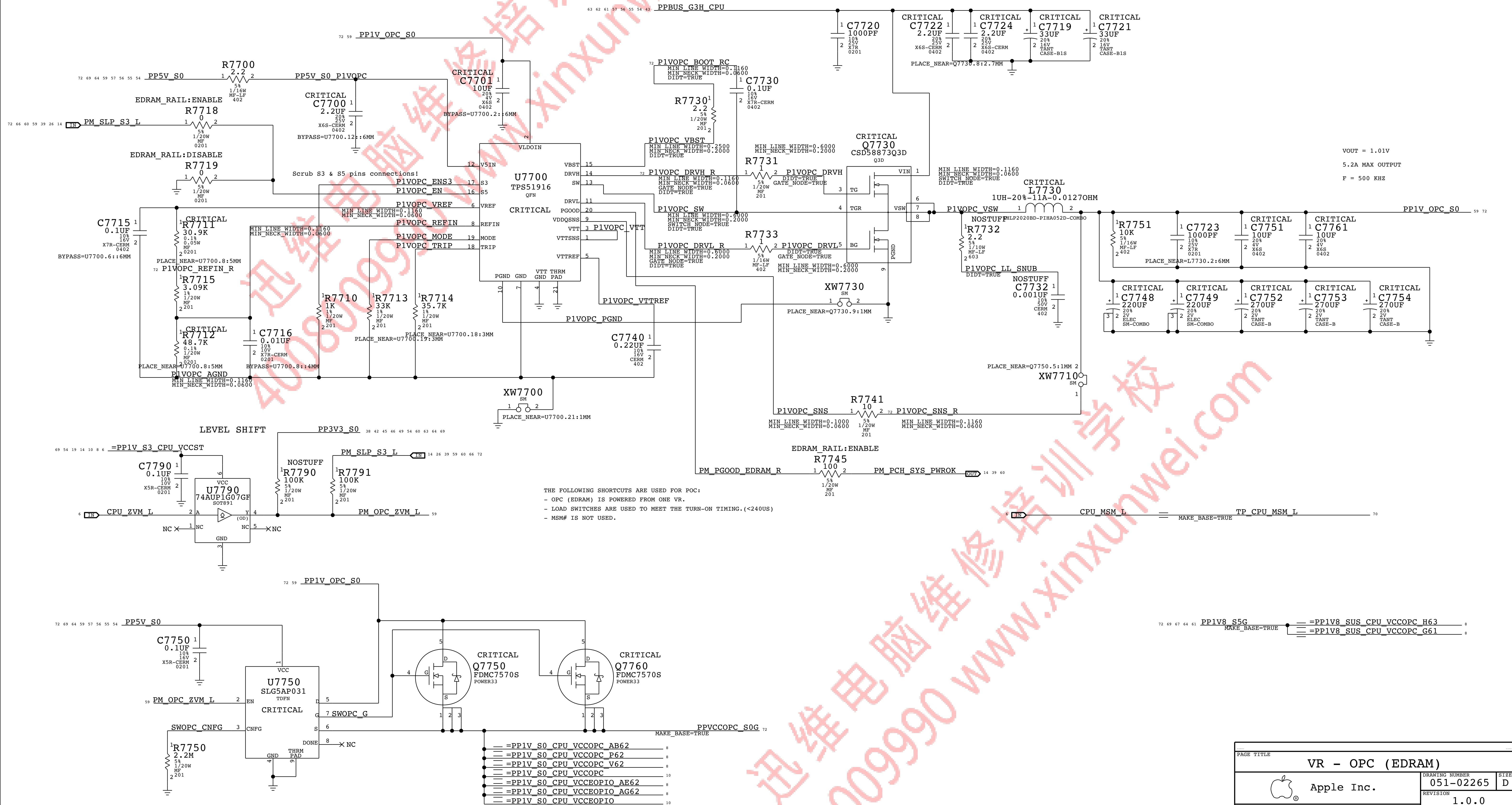
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| DESIGN: X502/MLB CATZ                                                                                              |                |           |
| LAST CHANGE: Mon Aug 8 12:54:34 2016                                                                               |                |           |
| PAGE TITLE                                                                                                         |                |           |
| GT IMVP VCCGT                                                                                                      |                |           |
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BOM\_COST\_GROUP=CPU & CHIPSET



|                                                                                                                                                                                                                                                                                                                          |                |
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| DESIGN: X502/MLB CATZ                                                                                                                                                                                                                                                                                                    |                |
| LAST CHANGE: Mon Aug 8 12:54:34 2016                                                                                                                                                                                                                                                                                     |                |
| PAGE TITLE                                                                                                                                                                                                                                                                                                               |                |
| VR - 5V S4, 3.3V S5                                                                                                                                                                                                                                                                                                      |                |
|  Apple Inc.                                                                                                                                                                                                                         | DRAWING NUMBER |
|                                                                                                                                                                                                                                                                                                                          | 051-02265      |
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1V S0 REGULATOR FOR EDRAM (ON-PACKAGE CACHE)

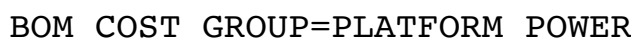


THE FOLLOWING SHORTCUTS ARE USED FOR POC:  
- OPC (EDRAM) IS POWERED FROM ONE VR.  
- LOAD SWITCHES ARE USED TO MEET THE TURN-ON TIMING. (<240US)  
- MSM# IS NOT USED.

|                                                                                                                                                                                                                                                                                                                    |  |                |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|-----------|
| PAGE TITLE                                                                                                                                                                                                                                                                                                         |  |                |           |
| VR - OPC (EDRAM)                                                                                                                                                                                                                                                                                                   |  |                |           |
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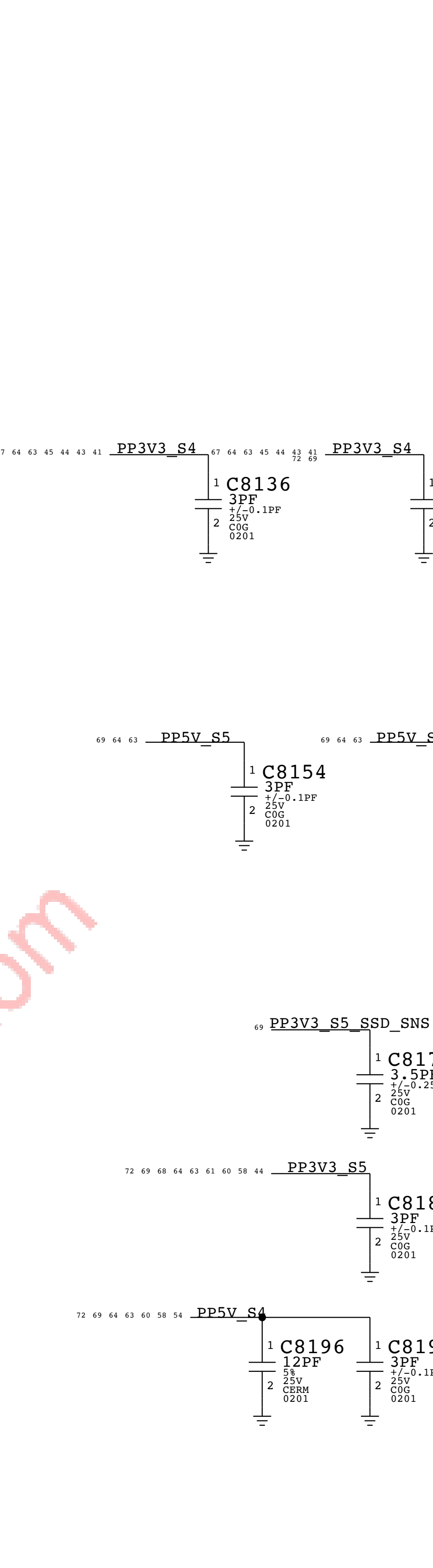
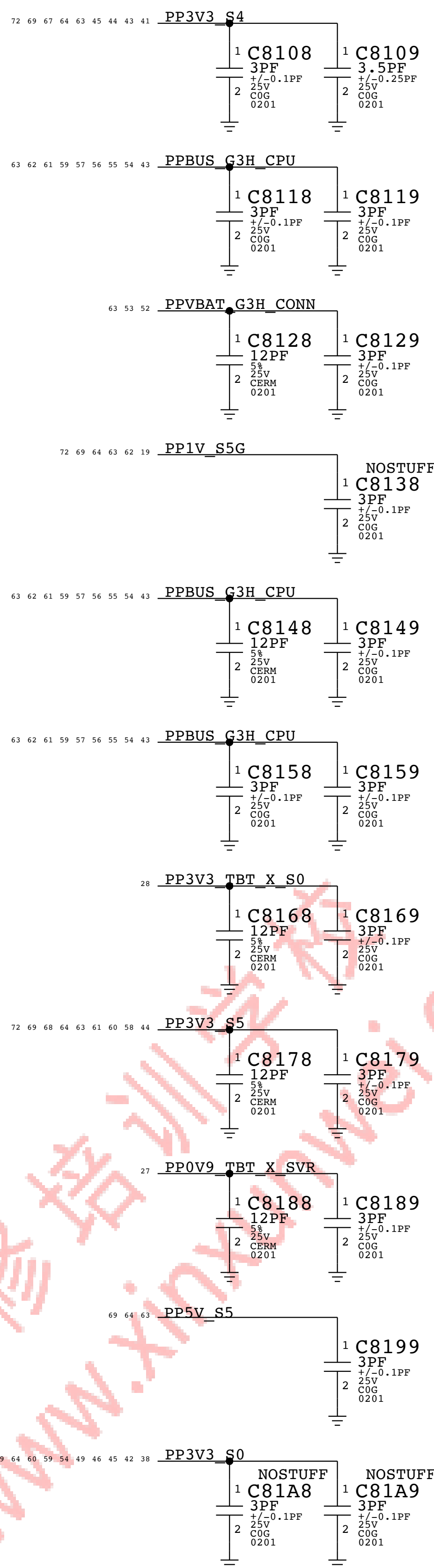
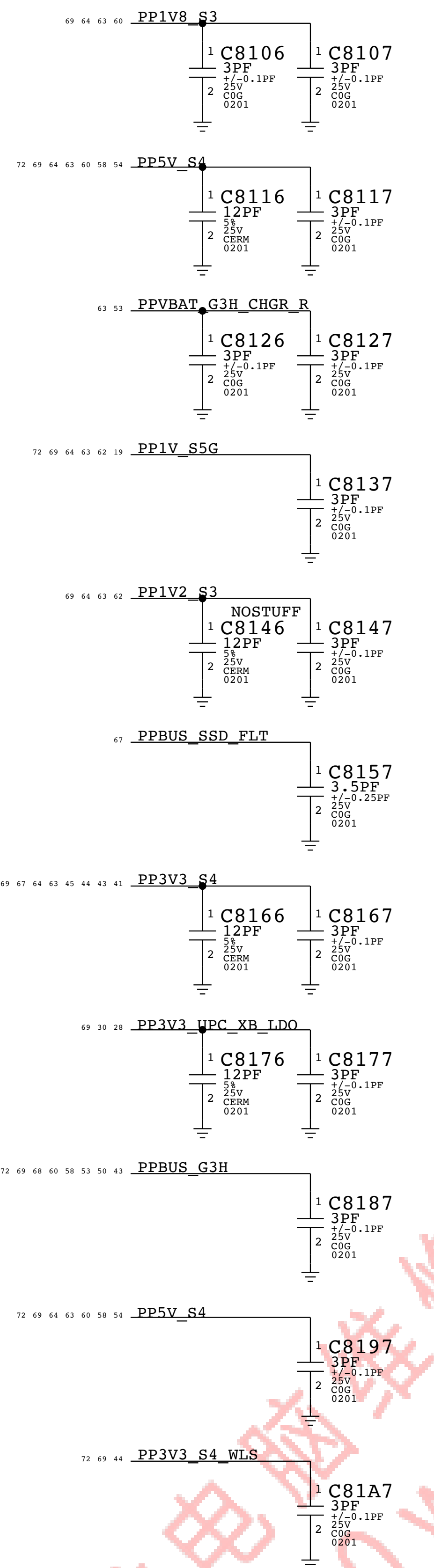
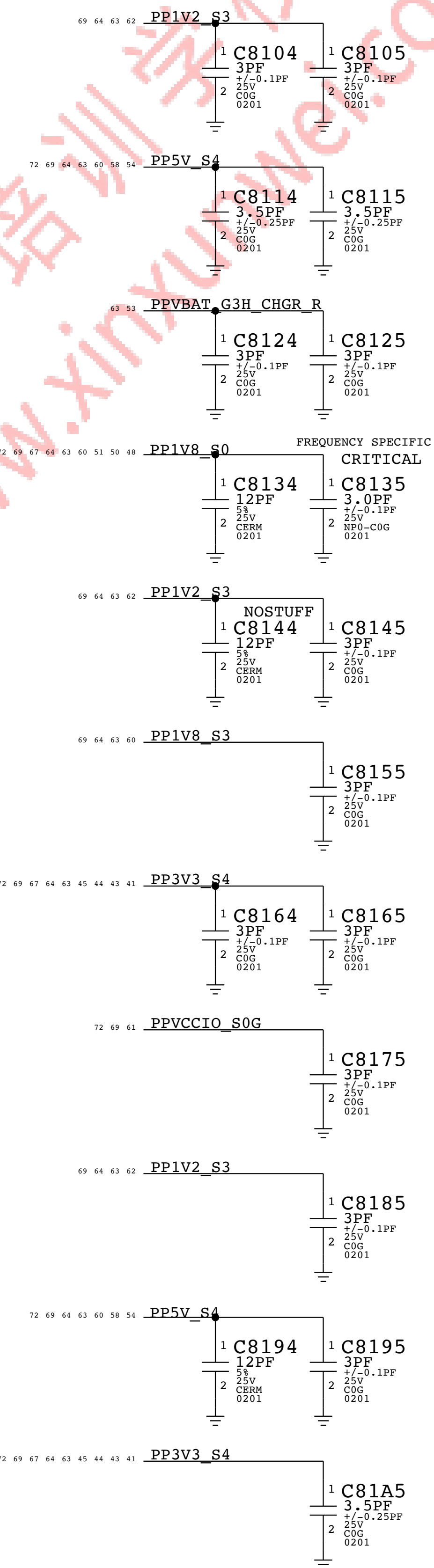
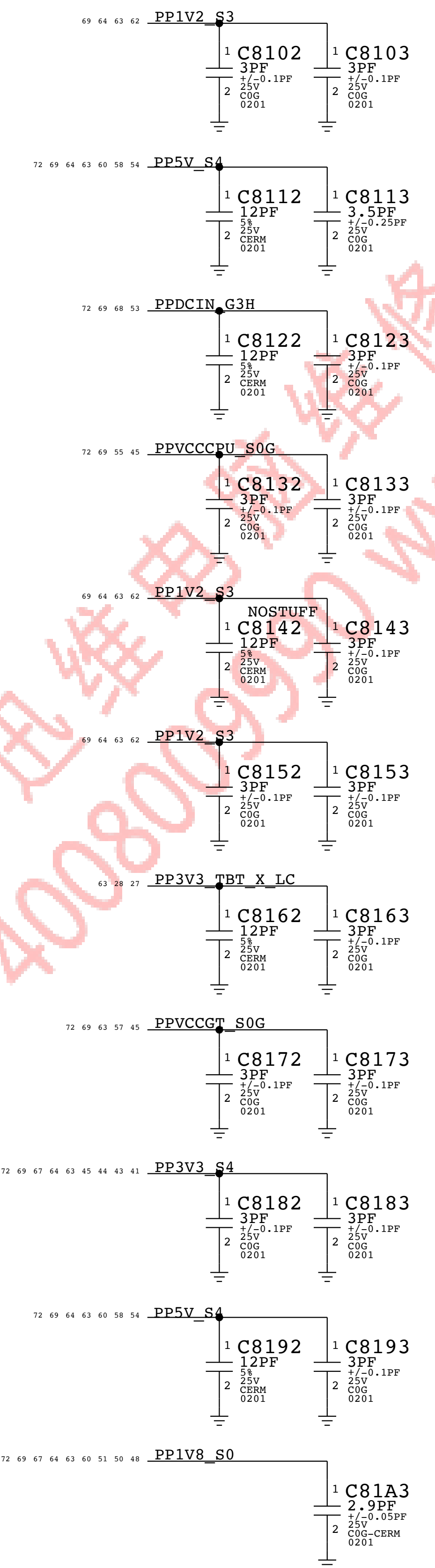
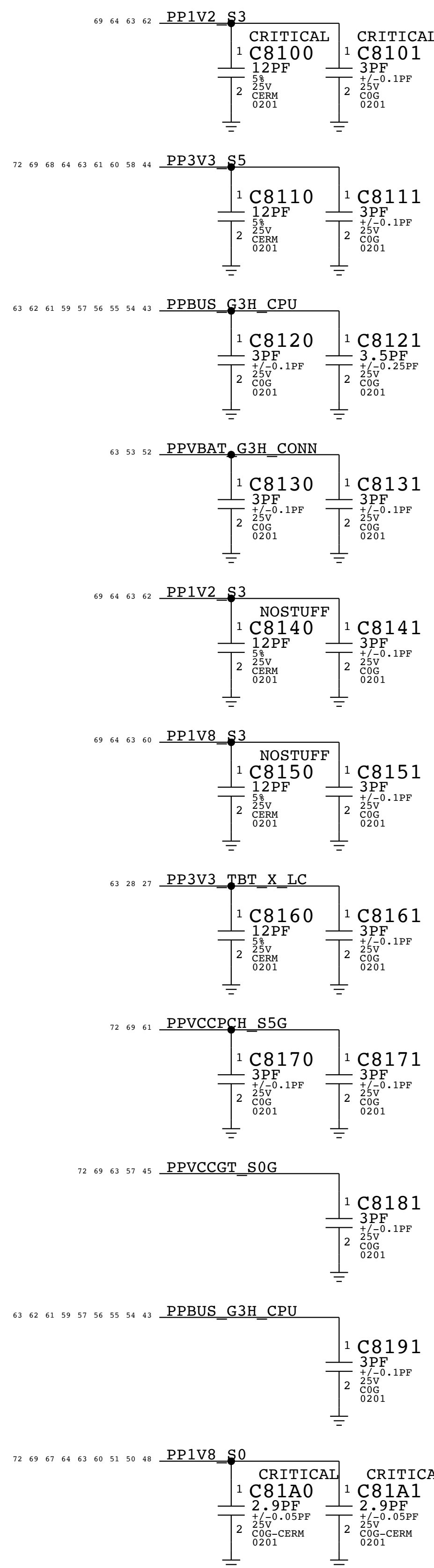
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| DESIGN: X502/MLB CATZ                                                                                              |                |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                               |                |           |
| PAGE TITLE                                                                                                         |                |           |
| RAIL DESENSE CAPS                                                                                                  |                |           |
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BOM\_COST\_GROUP=PLATFORM POWER

BOM\_COST\_GROUP=PLATFORM POWER

Page Notes

Power aliases required by this page:  
- PPVIN\_S0SW\_LCDBKLT\_FET (9-12.6V LCD BACKLIGHT INPUT)  
- PP5V\_S0\_BKLT (5V BACKLIGHT DRIVER INPUT)

D

C

B

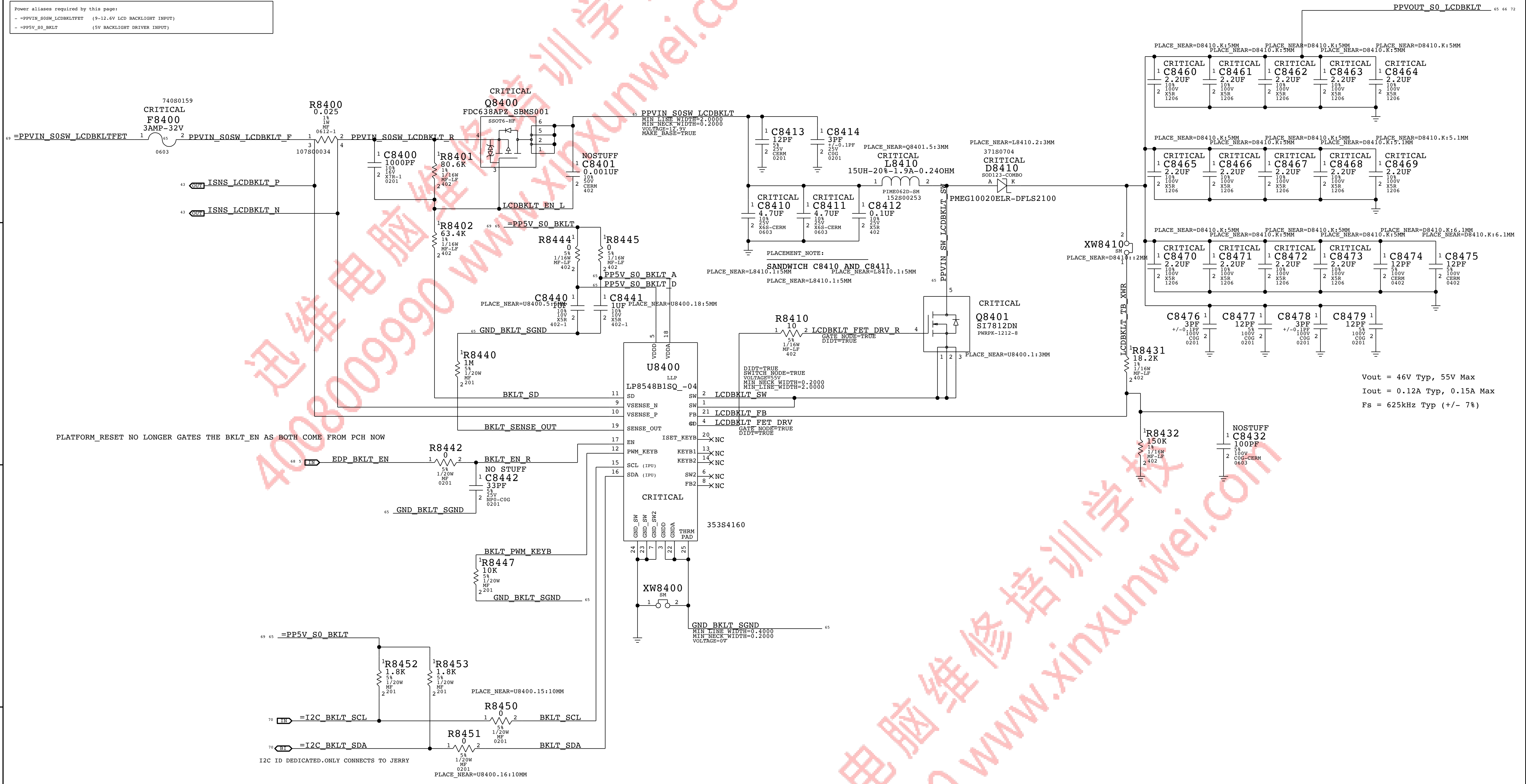
A

D

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A



PLATFORM\_RESET NO LONGER GATES THE BKLT\_EN AS BOTH COME FROM PCH NOW

Vout = 46V Typ, 55V Max  
Iout = 0.12A Typ, 0.15A Max  
Fs = 625kHz Typ (+/- 7%)

LINE WIDTHS

PBUS LINE WIDTHS

LCD BKLT LINE WIDTHS

- PP5V\_S0\_BKLT\_A MIN LINE WIDTH=2.0000 MIN NECK WIDTH=0.2000 VOLTAGE=5V
- PP5V\_S0\_BKLT\_D MIN LINE WIDTH=2.0000 MIN NECK WIDTH=0.2000 VOLTAGE=5V
- PPVIN\_S0SW\_LCDBKLT\_F MIN LINE WIDTH=2.0000 MIN NECK WIDTH=0.2000 VOLTAGE=12.9V
- PPVIN\_S0SW\_LCDBKLT\_R MIN LINE WIDTH=2.0000 MIN NECK WIDTH=0.2000 VOLTAGE=12.9V
- PPVIN\_S0SW\_LCDBKLT\_L MIN LINE WIDTH=2.0000 MIN NECK WIDTH=0.2000 VOLTAGE=12.9V
- LCD BKLT FET DRV MIN LINE WIDTH=2.0000 MIN NECK WIDTH=0.2000 VOLTAGE=5V GATE\_NODE=TRUE DIDT=TRUE
- PPVIN\_SW\_LCDBKLT\_SW MIN LINE WIDTH=2.0000 MIN NECK WIDTH=0.2000 VOLTAGE=55V SWITCH\_NODE=TRUE DIDT=TRUE
- PPVOUT\_S0\_LCDBKLT MIN LINE WIDTH=0.5000 MIN NECK WIDTH=0.2000 VOLTAGE=55V

BOM\_COST\_GROUP=DISPLAY

|                                                                                                                                                                                                                                                                                                                    |  |                   |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------|-----------|
| SYRC: 000000-0000                                                                                                                                                                                                                                                                                                  |  | SYRC: 000000-0000 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                         |  |                   |           |
| LCD Backlight Driver                                                                                                                                                                                                                                                                                               |  | DRAWING NUMBER    | 051-02265 |
| Apple Inc.                                                                                                                                                                                                                                                                                                         |  | REVISION          | 1.0.0     |
|                                                                                                                                                                                                                                                                                                                    |  | BRANCH            |           |
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|                                                                                                                                                                                                                                                                                                                    |  | SHEET             | 65 OF 73  |



# LCD PANEL INTERFACE (eDP) + Camera (MIPI)

LCD Panel HPD & AUX strapping

D

D

C

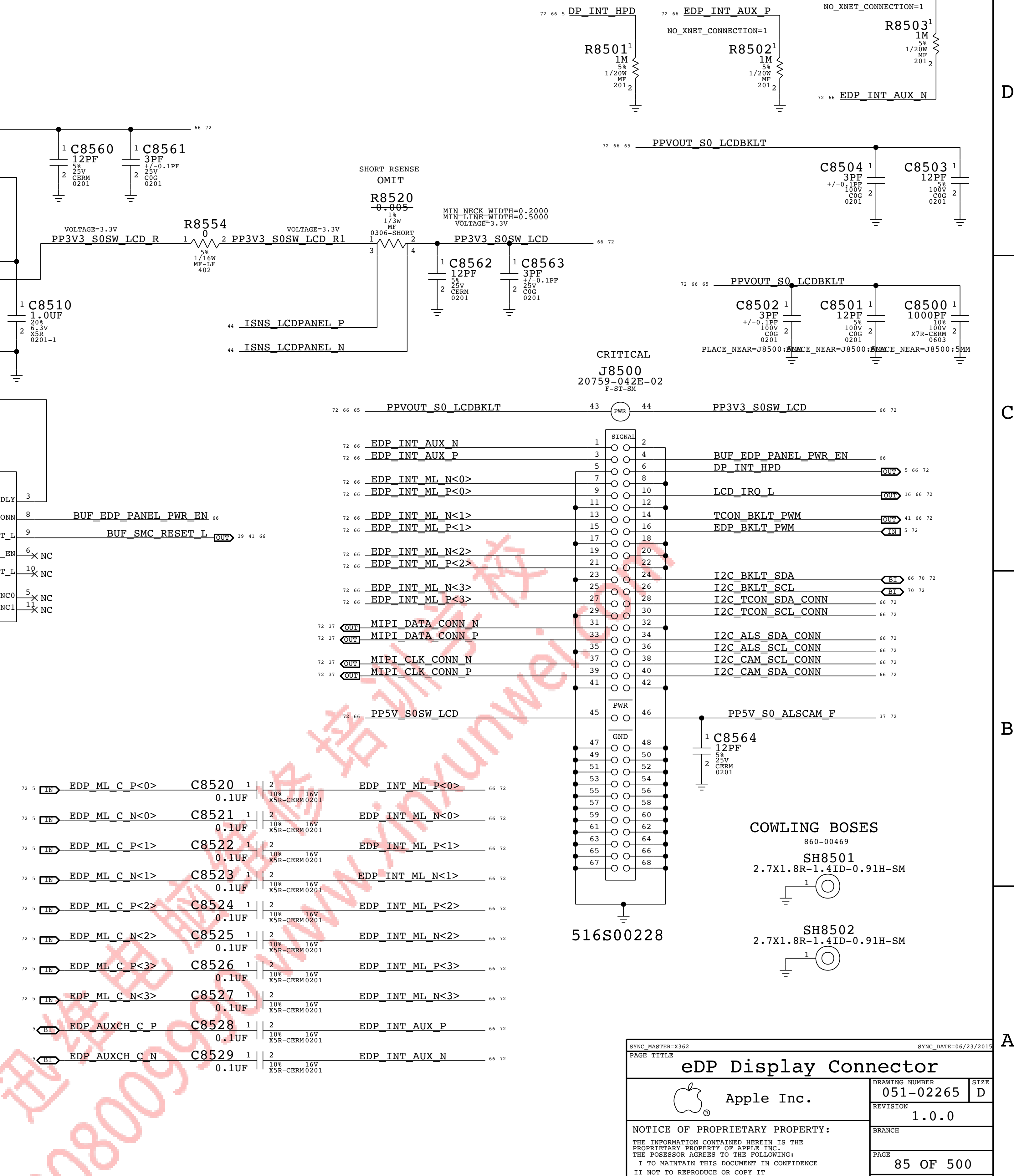
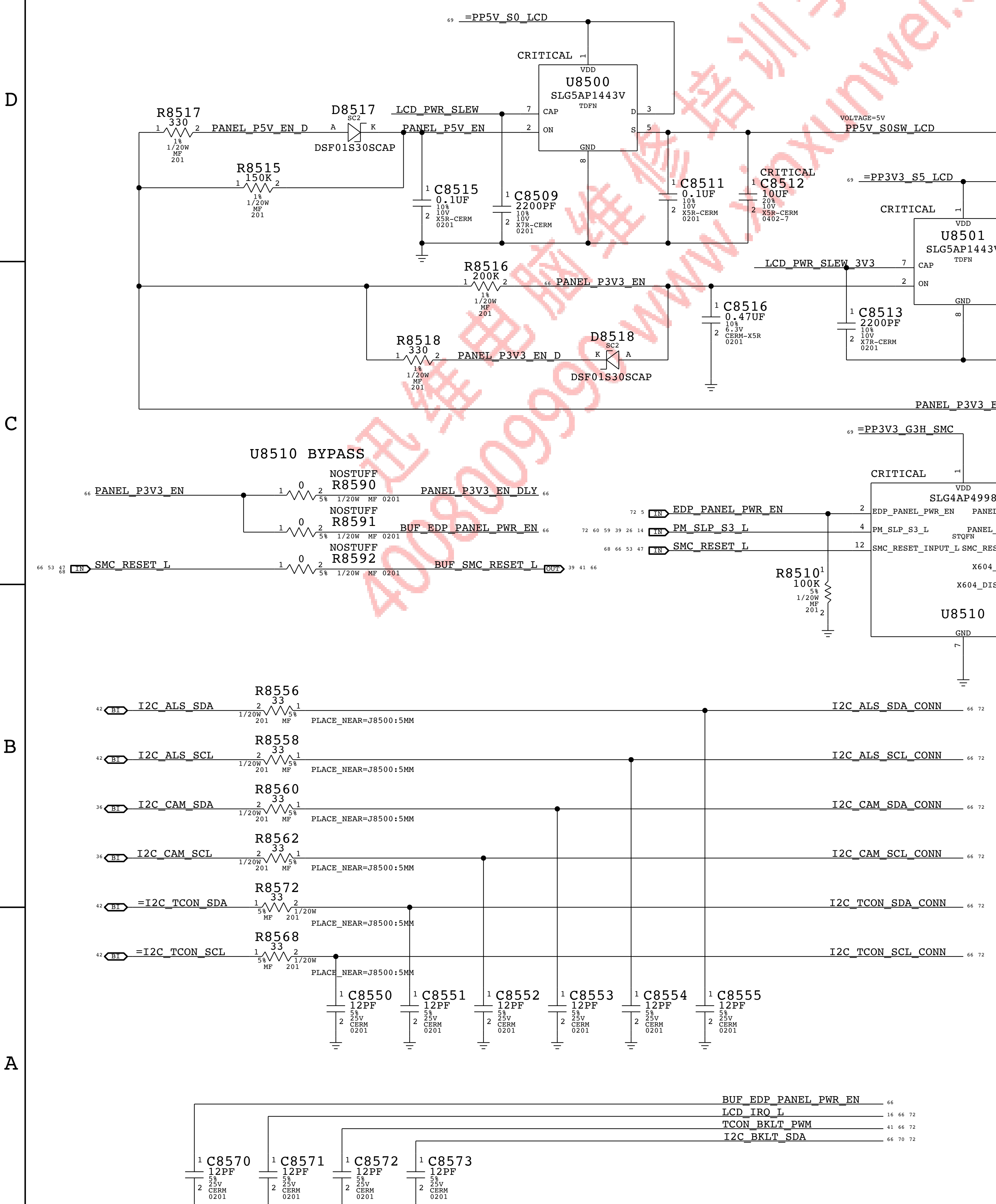
C

B

B

A

A



|                                                                                                                                                                                                                                                                                                                    |  |                       |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------|-----------|
| SYNC_MASTER=X362                                                                                                                                                                                                                                                                                                   |  | SYNC_DATE=06/23/2015  |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                         |  | eDP Display Connector |           |
|                                                                                                                                                                                                                                                                                                                    |  | DRAWING NUMBER        | 051-02265 |
|                                                                                                                                                                                                                                                                                                                    |  | REVISION              | 1.0.0     |
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|                                                                                                                                                                                                                                                                                                                    |  | PAGE                  | 85 OF 500 |
|                                                                                                                                                                                                                                                                                                                    |  | SHEET                 | 66 OF 73  |

BOM\_COST\_GROUP=DISPLAY



D

C

B

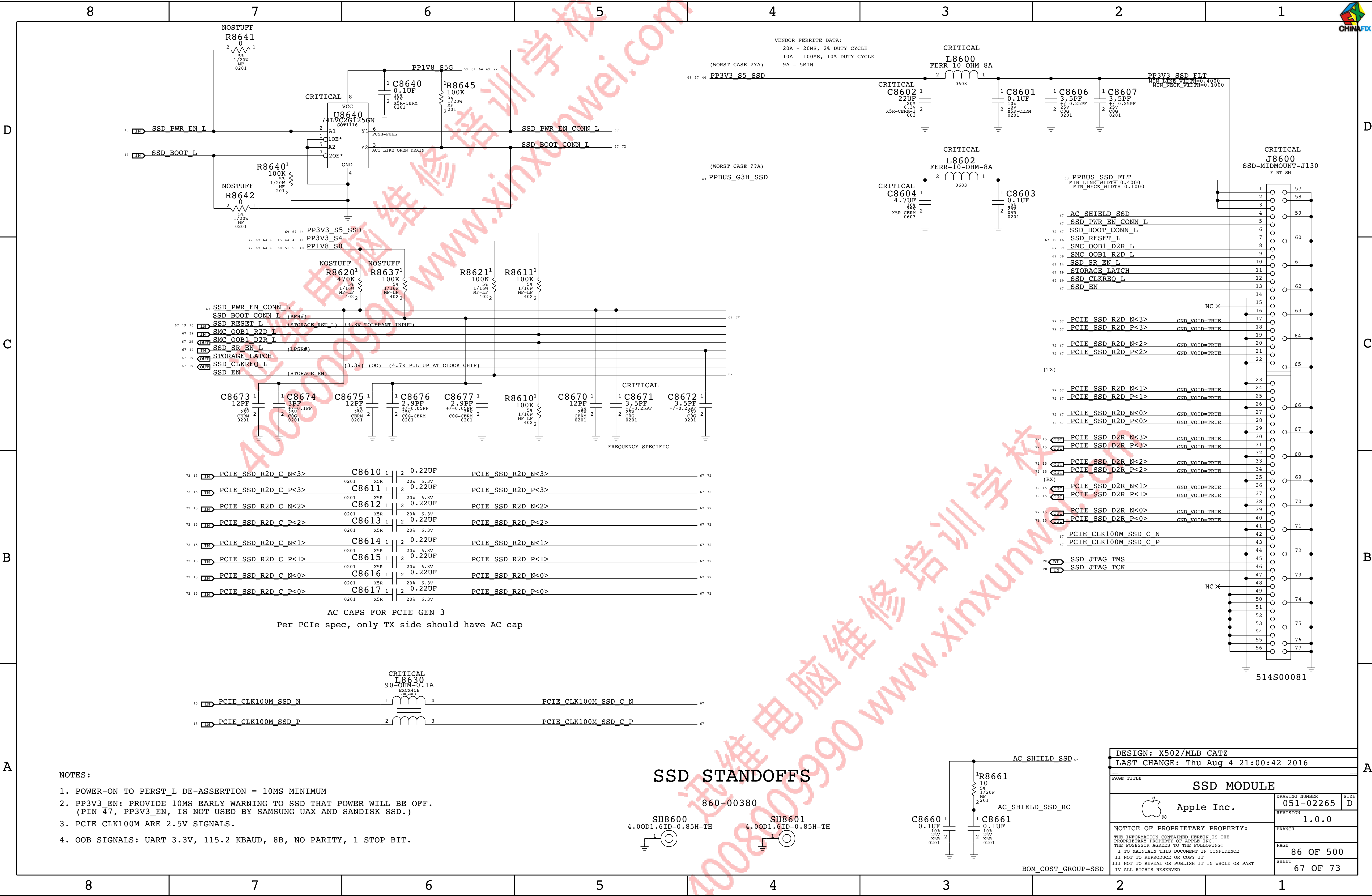
A

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NOTES:

- POWER-ON TO PERST\_L DE-ASSERTION = 10MS MINIMUM
- PP3V3\_EN: PROVIDE 10MS EARLY WARNING TO SSD THAT POWER WILL BE OFF. (PIN 47, PP3V3\_EN, IS NOT USED BY SAMSUNG UAX AND SANDISK SSD.)
- PCIe CLK100M ARE 2.5V SIGNALS.
- OOB SIGNALS: UART 3.3V, 115.2 KBAUD, 8B, NO PARITY, 1 STOP BIT.

|                                                                                                                    |                |           |
|--------------------------------------------------------------------------------------------------------------------|----------------|-----------|
| DESIGN: X502/MLB CATZ                                                                                              |                |           |
| LAST CHANGE: Thu Aug 4 21:00:42 2016                                                                               |                |           |
| PAGE TITLE                                                                                                         |                |           |
| SSD MODULE                                                                                                         |                |           |
|                                                                                                                    | DRAWING NUMBER | 051-02265 |
|                                                                                                                    | REVISION       | 1.0.0     |
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BOM\_COST\_GROUP=NO COST ITEMS

UNUSED GPIOs, HSIO

|    |    |                   |    |                |           |                      |
|----|----|-------------------|----|----------------|-----------|----------------------|
| 8  | NC | VCCPRIM_CORE_VID0 | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_VCCPRIM_CORE_VID0 |
| 8  | NC | VCCPRIM_CORE_VID1 | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_VCCPRIM_CORE_VID1 |
| 16 | NC | MLB_DEV_L         | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_MLB_DEV_L         |
| 16 | NC | I2C_UPC_SDA       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_I2C_UPC_SDA       |
| 16 | NC | I2C_UPC_SCL       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_I2C_UPC_SCL       |
| 13 | NC | PCH_BSSB_CLK      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_BSSB_CLK      |
| 13 | NC | PCH_BSSB_DATA     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_BSSB_DATA     |

|    |    |                       |    |                |           |                         |
|----|----|-----------------------|----|----------------|-----------|-------------------------|
| 16 | NC | TBT_X_DPMUX_SEL       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_TBT_X_DPMUX_SEL      |
| 16 | NC | TBT_T_CIO_PWR_EN      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_TBT_T_CIO_PWR_EN     |
| 16 | NC | TBT_T_USB_PWR_EN      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_TBT_T_USB_PWR_EN     |
| 16 | NC | TBT_T_PCI_RESET_L     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_TBT_T_PCI_RESET_L    |
| 16 | NC | TBT_T_DPMUX_SEL       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_TBT_T_DPMUX_SEL      |
| 16 | NC | =TBT_T_CLKREQ_L       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_TBT_T_CLKREQ_L       |
| 15 | NC | PCIE_CLK100M_TBT_T_N  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_CLK100M_TBT_T_N |
| 15 | NC | PCIE_CLK100M_TBT_T_P  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_CLK100M_TBT_T_P |
| 15 | NC | PCIE_TBT_T_D2R_N<0>   | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_TBT_T_D2R_N0    |
| 15 | NC | PCIE_TBT_T_D2R_P<0>   | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_TBT_T_D2R_P0    |
| 15 | NC | PCIE_TBT_T_R2D_C_N<0> | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_TBT_T_R2D_CN0   |
| 15 | NC | PCIE_TBT_T_R2D_C_P<0> | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_TBT_T_R2D_CP0   |
| 15 | NC | PCIE_TBT_T_D2R_N<1>   | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_TBT_T_D2R_N1    |
| 15 | NC | PCIE_TBT_T_D2R_P<1>   | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_TBT_T_D2R_P1    |
| 15 | NC | PCIE_TBT_T_R2D_C_N<1> | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_TBT_T_R2D_CN1   |
| 15 | NC | PCIE_TBT_T_R2D_C_P<1> | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_TBT_T_R2D_CP1   |

|    |    |                   |    |                |           |                      |
|----|----|-------------------|----|----------------|-----------|----------------------|
| 15 | NC | USB_CAMERA_DFR_N  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB2_03N          |
| 15 | NC | USB_CAMERA_DFR_P  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB2_03P          |
| 13 | NC | DEBUGUART_SEL_SOC | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_DEBUGUART_SEL_SOC |

|    |    |                      |    |                |           |                         |
|----|----|----------------------|----|----------------|-----------|-------------------------|
| 15 | NC | USB3_EXTB_D2R_N      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_EXTB_D2RN       |
| 15 | NC | USB3_EXTB_D2R_P      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_EXTB_D2RP       |
| 15 | NC | USB3_EXTB_R2D_C_N    | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_EXTB_R2DCN      |
| 15 | NC | USB3_EXTB_R2D_C_P    | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_EXTB_R2DCP      |
| 26 | NC | =DP_X_SRC_ML_P<3..0> | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_DP_X_SRC_ML_CP<3..0> |
| 26 | NC | =DP_X_SRC_ML_N<3..0> | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_DP_X_SRC_ML_CN<3..0> |
| 26 | NC | =DP_X_SRC_AUX_P      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_DP_X_SRC_AUXCHP      |
| 26 | NC | =DP_X_SRC_AUX_N      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_DP_X_SRC_AUXCHN      |

|    |    |             |    |                |           |             |
|----|----|-------------|----|----------------|-----------|-------------|
| 15 | NC | USB_EXT_A_N | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB2_01N |
| 15 | NC | USB_EXT_A_P | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB2_01P |

|    |    |          |           |
|----|----|----------|-----------|
| 15 | NC | USB2_05N | NO_TEST=1 |
| 15 | NC | USB2_05P | NO_TEST=1 |
| 15 | NC | USB2_06N | NO_TEST=1 |
| 15 | NC | USB2_06P | NO_TEST=1 |
| 15 | NC | USB2_07N | NO_TEST=1 |
| 15 | NC | USB2_07P | NO_TEST=1 |

|    |    |          |           |
|----|----|----------|-----------|
| 15 | NC | USB2_09N | NO_TEST=1 |
| 15 | NC | USB2_09P | NO_TEST=1 |
| 15 | NC | USB2_10N | NO_TEST=1 |
| 15 | NC | USB2_10P | NO_TEST=1 |

36 NC\_UPC\_XB\_I2C\_ADDR NO\_TEST=1

GND  
MIN LINE WIDTH=0.3000  
MIN WCK WIDTH=0.0850  
VOLTAGE=0V

Digital Ground

|    |    |               |    |                |              |    |    |
|----|----|---------------|----|----------------|--------------|----|----|
| 65 | BT | =I2C_BKLT_SCL | == | MAKE_BASE=TRUE | I2C_BKLT_SCL | 66 | BT |
| 65 | BT | =I2C_BKLT_SDA | == | MAKE_BASE=TRUE | I2C_BKLT_SDA | 66 | BT |

|    |    |                  |    |                |           |                     |
|----|----|------------------|----|----------------|-----------|---------------------|
| 16 | TP | BT_I2S_CLK       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_BT_I2S_CLK       |
| 16 | TP | BT_I2S_D2R       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_BT_I2S_D2R       |
| 16 | TP | BT_I2S_R2D       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_BT_I2S_R2D       |
| 16 | TP | BT_I2S_SYNC      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_BT_I2S_SYNC      |
| 16 | TP | CAM_GPIO3        | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CAM_GPIO3        |
| 6  | TP | CPU_AT5          | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_AT5          |
| 6  | TP | CPU_AU5          | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_AU5          |
| 6  | TP | CPU_AY4          | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_AY4          |
| 6  | TP | CPU_BB3          | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_BB3          |
| 6  | TP | CPU_BB5          | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_BB5          |
| 16 | TP | CPU_MSM_L        | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_MSM_L        |
| 59 | TP | CPU_NCTFVSS_A5   | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_NCTFVSS_A5   |
| 9  | TP | CPU_NCTFVSS_A70  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_NCTFVSS_A70  |
| 9  | TP | CPU_NCTFVSS_AV1  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_NCTFVSS_AV1  |
| 9  | TP | CPU_NCTFVSS_B71  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_NCTFVSS_B71  |
| 9  | TP | CPU_NCTFVSS_BA1  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_NCTFVSS_BA1  |
| 9  | TP | CPU_NCTFVSS_BA71 | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_NCTFVSS_BA71 |
| 9  | TP | CPU_NCTFVSS_BB70 | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_NCTFVSS_BB70 |
| 9  | TP | CPU_NCTFVSS_C1   | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_NCTFVSS_C1   |

|    |                    |    |                |           |                    |
|----|--------------------|----|----------------|-----------|--------------------|
| 13 | TP_PCH_CLKOUT_LPC1 | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_CLKOUT_LPC1 |
| 14 | TP_PCH_GPD7        | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPD7        |
| 14 | TP_PCH_GPP_D0      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_D0      |
| 14 | TP_PCH_GPP_D1      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_D1      |
| 16 | TP_PCH_GPP_D3      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_D3      |
| 16 | TP_PCH_GPP_D4      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_D4      |
| 5  | TP_PCH_GPP_E15     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_E15     |
| 14 | TP_PCH_GPP_F8      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_F8      |
| 14 | TP_PCH_GPP_F9      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_F9      |
| 14 | TP_PCH_GPP_F10     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_F10     |
| 14 | PCH_BT_ROM_BOOT    | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_F11     |
| 13 | PCH_SOC_DFU_STATUS | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_F18     |
| 13 | SOC_PANIC_L        | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_F19     |
| 13 | SOC_S2R_ACK_L      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_F20     |
| 13 | SOC_SLEEP_L        | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_GPP_F21     |
| 14 | TP_PCH_LANPHYPC    | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_LANPHYPC    |
| 14 | TP_PCH_PME_L       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_PME_L       |

|    |    |                   |    |                |           |                      |
|----|----|-------------------|----|----------------|-----------|----------------------|
| 14 | TP | PCH_SLP_WLAN_L    | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_SLP_WLAN_L    |
| 13 | TP | PCH_STRP_ESPI     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_STRP_ESPI     |
| 13 | TP | PCH_STRP_TLSCONF  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_STRP_TLSCONF  |
| 60 | TP | PMIC_PGC          | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PMIC_PGC          |
| 13 | TP | SPI_CS1_L         | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_SPI_CS1_L         |
| 13 | TP | SPI_CS2_L         | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_SPI_CS2_L         |
| 19 | TP | SYSCLK_CLK24M_SSD | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_SYSCLK_CLK24M_SSD |

|    |    |              |    |                |           |                 |
|----|----|--------------|----|----------------|-----------|-----------------|
| 15 | TP | USB3_03_D2RN | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_03_D2RN |
| 15 | TP | USB3_03_D2RP | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_03_D2RP |
| 15 | TP | USB3_03_R2DN | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_03_R2DN |
| 15 | TP | USB3_03_R2DP | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_03_R2DP |
| 15 | TP | USB3_04_D2RN | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_04_D2RN |
| 15 | TP | USB3_04_D2RP | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_04_D2RP |
| 15 | TP | USB3_04_R2DN | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_04_R2DN |
| 15 | TP | USB3_04_R2DP | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_USB3_04_R2DP |

|    |    |                   |    |                |           |                      |
|----|----|-------------------|----|----------------|-----------|----------------------|
| 14 | TP | PCH_SOC_FORCE_DFU | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_SOC_FORCE_DFU |
| 13 | TP | PCH_SOC_WDOG      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_SOC_WDOG      |
| 14 | TP | UPC_I2C_INT_L     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_UPC_I2C_INT_L     |

|    |    |                 |    |                |           |                    |
|----|----|-----------------|----|----------------|-----------|--------------------|
| 30 | TP | UPC_XB_SWD_DATA | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_UPC_XB_SWD_DATA |
|----|----|-----------------|----|----------------|-----------|--------------------|

|    |    |                 |    |                |           |                    |
|----|----|-----------------|----|----------------|-----------|--------------------|
| 29 | TP | UPC_XA_SWD_DATA | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_UPC_XA_SWD_DATA |
| 29 | TP | UPC_XA_SWD_CLK  | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_UPC_XA_SWD_CLK  |

|    |    |                    |    |                |           |                       |
|----|----|--------------------|----|----------------|-----------|-----------------------|
| 15 | TP | PCIE_CLK100M5P     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_CLK100M5P     |
| 15 | TP | PCIE_CLK100M5N     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCIE_CLK100M5N     |
| 15 | TP | PCH_CLKREQ05_L     | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_CLKREQ05_L     |
| 15 | TP | ITPXDP_CLK100MP    | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_ITPXDP_CLK100MP    |
| 15 | TP | ITPXDP_CLK100MN    | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_ITPXDP_CLK100MN    |
| 55 | TP | CPUVR_GH1          | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPUVR_GH1          |
| 6  | TP | CPU_RSVD_BB69      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_RSVD_BB69      |
| 6  | TP | CPU_RSVD_BB68      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_RSVD_BB68      |
| 6  | TP | CPU_RSVD_BA70      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_RSVD_BA70      |
| 6  | TP | CPU_RSVD_BA68      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_RSVD_BA68      |
| 6  | TP | CPU_RSVD_AW71      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_RSVD_AW71      |
| 6  | TP | CPU_RSVD_AW70      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_RSVD_AW70      |
| 6  | TP | CPU_RSVD_AK12      | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_CPU_RSVD_AK12      |
| 19 | TP | SPKR_ID0           | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_SPKR_ID0           |
| 60 | TP | PGOOD_PVCCIO       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PGOOD_PVCCIO       |
| 60 | TP | PGOOD_P1V00        | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PGOOD_P1V00        |
| 19 | TP | PCH_CLK32K_RTCX2   | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_CLK32K_RTCX2   |
| 19 | TP | PCH_CLK24M_XTALOUT | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_PCH_CLK24M_XTALOUT |
| 19 | TP | XDP_BPM_L<3>       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_XDP_BPM_L<3>       |
| 19 | TP | XDP_BPM_L<2>       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_XDP_BPM_L<2>       |
| 19 | TP | XDP_BPM_L<1>       | == | MAKE_BASE=TRUE | NO_TEST=1 | NC_XDP_BPM_L<1>       |

DESIGN: X502/MLB CATZ  
LAST CHANGE: Mon Aug 8 12:54:34 2016

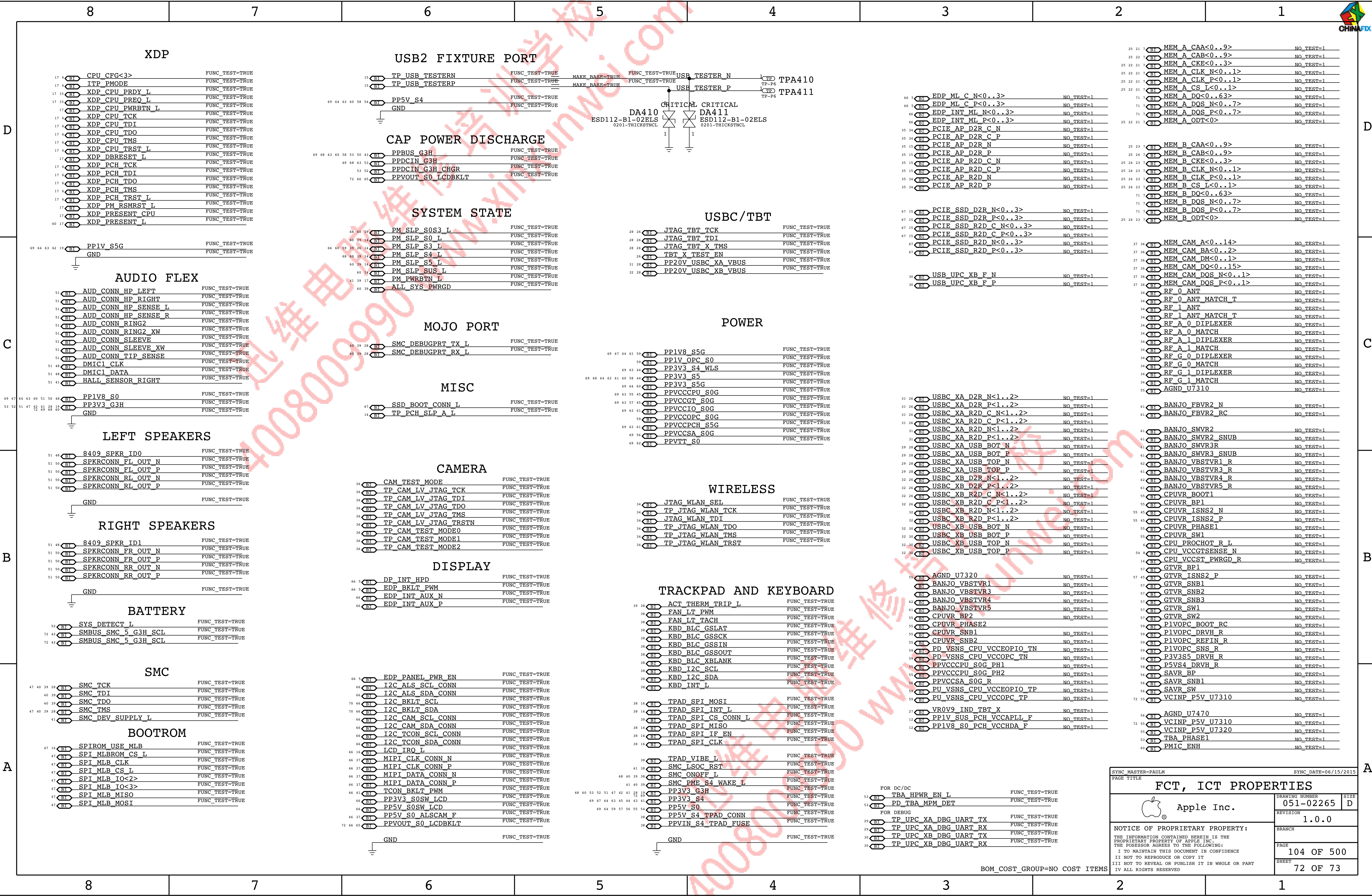
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BOM\_COST\_GROUP=NO COST ITEMS

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BOM\_COST\_GROUP=NO COST ITEMS



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